Dell EMC Solutions Enabler

CLI Reference Guide

9.2

September 2020
Rev. 01

DELL EMC

Notes, cautions, and warnings

NOTE: A NOTE indicates important information that helps you make better use of your product.

CAUTION: A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

WARNING: A WARNING indicates a potential for property damage, personal injury, or death.

© 2020 Dell Inc. or its subsidiaries. All rights reserved. Dell, EMC, and other trademarks are trademarks of Dell Inc. or its subsidiaries. Other trademarks may be trademarks of their respective owners.

PREFACE

As part of an effort to improve its product lines, Dell EMC periodically releases revisions of its software and hardware. Therefore, some functions described in this document might not be supported by all versions of the software or hardware currently in use. The product release notes provide the most up-to-date information on product features.

Contact your Dell EMC representative if a product does not function properly or does not function as described in this document.

NOTE: This document was accurate at publication time. New versions of this document might be released on Dell EMC Online Support (https://www.dell.com/support/home). Check to ensure that you are using the latest version of this document.

Purpose

This reference guide provides man pages for all the SYMCLI commands. It also includes information about environment variables, and options files.

Audience

This manual provides reference information for command-line users and script programmers that focus on configuring and managing storage on arrays.

Related documentation

The following documents provide additional Solutions Enabler information:

Dell EMC Solutions Enabler, VSS Provider, and SMI-S Provider Release Notes	Describes new features and any known limitations.
Dell EMC Solutions Enabler Installation and Configuration Guide	Provides host-specific installation instructions.
Dell EMC Solutions Enabler SRDF Family CLI User Guide	Describes how to configure and manage SRDF environments using SYMCLI commands.
Dell EMC Solutions Enabler Array Controls and Management CLI User Guide	Describes how to configure array control, management, and migration operations using SYMCLI commands for arrays running HYPERMAX OS and PowerMaxOS.
Dell EMC Solutions Enabler TimeFinder SnapVX CLI User Guide	Describes how to configure and manage TimeFinder SnapVX environments using SYMCLI commands.
EMC Solutions Enabler	Describes how to configure and manage TimeFinder Mirror, Clone, Snap and VP Snap environments using SYMCLI commands.

TimeFinder Family (Mirror, Clone, Snap, VP Snap) CLI User Guide	
Dell EMC Solutions Enabler SRM CLI User Guide	Provides Storage Resource Management (SRM) information related to various data objects and data handling facilities.
EMC VMAX All Flash and VMAX3 Family Security Configuration Guide	Describes how to securely deploy a VMAX3 Family (100K, 200K, 400K) or VMAX All Flash (250F, 450F, 850F, 950F) array with HYPERMAX OS.

Where to get help

EMC support, product, and licensing information can be obtained as follows:

NOTE:

To open a service request through Dell EMC Online Support (https://www.dell.com/support), you must have a valid support agreement. Contact your Dell EMC sales representative for details about obtaining a valid support agreement or to answer any questions about your account.

Product information

For documentation, release notes, software updates, or information about Dell EMC products, go to Dell EMC Online Support at https://www.dell.com/support/home.

Technical support

Dell EMC offers a variety of support options.

- Support by Product — Dell EMC offers consolidated, product-specific information on the Web through the Dell EMC Online Support site.

  The Support by Product web pages (https://www.dell.com/support/home then select **Product Support**) offer quick links to Documentation, White Papers, Advisories (such as frequently used Knowledgebase articles), and Downloads, as well as more dynamic content, such as presentations, discussion, relevant Customer Support Forum entries, and a link to Dell EMC Live Chat.

- Dell EMC Live Chat — Open a Chat or instant message session with an Dell EMC Support Engineer.

eLicensing support

To activate your entitlements and obtain your VMAX license files, visit the Service Center on Dell EMC Online Support (https://www.dell.com/support/home), as directed on your License Authorization Code (LAC) letter emailed to you.

- For help with missing or incorrect entitlements after activation (that is, expected functionality remains unavailable because it is not licensed), contact your Dell EMC Account Representative or Authorized Reseller.
- For help with any errors applying license files through Solutions Enabler, contact the Dell EMC Customer Support Center.
- If you are missing a LAC letter, or require further instructions on activating your licenses through the Online Support site, contact Dell EMC's worldwide Licensing team at licensing@emc.com or call:
  - North America, Latin America, APJK, Australia, New Zealand: SVC4EMC (800-782-4362) and follow the voice prompts.
  - EMEA: +353 (0) 21 4879862 and follow the voice prompts.

Your comments

Your suggestions help us improve the accuracy, organization, and overall quality of the documentation. Send your comments and feedback to: VMAXContentFeedback@emc.com

SYMCLI Commands

This chapter includes UNIX-style man pages for all SYMCLI commands. Commands are listed alphabetically.

symaccess	symacl	symapierr
symaudit	symauth	symbcv
symcfg	symcg	symchg
symchksum	symcli	symclient
symclone	symconfigure	symconnect
symdev	symdg	symdisk
symdm	symdrv	symevent
symfast	symhost	symhostfs
syminq	symioctl	symipsec
symlabel	symlmf	symlv
symmask	symmaskdb	symmdr
symmigrate	symmir	symntctl
symoptmz	sympart	sympd
symppath	symqos	symrcopy
symrdb	symrdf	symrecover
symreplicate	symreturn	symrpi
symrslv	symsan	symsg
symsnap	symsnapvx	symstar
symstat	symtier	symtw

symvg	

symaccess

Performs Auto-provisioning Group operations
on storage, initiator, and port groups. Allows you to create
and manage masking views.

SYNOPSIS

```
   symaccess -h

Storage Group

   symaccess -sid <SymmID> -name <Group_name> -type storage
       [-reserve_id ResvID[,ResvID[,ResvID...]]]

     create

     create -g <DgName> [-std] [-bcv] [-vdev] [-tgt]

     create -file <DeviceFileName> [src] [tgt]

     create devs <SymDevStart:SymDevEnd> |
       <SymDevName[,SymDevName[,SymDevName...]]>

     create sg <SgName>[,<SgName1>,<SgName2>,.,<SgNamen>]

     delete [-force][-noprompt]

   symaccess -sid <SymmID> -name <Group_name> -type storage

     rename -new_name <NewGroupName>

   symaccess -sid <SymmID> -name <GroupName> -type storage
       [-reserve_id ResvID[,ResvID[,ResvID...]]]
       [-ckd] [-celerra] [-rp]

     add -g <DgName> [-std] [-bcv] [-vdev] [-tgt] [-lun <addr>]

     add -file <DeviceFileName> [src] [tgt] [-lun <addr>]

     add devs <SymDevStart:SymDevEnd> [-lun <Addr>] |
       <SymDevName> [-lun <Addr>] |
       <SymDevName,SymDevName,SymDevName...>
       [-lun <Addr> | -lun <Addr,Addr,Addr...>]

     add sg <SgName>[,<SgName1>,<SgName2>,...,<SgNamen>]
       [-lun <Addr>]

   symaccess -sid <SymmID> -name <GroupName> -type storage
       [-reserve_id ResvID[,ResvID[,ResvID...]]] [-force]
       [-unmap [-celerra] [-rp]] [-ckd]

     remove -g <DgName> [-std] [-bcv] [-vdev] [-tgt]

     remove -file <DeviceFileName> [src] [tgt]

     remove devs <SymDevStart:SymDevEnd> | <SymDevName> |
       <SymDevName,SymDevName,SymDevName...>

     remove sg <SgName>[,<SgName1>,<SgName2>,...,<SgNamen>]

   symaccess -sid <SymmID> -target_sid <SymmID>

     copy -name <GroupName> -type storage
        [-reserve_id ResvID[,ResvID[,ResvID...]]]
```

```
   symaccess -sid <SymmID> [-offline] | -file <backup_filename>

     list -type storage
        [-devs <<SymDevStart>:<SymDevEnd> |
        <SymDevName> | <<SymDevName>,<SymDevName>...>>]
        [-name <GroupName>] [-v | -detail]

     show <GroupName> -type storage

Initiator Group

   symaccess -sid <SymmID> -name <GroupName> -type initiator
       [-consistent_lun]

     create [-bw_limit <MBperSec>]

     create -wwn <wwn> [-bw_limit <MBperSec>]

     create -iscsi <iscsi>

     create -file <InitiatorFileName> [-bw_limit <MBperSec>]

     create -ig <InitiatorGroupName>

     create -host <HostName>

   symaccess -sid <SymmID> -name <GroupName> -type initiator

     delete [-force] [-noprompt]

     rename -new_name <NewGroupName>

   symaccess -sid <SymmID>

     rename -wwn <wwn> -alias <alias>

     rename -iscsi <iscsi> -alias <alias>

     replace -wwn <wwn> -new_wwn <NewWWN> [-noprompt]

     replace -iscsi <iscsi> -new_iscsi <NewiSCSI> [-noprompt]

   symaccess -sid <SymmID> -name <GroupName> -type initiator

     add -wwn <wwn>

     add -iscsi <iscsi>

     add -ig <InitiatorGroupName>

     add -file <InitiatorFileName>

   symaccess -sid <SymmID> -name <GroupName> -type initiator

     remove -wwn <wwn> [-login]

     remove -iscsi <iscsi> [-login]

     remove -ig <InitiatorGroupName> [-login]

     remove -file <InitiatorFileName> [-login]

   symaccess -sid <SymmID> <-wwn <wwn> | -iscsi <iscsi>>
        [-dirport <Dir>:<Port> |
         -iscsi_dirport <Dir>:<Port> |
         -iqn <TargetIQN>] -login
```

```
      remove

   symaccess -sid <SymmID> -target_sid <SymmID>

      copy -name <GroupName> -type initiator

   symaccess -sid <SymmID> -wwn <wwn> | -iscsi <iscsi>

      set hba_flags <on <flag,flag,flag...> <-enable |-disable> |
        off [flag,flag,flag...]>

   symaccess -sid <SymmID> -name <GroupName> -type initiator

      set ig_flags <on <flag> <-enable |-disable> |
        off [flag]>

      set consistent_lun <on | off [-force]>

      set bw_limit <on <MBperSec> | off>

   symaccess -sid <SymmID> -wwn <wwn>

      set lockdown <on <fcid> | off>

   symaccess -sid <SymmID> -iscsi <iscsi>

      enable chap

      disable chap

      set chap -cred <Credential> -secret <Secret>

   symaccess -sid <SymmID> [-offline] | -file <backup_filename>

      list -type initiator [-wwn <wwn> | -iscsi <iscsi>]
            [-name <GroupName>] [-detail | -v]

      show <GroupName> -type initiator [-detail]

Port Group

   symaccess -sid <SymmID> -name <GroupName> -type port

      create

      create -dirport <Dir>:<Port>[,<Dir>:<Port>...]

      create -iscsi_dirport <Dir>:<Port>[,<Dir>:<Port>...]

      create -iqn <TargetIQN>[,<TargetIQN>...]

      delete [-force][-noprompt]

      rename -new_name <NewGroupName>

   symaccess -sid <SymmID> -name <GroupName> -type port
        [-celerra][-rp][-ckd]

      add -dirport <Dir>:<Port>[,<Dir>:<Port>...]

      add -iscsi_dirport <Dir>:<Port>[,<Dir>:<Port>...]

      add -iqn <TargetIQN>[,<TargetIQN>...]

   symaccess -sid <SymmID> -name <GroupName> -type port
        [-ckd][-force][-unmap [-celerra][-rp]]
```

```
      remove -dirport <Dir>:<Port>[,<Dir>:<Port>...]

      remove -iscsi_dirport <Dir>:<Port>[,<Dir>:<Port>...]

      remove -iqn <TargetIQN>[,<TargetIQN>...]

   symaccess -sid <SymmID> -target_sid <SymmID>

      copy -name <GroupName> -type port

   symaccess -sid <SymmID> [-offline] | -file <backup_filename>

      list -type port [-name <GroupName>] [-detail | -v]
            [-dirport <Dir>:<Port> |
             -iscsi_dirport <Dir>:<Port> |
             -iqn <TargetIQN>]

      show <GroupName> -type port

Masking View

   symaccess -sid <SymmID> view -name <ViewName>
       [-reserve_id ResvID[,ResvID[,ResvID...]]]
       [-ckd] [-celerra] [-rp]

      create -sg <StorageGroupName> -pg <PortGroupName>
        -ig <InitiatorGroupName> [-lun <Addr>]

      create <-wwn <wwn> | -iscsi <iscsi>>
         -dirport <Dir>:<Port>[,<Dir>:<Port>...]
        < devs <SymDevStart:SymDevEnd> [-lun <Addr>] >  |
        < <SymDevName> [-lun <Addr>] > |
        < <SymDevName,SymDevName,SymDevName...>
        [ -lun <Addr,Addr,Addr...>] >

      create <-wwn <wwn> | -iscsi <iscsi>>
        -dirport <Dir>:<Port>[,<Dir>:<Port>...]
        <-g <DgName> [-std] [-bcv] [-vdev] [-tgt]> |
        <-file <DeviceFileName> [src] [tgt]>

   symaccess -sid <SymmID> view -name <ViewName>
       [-reserve_id ResvID[,ResvID[,ResvID...]]]

      delete [-unmap [-celerra][-rp]][-ckd][-noprompt]

      rename -new_name <NewViewName>

   symaccess -sid <SymmID> -target_sid <SymmID>

      copy -name <ViewName> view [-ckd] [-celerra] [-rp]
         [-reserve_id ResvID[,ResvID[,ResvID...]]]

   symaccess -sid <SymmID> [-offline] | -file <backup_filename>

      list view [-name <ViewName>] [-v] [-detail [-mb | -gb | -tb]]

      show view <ViewName> [-ig <ChildInitiatorGroupName>]
         [-detail] [-mb | -gb | -tb]

CHAP on Director Port

   symaccess -sid <SymmID>
        <-dirport <Dir>:<Port> |
         -iscsi_dirport <Dir>:<Port> |
         -iqn <TargetIQN>>

      set chap -cred <Credential> -secret <Secret>
```

```
   symaccess -sid <SymmID>
        [-dirport <Dir>:<Port> |
         -iscsi_dirport <Dir>:<Port> |
         -iqn <TargetIQN>]

     enable chap

     disable chap

     delete chap

   symaccess -sid <SymmID> | -file <backup_filename>

     list chap [-dirport <Dir>:<Port> |
                -iscsi_dirport <Dir>:<Port> |
                -iqn <TargetIQN>] [-v]

Miscellaneous

   symaccess discover hba [-rename] [-v]

   symaccess list hba [-v]

   symaccess -sid <SymmID>
     -file <BackupFileName> [-noprompt]

     backup  [-symforce]

     restore [-remove_ckd] [-disassociate] [-force]

   symaccess -sid <SymmID> | -file <backup_filename> [-log]

     verify

   symaccess -sid <SymmID> [-offline] | -file <backup_filename>

     list [-name <GroupName>] [-v | -detail]

     list devinfo [-mb | -gb | -tb] [-ig <InitiatorGroupName> [-detail]]

   symaccess -sid <SymmID>

     list assignment [-v] -devs <<SymDevStart>:<SymDevEnd> |
         <SymDevName> | <<SymDevName>,<SymDevName>...>>

     list no_assignments [-dirport <Dir>:<Port>]

   symaccess -sid <SymmID> -wwn <wwn> | -iscsi <iscsi>

     list logins [-v] [-dirport <Dir>:<Port> |
                       -iscsi_dirport <Dir>:<Port> |
                       -iqn <TargetIQN>]

DESCRIPTION

     The symaccess command provides the ability to perform the
     following actions:

      - Create initiator, port, and storage groups.

      - Create masking views, each one containing a single
        initiator, port, and storage group.

      - Delete a masking view.

      - Delete initiator, port, and storage groups when they
```

are not part of a masking view.

- Rename masking views or initiator, port, or storage groups.

- Associate an ASCII name with the initiator as a convenience. To NULL the alias, use a slash (/) as input.

- Add or remove devices for a specifed storage group.

- Add or remove front-end director-ports for a specified port group.

- Add or remove initiators for a specified initiator group.

- Copy masking views or initiator, port, or storage groups from one Symmetrix array to another.

- Replace the host HBA without losing established permissions.

- Set HBA port flags on a per initiator basis. This feature allows the user to change some attributes for a different host type on the FA or SE for the specified initiator.

- Set the consistent LUN flag for an initiator group to force that any device masked to this group has the same LUN for all ports.

- Display the login history table.

- Associate the Fibre Channel ID (FCID) of a switch in a fabric to the path from a host HBA to a Symmetrix array. This further restricts the path by which a host can connect to a Symmetrix array.

- Set, enable, disable, or delete CHAP credentials from the database for either the director/port or a specified iSCSI initiator.

- Back up the Auto-provisioning Group data to a user-named file on the host.

- Restore the Auto-provisioning Group data from a backup file stored on the host.

- List the host HBA information.

- List the group information.

- List the view information.

- Verifies that the AutoProvisioning Database is consistent.

ARGUMENTS

add	Adds elements to the specified group.
backup	Creates a file containing all of the group and view information currently on the array.
copy	Copies views or groups from one Symmetrix

array to another.

create — Creates a view or group of the specifed type.

delete — Deletes the masking view or security information that was previously set for either a director/port or an iSCSI initiator.

disable — Disables security information that was previously set for a director/port or an iSCSI initiator.

discover — Discovers the WWN or iSCSI names of the HBAs on the host which has paths to the Symmetrix and writes the ASCII alias names to the login history table (if empty).

enable — Enables security information that was previously set for a director/port or an iSCSI initiator.

list — Lists the group, view, or security information.

remove — Removes elements from the specified group.

rename — Renames the ASCII name of a group or view, or renames the alias for the specifed initiator within a group and the login history table.

replace — Replaces the WWN or iSCSI name within an initiator group with the specified new WWN or iSCSI name.

restore — Restores all of the group, view, and security information from the specified backup file.

set — Sets initiator attributes or CHAP credentials.

show — Shows detailed information about the groups or views.

verify — Checks the Auto-provisioning database to verify that it is consistent.

KEYWORDS

assignments — Lists the currently assigned initiators for the specified devices.

bw_limit — Sets the bandwidth limits in MB per second for an initiator group.

chap — Specifies the iSCSI CHAP credential.

consistent_lun — Sets the consistent LUN for the specified initiator group.

devinfo — Lists the device information by initiator group.

devs — Specifies devices to be added or removed.

hba — Specifies the WWN or iSCSI name of the HBA on the host.

hba_flags Sets the HBA port settings for an initiator within a group for any settings that should differ from the current settings on the port.

ig_flags Sets the port settings for an initiator group for any settings that should differ from the current settings on the port.

lockdown Sets the FCID value for an initiator within a group.

logins Specifies the entries in the login history table.

no_assignment
Lists devices that are mapped, but not yet assigned within a masking view.

sg Indicates a list of storage group names.

src Limits the action to the source devices in a device file.

tgt Limits the action to the target devices in a device file.

view Indicates that the action will be performed on a view.

OPTIONS

-bcv Limits the action to the BCV devices of a device group.

-bw_limit Sets the bandwidth limits in MB per second when creating an initiator group.

-celerra Required to map and unmap Celerra devices.

-ckd Allows CKD devices to be masked. By default, CKD devices are blocked from masking but will be allowed if the devices are already mapped.

-consistent_lun
Sets the consistent LUN for the initiator group being created.

-detail Provides detailed information for the specified group or masking views.

-devs Applies Symmetrix device names to the action.

-dirport Specifies the director and port number.

-disable Disables the overridden HBA port flags on a per initiator basis.

-disassociate
Disassociates the storage group from a FAST policy if the contents of the group would contain invalid devices for FAST.

-enable Enables the overridden HBA port flags on a per initiator basis.

-file Specifies the filename. The file option can be used to specify a device, initiator, or backup

file. -f is synonymous with -file.

-force Forces the deletion of the storage, port, or initiator group with or without members. For a storage group, it allows the removal of invalid devices from the storage group. For a restore, it forces the restore of a backup file.

-g Applies a device group name to the command.

-h Provides brief, online help.

-host Host name.

-ig Specifies the initiator group name. The -ig option is synonymous with -initgrp.

-iqn Specifies the iSCSI target IQN name.

-iscsi Specifies the iSCSI name.

-iscsi_dirport Specifies the SE director and the iSCSI target virtual port number.

-log Reports all of the inconsistencies found in the AutoProvisioning Database in the SYMAPI log.

-lun Specifies the LUN addresses to be used for the devices being added to a storage group, to the host HBA, or to a view. Supply a single LUN address for all devices to be added, or provide a list of LUN addresses equal to the number of device ranges in the list. For storage groups, LUN values may only be supplied after a storage group belongs to a view.

-name Specifies the group or view name.

-new_name Specifies the new name for the group or view.

-noprompt Requests that no prompts are returned after the command is entered. The default is to prompt the user for confirmation.

-pg Specifies the name of the port group. -pg is synonymous with -portgrp.

-remove_ckd Skips over all CKD devices within the backup, which allows the backup to be restored if the CKD devices are no longer mapped.

-rename Forces the hostname/adapter or hostname/IP to be written out to both the login history table and the specified initiator within a group, even if one is present. Overwrites any existing alias in the record.

-reserve_id Specifies the device reservation IDs for the devices in the operation.

-rp Required to map and unmap devices tagged for RecoverPoint use.

-secret Designates the secret value associated with the CHAP protocol's authentication data.

-sg Specifies the name of the storage group when creating a view. -sg is synonymous with -storgrp.

-sid Specifies the unique Symmetrix ID.

-std Limits the action to the source devices of a device group.

-symforce Forces the operation to execute when normally it would be rejected. On backup, it causes an inconsistent masking database to be backed up.
IMPORTANT: Use extreme caution with this option.

-target_sid Specifies the unique target Symmetrix ID.

-tgt Limits the action to the target devices of a device group.

-type Specifies one of the following group types:

Type	Description
storage	Indicates a storage group. It may contain up to 4,000 Symmetrix devices.
initiator	Indicates an initiator group. It may contain up to 32 fibre initiators, 8 iSCSI names, the name of another initiator group, or a combination of them.
port	Indicates a port group. It may contain any number of valid front-end ports.

-unmap Unmaps devices from ports.
When a device is removed from a storage group that is part of a view, the device will get unmapped from the ports contained in the port group of that view.
When a port is removed from a port group that is part of a view, all devices in the storage group associated with the view will be unmapped from this port.
When a view is dissolved, all devices in the storage group associated with the view will be unmapped from all ports in the port group of that view.

If the devices are mapped to the same ports through other views, those mappings will remain in place.

-v Provides a more detailed, verbose listing.

-vdev Limits the action to the VDEV devices in a device group.

-wwn Specifies a World Wide Name. Only one WWN can be provided on the command line. Some commands allow a file to be taken in as input. Multiple WWNs can be provided in this file.

PARAMETERS

Addr The LUN address to be used for the corresponding device.

BackupFileName The backup file name to which operations will be performed.

ChildInitiatorGroupName
The child initiator group name. Changes the output to diplay the initiator and LUN information for the child initiator group.

Credential The CHAP protocol’s credential name. For Microsoft users, the string should be between 8 and 256 characters.

DeviceFileName Name of the file where devices are listed. The device file can contain devices (SymDevNames) separated by new lines or device pairs (SymDevNames) listing a pair each line (the source device, followed by a target device).

Examples of device file format:

Example 1:

```
0026  0029
0015  0016
```

Example 2:

```
0001
0002
```

DgName The device group name.

Dir Applies a director number designation.

fcid A Fibre Channel ID associated with the switch.

flag Specifies the overridden HBA port flags or initiator group port flags from the following values in []:

Supported HBA port flags:

- Common_Serial_Number [C]
- Disable_Q_Reset_on_UA [D]
- Environ_Set [E]
- Avoid_Reset_Broadcast [ARB]
- AS400 [AS4]
- OpenVMS [OVMS]
- SCSI_3 [SC3]
- SPC2_Protocol_Version [SPC2]
- SCSI_Support1 [OS2007]

Supported initiator group port flags:

- Volume_Set_Addressing [V]

- Common_Serial_Number [C]
- Disable_Q_Reset_on_UA [D]
- Environ_Set [E]
- Avoid_Reset_Broadcast [ARB]
- AS400 [AS4]
- OpenVMS [OVMS]
- SCSI_3 [SC3]
- SPC2_Protocol_Version [SPC2]
- SCSI_Support1 [OS2007]

GroupsAndViewName
The group and view name.

GroupName The group name.

InitiatorFileName
The file which contains initiator names. The format of the file is each initiator on a new line which starts with 'WWN:' or 'iSCSI:' or 'IG:' depending on the type of the initiator or initiator group name. Any line which starts with '#' will be considered a comment line.

Example of initiator file format:

```
WWN:10000000c94ef69c
iSCSI:iscsiname
IG:IgName
#WWN:10000000c94ef69d
```

InitiatorGroupName
The initiator group name.

iscsi An iSCSI name.

MBperSec Initiator Group Banwidth Limits in MB/Sec.

NewGroupName The new group name.

NewViewName The new masking view name for a rename operation.

off Turn feature off.

on Turn feature on.

Port Applies a port number designation.

PortGroupName The port group name.

ResvID The device reservation ID.

Secret The CHAP protocol's secret value, which is a user-defined string of up to 32 ASCII characters or 64 binary characters. Binary values should be prefixed with the string 0X. Microsoft users must specify between 12 and 16 characters.

SgName The storage group name.

StorageGroupName
The storage group name.

SymDevEnd The last Symmetrix device name in a sequence, such as 00B6.

SymDevName	The Symmetrix device name.
SymDevStart	The first Symmetrix device name in a sequence, such as 001C.
SymmID	The 12-digit ID of the Symmetrix array.
TargetIQN	The iSCSI target IQN name.
ViewName	The masking view name.
wwn	The World Wide Name.

EXAMPLES

To create an initiator group named initexample, and to add WWN initiator 210000e08b04daac to it, enter:

```
symaccess -sid 234 -type initiator -wwn 210000e08b04daac
   -name initexample create
```

To create a storage group named storexample, and to add device 0026 to it, enter:

```
symaccess -sid 234 -type storage devs 0026
   -name storexample create
```

To create a port group named portexample, and to add director 7E and port 1 to it, enter:

```
symaccess -sid 234 -type port -dirport 7E:1
   -name portexample create
```

To add devices to a storage group named storexample, enter:

```
symaccess -sid 234 -type storage -name storexample
   add devs 0027
```

To add storage groups storgrp_1 and storgrp_2 to a storage group named storgrp_3 enter:

```
symaccess -sid 234 -type storage -name storgrp_3
   add -sg storgrp_1,storgrp_2
```

To remove WWN initiator 210000e08b04daac from an initiator group named initexample, enter:

```
symaccess -sid 234 -type initiator -wwn 210000e08b04daac
   -name initexample remove
```

To delete an initiator group named initexample, enter:

```
symaccess -sid 234 -type initiator -name initexample
   delete
```

To list all initiator, port and storage groups, enter:

```
symaccess -sid 234 list
```

To create a view named viewexample, containing initiator group initexample, port group portexample, and storage group storexample, enter:

```
symaccess -sid 234 -name viewexample -sg storexample
   -pg portexample -ig initexample create view
```

To delete a view named viewexample, enter:

```
symaccess -sid 234 -name viewexample delete view
```

To rename a view named viewexample to mvexample, enter:

```
symaccess -sid 234 rename view -name viewexample
  -new_name mvexample
```

To create a view, storage, port and initiator group named TEST, and to add devices to storage group TEST, initiators to initiator group TEST, and director ports to port group TEST, enter:

```
symaccess -sid 234 create view -name TEST -wwn
   210000e08b04daac -dirport 7E:1 devs 0026
```

To backup groups and views of Symmetrix ID 234 to a file backup_from_lab, enter:

```
symaccess -sid 234 backup -file backup_from_lab
```

To restore group and view information in file backup_from_lab to Symmetrix ID 234, enter:

```
symaccess -sid 234 restore -file backup_from_lab
```

symacl

Sets up or updates Symmetrix access control information.

SYNOPSIS

```
symacl -h

symacl <-file <CommandFile> | 'redirect stdin'>
       [-v | -noecho]

   preview

symacl -sid <SymmID> [-v | -noecho]
      <-file <CommandFile> | 'redirect stdin'>

   prepare

   commit [-force] [-restore]

symacl

   release -sid <SymmID> [-noprompt]

symacl [-sid <SymmID | ALL>]

   list [-v]

   list [-accpool | -accgroup | -acl]

   show accpool <PoolName> [-acl]

   show accgroup <GroupName> [-acl]

symacl

   backup -sid <SymmID> -file <CommandFile>

symacl -unique [-passphrase [<PassPhrase> |
                -file <PassFile>]] [-force]
```

DESCRIPTION

The symacl command allows the user to set up or update Symmetrix access control information. All information regarding access control of Symmetrix devices is stored within the Symmetrix array.

A lock is taken out by the specified Symmetrix during an access control change session. Only one access control session can be active within a Symmetrix array at any one time. When making changes to the access control database, the host making the change must have the ADMIN privilege and the caller must supply an ADMIN PIN. The verbose list option can help determine if a host has such an ADMIN privilege. The verbose list option can be used to check if a Symmetrix array was configured for access control.

To execute a change to the Symmetrix access control information, you need to enter the changes in a command file (CommandFile) and execute the following operations:

- preview
- prepare
- commit

The preview argument is used after you first create the command file. It verifies the syntax and correctness of the contents of the entries in the command file.

The prepare argument performs the preview checks and also verifies the appropriateness of the requested access control modifications against the current state of the Symmetrix array.

The commit argument performs both the preview and prepare checks and then commits the contents of the command file to the Symmetrix access control database.

Note: It is not mandatory to execute a preview or prepare action before a commit. Use these actions in the debug of the command file entries or to ensure the commit action is not rejected. If you are the security administrator and you intend to release a lock on the command file session, you must either set the environment variable SYMCLI_ACCESS_PIN to your access ID or enter your PIN every time symacl prompts you.

The command file format contains various command entries, terminated with a semicolon (;). The commands are parsed case insensitive, but the data with the commands is parsed case sensitive.

The following are various types of changes possible in the command file:

- Create new access groups
- Add and remove access IDs to access groups
- Move an access ID from one group to another
- Remove access IDs from access groups
- Create new device pools
- Add and remove devices to device pools
- Delete device pools and groups
- Add ACEs to grant access
- Remove ACEs to deny access

Optionally on UNIX platforms, you can redirect a number of command operations to stdin to save keystroke entries and avoid using the command file.

The backup operation saves the contents of the access control database in the file specified by the file argument. The file must not previously exist. The backup file created is compatible for use with the symacl utility.

The restore operation replaces the contents of the access control database with the contents of the file specified by the file argument.

By default, in the client/server mode, the access ID displayed when using the -unique option belongs to the server host. If the option SYMAPI_CLIENT_SIDE_ACCESS_ID is enabled in the options file on the client, the access ID of the client host is displayed.

ARGUMENTS

backup — Backs up the access control database for the specified Symmetrix array to the specified file.

commit	Commits the changes defined in the command file into the specified Symmetrix array.
list	Lists all ACEs, device (access) pools, or access groups.
prepare	Performs the preview checks and also verifies the appropriateness of the requested access control modifications against the current state of the Symmetrix array.
preview	Verifies the syntax of the changes specified in the command file.
release	Releases any pending access control session lock and aborts the access control session.
show	Shows detailed information about the access group or pool.

KEYWORDS

accgroup	Returns the access IDs of a specified access group.
accpool	Returns the Symmetrix devices in a specified access-controlled device pool.

OPTIONS

-accgroup	Returns the access group information associated with a specific Symmetrix array.
-accpool	Returns the access pools associated with a specific Symmetrix array.
-acl	Displays access control entries in the list and show argument output. In addition, it can display all access control entries for a specified access group.
-file	Specifies the command file containing the access control changes to be processed.
-force	Forces a commit action, even if there are non-fatal errors encountered in the prepare stage. Use this option with discretion.
-h	Provides brief, online help information.
-noecho	Blocks the printing of session status and progress messages during the access control change session’s preview, prepare, and commit actions. Cannot be used with the -v option.
-noprompt	Suppresses the automatic reply (prompt) to the user for confirmation before executing the indicated operation, when used with the release action.
-passphrase	Specifies the passphrase used to generate the encrypted 24-digit access ID. The passphrase needs to be between 4-1000 characters and can contain following

characters:
a-z A-Z 0-9 _ ! @ # $ % ^ & * () - . and a space character.
If a passphrase is not supplied on the command line, it must be provided using -file option.

-restore — Replaces the contents of the access control database with the contents of the specified file.

-sid — Specifies the Symmetrix ID whose access control information is read or modified. When ALL is specified, the action is directed to all Symmetrix arrays.

-unique — Returns an encrypted 24-digit access ID for the host machine or operating node.

-v — Echoes the contents of the command file to the output terminal. Cannot be used with the -noecho option. When used with list, a more detailed, verbose listing is provided. Use the verbose list option to verify the specified Symmetrix complies with the requirements for host-based access control changes.

PARAMETERS

CommandFile — The command file name. The command file contains a set of access control command entries.

GroupName — The access control group name of some common users or hosts (31 character maximum).

Passfile — The passphrase file name. The passphrase file name contains the passphrase used for generating the 24-digit access ID. The passphrase must be between 4-1000 characters and can contain following chars: a-z A-Z 0-9 _ ! @ # $ % ^ & * () - . and a space character.

PoolName — The pool name of a specific set of devices to be protected.

redirect stdin — The command line entries passed to stdin. Optionally, on UNIX platforms, you can redirect a number of command operations to stdin to save keystroke entries and avoid using a command file.

For example, to prepare a series of symacl commands on the command line to be redirected to stdin, use the following syntax:

```
symacl -sid SymmID prepare <<DELIM
create accgroup foo...;
add host accid...;
add user accid...;
add user accid...;
DELIM
```

```
     SymmID          The 12-character ID that specifies the
                     Symmetrix array.

COMMAND FILE SYNTAX

     The following illustrateS the syntax for command file
     entries:

To create a new pool:

        create accpool <PoolName>;

To add devices to a pool:

        add dev <StartDevName>[:<EndDevName>] to
          accpool <PoolName>;

To remove devices from a pool:

        remove dev <StartDevName>[:<EndDevName>] from
          accpool <PoolName>;

To delete a pool:

        delete accpool <PoolName> [remove_aces=true];

To create a new access group:

        create accgroup <GroupName>

To add an access ID to an access group:

        add user accid <Id> name <IdName> to accgroup AdminGrp;
        add host accid <Id> name <IdName> to
          accgroup <GroupName>;
        add restored accid <Id> name <IdName> to
          accgroup <GroupName>;
        add default accid name <IdName> to accgroup <GroupName>;

To remove an access ID from an access group:

        remove accid name <IdName> from accgroup <GroupName>;

To move an access ID to an access group:

        move accid name <IdName> to accgroup <GroupName>;

To delete an access group:

        delete accgroup <GroupName> [remove_aces=true];

To grant an access control entry:

        grant access=<AccessType,...> to accgroup <GroupName>
          for <accpool <PoolName>> | ALL | <NON-POOLED devs>;

To remove access control entries:

        remove access=<AccessType,...> from
          accgroup <GroupName> for
          <accpool <PoolName>> | ALL | <NON-POOLED devs>;
        remove aces from accgroup <GroupName>;
        remove aces from accpool <PoolName>;

COMMAND FILE KEYWORDS

     ALL             When used with the grant command,
```

creates an ACE for all devices in the Symmetrix array regardless of whether they are already part of a device pool.

When used with the remove access command, removes the ACE for all devices.

When ALL is specified, the PoolName is not entered as part of the command.

NON-POOLED When used with the grant command, creates an ACE for all devices in the Symmetrix array that are not currently part of a device pool.

When used with the remove access command, removes the ACE.

remove_aces=true
When deleting a pool or group, this command option removes any corresponding access control entries. If this action is not done, ACEs must be removed before the pool or group can be deleted.

COMMAND FILE PARAMETERS

accessType The type of access desired for the pool or group. Possible values are:

Access Type	
ADMIN	ECC
ADMINRD	ERASE
ALL	OPTMZR
BASE	POWRPATH
BASECTRL	QOS
BCV	RCOPY
CACHCTRL	RDF
CFGDEV	RPA
CFGSYM	SDDF
CHECKSUM	SDR
CREATEDV	SNAP
DIRCTRL	VLOGIX

Note: ALL, CFGSYM, CREATEDV, DIRCTRL, POWRPATH, and VLOGIX access types can be specified only for all the devices in a Symmetrix array or all NON-POOLED devices not associated with an access control device pool. These types cannot be associated with access control device pools.

CHECKSUM access type is not available on Enginuity 5977 and higher.

EndDevName The last Symmetrix device name in a sequence (such as 02C).

GroupName The name of the access group (no spaces, case sensitive; alphanumeric characters, plus underscore, and dash). A maximum of 8 characters for Enginuity 5771 and lower; a maximum of 31 characters for Enginuity 5772 and higher.

Id	The unique ID. If creating a host-based access ID, the ID is obtained by using the -unique option of symacl. If creating a new user-based access ID for the AdminGrp, the ID, which is assigned by the access control administrator, must be between four and twelve characters long.
IdName	The name of the access ID (no spaces, case sensitive; alphanumeric characters, plus underscore, and dash). A maximum of 8 characters for Enginuity 5876 and lower; a maximum of 31 characters for Enginuity 5977 and higher.
PoolName	The name of the device pool (no spaces, case sensitive; alphanumeric characters, plus underscore, and dash). A maximum of 8 characters for Enginuity 5771 and lower; a maximum of 31 characters maximum for Enginuity 5772 and higher.
StartDevName	The first Symmetrix device name in a sequence (such as 00C).

RETURN CODES

Code #	Code Symbol
0	CLI_C_SUCCESS
1	CLI_C_FAIL
2	CLI_C_DB_FILE_IS_LOCKED
19	CLI_C_GK_IS_LOCKED All GateKeepers to the Symmetrix array are currently locked.

EXAMPLES

The following examples create pools and groups, add devices and IDs, and assign access control entries to those groups and pools. It also sets up default access for those hosts that are not yet or never will be registered.

Pool Examples

To create an access pool named poola using a command file, enter:

```
symacl -sid 12345 -file add_new_pool.cmd commit
```

Where add_new_pool.cmd command file contains:

```
create accpool poola;
```

To add devices to an access pool using a command file, enter:

```
symacl -sid 12345 -file add_new_devices.cmd commit
```

Where add_new_devices.cmd contains:

```
add dev 0A:0B to accpool poola;
```

To remove devices from an access pool using a command file, enter:

```
symacl -sid 12345 -file remove_devices.cmd commit
```

Where remove_devices.cmd contains:

```
remove dev 0A:0B from accpool poola;
```

To delete an access pool and all the ACEs associated with the access pool using a command file, enter:

```
symacl -sid 12345 -file delete_pool_aces.cmd commit
```

Where delete_pool_aces.cmd contains:

```
delete accpool poola remove_aces;
```

To delete an access pool using a command file, enter:

```
symacl -sid 12345 -file delete_pool.cmd commit
```

Where delete_pool.cmd contains:

```
delete accpool poola;
```

Group Examples

To create an access group using a command file, enter:

```
symacl -sid 12345 -file add_new_group.cmd commit
```

Where add_new_group.cmd contains:

```
create accgroup groupa;
```

To add a user access ID to an access group using a command file, enter:

```
symacl -sid 12345 -file add_new_id_to_grp.cmd commit
```

Where add_new_id_to_grp.cmd contains:

```
add user accid my_pin name admin1 to accgroup groupa;
```

To add a host access ID to an access group using a command file, enter:

```
symacl -sid 12345 -file add_new_id_to_grp.cmd commit
```

Where add_new_id_to_grp.cmd contains:

```
add host accid 12345678-34567890-08974321
  name nodea to accgroup groupa;
```

To add the default access ID to an access group using a command file, enter:

```
symacl -sid 12345 -file add_new_id_to_grp.cmd commit
```

Where add_new_id_to_grp.cmd contains:

```
add default accid name unknown to accgroup groupa;
```

To remove an ID from an access group using a command file, enter:

```
      symacl commit -sid 12345 -file remove_id_from_grp.cmd

   Where remove_id_from_grp.cmd contains:

      remove accid name nodea from accgroup groupa;

   To move an ID to an access group using a command
   file, enter:

      symacl commit -sid 12345 -file move_id_to_group.cmd

   Where move_id_to_group.cmd contains:

      move accid name nodea to accgroup groupa;

   To delete an access group, enter:

      symacl -sid 12345 -file delete_group.cmd commit

   Where delete_group.cmd contains:

      delete accgroup groupa;

   To delete an access group and corresponding ACEs
   (if any exist) using a command file, enter:

      symacl -sid 12345 -file del_grp_and_aces.cmd commit

   Where del_grp_and_aces.cmd contains:

      delete accgroup groupa remove_aces;

Add ACE Examples

   To add an ACE, granting ADMIN privilege, using a
   command file, enter:

      symacl -sid 12345 -file add_acl.cmd commit

   Where add_acl.cmd contains:

      grant access=ADMIN to accgroup groupa
        for accpool poola;

   To add an ACE for all Symmetrix devices regardless of
   whether they are already in a pool and grant BASE access
   using a command file, enter:

      symacl -sid 12345 -file grant_all_devs_acl.cmd commit

   Where grant_all_devs_acl.cmd contains:

      grant access=BASE to accgroup groupa for ALL devs;

   To add an ACE granting BASE access for all Symmetrix
   devices that do not belong to an access pool, using a
   command file, enter:

      symacl -sid 12345 -file add_not_in_pool_devs_acl.cmd
        commit

   Where add_not_in_pool_devs_acl.cmd contains:

      grant access=BASE to accgroup groupa for
        NON-POOLED devs;
```

Remove ACE Examples

To remove an ACE using a command file, enter:

```
symacl -sid 12345 -file remove_acl.cmd commit
```

Where remove_acl.cmd contains:

```
remove access=ADMIN from accgroup groupa for
  accpool poola;
```

To remove all ACEs for groupa using a command file, enter:

```
symacl -sid 12345 -file remove_aces_for_group.cmd
  commit
```

Where remove_aces_for_group.cmd contains:

```
remove aces from accgroup groupa;
```

To remove all ACEs for poola using a command file, enter:

```
symacl -sid 12345 -file remove_aces_for_pool.cmd
  commit
```

Where remove_aces_for_poola.cmd contains:

```
remove aces from accpool poola;
```

To remove an ACE setup of BASE access for all Symmetrix devices regardless of whether they are already in a pool, using a command file, enter:

```
symacl -sid 12345 -file rem_all_devs_acl.cmd commit
```

Where rem_all_devs_acl.cmd contains:

```
remove access=BASE from accgroup groupa for ALL devs;
```

To remove an ACE setup of BASE access for all Symmetrix devices not already in a pool using a command file, enter:

```
symacl -sid 12345 -file rem_not_in_pool_devs_acl.cmd
  commit
```

Where rem_not_in_pool_devs_acl.cmd contains:

```
remove access=BASE from accgroup groupa
  for NON-POOLED devs;
```

symapierr

Translates a SYMAPI return code to a string.

SYNOPSIS

symapierr [-h] <ErrorCode>

DESCRIPTION

Returns a string with a detailed description of any return code generated by any SYMAPI function.

ARGUMENTS

None.

OPTIONS

-h Provides brief, online help information.

PARAMETERS

ErrorCode A numerical representation of an error.

RETURN CODES

Code #	Code Symbol
0	CLI_C_SUCCESS
1	CLI_C_FAIL

EXAMPLES

To return a string for error number 10, enter:

```
symapierr 10
```

The following will be output:

```
SYMAPI Error Symbol : SYMAPI_C_NO_DEVS_FND_UPGRADE
SYMAPI Error Message: No Symmetrix devices found with
                      microcode version 5x63 or up.
```

symaudit

Allows the user to extract records from a Symmetrix audit log file to determine what application on what host initiated actions that directed Symmetrix arrays behavior.

Provides a monitor option for displaying the records as they are written to the log file. Provides the ability to determine the date and time of the current log file data and its size.

SYNOPSIS

```
symaudit -h

symaudit list -sid <SymmID> [-text | -v]
   [-function_class [-exclude]
     <ClassName>[, <ClassName>, ...] ]
   [-action_code [-exclude]
     <ActionName>[, <ActionName>, ...] ]
   [-host <HostName>]
   [-vendor_id <VendorId>]
   [-application_id
     <ApplId>[, <ApplId>, ...] ]
   [-activity_id <ActivityId>]
   [-symdev_range <SymDevStart>:<SymDevEnd>]
   [-start_date <date_time>] [-end_date <date_time>]
   [-record_num <RecordNumber>]
   [-n <RecordCount>]
   [-last_n <RecordCount>]
   [-user <UserName>]

symaudit monitor -sid <SymmID> [-text | -v]
   [-i Interval] [-c Count]

symaudit show -sid <SymmID>
```

DESCRIPTION

The symaudit command is used to retrieve information from the Symmetrix audit log file. Data is written to the audit file during control operations initiated by host applications. The audit file correlates activity from all hosts into one file.

The symaudit command can filter the extracted data through the use of options that specify match criteria. The options include host name, application name, function class, and action code. A combination of filters can be used.

The monitor action polls the Symmetrix for new audit log records every Interval, defined in seconds, until the iteration Count is satisfied or the program is stopped. The command is run in the foreground. Verbose mode (-v) provides more detailed output.

ARGUMENTS

list	Lists the extracted audit log records.
monitor	Monitors the Symmetrix array for new audit log data in real time.
show	Shows the time period and quantity of data

in the audit log file.

OPTIONS

-action_code Filters the audit log records so that only the records containing the specified action code return.

-activity_id Filters the audit log records to only show records with the matching Activity ID.

-application_id
Filters the audit log records so that only the records generated by the specified application return.

-c Specifies the number (count) of times to poll for data. If this option is not specified, the audit log is polled continuously.

-end_date Indicates the date and time of the last audit log record to display. The format is [mm/dd[/yy]]:[hh:mm[:ss]]. If only the hh:mm is provided, the current day will be assumed. If only mm/dd is provided, the current year is assumed. A four-digit year can also be specified. If no time is specified, it will default to 0:0:0, the very beginning of the day. If the end_date and the -n options are omitted, the output continues until the end of file.

-exclude Filters out, or excludes, records that match the specified values.

-function_class
Filters the audit log records so that only the records belonging to the specified function_class return.

-h Provides brief online help information.

-host Filters the audit log records so that only the records generated from the specified host return.

-i Specifies the repeat interval in seconds.
The default interval is 30 seconds.
The minimum interval is 5 seconds.
For passive actions the minimum interval is 15 seconds. Passive actions are actions that do not acquire an exclusive lock.

-last_n Specifies the number of most recent records to display.

-n Specifies the number of records to display.

-record_num Indicates at which record number in the audit log to start processing.

-sid Specifies the Symmetrix ID of the Symmetrix audit log file to process.

-start_date Indicates the date and time of the first audit log record to display. Format is

[mm/dd[/yy]]:[hh:mm[:ss]]. If only the hh:mm is provided, the current day will be assumed. If only mm/dd is provided, the current year is assumed. A four-digit year can also be specified. If no time is specified, it will default to 0:0:0, the very beginning of the day.

-symdev_range — Filters the audit log records so that only the records containing the name of a symdev within the indicated range in the text field are returned.

-text — Indicates that the text associated with the audit log record should be displayed.

-user — Filters the audit log records so that only the records containing the specified user name return.

-v — Provides a more detailed, verbose listing.

-vendor_id — Filters the audit log records so that only the records containing the specified vendor_id are returned.

PARAMETERS

ActionName — The name of a control action associated with an audit log entry. These are not case sensitive. Possible actions are:

Action Names		
Abort	AbortSnapshot	AbrtSnap
Acquire	Activate	Add
AllcStop	AllcStrt	AllocateStart
AllocateStop	Analyze	ArchiveLog
ArchLog	Assoc	Associate
AuthControl	AuthCtrl	BegBckup
BeginBackup	BeginRestore	BeginSnapshot
BegRestr	BegSnap	Bind
BlkSIO	Block	BlockDirectIO
BluFlash	Break	Cancel
CancelImageRefresh		Checkpoint
ChkPt	Cleanup	ClearStats
ClrStats	CnImgRef	CodeLoad
Commit	CompressStart	CompressStop
CompStop	CompStrt	Configur
Configure	Connect	Convert
CPAnalyz	CPAnalyze	Create
CreatePair	CrtPair	CrtRestrTDev
CrtRTDev	Cutover	Deactiv
Deactivate	Delete	DeletePair
DelPair	Denied	Disable
Disassoc	Disassociate	DisConn
Disconnect	Enable	EndBackup
EndBckup	EndRestore	EndRestr
EndSnap	EndSnapshot	EnvExpan
EnvExpand	EnvRemov	EnvRemove
EnvRestore	EnvRestr	EnvSetup
Erase	Expand	ExpandDB
Failback	Failover	FASTMove
FASTSchd	FASTSchedule	FASTSwap
FileTrf	FrAlStop	FrAlStrt
FreeAllStart	FreeAllStop	FreeStart
FreeStop	FreeStrt	Freeze

```
FSnapRes        FullEsta        FullEstablish
FullRestore     FullRstr
FullSnapRestore                 GCMOff
GCMOn           GenSwapList     GnSwpLst
HalfDeletePair  HalfMovePair    HalfSwap
Halt            HBAstate        HDelPair
HMovPair        Hold            HostIO
HSwap           ImageRefresh    ImgRef
IncEsta         IncEstablish    IncRestore
IncRstr         IncSnapRestore  Init
Initialize      Invalidate      Invldate
ISnapRes        Isolate         LEDOn
Link            Lock            Login
Logout          Map             Merge
Migrate         Modify          MovePair
MovPair         MSCCleanup      MSCClnup
MSCSwit         MSCSwitch       NotRdy
NotReady        Offline         Online
OptMove         OptmzrMove      OptmzrSchedule
OptmzrSwap      OptSchd         OptSwap
ORMOff          ORMOn           ORMSys
Other           PerstOff        PerstOn
PhoneHome       PhonHome        Pin
PortAccess      PortAccs        PowerPath
Prepare         Protect         PwrPath
R1Update        RclmStop        RclmStrt
RdfSet          Ready           ReadyTgt
Reassoc         Reassociate     Rebind
Reboot          ReclaimStart    ReclaimStop
Reconfig        Reconfigure     Recover
Recreate        Refresh         Relabel
Release         Relink          Relocated
Relocted        Remove          Rename
Repair          Replace         Reserve
Reset           Restart         RestrCleanup
RestrPrepare    Resume          Revert
Rollback        RstrClup        RstrPrep
RWDisabl        RWDisable       RWEnable
Scan            Set             SetAcpD
SetAcpDisk      SetAsync        SetCmpat
SetCompatibilityID              SetCopy
SetCpy          SetGCMOff       SetGCMOn
SetLEDBlueFlash                 SetLEDOn
SetLEDSlowBlink                 SetMobID
SetMobilityID   SetNoCopy       SetNoCpy
SetORMOff       SetORMOn        SetORMSystem
SetPersistentOff
SetPersistentOn                 SetSec
SetSecure       SetTTL          SloBlink
SNAPCrea        SNAPCreate      Sparing
Split           SPPEstab        SPPEstablish
STARdis         STARdisable     STARena
STARenable      STARref         STARrefresh
Start           Stop            Suspend
Swap            Switch          Sync
SyncStart       SyncStop        SyncStrt
SysCalls        Terminat        Terminate
Thaw            TokenMgt        Unbind
UnblkSIO        Unblock
UnblockDirectIO
UncompressStart                 UncompressStop
UncoStop        UncoStrt        UnHold
Unlink          Unlock          Unpin
UnProtct        Unprotect       Validate
VolAcces        VolAccess       VTOC
WRDisabl        WRDisable
```

ActivityId — The Activity ID associated with the performed action in the audit log.

ApplId — The name of an application whose activity generated audit log entries.

ClassName — The name of a functional class area. These values are not case sensitive. Valid class names are:

Class Names		
Base	BCV	CfgChg
CGRDF	Checksum	Clone
DDF	DevMask	Dir
Erase	Maint	N/A
Migrate	Optmzr	Other
QoS	RCopy	RDF
Recovery	ResvCtrl	Security
Snap	UserFunc	Worm
Fast		

HostName — The name of the host system whose application generated the audit log entry.

RecordCount — A count of the number of audit log records that should be returned.

RecordNumber — A record sequence number that is within the audit log file’s current range.

SymDevEnd — The last Symmetrix device name in a range that should be used to filter log file entries.

SymDevStart — The first Symmetrix device name in a range that should be used to filter log file entries.

SymmID — The 12-digit ID of the Symmetrix array.

VendorId — The name of the vendor that produced the application whose activity generated audit log entries.

RETURN CODES

Code #	Code Symbol
0	CLI_C_SUCCESS
1	CLI_C_FAIL
19	CLI_C_GK_IS_LOCKED All GateKeepers to the Symmetrix array are currently locked.

EXAMPLES

To show the time period and information for a specific array’s audit log, enter:

```
symaudit -sid 04 show
```

The following output returns:

```
                A U D I T   L O G   D A T A

   Symmetrix ID            : 000192606204

   Starting date           : 04/26/2011 12:55:39
   Ending date             : 05/11/2011 13:40:37

   Starting record number  :  175500
   Ending record number    :  237198
   Total record count      :   61699
```

To list detailed audit log entries for a specific array within a certain time period, enter:

```
symaudit -sid 04 list -v -start_date 5/11:9:40
  -end_date 5/11:9:45
```

The following output returns:

```
                A U D I T   L O G   D A T A

  Symmetrix ID            : 000192606204

    Record Number         :   237178
      Records in Seq      :        1
      Offset in Seq       :        1
      Time                : 05/11/11 09:42:37
      Vendor ID           : EMC Corp
      Application ID      : SYMAUTH
      Application Version : 7.3.0.0
      API Library         : SEK
      API Version         : V7.3.0.0 (Edit Level: 1207)
      Host Name           : host_196
      OS Name             : SunOS
      OS Revision         : 5.8Generic
      Client Host         :
      Process ID          : 00001235
      Task ID             : 00000001
      Function Class      : ACCESS
      Action Code         : Set
      Text                : Starting a User Authorization
                            operation to modify settings:
                            Enforcement Policy [enforce]
      Username            : H:host_196\\ruggip
      Activity ID         : SE57a9e3d8d8

    Record Number         :   237179
      Records in Seq      :        1
      Offset in Seq       :        1
      Time                : 05/11/11 09:42:37
      Vendor ID           : EMC Corp
      Application ID      : SYMAUTH
      Application Version : 7.3.0.0
      API Library         : SEK
      API Version         : V7.3.0.0 (Edit Level: 1207)
      Host Name           : host_196
      OS Name             : SunOS
      OS Revision         : 5.8Generic
      Client Host         :
      Process ID          : 00001235
      Task ID             : 00000001
      Function Class      : ACCESS
      Action Code         : Set
      Text                : The User Authorization modify
                            settings operation SUCCEEDED
      Username            : H:host_196\\ruggip
```

```
      Activity ID         : SE57a9e3d8d8
```

To list all audit log entries matching several functional classes, enter:

```
symaudit -sid 04 list -function_class BCV, CGRDF, RDF
```

To list all audit log entries which are not of several action code types, enter:

```
symaudit -sid 04 list -action_code -exclude Init, Add
```

To list audit log entries made by a certain user from a certain host, within a given record range, enter:

```
symaudit -sid 04 list -user root -host myHost
  -record 200 -n 100
```

symauth

Sets up or updates Symmetrix user authorization information.

SYNOPSIS

```
symauth -h

symauth [-sid <SymmID>] [-noprompt]

   enable

   disable

symauth [-sid <SymmID>] [-noprompt]

   set enforcement [advise | enforce]

symauth -sid <SymmID> [-noprompt]

   set secure_reads [enable | disable]

symauth [-sid <SymmID>]

   list

   list -users [-by_domain | -by_role | -by_user]
        [-current_user]

symauth

   list -roles

symauth

   list -components

symauth [-sid <SymmID>]

   list -hosts

symauth

   show -username

symauth [ -sid <SymmID> ] [-v | -noecho] [-noprompt]
     [-file <CommandFile> | 'redirect stdin']

   preview

   commit

symauth -sid <SymmID>

   backup -file <BackupFile>

   commit -restore -file <BackupFile> [-noprompt]

symauth -sid <SID> commit -acl_import -file <AclBackupFile>

symauth -unique [-passphrase [<PassPhrase> |
                 -file <PassFile>]] [-force]
```

DESCRIPTION

This command allows the user to set up or modify Symmetrix

user authorization information. All Symmetrix information authorization information for a Symmetrix array are stored on the array itself.

This mechanism allows an authorization role to be assigned to a user or group (of users). This role then controls access to the array and the types of management functions that can be performed on it.

The following roles are supported:

None	No access allowed.
Monitor	Ability to perform read-only / view operations.
PerfMonitor	Ability to perform read-only / view operations and set performance monitors in UniSphere for Vmax with Performance Analyzer product.
Auditor	Ability to view security information and the Symmetrix audit log.
LocalRep	Ability to perform Local Replication operations. This does not grant the right to create secure snapshots.
RemoteRep	Ability to perform Remote Replication operations.
DeviceManage	Ability to perform Device Management operations (e.g.: device controls).
StorageAdmin	Ability to perform any storage management operations.
SecurityAdmin	Ability to perform security operations.
Admin	Ability to perform any storage or security operations.

A role assignment can grant access to either:
- The entire Symmetrix array.
- One or more Storage Groups on the aray, along with the devices they contain.

Only the LocalRep, RemoteRep and DeviceManage roles can be assigned to Storage Groups.

To change the Symmetrix user authorization information, enter the changes in a command file (CommandFile) and execute the preview and commit operations against the file.

The preview operation can be used after you first create the command file. It verifies the syntax and validity of the entries in the command file.

The commit operation performs the preview checks and then commits the contents of the command file to the Symmetrix user authorization database.

> Note: It is not necessary to execute a preview action before a commit as the preview action is automatically performed during a commit.

The CommandFile is used to specify several commands, each

of which is terminated with a semicolon (;). With the exception of the names that can be used in the file (User, Group, StorageGroup etc.), the commands in the file are not case sensitive.

The following operations can be specified in the CommandFile:

- assign/reassign a user or group to one or more roles
- delete the role assigned to a user or group
- remove a role assigned to a user or group if there are multple roles assigned
- set the enforcement mode
- set the secure_reads mode

The enforcement and secure_reads modes can also be set directly on the command line.

On UNIX platforms, you can redirect commands from stdin (standard input) instead of using a command file by using the following syntax:

```
symauth -sid <SymmID> preview|commit <<DELIM
assign user testuser to role monitor;
assign user testadmin to role admin;
DELIM
```

The backup operation saves the contents of the user authorization database from a Symmetrix array to the specified file.

The restore operation re-initializes the user authorization database on a Symmetrix array from a previously generated backup file. The specified file should have been created by an earlier backup operation - from the same or a different Symmetrix array. If the restoration image does not assign a role of Admin or SecurityAdmin to you, the final step in which Authorization is re-enabled for the array will fail.

Note: In this case, you may need to assign yourself one of those roles and then manually enable User Authorization. Alternatively, you can have someone else who does have Admin or SecurityAdmin privileges perform that operation.

The ’show -username’ operation displays the user and group name that will be used for the currently logged in user.

If user authorization is disabled, users are granted full rights to the Symmetrix array. Otherwise, rights for users are calculated as follows.

Authorization entries contain either a User or Group name and a corresponding Role that corresponds to a set of rights. These names, as described below, can be either fully qualified or unqualified.

An example of an entry with a fully qualified name:

```
User    H:host1\\joe      Admin
Group   D:domain1\\sales  StorageAdmin
```

Examples of entries with unqualified names:

```
User    H:*\\joe          Monitor
Group   H:host1\\*        Monitor
User    joe               Monitor
```

Rights present in any User or Group entries that match

a user's identity are granted to the user.

Unqualified User entries will only be used if there are no fully qualified User entries that match the user.

Unqualified Group entries will only be used if there are no fully qualified Group entries that match the user.

By default, in the client/server mode, the access ID displayed when using the -unique option belongs to the server host. If the option SYMAPI_CLIENT_SIDE_ACCESS_ID is enabled in the options file on the client, the access ID of the client host is displayed.

ARGUMENTS

backup	Backs up the user authorization database for the specified Symmetrix array to the specified file.
commit	After verification, commits the changes defined in the CommandFile and updates the authorization data on the Symmetrix array. With the -restore option, a previously generated backup image is restored to the Symmetrix array.
disable	Disables user authorization.
enable	Enables user authorization.
list	Lists user authorization information.
preview	Verifies the syntax of the CommandFile.
set	Sets the Symmetrix authorization mode.
show	Shows your current UserName.

KEYWORDS

enforcement — If authorization is enabled, this controls how the authorization rules are applied. There are two values:

Value	Description
enforce	Authorization rules are enforced. If a user does not have the necessary rights (role), operations will fail. This is the default setting.
advise	Authorization rules are checked but not enforced. Operations will succeed if the user does not have the necessary rights (role).

In both cases, authorization errors are written to both the SYMAPI log file and the Symmetrix Audit log.

secure_reads — If authorization is enabled, this controls visibility of authorization rules. There are two values:

Value	Description
enable	Users may only view authorization rules that apply to them. Only users with SECURITY_VIEW permissions (AUDITOR role) can view the full set of rules.
disable	Authorization rules are visible to all users. This is the default setting.

OPTIONS

-acl_import	Specifies that Host ID values should be loaded from an existing Symmetrix ACL backup file. From this backup, Access Groups and Access IDs are interpreted as RBAC Host Names and Host IDs. Any Host Names that are already defined within the RBAC database are ignored.
-by_domain	Used in conjunction with list and -users. This option causes the list of known users (and groups) to be sorted by their domain.
-by_role	Used in conjunction with list and -users. This option causes the list of known users (and groups) to be sorted by their assigned role.
-by_user	Used in conjunction with list and -users. This option causes the list of known users (and groups) to be sorted by their UserName.
-components	Lists the component types that are supported. Only used with list.
-current_user	Filter the list -users results to list only the authorization roles for the user making the call.
-file	Specifies an existing file to be processed. Either: [a] A command file to be processed for changes to the RBAC database. [b] A file holding the passphrase to be used when generating a Host ID value for some host. [c] An existing Symmetrix ACL backup file to use when when importing Host ID values to the RBAC database.
-force	If a Host ID (or symacl Access ID) is already defined for the host in question, this flag is needed to overwrite or replace that value.
-h	Provides brief online help information.
-hosts	Lists the hosts that have Host IDs assigned with them.
-noecho	Blocks the printing of session status and progress messages during preview and commit actions. Cannot be used with the -v option.
-noprompt	Requests that no prompts are required

after the command is entered. The default is to prompt the user for confirmation.

-passphrase — Specifies the passphrase used in the generation of a (new) Host ID. An alternative to supplying a passphrase on the command line is to provide it in a file on disk (the -file option).

-restore — Replaces the contents of the user authorization database from the specified file.

-roles — Lists the various user authorization roles available on a Symmetrix array along with a short description of that role. Only used with list.

-sid — Specifies the Symmetrix ID for which to read or modify authorization information.

-unique — Returns a secure 24-digit Host ID for the host being executed on. If necessary, one is defined \226 using either a supplied passphrase or certain hardware characteristics of the host.

-username — Displays the currently logged on UserName and GroupName. Only used with the show argument.

-users — Lists the users and groups currently defined on the Symmetrix array along with their corresponding role. Only used with the list argument.

-v — Echoes the contents of the CommandFile to the output terminal when used with a preview or a commit action. Cannot be used with the -noecho option.

PARAMETERS

AclBackupFile — A pathname to an existing Symmetrix ACL (symacl) backup file.

BackupFile — Name of a backup image generated by the backup command.

CommandFile — Name of the file containing a set of authorization commands.

PassFile — A pathname to a file that contains the passphrase to be used to generate a new 24-digit Host ID. Refer to the description of the PassPhrase parameter for the valid syntax.

PassPhrase — The passphrase that will be used to generate a new 24-digit Host ID. The passphrase needs to be between 4-1000 characters and can contain the following characters:
a-z A-Z 0-9 _ ! @ # $ % ^ & * ( ) - . along with a space character

redirect stdin — Command line entries passed to stdin. Optionally, on UNIX platforms, you can

redirect a number of command operations to stdin to save keystroke entries and avoid using a command file.

For example, use the following syntax:

```
symauth -sid SymmID preview <<DELIM assign
user testuser role...;
DELIM
```

SymmID — The 12-character ID that specifies the Symmetrix array.

COMMAND FILE SYNTAX

The following are the possible command syntaxes for the <CommandFile> entries.

Associate a Host ID with a host name:

```
assign host <HostName> to hostid <HostID>;
```

Re-assign a host name to a different Host ID:

```
reassign host <HostName> to hostid <HostID>;
```

Disassociate a Host ID from a host name:

```
delete host <HostName>;
```

Rename a host name in an existing Host ID to name Association:

```
rename host <HostName> to <NewHostName>
```

Assign a user or group to an authorization role for the entire Symmetrix array:

```
assign user <UserName> to role <RoleName>;
assign group <GroupName> to role <RoleName>;
```

Re-assign a user or group to a different authorization role for the entire Symmetrix array:

```
reassign user <UserName> to role <RoleName>;
reassign group <GroupName> to role <RoleName>;
```

Delete a user or group authorization role for the entire Symmetrix array:

```
delete user <UserName>;
delete group <UserName>;
```

Assign, re-assign or delete a user or group authorization role for a Symmetrix sub-component:

```
assign user <UserName> to role <RoleName>
        for <Comp> <CompName>;
assign group <GroupName> to role <RoleName>
        for <Comp> <CompName>;
reassign user <UserName> to role <RoleName>
        for <Comp> <CompName>;
reassign group <GroupName> to role <RoleName>
        for <Comp> <CompName>;
delete user <UserName> for <Comp> <CompName>;
delete group <UserName> for <Comp> <CompName>;
```

Anywhere <RoleName> is shown above it can be subsituted with up to four roles. The roles are added with "+" between them:

```
assign user <UserName> to role <RoleName>+<RoleName>;
```

With multiple roles, the assign command can assign additional authorization roles to existing roles for users and groups. The syntax of the assign command is the same as when a new assignment is made:

```
assign user <UserName> to role <RoleName>+<RoleName>;
```

With multiple roles, individual roles can be removed using the remove command:

```
remove user <UserName> from role <RoleName>;
```

Set the authorization enforcement policy:

```
set enforcement [advise | enforce];
```

Set the secure_reads policy:

```
set secure_reads [enable | disable];
```

COMMAND FILE PARAMETERS

HostID — A 24-digit Host ID - as returned by the 'symauth -unique' command for a host.
e.g.: 01234567-89ABCDEF-01234567

HostName — The name of a host - as used in one or more H: style RBAC rules. Case does not matter.
e.g.: Host1234

NewHostName — For rename Host Name operations, the new Host Name - the one that <HostName> should be changed to.

UserName — The name of a user.
A name can consist of 3 fields:
<Type>:<Qualifier>\\<Name>

<Type>	The type of name - how it was authenticated to the system.
<Qualifier>	The host or domain name that the name was authenticated on.
<Name>	The user name.

A <Qualifier> or <Name> consisting of a single '*' character is a wildcard - which will match any host/domain or user name.

A fully qualified name is one in which all three fields are present - with no implicit or explicit wildcard characters.

Examples of fully qualified names are:

H:host\\joe	User "joe" logged into the specified host.
D:domain\\joe	User "joe" logged in through the specified Windows domain.
domain\\joe	Interpreted the same as

"D:domain\\joe".

Examples of partially qualified names are:

H:host*	Any user logged onto the specified host.
D:*\\joe	User 'joe' logged in through any Windows domain.
D:\\joe	Interpreted the same as "D:*\\joe".
joe	User "joe" regardless of how they have logged in - any host or domain.
*	Any user.

The following names indicate alternate authentication mechanisms - and are only relevant for users logged on through Unisphere for VMAX.

C:host\\joe	User "joe" authenticated through the built-in Unisphere user database on the specified host.
L:host\\joe	User "joe" authenticated through the LDAP server on the specified host.

Spaces can be included in the name by quoting ("") the entire name.

```
"H:host1\\User 123"
```

The fully qualified UserName and GroupName of the user running the symauth command can be obtained via:

```
symauth show -username
```

GroupName — The name of a group. The syntax is the same as for UserName above - but identifies a group instead of a user.

RoleName — The name of the role to assign to a user or group. The following roles are supported:

- Admin
- SecurityAdmin
- Auditor
- StorageAdmin
- Monitor
- PerfMonitor
- LocalRep
- RemoteRep
- DeviceManage
- None

Role names are not case sensitive. Up to four roles can be defined. The roles are separated by "+".

Comp — A type of component : StorGrp (Storage Group).

Only the LocalRep, RemoteRep and DeviceManage roles can be assigned to a

```
                Storage Group component.

     CompName   For a StorGrp Component, the name of a
                Storage Group.  A simple wildcard syntax
                is supported, allowing one role assignment
                to apply to multiple Storage Groups.

                The following syntax is supported:
                  *             Any 0-or-more characters
                  ?             Any 1 character
                  +             0-or-more repeats of the
                                prior character patern
                  [a-zA-Z0-9]   Any of these characters
                  [!0-9]        Any character other than
                                these
                Characters are case sensitive.

                A few examples:
                  CompName      Matches Storage Groups:
                  --------      --------
                  home_123      home_123
                  home*         home
                                home_east
                                home_west
                  *_home        east_home
                                west_home
                  ????_sales    east_sales
                                1234_sales
                  [A-Z][a-z]+_* East_123
                                West_home

RETURN CODES

     Code #       Code Symbol
     ------       -----------

       0          CLI_C_SUCCESS

       1          CLI_C_FAIL

EXAMPLES

     To enable user authorization, enter:

        symauth -sid 0123 enable

     To disable user authorization, enter:

        symauth -sid 0123 disable

     To list the overall authorization state, enter:

        symauth -sid 0123 list

     To list the supported roles, enter:

        symauth list -roles

     To list the authorization roles assigned to users/groups,
     sorted in some manner, enter one of:

        symauth -sid 0123 list -users
        symauth -sid 0123 list -users -by_role
        symauth -sid 0123 list -users -by_domain
        symauth -sid 0123 list -users -by_user

     To list only the authorization roles assigned to the
```

current user, enter:

```
symauth -sid 0123 list -users -current_user
```

To display the UserName and GroupName for the current user, enter:

```
symauth show -username
```

The following examples show how to change user authorizations using a command file.

To add an authorization role for a user or group using a command file, enter:

```
symauth -sid 0123 -file assign_user.cmd commit
```

Where assign_user.cmd contains commands similar to:

```
assign user  H:venus\John     to role Admin+Auditor;
assign group D:Corp\Sales     to role StorageAdmin;
assign user  Smith            to role StorageAdmin;

assign user  H:mars\Mary      to role RemoteRep
                                  for StorGrp Home_*;
```

To replace the authorization role for a user or group using a command file, enter:

```
symauth -sid 0123 -file reassign_user.cmd commit
```

Where reassign_user.cmd contains commands similar to:

```
reassign user  H:venus\John   to role StorageAdmin;
reassign group D:Corp\Sales   to role Monitor;
reassign user  Smith          to role LocalRep+Auditor;

reassign user  H:mars\Mary    to role RemoteRep+Auditor
                                  for StorGrp Home_*;
```

To delete the authorization role assigned to a user or group using a command file, enter:

```
symauth -sid 0123 -file del_user.cmd commit
```

Where del_user.cmd contains commands similar to:

```
delete user  H:venus\John;
delete group D:Corp\Sales;
delete user  Smith;

delete user  H:mars\Mary          for StorGrp Home_*;
```

To remove some authorization role(s) from a set of roles currently assigned to a user or group using a command file, enter:

```
symauth -sid 0123 -file remove_user.cmd commit
```

Where remove_user.cmd contains commands similar to:

```
remove user  H:venus\John   from role RemoteRep;
remove group D:Corp\Sales   from role LocalRep;
remove user  Smith          from role LocalRep+Auditor;

remove user  H:mars\Mary    from role RemoteRep
                                  for StorGrp East_*;
```

One or more commands, of the same or different types, can be included within a command file.

```
symbcv

        Performs support operations on one or more
        Symmetrix BCV (Business Continuance Volume)
        devices.

        SYNOPSIS

             symbcv -h

             symbcv [-sid <SymmID>] [-offline] [-v]
                  [-resv | -emulation] [-i <Interval>] [-c <Count>]
                  [-mb | -gb | -tb]

                list pd

                list [dev]

             symbcv -g <DgName> [-offline] [-v]

                associate pd <PdevName> [<LdevName>]

                associateall [pd | -host <HostName>] [-sid <SymmID>]
                  [-SA <# | ALL>] [-p <#>] [-N <#>]
                  [-cap <#> [-captype <mb> | <cyl>]]
                  [-R1 | -NOR1] [-R2 | -NOR2]
                  [-sel_rdfg <SelRdfGrpNum>]
                  [-devs <<SymDevStart>:<SymDevEnd> | <SymDevName>
                      [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]>]

                disassociate pd <PdevName> [-force]

                move pd <PdevName> <DestDgName> [-force] [-rename]

                copy pd <PdevName> <DestDgName> [-force] [-rename]

             symbcv -g <DgName> [-offline] [-sid <SymmID>]
                  [[-rdf [-bcv]] | [-rrdf] | [-hop2]]
                  [-rdfg <GrpNum> [-remote_rdfg <RemoteGrpNum>]]
                  [-v]

                associate dev <SymDevName> [<LdevName>] [-rdfg <GrpNum>]
                  [-remote_rdfg <RemoteGrpNum>]

                disassociate dev <SymDevName> [-force]

                move dev <SymDevName> <DestDgName> [-force] [-rename]

                copy dev <SymDevName> <DestDgName> [-force] [-rename]

             symbcv -g <DgName> [-offline]
                  [[[-rdf] [-bcv]] | [-rrdf]] [-force] [-v]

                disassociate ld <LdevName>

                copy ld <LdevName> <DestDgName>[-rename]

                move ld <LdevName> <DestDgName>[-rename]

             symbcv -g <DgName> [-offline] [-force]
                  [-SA <# | ALL>] [-p <#>] [-N <#>]
                  [-cap <#> [-captype <mb> | <cyl>]]
                  [-R1 | -NOR1] [-R2 | -NOR2]
                  [[[-rdf] [-bcv]] | [-rrdf]]
                  [-remote_rdfg <RemoteGrpNum>] [-SA <#|ALL>] [-p <#>]
                  [-sel_rdfg <SelRdfGrpNum>]
```

```
      [-devs <<SymDevStart>:<SymDevEnd> | <SymDevName>
          [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]>]
      [-v]

   associateall [devs | -host <HostName>]

   rmall

   copyall <DestDgName> [-rename]

   moveall <DestDgName> [-rename]

symbcv -cg <CgName> [-offline] [-v]

   associate pd <PdevName> [<LdevName>]

   associateall [pd | -host <HostName>] [-sid <SymmID>]
     [-SA <# | ALL>] [-p <#>] [-N <#>]
     [-cap <#> [-captype <mb> | <cyl>]]
     [-R1 | -NOR1] [-R2 | -NOR2]
     [-sel_rdfg <SelRdfGrpNum>]
     [-devs <<SymDevStart>:<SymDevEnd> | <SymDevName>
         [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]>]

   disassociate pd <PdevName> [-force]

   move pd <PdevName> <DestCgName> [-force]

   copy pd <PdevName> <DestCgName> [-force]

symbcv -cg <CgName> [-offline] -sid <SymmID>
     [[-rdf [-bcv]] | [-rrdf] | [-hop2]]
     [-rdfg <GrpNum> [-remote_rdfg <RemoteGrpNum>]]
     [-v]

   associate dev <SymDevName> [<LdevName>]

   disassociate dev <SymDevName> [-force]

   move dev <SymDevName> <DestCgName> [-force]

   disassociate ld <LdevName> [-force]

   move ld <LdevName> <DestCgName> [-force] [-rename]

   copy ld <LdevName> <DestCgName> [-force] [-rename]

symbcv -cg <CgName> [-offline] [-sid <SymmID>]
     [-SA <# | ALL>] [-p <#>] [-N <#>]
     [-cap <#> [-captype <mb> | <cyl>]]
     [-R1 | -NOR1] [-R2 | -NOR2]
     [[-rdf [-bcv]] | [-rrdf] | [-hop2]]
     [-rdfg <GrpNum>] [-remote_rdfg <RemoteGrpNum>]
     [-sel_rdfg <SelRdfGrpNum>] [-force]
     [-devs <<SymDevStart>:<SymDevEnd> | <SymDevName>
         [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]>]
     [-v]

   associateall [devs | -host <HostName>]

   moveall <DestCgName>

   copyall <DestCgName>

   rmall
```

DESCRIPTION

The symbcv command performs operations on a BCV device. The BCV device can be addressed by its physical (host) name or by its device Symmetrix name.

Note that in certain versions of Enginuity, the BCV device must be local to this host before it can be associated with a group. These operations include:

- Associating a BCV device with a device group.
- Associating all devices in a Symmetrix array with a device group.
- Listing the BCV devices.
- Disassociating a BCV device from a device group.
- Associating a BCV device with a composite group.
- Associating all devices in a Symmetrix array with a composite group.
- Disassociating a BCV device from a composite group.

ARGUMENTS

associate	Associates a Symmetrix BCV device with an existing group. Note that add can be substituted for associate.
associateall	Associates all BCV devices with an existing group. Associateall only acts on BCV devices that are not associated with a group. The dev keyword specifies all Symmetrix devices, regardless of whether they are visible to the host. Note that addall can be substituted for associateall.
copy	Copies one BCV device from one existing group to another existing group. The source and destination groups can be different group types.
copyall	Copies all BCV devices from one existing group to another existing group. The source and destination groups can be different group types.
disassociate	Disassociates a BCV device from a group. The group must exist, the BCV device must have been previously associated with the group, and the BCV device must be in a state that allows it to be disassociated. Note that remove can be substituted for disassociate.
list	Lists all BCV devices that are configured on the Symmetrix arrays attached to this host, when used with the dev keyword (default). Lists all BCV devices that are visible to this host, when used with the pd keyword.
move	Moves one BCV device from one existing group to another existing group. The source and destination groups can be different group types.
moveall	Moves all BCV devices from one existing group to another existing group. The source and destination groups can be different group types.

rmall — Removes all BCV devices from an existing group.

KEYWORDS

dev — Indicates a Symmetrix device name.

devs — Indicates multiple Symmetrix device names.

ld — Indicates a logical device name.

pd — Indicates a physical device name.

OPTIONS

-bcv — Indicates that the remotely attached BCV will be paired with the remote mirror of a locally attached BCV RDF device. This option can only be used in conjunction with the -rdf option.

-c — Indicates the number (count) of times to display. If this option is not specified and an interval (-i) is specified, statistics will display continuously.

-cap — Sets a minimum device size to the selection criteria of devices.

-captype <mb | cyl> — Specifies the units of capacity, either megabytes or cylinders. The default is mb.

-cap — Sets the BCV device capacity (size) in MB.

-cg — Specifies the composite group name. Note that the -cg and -g options cannot be used at the same time.

-devs — Specifies the ranges of BCV devices to add, remove, and move. For example, to associate BCV devices 00A to 00D, specify a range of 00A:00D.

-emulation — Lists BCV devices that are clone emulated.

-force — Forces BCV device(s) to be disassociated or moved from a group without querying the Symmetrix array for the device’s BCV pair state(s).

-g — Specifies the device group name. Note that the -cg and -g options cannot be used at the same time.

-h — Provides brief, online help information.

-hop2 — Indicates that the device is two hops away. For add and addall operations, if -hop2 is specified, both -rdfg and -remote_rdfg must also be specified.

-host — Limits devices associated to those mapped to the host’s front-end directors.

-i — Sets the repeat interval in seconds. The default interval is 30 seconds.

The minimum interval is 5 seconds. For passive actions the minimum interval is 15 seconds. Passive actions are actions that do not acquire an exclusive lock.

-N — Sets the number of BCV devices to add, remove, or move to/from a specified group.

-NOR1 — Indicates that BCV RDF R1 devices should not be associated through the SRDF links.

-NOR2 — Indicates that BCV RDF R2 devices should not be associated through the SRDF links.

-offline — Obtains information from the Symmetrix host configuration database.

-p — Supplies the front-end (SCSI or Fibre) director port number only to selected devices that are physically connected through this director port. All ports are selected by default.

-R1 — Indicates that only BCV RDF R1 devices should be associated.

-R2 — Indicates that only BCV RDF R2 devices should be associated.

-resv — Lists BCV devices that have SCSI reservations.

-rdf — Indicates that the BCV is being remotely associated with the group. This option can be used when the BCV device is reachable by the SRDF links and the group is an RDF group.

-rdfg — Indicates the Symmetrix RA (RDF) group number through which the remotely-associated BCV device is reached. It must be the same RA group of the group. This option can only be used with the -rdf or -rrdf options.

-remote_rdfg — Indicates the Symmetrix RA (RDF) group number through which the remotely-associated RBCV device or 2BCV is reached. This option can only be used with the -rrdf and the -hop2 options.

-rename — Renames the BCV device(s)to the default names when they are moved from their current group to the destination group. By default, they will retain their current logical names.

-rrdf — Indicates that the BCV is being remotely associated with a remote BCV in the group. This option can be used when the BCV device, reachable by the SRDF links, is two hops away. The group must be an RDF group.

-SA — Supplies the front-end (SCSI or Fibre) director number.

-sel_rdfg — Indicates the Symmetrix RA (RDF) group number of the devices to be added by the

an associateall operation.

-sid — Supplies the unique Symmetrix ID.

-v — Provides a more detailed, verbose listing.

PARAMETERS

Count — Number of iterations to execute before exiting.

DestCgName — Destination composite group in which to move the BCV device(s).

DestDgName — Destination device group in which to move the BCV device(s).

DgName — Device group name.

GrpNum — RDF (RA) group number.

HostName — Host name.

Interval — Interval between polls, in seconds.

LdevName — BCV logical device name, either named by the user or automatically assigned when a BCV device is associated with a group.

PdevName — Physical device (host) name for the device, such as /dev/rdsk/c2t0d2s2

RemoteGrpNum — Remote RDF (RA) group number.

SelRdfGrpNum — Symmetrix RA (RDF) group number of the devices to be added when using an associateall operation.

SymDevName — Symmetrix device name, unique to each Symmetrix array, such as 01C.

SymDevEnd — Symmetrix device name, ending the contiguous range of selected devices, such as 00C.

SymDevStart — Symmetrix device name, starting the contiguous range of selected devices, such as 00C.

SymmID — 12-digit ID of the Symmetrix array.

RETURN CODES

Code #	Code Symbol
0	CLI_C_SUCCESS
1	CLI_C_FAIL
2	CLI_C_DB_FILE_IS_LOCKED
19	CLI_C_GK_IS_LOCKED All GateKeepers to the Symmetrix array are currently locked.

EXAMPLES

To associate a BCV device with device group ProdDB, enter:

```
symbcv -g ProdDB associate pd /dev/rdsk/c2t0d2s2
```

To associate a BCV device with device group ProdDB, and name it ProdBCV001, enter:

```
symbcv -g ProdDB associate dev 00C ProdBCV001
```

To associate all BCV devices (that are not already associated with any group and are visible to this host) on the specified Symmetrix array with device group ProdDB, enter:

```
symbcv -g ProdDB -sid 87 associateall
```

To list all BCV devices that are visible to a host, enter:

```
symbcv list pd
```

To list all BCV devices regardless of whether they are visible to a host, enter:

```
symbcv list
```

To disassociate a BCV device from device group ProdDB, enter:

```
symbcv -g ProdDB disassociate ld ProdBCV001
```

To associate a BCV device with composite group MyCg, enter:

```
symbcv -cg MyCg associate pd /dev/rdsk/c3t0d2s2
```

To associate a BCV device with composite group MyCg, enter:

```
symbcv -cg MyCg associate dev 00C -sid 87
```

To associate all BCV devices (that are not already associated with any group and are visible to this host) on the specified Symmetrix array with composite group MyCg, enter:

```
symbcv -cg MyCg -sid 87 associateall
```

symcfg

Discovers or displays Symmetrix configuration information. Refreshes the host's Symmetrix database file or removes Symmetrix information from the database file. Rebuilds the set of devices known to the local host.

It can also be used to:

- View or release a hanging Symmetrix exclusive lock.
- Set online or offline one RDF (RA) director.
- Set online or offline one RA or FA director port.
- Display available network services entered in the network service file.
- Display existing UNIX gatekeeper and database semaphores.
- Display application and host registration information.
- Display feature registration information.
- Display host port connection information.
- Display mainframe CU image information.
- Display the state of major components in a configuration (environment data).
- Manage an authorization file.
- Display the list of Enginuity patches installed on the Symmetrix array.
- Verify the state of Virtual Provisioning thin devices.
- Change the lockbox password used to access it in attended mode.
- Reset the Stable System Values (SSVs) saved in the lockbox.
- Verify the Stable System Values (SSVs) saved in the lockbox.
- Display a list of Guest OS containers on the Symmetrix array.
- Display the configuration of a specific Guest OS container on a Symmetrix array.
- Display IP Interface, iSCSI Target and IP Route Information.
- Support storage container feature.
- Display or configure vWitness definitions.
- Display Efficiency reports for the Symmetrix array or SRP.
- Enable or disable array attributes powerpath initiator and host registration on the array
- Display powerpath host registration information.
- Configure Remote Machine Table (RMT) used for RDF configuration through RE directors.
- Display Remote Machine Table (RMT) entries.
- Configure IP Interface on RE and SE director.
- Support Snapshot Protection Policy feature.

SYNOPSIS

```
symcfg -h

symcfg -version | -kit | -db

symcfg

   discover [-pdev [-sid <SymmID>] | -sid <SymmID>]
     [-cache | -nocache]
```

```
   scan

symcfg

   sync [-sid <SymmID>] [-rdf | -bcv | -local | -dirsts |
     -snap | -cfgmgr | -rcopy | -env_data | -sg |
     -fast | -tier | -vpdata | -masking]

   remove [-sid <SymmID>] [-noprompt]

symcfg -sid <SymmID> [-force] [-noprompt]

   release -lockn <# | RDFA | RDF | SRDF_MSCS | GNS |
                       FAST | SL | ALL>

symcfg -sid <SymmID> [-i <Interval>] [-c <Count>]
     <-pool <PoolName>                |
      -g <DgName>                     |
      -sg <SgName>                    |
      -cg <CgName>                    |
      -devs <<SymDevStart>:<SymDevEnd> | <SymDevName>
         [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]>>

   verify -tdev
     <-bound | -binding | -allocating | -deallocating |
      -unbound | -unbinding | -reclaiming | -compressing |
      -uncompressing | -freeingall>

symcfg -sid <SymmID> [-i <Interval>] [-c <Count>]
     <-pool <PoolName>                |
      -devs <<SymDevStart>:<SymDevEnd> | <SymDevName>
         [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]>>

   verify -datadev
     <-draining | -drainwait | -disabled | -enabled |
      -deactivated | -nonpooled | -balancing>

symcfg -sid <SymmID> [-i <Interval>] [-c <Count>]
     -pool <PoolName>

   verify -poolstate
     <-enabled | -disabled | -balancing>

symcfg [-sid <SymmID>] [-offline]

   list [-DA  <# | ALL>] [<-v | -port [-p <#>]>]

   list [-DX  <# | ALL>] [<-v | -port [-p <#>]>]

   list [-dir <# | ALL>] [-address [-available] [-fibre]]
        [-p <#>]

   list [-dir <# | ALL>] [-v [-ALL]] [-p <#>]

   list [-dir <# | ALL>] [-port [-[no]virtual] [-p <#>]]

   list [-EF  <# | ALL>] [<-v | -port [-p <#>]>]

   list [-EF  <# | ALL> [-p <#>]] [-address [-available]]

   list [-FA  <# | ALL> [-p <#>]] [-address [-available]]

   list [-FA  <# | ALL>]
        [<-v [-ALL] | -port [-[no]virtual] [-detail]
        [-p <#>]>]
```

```
   list [-FN  <# | ALL> [-p <#>]] [-address [-available]]

   list [-FN  <# | ALL>]
        [<-v [-ALL] | -port [-[no]virtual] [-detail]
        [-p <#>]>]

   list [-RA  <# | ALL> [<-v | -p <#>>] [-switched]]
        [-rdfg <# | ALL>]

   list [-RE  <# | ALL>] [<-v | -port [-detail] [-p <#>]>]

   list [-RF  <# | ALL>] [<-v | -port [-p <#>]>]

   list [-SE  <# | ALL>] [<-v | -port [-detail] [-p <#>]>]
        [-address [-available]] [-iscsi_port <#>]

symcfg [-sid <SymmID>] [-offline]

   list -port -free

        [[-slot <#>] [-dx | -fa | -fcoe | -re | -rf | -se | -fn]
         [-speed <#>]] | [-dir <#>]

symcfg [-sid <SymmID>]

   list -host_cache

   list -memory [-sid <SymmID>] [-offline]

   list -rdfg <# | ALL> [-dynamic | -static] [-offline]
     [-rdfa | -detail | -metro]

   list -status

   list -upatches

   list -sg_compression [-by_compressibility] [-all]
     [-srp <SRPName>] [-gb | -tb]

   list -vasa -rdfg <# | ALL>

symcfg [-sid <SymmID>] [-v]

   list [-LRU <# | ALL>] [-offline]

   list [-lock]

   list [-lockn <# | RDFA | RDF | SRDF_MSCS | GNS |
                    FAST | SL | ALL>]

   list [-ssid] [-offline]

   list [-connections [-sorthost] [-capacity] [-offline]
        [-ipv6]]

   list [-applications [-client] [-host <HostName>]
         [-offline]]

   list [-features [-class <ClassName>]
        [-disabled | -enabled]
        [-blocked | -unblocked]]

   list [-cuimage]

   list [-env_data [-service_state [not]degraded |
        [not]failed | [not]normal][-offline]]
```

```
    list -ppreg [-host <Hostname>]

    list -container [-v] [-type <ContainerType>]
         [-dir <# | ALL>]

    show -container <ContainerName>

    show -applications <AppID> [-client] [-host <HostName>]

    show -cuimage <CuImage_Number> [-ssid_num <SSID>]

    show -env_data <BayName> [-offline]

 symcfg [-sid <SymmID>] [-offline]

    list [-bay_info]

 symcfg [-v]

    list [-services] [-offline]

    list [-semaphores]

 symcfg [-sid <SymmID>] [-offline] [-mb | -gb | -tb]
        [-i <Interval>] [-c <Count>]

    list [-pool [-snap | -rdfa_dse [-rdfg <GrpNum>]] [-v]
      [-fba] [-ckd] [-ckd3390] [-ckd3380] [-as400] [-all]]

    list [-pool -thin [-fba] [-ckd] [-all] [-v]]

    list [-pool -thin -detail [-fba] [-ckd]]

    list [-savedev [-fba] [-ckd3390] [-ckd3380] [-as400]
      [-nonpooled]
      [-devs <<SymDevStart>:<SymDevEnd> | <SymDevName>
         [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]>]

    list [-datadev [-fba] [-ckd] [-nonpooled]
      [-devs <<SymDevStart>:<SymDevEnd> | <SymDevName>
         [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]>]

    list [-tdev
      [-devs <<SymDevStart>:<SymDevEnd> | <SymDevName>
         [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]>]
      [-pool <PoolName> | [-fba] [-ckd3390]
         [-bound | -unbound]]
      [-sg <SgName>] [-detail | -tier]]

    show -pool <PoolName> <-snap | -rdfa_dse | -thin>
      [-all] [-fba | -ckd3390 | -ckd3380 | -as400]

    show -pool <PoolName> -thin -detail [-all]

symcfg [-sid <SymmID>] [-offline]

    list -sl [-detail] [-by_resptime] [-all] [-v]
      [-fba|-ckd] [-gb | -tb]

    list -srp [-detail] [-rdfa_dse] [-v] [-fba|-ckd]
      [-gb | -tb]

    list -srp -demand [-type <sl | sg>] [-detail] [-fba|-ckd]
      [-gb | -tb]

    list -srp -sized
```

```
    list -tdev -srp
      [-devs <<SymDevStart>:<SymDevEnd> | <SymDevName>
         [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]>]
      [-sg <SgName>] [-fba|-ckd] [-gb | -tb]

    list -demand [<-detail | -md | -v>] [-gb | -tb]

    show -sl <SLName> [-fba|-ckd] [-gb | -tb]

    show -srp <SRPName> [-detail] [-fba|-ckd] [-gb | -tb]

symcfg [-sid <SymmID>] [-offline]

    list -policy -type <snap> [-detail] [-v]

    show -policy <PolicyName> -type <snap>

symcfg [-sid <SymmID>] [-offline]

    list -ip_interface [-SE <#|ALL>] [-p <#>] [-by_ip]

    list -ip_interface [-RE <#|ALL>] [-p <#>] [-by_ip]

    list <-SE <#|ALL>>
      <-iscsi_tgt
        [-iqn <TargetIQN> | -iscsi_port <#> | -bootstrap]
        [-by_iqn] [-detail]>

    list -route [-SE <#|ALL>] [ -ipv4 | -ipv6 ] [-v]

    list -route [-RE <#|ALL>] [ -ipv4 | -ipv6 ] [-v]

    list [-SE  <# | ALL>]
      [<-v | -port [-detail] [-p <#>]>]
      [-address [-available]] [-iscsi_port <#>]

symcfg [-sid <SymmID>] [-offline]

    list -ficon_split [-address [-available]] [-v]

    show -ficon_split <FiconSplitName>

symcfg [-sid <SymmID>] [-offline]

    list -efficiency

    list -srp -efficiency

 symcfg -RA <#> [-p <#>] -sid <SymmID> [-noprompt] [-v]

    online

    offline

 symcfg -EF <#> -p <#> -sid <SymmID> [-noprompt] [-v]

    online

    offline

 symcfg -FA <#> -p <#> -sid <SymmID> [-noprompt] [-v]

    online

    offline

 symcfg -FA <#> -p <#> -sid <SymmID> [-noprompt]
```

```
   set -fa_loop_id <0-125>

   enable -port_flag <<flag>,<flag>,..>

   disable -port_flag <<flag>,<flag>,..>

   set -topology <TopoType>

symcfg -FN <#> -p <#> -sid <SymmID> [-noprompt]

   enable -port_flag ShowACLX

   disable -port_flag ShowACLX

symcfg -SE <#> <-p <#> | -iscsi_port <#> | -iqn <TargetIQN>>
       -sid <SymmID> [-noprompt] [-v]

   online

   offline

symcfg -FN <#> -p <#> -sid <SymmID> [-noprompt] [-v]

   online

   offline

   set -topology <TopoType>

symcfg -RF <#> -p <#> -sid <SymmID> [-noprompt]

   set -topology <TopoType>

symcfg

   authorization list <-vmware | -hyperv | -smi | -snmp>
      [-v]

   authorization <add | update> -vmware -host <HostName>
      -username <UserName> [-password <PassWord>]
      [-namespace <NameSpace>] [<-vmport | -port> port]

   authorization <add | update> <-hyperv | -smi>
      -host <HostName> -username <UserName>
      [-password <PassWord>] [-namespace <NameSpace>]
      [-port port]

   authorization <add | update> -snmp -host <HostName>
      -username <UserName> [-password <PassWord>]
      [-namespace <NameSpace>] [-port port]
      [-key <PrivKey>]

   authorization delete -host <HostName>
      <-vmware | -hyperv | -smi | -snmp>
      [-namespace <NameSpace>]

symcfg [-sid <SymmID>] [-i <Interval>] [-c <Count>]
       [-percent <1-100> -action <ScriptFile> [-norepeat]]
       [-snap | -rdfa_dse | -thin] [-pool <PoolName>]
       [-mb | -gb | -tb]

   monitor

symcfg -sid <SymmID>

   set -led < on | blue_flash >
```

```
    set -led < on | slow_blink > -bay_name <BayName>

symcfg -lockbox [-password <PassWord>]

    reset -ssv

    setpw [-new_password <NewPassWord>]

symcfg -lockbox

    verify -ssv

symcfg -sid <SymmID> -sc

    create -name <StorageContainer>
      -type vvols
      [-description <Description>]

    delete -sc_name <StorageContainer>

    set -sc_name <StorageContainer>
      <-description <Description>>

symcfg -sid <SymmID> -sc -sc_name <StorageContainer>
       [-noprompt]

    add -sresource <StorageResourceName>
      <-sl <SLName> [-wl <WorkloadName>]>
      [-srp <SRPName>] [-nocompression]
      <-subscribed_max <GB>>

    remove -sresource <StorageResourceName>

    set -sresource <StorageResourceName>
      <-subscribed_max <GB>>

symcfg -sid <SymmID> -sc [-offline] [-tb]

    list [-v] [-detail]

    show -sc_name <StorageContainer> [-detail]

symcfg -sid <SymmID>

    create -policy <PolicyName> -type <snap>
      -interval [[<D>:]<HH>:]<MM>
      -max_count <#> [-secure <yes | no>]
      [-offset [[<D>:]<HH>:]<MM>]
      [-critical_compliance <#>]
      [-warning_compliance <#>]

    delete -policy <PolicyName> -type <snap>

    set -policy <PolicyName> -type <snap>
      <[-interval [[<D>:]<HH>:]<MM>]
       [-max_count <#>]
       [-offset [[<D>:]<HH>:]<MM>]
       [-critical_compliance <#> | -no_critical_compliance]
       [-warning_compliance <#> | -no_warning_compliance]
       [-new_name <SnapSLName>]>

    suspend -policy <PolicyName> -type <snap> -sg <SGName>

    suspendall -policy <PolicyName> -type <snap>

    suspendall -policy -type <snap> -sg <SGName>
```

```
   resume -policy <PolicyName> -type <snap> -sg <SGName>

   resumeall -policy <PolicyName> -type <snap>

   resumeall -policy -type <snap> -sg <SGName>

symcfg -sid <SymmID>

   add     -witness <WitnessName> -location <DNSorIPAddr>
   remove  -witness <WitnessName>
   enable  -witness <WitnessName>
   disable -witness <WitnessName> [-force] [-symforce]

   enable   -ppath_initiator_registration
   disable  -ppath_initiator_registration

   enable   -ppath_host_registration
   disable  -ppath_host_registration

   set      -ppath_initiator_registration
   reset    -ppath_initiator_registration

   set      -ppath_host_registration
   reset    -ppath_host_registration

   set      -sl_rt_multiplier <none | low | default>

symcfg
   list    -witness [-sid <SymmID>] [-offline] [-v]
   show    -witness <WitnessName> [-sid <SymmID>] [-offline]

symcfg -sid <SymmID> -rmt

   create -remote_sid <SymmID>
     -remote_dir <#> -remote_p <#>
     <[-ipv4_address <IPAddress>]
      [-ipv6_address <IPAddress>]>

   modify -remote_sid <SymmID>
     -remote_dir <#> -remote_p <#>
     <[-ipv4_address <IPAddress>]
      [-ipv6_address <IPAddress>]>

   delete -remote_sid <SymmID>
     -remote_dir <#> -remote_p <#>
     <[-ipv4_address <IPAddress>]
      [-ipv6_address <IPAddress>]>

symcfg [-sid <SymmID>]
   list -rmt

symcfg -RE <#> -p <#>
       -sid <SymmID> [-noprompt] [-v] -ip_interface

   create -ip_address <IPAddress> -ip_prefix <IPPrefix>
     [-default_gateway <DefaultGateway>] [-mtu <MTU>]

   modify -ip_address <IPAddress>
     <[-new_ip_address <IPAddress>]
      [-ip_prefix <IPPrefix>] [-mtu <MTU>]>

   delete -ip_address <IPAddress>

symcfg -SE <#> -p <#>
       -sid <SymmID> [-noprompt] [-v] -ip_interface
```

```
   create -ip_address <IPAddress>
     -ip_prefix <IPPrefix> -network_id <NetworkId>
     -vlan_id <VlanId> [-mtu <MTU>]

symcfg -SE <#>
       -sid <SymmID> [-noprompt] [-v] -ip_interface

   modify -ip_address <IPAddress>
     -network_id <NetworkId>
     <[-new_network_id <NetworkId>]
      [-new_ip_address <IPAddress>]
      [-ip_prefix <IPPrefix>]
      [-mtu <MTU>]>

   delete -ip_address <IPAddress>
     -network_id <NetworkId>
```

DESCRIPTION

The symcfg command is used to discover a Symmetrix configuration, refresh the host's Symmetrix configuration database file, and display configuration information about the Symmetrix arrays and any of its directors attached to the host.

It includes the ability to rebuild the set of physical disks that are known to the host: (symcfg scan).

Directed-Symmetrix discovery is performed by specifying the SymmID, and if Symmetrix devices are freshly mapped to the host (and any host OS required scan operations are performed), the -pdev flag can be used to bring the new pdevs into the configuration database.

It can be used to view whether the specified Symmetrix array(s) have an exclusive Symmetrix lock. You can release a lock if it is determined to be hanging.

The symcfg command can be used to set one RDF RA director or one RA or FA port on a locally attached Symmetrix array to either online or offline. The symcfg command can also be used to display the LRU cache management configuration.

The symcfg command can also be used to list the services entered in the network services file (netcnfg) or list all the UNIX gatekeeper, database, and lock file semaphores. It can display application and host registration, and port connection information. The symcfg command can also list feature registration data, including usage information and capacity limits.

In addition, some arrays require authorization information to access the array. The symcfg authorization command is used to supply this information for use in subsequent discovery operations. The symcfg authorization command allows you to list, add, update, or delete this connectivity information. The update option allows you to update the password of an existing entry.

If a configuration has devices mapped to EF (FICON) front-end director, the CU image information can be viewed.

Larger Symmetrix arrays contain a system bay and a set of drive bays. Environmental data and status for internal modules can be viewed/synched using the -env_data option.

This requires Enginuity 5771 or above.

When the host is upgraded or when the lockbox file has been moved to another host, the host fingerprint may no longer match and the Stable System Values (SSVs) saved in the lockbox. They must be reset before the lockbox being accessible from Solution Enabler. To reset the SSVs in the lockbox, the password for the lockbox is required. To better protect the contents in the lockbox, a new password can also be set after its first creation with a default password.

ARGUMENTS

add — Adds the Storage Container Resources to the specified Storage Container.
Add a vWitness to the array.

authorization — Supplies and manages the connectivity information required to communicate with either:

- certain storage arrays
- the Virtual Infrastructure Service
  (Virtual Center Server or ESX).

The symcfg authorization command allows you to list, add, update, and delete this connectivity information.

This connectivity information consists of the following data:

Information	Option to Use
agent address	[-host]
user name	[-user]
password	[-password]
port	[-port]
namespace	[-namespace]
SNMPv3 passphrase	[-key]

The agent address, user name, and password data information is required. While the the password can be entered via the command line option it is recommended to not specify the option on the command line and have the command prompt for the password. This is a more secure way of entering the password. The port is useful for storage arrays.

The optional namespace parameter, if present, qualifies the agent address in some way. If it was provided when authorization information was originally defined, it must be provided during any subsequent update or delete operations.

The SNMPv3 passphrase is only used for authorization records that will be used by SNMP clients to authorize them to receive SNMP traps. While the SNMPv3 passphrase can be entered via the command line option, it is recommended to not specify the option on the command line and have the command prompt for the passphrase. This is a more secure

way of entering the password.

An authorization record is considered a match if the type, hostname, username, and namespace match the criteria provided for adding, updating, or deleting a record.

create — Used with -rmt to create a new entry in the Remote Machine Table.

Maximum of 32 Remote Machine entries can be created on the specified Symmetrix. The remote machine entry can be used for Fibre (RF), GigE (RE) or both. The entry used for Fiber (RF) can be reused for GigE (RE).

Used with -ip_interface to create IP interface for a specified RE or SE director port.

Used with -policy to create a Snapshot Protection Policy. A maximum of 20 Snapshot Protection Policies can be created on the specified Symmetrix array.

delete — Deletes the specified Storage Container.

Used with -rmt to delete remote target from the RMT entry. The RMT entry will be automatically deleted when the last remote target gets deleted

Used with -ip_interface to delete an IP address on the specified RE or SE director port. For SE director port, additional network ID option is needed.

Used with -policy to delete the specified Snapshot Protection Policy.

disable — Disables the specified vWitness on the array.

Disables powerpath initiator and host registration on the array.

Used with -port_flag option to disable the FA|FN port flags.

discover — Scans all devices on the host looking for Symmetrix devices and builds (or rebuilds) the Symmetrix host database. If the Symmetrix array is reconfigured by adding or removing devices that it sees, then run the discover command before running sync to obtain accurate information.

This command interrogates all SCSI devices and can take a significant amount of time to complete. If you need to update the device configuration status information, the sync action is more efficient.

The -cache option attempts to recover the results of a previous SCSI interrogation from a running base daemon for increased speed. Configuration data

is always retrieved from cache, if possible.

Note: If you had previously run discover and had subsequently removed Symmetrix array(s), a later execution of discover will not remove from the the database information relating to the removed Symmetrix array(s).

For selected Symmetrix array discovery, use the -sid option.

enable — Enables the specified vWitness on the array.

Enables powerpath initiator and host registration on the array.

Used with -port_flag option to enable the FA|FN port flags.

list — Lists brief or detailed information about the Symmetrix configuration. Can also be used to view whether one or more Symmetrix arrays have an exclusive lock, host registration data, and application and feature registration data.

Using the -lock option with list, you can view whether one or more Symmetrix arrays have an external lock held. By default, only lock 0 is checked. The -lockn ALL command checks for all Symmetrix external locks known to SYMAPI. The -lockn RDF command checks for locks specific to RDF. The -lockn GNS command lists the GNS specific locks. The -lockn SRDF_MSCS lists the SRDF/CE for MSCS specific locks. The -lockn FAST command lists the FAST Hint Manager specific locks.

In addition, the list action can list network services available and the state of gatekeeper and database semaphores. The -rdfg option displays a list of RDF groups for the specified Symmetrix array.

Lists the Storage Container defined on the Symmetrix array.

Lists the Efficiency reports for the Symmetrix array or SRP.

Lists the vWitness definitions on the array. The -v option provides information in the same format as the show command.

Lists the Remote Machine Table entries.

Lists the Snapshot Protection Policies defined on the Symmetrix array.

modify — Used with -rmt to modify IP addresses; IPv4 and/or IPv6 of an existing RMT entry. It can also be used to add remote target.

	Used with -ip_interface to modify IP interface properties such as IP address, Maximum Transmission Unit (MTU), netmask or IP prefix length and network ID.
monitor	Checks the total percent full of the devices in SNAP, RDFA_DSE or THIN pools and can optionally execute a script file if a specified percentage is encountered.
offline	Sets one RDF RA director or one RA or FA director port on a locally-attached Symmetrix array to the offline status. Note: Use caution when applying this action.
online	Sets one RDF RA director or one RA or FA director port on a locally-attached Symmetrix array to the online status. Note: Use caution when applying this action.
release	Releases any existing Symmetrix exclusive locks from the specified Symmetrix array(s). WARNING: Use this action ONLY if you are SURE that no operation using these locks is currently in progress.
remove	Removes all information about the specified Symmetrix array from the host database. If there are more than one Symmetrix array attached, then information and definitions about all arrays are removed. Remvoes the Storage Container Resources from the specified Storage Container. Removes the specified vWitness from the array.
reset	Resets the Stable System Values (SSVs) saved in the lockbox. This operation is required when the host fingerprint has changed, which usually occurs when the host is upgraded or the lockbox file has been moved to another host.
resume	Resumes the specified suspended Snapshot Protection Policy associated with an SG.
resumeall	When used with -policy <PolicyName> and the -sg option is not specified, resumes the specified suspended Snapshot Protection Policy. When used with both -policy -type snap and -sg <SgName>, resumes all suspended Snapshot Protection Policies that are associated with an SG.
scan	Results in a list of devices accessible to the host system. It should be initiated when the set of devices that a host can access has changed. Follow this action with a discover if the device changes are associated with Symmetrix devices. Also see the -pdev option.

set	Used with -led to change the state of system bay or drive bay LEDs. Used with -sc to set the Storage Container description or set the maximum amount of subscribed storage in GB that can be provisioned on the Storage Container. Used with -fa_loop_id to set the FA port address. Used with -policy to modify the attributes of the specified Snapshot Protection Policy.
setpw	Sets a new password for the lockbox.
show	Shows detailed application registration data, CU image definitions, environment data, or a SAVE device pool for a specific system or drive bay, or Storage Container, or Efficiency reports or vWitness definitionsn, or Snapshot Protection Policy.
suspend	Suspends the specified Snapshot Protection Policy associated with an SG.
suspendall	When used with -policy <PolicyName> and the -sg option is not specified, suspends the specified Snapshot Protection Policy. When used with both -policy -type snap and -sg <SgName>, suspends all Snapshot Protection Policies that are associated with an SG.
sync	Refreshes the Symmetrix configuration database file with data from the arrays. The Symmetrix configuration must have been previously discovered, using the discover action. If you reconfigure your Symmetrix array by adding or removing devices that the host sees, you need to run a discover before a symcfg sync to obtain accurate information.
verify	Without the -tdev, -datadev or -lockbox and -ssv flags, verifies whether the Symmetrix configuration and the Symmetrix configuration database file are in sync. With the -tdev flag, verifies whether one or more named device(s) (with the -devs flag), standard devices in a composite group (with the -cg flag), devices in a storage group (with the -sg flag), or all Symmetrix thin devices in a named pool are in a given state. With the -datadev flag, verifies whether one or more named device (with the -devs flag) or all Symmetrix data devices in a named pool are in a given state. With the -lockbox and -ssv flags, verifies whether the Stable System Values (SSVs) in the lockbox are consistent with the current system values, or if they have to be reset.

OPTIONS

-action Selects a script to run when the specified percent value is encountered. The full pathname to the action script needs to be specified. The first argument passed to the script is automatically set to the percent value. This option requires the -percent option and is only valid with the monitor command.

-address Lists the Vbus, TID, and LUN addresses associated with devices mapped to the front-end directors.

-all If used with list -v on a front-end director, all port flags will be listed. If used with pool operations, both active and inactive information about the pools will be included in the display.

If used with 'list -sl', it specifies that all the Service Levels configured in the array be listed.

If used with '-sg_compression', it specifies that the data compressibility report will include compression enalbed SGs.

-allocating Specifies that the applicable Symmetrix thin devices must be in the "allocating" state for the verification to return success.

-applications Lists the application registrations sorted by Symmetrix ID. Only those applications that have been run will be listed.

-as400 Specifies pools or SAVE devices with emulation type AS400.

-available Requests the next available Vbus, TID, or LUN address be appended to the output list. Used with the -address option.

-balancing Verifies devices that are balancing.

-bay_info Displays the name of each system and drive bay along with bay location information and bay LED state.
Requires Enginuity level 5875 and higher.

-bay_name Specifies the system bay or drive bay for control of the LED.

-bcv Refreshes (synchronizes) the configuration database file with BCV information from the Symmetrix array.

-binding Specifies that the applicable Symmetrix thin devices must be in the "binding" state for the verification to return success.

-blocked Displays only blocked features.
Not compatible with -unblocked.

-bootstrap When listing, lists the bootstrap iSCSI target configured on the Symmetrix if one exists.

-bound When listing, lists devices that are bound to a thin pool. When verifying, specifies that the applicable Symmetrix thin devices must be in the "bound" state in order for the verification to return success.

-by_compressibility
Specifies that the data compressibility report will be sorted in descending order.

-by_ip When listing IP addresses configured on a specified director port, display the list in sorted order by IP addresses followed by director and port.

-by_iqn When listing iSCSI targets, display the list in sorted order by iSCSI target IQN name.

-by_resptime When listing Service Levels (SLs), display the list of Service Levels in sorted order by response time, it can be abbreviated to 6 characters including the '-' character.

-c Specifies the number (count) of times to execute an exclusive lock or thin device verification operation. If this option is not specified but an interval (-i) is specified, the program will loop continuously. The looping may be terminated during verification if all devices enter the requested state.

-cache Attempts to recover the results of a previous SCSI interrogation from a running base daemon for increased speed. Configuration data is always retrieved from cache, if possible.

-capacity Shows the total Symmetrix storage space connected to each registered host, based on devices mapped to director/port. For a list of masked devices per host, use the symmaskdb list capacity command.

-cfgmgr Refreshes the SYMAPI configuration database file with disk space and Symmetrix configuration metrics gathered from the Symmetrix configuration server.

-class Displays only features in the specified class.

-client Directs the application registration list and show commands to reference the client application table, instead of referencing the SYMAPI-generated application table.

-ckd3380 Specifies pools or SAVE devices with emulation type CKD3380.

-ckd3390 Specifies pools or SAVE devices with emulation type CKD3390.

-compressing Specifies that the applicable Symmetrix thin devices must be in the "compressing" state for the verification to return success.

-connections Applies the list action to display the host-to-Symmetrix connectivity, sorted by Symmetrix ID. Only those hosts that have at least one registered application will be listed.

-container Applies the list or show action to display configuration information on Guest OS containers on the specified Symmetrix.

When used with the show command, this option takes an optional argument to specify the name of the container to be displayed.

-critical_compliance
Specifies the critical compliance count attribute of the specified Snapshot Protection Policy. The compliance count cannot be set to 0 and must be less than or equal to the maximum count of the Snapshot Protection Policy. If the warning compliance count is also set, the critical compliance count must be less than or equal to that.

-CUimage Lists or shows mainframe CU image information.

-DA Limits the action to a disk director number. To select all disk director numbers, specify ALL.

-datadev Displays information about or verifies the DATA devices.

-db Displays Symmetrix configuration database information.

-deactivated Specifies that the applicable Symmetrix data devices must be in the "deactivated" state for the verification to return success.

-deallocating Specifies that the applicable Symmetrix thin devices must be in the "deallocating" state for the verification to return success.

-default_gateway
Specifies the gateway address.

-demand Specifies a demand report be generated and and can be abbreviated to 4 characters.

-description Specifies the Storage Container description.

-detail Displays detailed information for thin devices, RDF groups, pool, SRP or Service Level. Used with -port in order to request that FA-specific or SE-specific flag settings be displayed.

-devs Specifies a set of Symmetrix device ranges

and/or individual Symmetrix devices.

-dir — Limits the action to a director number. To select all director numbers, specify ALL.

-dirsts — Refreshes the configuration database file with director status information from the specified Symmetrix array.

-disabled — When displaying features, shows only disabled features. Not compatible with -enabled.

When verifying thin devices, specifies that the applicable Symmetrix data devices must be in the "disabled" state in order for the verification to return success.

-draining — Specifies that the applicable Symmetrix data devices must be in the "draining" state in order for the verification to return success.

-drainwait — Specifies that the applicable Symmetrix data devices must be in the "drainwait" state in order for the verification to return success.

-DX — Limits the action to the DX director. To select all DX director numbers, specify ALL.
Use with -free in order to restrict the display of free ports to those which support the DX interface type.

-dynamic — Lists dynamic RDF groups.

-EF — Limits the action to a FICON (Fibre-ESCON) director number. To select all FICON director numbers, specify ALL.

-enabled — Display only enabled features. Not compatible with -disabled.

-env_data — Displays or synchronizes the status of the major hardware modules comprising a Symmetrix system with Enginuity level 5771 or above. Verbose or summarized formats are available when displaying data.

-FA — Identifies the front-end (Fibre) director number. Use ALL to return data for all available Fibre front-end directors.
Use with -free in order to restrict the display of free ports to those which support the FA interface type.

-fa_loop_id — FA port address, between 0 and 125.

-fast — Refreshes (synchronizes) the configuration database file with FAST information gathered from the Symmetrix array.

-fba — Specifies pools or SAVE devices with emulation type FBA.

-fcoe Use with -free in order to restrict the display of free ports to those which support the Fibre Channel Over Ethernet interface type.

-features Lists the feature registrations and usage data. Where appropriate, capacity types and limits are also displayed. Usage information is displayed when the -v option is provided.

-fibre Confines the front-end information output to Fibre directors only. Used with the -address option.

-ficon_split Lists/Shows information for FICON Splits.

-force Forces the release of a Symmetrix external lock (EMC use only). Currently, releasing lock 15 (Symmetrix configuration lock) requires the use of this option.

CAUTION: Use this action ONLY if you are SURE that no operations using these locks are in progress.

Forces the disabling of a vWitness if it is in use and if there is an alternate Witness

-free Use with -port in order to request a display of ports which are currently available to be associated with a director.

-freeingall Specifies that the applicable Symmetrix thin devices must be in the "freeingall" state for the verification to return success.

-gb Lists capacity in gigabytes.

-h Provides brief online help information.

-host Lists only application information for the specified host or specifies a host name during authorization actions.

-host_cache Lists detailed information about host cache cards for each host associated with the specified Symmetrix. A list of devices registered for control by each host cache card is provided.

-hyperv Indicates that the supplied information is for Hyper-V Virtual Infrastructure Service. Used with the authorization action.

-i Specifies the repeat interval in seconds to execute an exclusive lock or thin device verification operation.
The default interval is 30 seconds.
The minimum interval is 5 seconds.
For passive actions the minimum interval is 15 seconds. Passive actions are actions that do not acquire an exclusive lock.

-interval Specifies the interval in minutes to take

the snapshots for a specified Snapshot Protection Policy. The format must be in minutes, HH:MM or D:HH:MM. The only allowable interval values in minutes are: 10, 12, 15, 20, 30, 60, 120, 180, 240, 360, 480, 720, 1440 and 10080 minutes.

-ip_address — Specifies a valid IPv4 or IPv6 address.

-ipv4_address — Specifies a valid IPv4 address.

-ipv6 — Specifies that the output display should not truncate IPv6 node names or addresses.

-ipv6_address — Specifies a valid IPv6 address.

-ip_interface — Specifies the operation to configure IP Interface on RE or SE director port. -ip_interface can be abbreviated to 3 characters including the '-' character. The IP interface consists IP address, Maximum Transmission Unit (MTU), netmask or IP prefix length, network ID, vlan ID and default gateway. The IP address could be IPv4 or IPv6 format.
The network ID and vlan ID are not applicable to RE director port.
The default gateway is not applicable to SE director port.

-ip_prefix — Specifies the IPv4 or IPv6 prefix length of IP interface.

-iqn — Only the iSCSI targets configured with the specified IQN name will be displayed.

-iscsi_port — Only the iSCSI targets configured on that iscsi virtual port will be displayed.

-iscsi_tgt — Only the iSCSI targets configured on the Symmetrix will be displayed.

-key — The passphrase associated with the user supplied by the -username option. The option is optional. If it is not specified on the command line the passphrase is prompted for. The characters entered are not displayed for additional security.

-kit — Lists the SYMAPI kit details.

-local — Refreshes (synchronizes) the configuration database file with local Symmetrix information.

-led — Used with the set keyword to change the state of system bay or drive bay LEDs.

-lock — Displays whether the Symmetrix array has an exclusive lock.

-location — Species the location of the vWitness

-lockbox — Manages accessibilty of the lockbox from Solution Enabler:

- reset SSVs saved in the lockbox.
- set a new password for the lockbox.
- verify SSVs saved in the lockbox.

-lockn
Specifies the Symmetrix external lock (SEL) number. You can choose to set a specific lock number to return, and if this option is not specified, the lock number defaults to 0. Optionally, you can return only RDF locks by specifying RDF, RDFA locks by specifying RDFA, SELs used by GNS by specifying GNS, or the SRDF/CE for MSCS locks by specifying SRDF_MSCS. To list all locks, specify ALL.

-LRU
Lists the cache-slot allocation and allocation percentage of a specified LRU cache management group number, when used with the list action. Use ALL to list all the LRUs.

-masking
Refreshes (synchronizes) the configuration database file with masking information gathered from the Symmetrix array.

-max_count
Specifies the maximum number of snapshots that should be maintained for a specified Snapshot Protection Policy. The maximum count must be betweeen 1 to 1024.

-mb
Lists capacity in megabytes.

-md
Lists meta data usage in the demand report.

-memory
Displays information about the memory boards.

-metro
Displays RDF Metro information.

-mtu
Specifies the maximum transmission unit (mtu) value.

-name
Specifies the Storage Container name that will be created.

-namespace
Specifies a namespace with authorization operations.

-new_ip_address
Specifies new IPv4 or IPv6 address to be modified for IP Interface on RE or SE director port.

-new_name
Specifies a new Snapshot Protection Policy name.

-new_password
The new password for the lockbox. This option has to be used in combination with the -lockbox option. If it is not specified on the command line the new password is prompted for. The characters entered are not displayed for additional security.

-new_network_id
Specifies new network ID to be modified for IP Interface on SE director port.

-network_id Specifies the network id for iSCSI IP interface.

-nocache Bypasses the cache and rescans the devices for configuration information.

-nocompression Disables the compression setting of the the storage container resource when adding to the storage container.

-nonpooled Lists or verifies devices that are ready to be assigned to a pool.

-noprompt Requests that no prompts are returned after the command is entered. The default is to prompt the user for confirmation.

-norepeat Specifies that the action script should be run only once if the threshold has been met. Used with the action script option on the monitor command.

-novirtual Use with -port in order to restrict the display to physical ports only.

-no_critical_compliance
Specifies that no critical compliance count should be used.

-no_warning_compliance
Specifies that no warning compliance count should be used.

-offline Sets the operation to work in offline mode, utilizing the host configuration database exclusively.

-offset Specifies the exact time the snapshots will be taken for a specified Snapshot Protection Policy. The offset must be less than the interval of the Snapshot Protection Policy. The format must be in minutes, HH:MM or D:HH:MM

-p Identifies a specific port number to be used.

-password The password associated with the user supplied by the -username option. The option is optional. If it is not specified on the command line the password is prompted for. The characters entered are not displayed for additional security.

When this option is used in combination with -lockbox option, the input password is for the lockbox. If it is not specified on the command line the password is prompted for as above.

-pdev For a discover operation, this limits the work performed to collect only the pdev information. In cases where the Symmetrix configuration was changed, a full discovery is performed. When used with the -sid option, only pdevs for the specified Symmetrix array are updated.

-percent Causes the action script to be executed when the percent full argument is encountered.

-policy Specifies the Snapshot Protection Policy.

-pool Specifies the pool type.

When used with the monitor command, this option takes an argument to specify the pool to be monitored.

-poolstate Verifies the state of the named thin pool.

-port Requests a display of port-level information.

When used with the authorization command, this takes an argument to provide the port at which the agent is listening.

When used with -free, lists information about ports available to be associated with a director.

-port_flag FA port flag name. FA port flags include VSA, NonPart, ACLX, OVMS, or ShowACLX.

Only ShowACLX port flag is supported for FN.

-ppath_host_registration
Specifies array attribute powerpath host registration to be enabled or disabled.

-ppath_initiator_registration
Specifies array attribute powerpath initiator registration to be enabled or disabled.

-ppreg Displays powerpath host registration records.

-RA Limits the action to an RDF director number. To select all RDF director numbers, specify ALL.

-rcopy Refreshes (synchronizes) the configuration database file with remote copy session information gathered from the Symmetrix array.

-rdf Refreshes (synchronizes) the configuration database file with RDF information from the Symmetrix array(s) and the attached remote Symmetrix array(s).

-rdfa Displays SRDF/Asynchronous information.

-rdfg If given an RDF group number, the particular RDF group is displayed; if ALL is specified, all RDF groups are displayed.

-rdfa_dse Displays SRDF/Asynchronous DSE pool information.

When used with ’list -srp’, display only SRPs that can be used for RDFA DSE spillover.

-RE Limits the action to an RDF Gig-E director number. To select all RDF Gig-E director

numbers, specify ALL.
Use with -free in order to restrict the display of free ports to those which support the RDF Gig-E interface type.

Use with -ip_interface to configure IP Interface on RE director port.

-reclaiming Verifies devices that are currently being reclaimed from a thin pool.

-remote_dir Specifies the remote RE director number.

-remote_p Specifies the remote RE director port number.

-remote_sid Specifies the unique Symmetrix ID of remote array.

-RF Limits the action to an RDF Fibre director number. To select all RDF Fibre director numbers, specify ALL.
Use with -free in order to restrict the display of free ports to those which support the RDF Over Fibre interface type.

-rmt Specifies the operation to manage or list Remote Machine Table (RMT). Each operation specifies RMT entry options, such as remote machine serial number, remote RE director number, port number and IPV4 and/or IPv6 address.

-route Lists the IP routes configured on an array.

-savedev Displays information about the SAVE devices.

-sc_name Specifies the Storage Container name.

-SE Limits the action to a Gig-E director number. To select all Gig-E director numbers, specify ALL.
Use with -free in order to restrict the display of free ports to those which support the SE interface type.

Use with -ip_interface to configure IP Interface on SE director port

-semaphores Displays gatekeeper, database, and lock file semaphores.

-service_state Limits the display of environment data to include only modules that are in the specified service state.

-services Displays configured network services.

-sc Specifies the operation for Storage Container support.

-secure Specifies the snapshots generated by the Snapshot Protection Policy should be secure.

-sg Refreshes (synchronizes) the configuration

database file with Storage Group (SG) information gathered from the Symmetrix array or specifies the name of the storage group to use.

-sg_compression
Display the data compressibility for the storage group.

-sid Specifies the unique Symmetrix ID.

-sized Display information on what the SRP is sized for.

-sl Display information about the Service Levels (SLs). Or specifies the Storage Container Resource Service Level name.

-sl_rt_multiplier
Specifies array attribute for the Service Level response time multiplier.

-slot Use with -free in order to restict the display of free ports to those which reside on a specified slot.

-snap Refreshes (synchronizes) the configuration database file with updated TimeFinder/Snap information when used with sync.
If used with pools, displays Snap pool information.

-snmp Indicates that the supplied information is for SNMPv3 authentication. This record requires a passphrase, which will be prompted for if it is not included on the command line.

-sorthost Sorts the -connections list by host, rather than by Symmetrix ID.

-speed Use with -free in order to restrict the display of free ports to those of a specified maximum speed (Gb/sec).

-sresource Specifies the Storage Container Resource name.

-srp Display information about the SRPs. Or specifies the Storage Container Resource SRP name. Or Specifies the SRP for the data compressibility report.

-ssid Displays MVS subsystem information.

-ssid_num Specifies a SSID number to qualify a specific CU image, as CU image numbers are only unique relative to individual FICON Splits.

-ssv Indicates to reset or verify the Stable System Values (SSVs) saved in the lockbox.

-static Lists static RDF groups.

-status Displays status information referring to whether the configuration has

changed and if the Symmetrix array was discovered during the last discover action.

-subscribed_max
Specifies the maximum amount of subscribed storage in GB that can be provisioned on the Storage Container.

-switched Displays the local and remote Symmetrix arrays, their RDF directors, and RA groups connected in the open RDF switch fabric.

-symforce Forces the disabling of a vWitness if it is in use and there is no alternate Witness

-tb Lists capacity in terabytes.

-tdev Displays information about thin devices.

-thin Displays Virtual Provisioning thin pool information.

-tier Refreshes (synchronizes) the configuration database file with tier information gathered from the Symmetrix array.

-type When used with -sc, specifies the Storage Container type. When used with -policy, specifies the type of Snapshot Protection Policy. When used with -container, specifies the embedded guest type.

-topology Use with set, to set the director port topology. This Option takes topology type as an argument. Valid topology types are FABRIC, AL and P2P.

-unblocked Displays only unblocked features. Not compatible with -blocked.

-unbinding Verifies devices that are currently unbinding from a thin pool.

-unbound Lists or verifies devices that are ready to be bound to a thin pool.

-uncompressing Specifies that the applicable Symmetrix thin devices must be in the "uncompressing" state for the verification to return success.

-upatches Lists all Enginuity patches on this Symmetrix array.

-username Indicates the name of the user to authorize.

-v Provides a more detailed, verbose listing.

-vasa Displays VASA RDF group(s).

-version Displays SYMCLI/SYMAPI version information.

-virtual Use with -port in order to restrict the display to virtual ports only.

-vlan_id Specifies Virtual LAN Identifier for

iSCSI IP interface.

-vmware — Indicates that the supplied information is for a VMWare Virtual Infrastructure Service. Used with the authorization action.

-vpdata — Refreshes (synchronizes) the configuration database file with virtual provisioning information gathered from the Symmetrix array.

-warning_compliance — Specifies the warning compliance count attribute of the specified Snapshot Protection Policy. The compliance count cannot be set to 0 and must be less than or equal to the maximum count of the Snapshot Protection Policy. If the critical compliance count is also set, the warning compliance count must be greater than or equal to that.

-wl — Specifies the workload name for the Storage Container Resource.

-witness — Specifies the name of the vWitness.

PARAMETERS

ACLX — When enabled, allows storage provisioning using Auto-provisioning Groups.

ARB — When enabled, a SCSI bus reset only occurs to the port that received the reset (not broadcast to all channels).

AppID — The application ID.

BayName — The Symmetrix cabinet name.

BayName must conform to one of the patterns below. These will be accepted in either upper or lower case.

SB-x where x is a system bay number
DB-xy where x is a drive bay index and y is the bay side (A, B, C, D)

These forms are consistent with bay names displayed in output returned by the "symcfg -env_data" and "symcfg -bay_info" list commands.

ClassName — The class of features list. This string must match what is displayed as the feature class in the normal feature output. Class names with spaces must be enclosed in quotes.

ContainerName — The name of the container to be displayed.

ContainerType — The type of the container to be displayed.

D — When Disable_Q_Reset_on_UA(D) flag is enabled, a Unit Attention (UA) that is propagated from another director does not flush the queue for this device

on this director. Used for hosts that do not expect the queue to be flushed on a 0629 sense (only on a Hard Reset).

DefaultGateway The gateway or router address for a front-end RE port.

Description The Storage Container description.

DNSorIPAddr The DNS location or IP address of the vWitness.

E When Environ_Set(E) flag is enabled, it turns on the environmental error reporting by the array to the host on the specific port.

FiconSplitName The name associated with a FICON Split.

flag The FA port flags from the following values in []:

Port Flag	Value Indicator
Volume_Set_Addressing	[V]
Non_Participating+	[NP]
Access_Logix	[ACLX]
OpenVMS	[OVMS]
Show_ACLX_Device	
Soft_Reset	[S]
Environ_Set	[E]
Disable_Q_Reset_on_UA	[D]
SCSI_3	[SC3]
SPC2_Protocol_Version	[SPC2]
SCSI_Support1	[OS2007]
Avoid_Reset_Broadcast	[ARB]

HostName The host name.

IPAddress A valid IPv4 or IPv6 address is expected.

IPPrefix IPv4 or IPv6 prefix length of IP Interface and IP Route.
IPv4 prefix length range: 1-32
IPv6 prefix length range: 1-128

MTU The mtu field allows the user to specify MTU (maximum transmission unit) value in the range of 1200-9000 for the IP interface. The default value is 1500.

NameSpace The namespace used with Authorization actions.

NetworkId It is used to support multiple IP addresses with same IP subnet on a SE director. Valid values: 1 to 16383.

NewPassWord The new password for the lockbox.

NP When Non_Participating(NP) flag is enabled, the Fibre Channel director only uses hard-assigned addressing when it initializes on the loop. Otherwise, soft-assigned addressing is used during loop initialization (the default).

OS2007 HP_UX & Win Longhorn specific setting.

OVMS Enabled for an OpenVMS fibre connection.

PassWord — The password associated with the UserName supplied during authorization actions.

When used in combination with -lockbox option, the input password is for the lockbox.

PolicyName — The Snapshot Protection Policy name.

PoolName — The device pool name.

PrivKey — The privacy key to be used as the SNMPv3 security passphrase.

S — When SoftRst flag enabled for a Bull/GCOS-7 host, the array port supports the SCSI Soft Reset option.

ScriptFile — The full pathname of a script file to be executed.

SC3 — When SCSI3 flag is enabled, the Inquiry data is altered when returned by any device on the port to report that the array supports SCSI 3 protocol. When this flag is disabled, the SCSI 2 protocol is supported.

SgName — The storage group name.

ShowACLX — Enabled/Disabled, to make the ACLX device visible or to remove visibility from the ACLX device respectively. By default all ACLX enabled ports will have the ShowACLXDevice attribute disabled.

SLName — The Performance Service Level name.

SPC2 — SPC-2 in inquiry data.

SRPName — The SRP name.

SSID — The subsystem number.

StorageContainer — The Storage Container name.

StorageResourceName — The Storage Container Resource name.

SymDev — A single Symmetrix device name, such as 008A.

SymDevStart — The first Symmetrix device name in a sequence, such as 001C.

SymDevEnd — The last Symmetrix device name in a sequence, such as 00B6.

SymmID — The 12-digit ID of the Symmetrix array.

TargetIQN — The iSCSI target IQN name.

TopoType — The Topotype allows the user to set the director port topology.

UserName — The username supplied during authorization actions.

```
     V              When Volume_Set_Addressing(V) is enabled for octal
                    addressing for HP-UX hosts.

     VlanId         Virtual LAN Identifier. Max value is 4094.

     WorkloadName   The Workload name.

     WitnessName    The name of vWitness

RETURN CODES

     Code #    Code Symbol
     ------    -----------

       0       CLI_C_SUCCESS

       1       CLI_C_FAIL

       2       CLI_C_DB_FILE_IS_LOCKED

      19       CLI_C_GK_IS_LOCKED
               All GateKeepers to the Symmetrix
               array are currently locked.

Return codes for symcfg -lock

      16       CLI_C_SYM_NOT_ALL_LOCKED
               Only returned if not all the
               targeted Symmetrix arrays
               currently have an exclusive
               Symmetrix lock.

      17       CLI_C_SYM_NONE_LOCKED
               Only returned if none of the
               targeted Symmetrix arrays
               currently has an exclusive
               Symmetrix lock.

Return codes for symcfg verify

      24       CLI_C_NOT_IN_SYNC
               The Symmetrix configuration
               and the symapi database file
               are not in sync. You should
               run either a discover or a
               full sync.

EXAMPLES

     To discover all Symmetrix arrays connected to this host,
     and to build or rebuild the Symmetrix configuration
     database file from information gathered, enter:

        symcfg discover

     To display information about the attached Symmetrix
     arrays, enter:

        symcfg list

     To display detailed information about the attached
     Symmetrix arrays and their directors, enter:

        symcfg list -v -dir all

     To display detailed information about the attached
     Symmetrix arrays and director 1B, enter:
```

```
   symcfg list -v -dir 1B
```

To display information about all front-end directors for the specified Symmetrix array, enter:

```
   symcfg list -SA ALL -sid 710
```

To list information about all registered hosts connected to the specified Symmetrix array, enter:

```
   symcfg list -connections -sid 010000658710
```

To list information about all registered applications on every locally-attached Symmetrix array, enter:

```
   symcfg list -applications
```

To list all configured network services in the network services file, enter:

```
   symcfg list -services
```

To list all gatekeeper and database semaphores, enter:

```
   symcfg list -semaphores
```

To verify whether the Symmetrix 0098 configuration and the Symmetrix configuration database are in sync, enter:

```
   symcfg verify -sid 0098
```

To display the content of CU image 0x00, enter:

```
   symcfg -cumage show 0
```

To display the status of modules comprising a storage array running Enginuity 5771 or above that are in a failed state, enter:

```
    symcfg -env_data list -service_state failed
```

```
symcg

        Performs operations on a Symmetrix composite
        group (CG).

        SYNOPSIS

             symcg -h

             symcg [-i <Interval>] [-c <Count>] [-v]

                activate <CgName> [-noprompt]

                activateall [-noprompt]

                create <CgName>
                  [-type REGULAR | RDF1 | RDF2 | RDF21 | ANY]
                  [-apidb | -rdf_consistency]

                delete <CgName> [-force] [-symforce]

                export <CgName> [-file <FileName>] [-rdf]
                  [-grpfile <GrpDbFileName>]

                exportall [-file <FileName>] [-rdf]
                  [-grpfile <GrpDbFileName>]

                import <CgName> [-file <FileName>]
                  [-apidb | -rdf_consistency] [-rename]

                importall [-file <FileName>]
                  [-apidb | -rdf_consistency]

                list [-offline] [-v [-mb | -gb | -tb]]
                  [-apidb | -rdf_consistency]
                  [-grpfile <GrpDbFileName>]

                list [-inactive]

                release <CgName> [-force] [-noprompt]
                  [-sid <SymmID>] [-lock <#>]

                rename <OldCgName> <NewCgName>

                show <CgName> [-inactive] [-offline | -lock]
                  [-mb | -gb | -tb] [-grpfile <GrpDbFileName>]

             symcg list -novalidate
                  [-apidb | -rdf_consistency]
                  [-grpfile <GrpDbFileName>]

             symcg -cg <CgName> [-i <Interval>] [-c <Count>] [-v]

                add dg <DgName>[,<DgName1>,<DgName2>,...]

                add [pd] <PdevName> [<LdevName>]

                copy ld <LdevName> <DestCgName> [-force] [-rename]

                move ld <LdevName> <DestCgName> [-force] [-rename]

                rename ld <OldLdevName> <NewLdevName>

                remove ld <LdevName> [-force] [-symforce]

                remove [pd] <PdevName> [-force] [-symforce]
```

```
    set <-name [<Name>] | -recovery_rdfg <GrpNum>>
      [-rdfg <<SymmID>:<<GrpNum>[,<GrpNum>,...]|all>[,...] |
        name:<RdfGroupName>[,<RdfGroupName>]>]

    show ld <LdevName>

 symcg -cg <CgName> -sid <SymmID>
      [-i <Interval>] [-c <Count>] [-v]
      [-rdf | -hop2]
      [-rdfg <GrpNum> [-remote_rdfg <RemoteGrpNum>]]

    add dev <SymDevName> [LdevName] [-vdev | -tgt]

    remove dev <SymDevName> [-force] [-symforce]
      [-vdev | -tgt]

    remove dg <DgName>[,<DgName1>,<DgName2>,...]

 symcg -cg <CgName> [-i <Interval>] [-c <Count>] [-v]
      [-sid <SymmID>]
      [-SA <# | ALL>] [-P <#>] [-N <#>]
      [-cap <#> [-captype <mb> | <cyl>]]
      [-rdf | -hop2] [-vdev | -tgt]
      [-rdfg <GrpNum> [-remote_rdfg <RemoteGrpNum>]]
      [-sel_rdfg <SelRdfGrpNum>]
      [-devs <<SymDevStart>:<SymDevEnd> | <SymDevName>
          [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]>]

    addall [pd | devs [-rdf | -hop2]]
      [-R1 | -R2 | -R21 | -noRDF] [-v]

    rmall [-rdf | -hop2] [-force] [-symforce]
      [-R1 | -R2 | -R21 | -noRDF]

 symcg -cg <CgName> [-i <Interval>] [-c <Count>] [-v]
      [-sid <SymmID>]
      [-SA <# | ALL>] [-P <#>] [-N <#>]
      [-cap <#> [-captype <mb> | <cyl>]]
      [<-vdev | -tgt> [-hop2] | -rvdev | -rtgt]
      [-rdfg <GrpNum> [-remote_rdfg <RemoteGrpNum>]]
      [-sel_rdfg <SelRdfGrpNum>]
      [-devs <<SymDevStart>:<SymDevEnd> | <SymDevName>
          [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]>]

    copyall <DestCgName> [-force] [-symforce] [-rename]
      [-R1 | -R2 | -R21 | -noRDF]

    moveall <DestCgName> [-force] [-symforce] [-rename]
      [-R1 | -R2 | -R21 | -noRDF]

symcg -cg <CgName> [-i <Interval>] [-c <Count>]
      [-noprompt] [-v]
      [-rdfg name:<Name>] [-hop2] [-star]

    enable

    disable [-force]

symcg -cg <CgName> [-i <Interval>] [-c <Count>]
      [-noprompt] [-v] [-force]
      [-bcv | -vdev | -tgt] [-rp] [-star][-sid <SymmID>]
      [-celerra]

    rw_enable

    write_disable
```

```
symcg -cg <CgName> [-i <Interval>] [-c <Count>]
      [-noprompt] [-v] [-force]
      [-bcv | -vdev | -tgt] [-star][-sid <SymmID>]

    host_active

    pin

    relabel

    unpin

symcg -cg <CgName> [-i <Interval>] [-c <Count>] [-v]
      [-noprompt] [-force]
      [-bcv [-hop2] | -rbcv | -brbcv | -rrbcv |
       -vdev [-hop2] | -rvdev | -tgt [-hop2] | -rtgt ]
      [-rp] [-star] [-sid <SymmID> |
      -rdfg <<SymmID>:<<GrpNum>[,<GrpNum>,...]|all>[,...] |
        name:<RdfGroupName>[,<RdfGroupName>]>] [-celerra]

    not_ready [-metro]

    ready [-metro]

    hold

    compress [-stop]

    uncompress [-stop]

    bind -pool <PoolName>

    unbind

    rebind -pool <PoolName>

    allocate [-persistent]

    allocate -stop

    free [-all]

    free [-all] -stop

    reclaim [-persistent]

    reclaim -stop

    set -persistent

    unset -persistent

    set -orm < system | on | off>

    reset -identity

symcg -cg <CgName> [-i <Interval>] [-c <Count>] [-v]
      [-noprompt] [-symforce]
      [-bcv [-hop2] | -rbcv | -brbcv | -rrbcv |
       -vdev [-hop2] | -rvdev | -tgt [-hop2] | -rtgt ]
      [-rp] [-star] [-sid <SymmID> |
      -rdfg <<SymmID>:<<GrpNum>[,<GrpNum>,...]|all>[,...] |
        name:<RdfGroupName>[,<RdfGroupName>]>] [-celerra]

    unhold

    set -gcm
```

```
   unset -gcm

symcg -cg <CgName> [-i <Interval>] [-c <Count>]
      [-noprompt] [-v]
      -sid <SymmID>
      -devs <<SymDevStart>:<SymDevEnd> | <SymDevName>
        [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]> |
        -file <FileName>
      -stg_rdfg <GrpNum>[,<GrpNum>]
      -cg_rdfg <CgGrpNum>[,<CgGrpNum>]
     [-stg_r21_rdfg <R21GrpNum> -cg_r21_rdfg <CgR21GrpNum>]

   modify -add [-force]

   modify -remove

symcg -cg <CgName> [-i <Interval>] [-c <Count>]
     [-noprompt] [-v] [-force]

   modify -recover
```

DESCRIPTION

The symcg command performs the following operations specific to composite groups: creating a new composite group; deleting, exporting, importing, listing, or showing information about a composite group; adding devices to a composite group; removing devices from a composite group; and enabling and disabling RDF consistency on a composite group.

The symcg command also performs the following control actions on some or all of the devices in a composite group: write_disable, rw_enable, ready, not_ready, hold, unhold, pin, unpin, compress, uncompress, relabel, set, unset, bind, unbind, rebind, allocate, free and reclaim. By default the actions will only be applied to the standard devices in the group. The -bcv and -tgt switches must be specified to operate on those types of devices.

ARGUMENTS

allocate	Allocates storage in the thin pool.
activate	Activates a specified composite group.
activateall	Activates all of the inactive composite groups. Imports to Group Name Services (GNS).
add	Adds a Symmetrix device or an existing device group to an existing composite group.
addall	Adds all Symmetrix devices that are visible to this host, are not already members of a composite group, and belong to a specified Symmetrix array to the specified existing composite group. When the dev keyword is specified, devices that are not visible to the host are also added.
bind	Binds the thin device(s) to the thin pool.
compress	Starts data compression on thin device(s). When combined with the -stop option,

data compression is stopped.

copy	Copies one device from one existing composite group to another existing composite group. The source and destination composite groups must have compatible types.
copyall	Copies all specified devices from one existing composite group to another existing composite group. The source and destination composite groups must have compatible types.
create	Creates an empty composite group. If no type is specified, then the group will become a REGULAR group.
delete	Deletes an existing composite group. If the composite group has members, the command will fail unless the -force option is used. If the -force option is specified, the devices that are members of the group are removed and the group is deleted. If RDF consistency is enabled and cannot be disabled, the command will fail unless the -symforce option is used.
disable	Disables RDF consistency for all device(s) in the CG or for all devices(s) in the rdfg name.
enable	Enables RDF consistency for all device(s) in the CG or for all device(s) in the rdfg name.
export	Exports the contents of a composite group to a text file, which can later be used to import the composite group.
exportall	Exports the contents of all composite groups to a text file, which can later be used to import all of the composite groups.
free	Frees storage in the thin pool.
hold	Holds all devices in the composite group. By default, all devices in the STD device list are acted upon.
host_active	Sets the host active mode on device(s). The device(s) must be in a host passive mode for this operation to succeed. If a device is host visible, the command will fail unless the -symforce option is used.
import	Imports the composite group described by a text file that was created by the export action.
importall	Imports all of the composite groups described by a text file that was created by

	the exportall action.
list	Lists all of the composite groups that have been created for this host. If the -inactive option is specified, all of the inactive composite groups will be listed.
modify	Moves devices between the staging area and the RDF Consistency enabled CG, and updates the CG definition to reflect the change.
move	Moves one device from one existing composite group to another existing composite group. The source and destination composite groups must have compatible types.
moveall	Moves all specified devices from one existing composite group to another existing composite group. The source and destination composite groups must have compatible types.
not_ready	Makes all of the devices in the composite group not ready. By default, all devices in the STD device list are acted upon.
pin	Sets the device(s) to the user pinned state. User pinned device(s) will not be moved via FAST controller, but they can be moved via Optimizer or Symmigrate.
ready	Makes all devices in the composite group ready. By default, all of the devices in the STD device list are acted upon.
rebind	Rebinds the device(s) to the thin pool.
reclaim	Reclaims storage from the thin pool.
relabel	Applies the defined label to the device. Please refer to the symlabel command to learn how to define a device label.
release	Releases the Device External Lock (DEL) associated with the devices within a composite group.
remove	Removes a Symmetrix device or device group from an existing composite group. If RDF consistency is enabled, the command will fail unless the -force option is used. If RDF consistency cannot be disabled, the command will fail unless the -symforce option is used.
rename	Renames a logical device.
reset	Sets the device to its original identity when combined with -identity option.
rmall	Removes all Symmetrix devices from an existing composite group.

If RDF consistency is enabled, the command will fail unless the -force option is used.

If RDF consistency cannot be disabled, the command will fail unless the -symforce option is used.

rw_enable — Enables all devices in the composite group for reads and writes. By default, all devices in the STD device list are acted upon.

set — Associates a logical name or a STAR recovery RDF group number with an RDF (RA) group(s) when combined with the -name or -recovery_rdfg option. Sets the persistent indicator for allocations on thin device(s) when combined with the -persistent option. Sets the Optimized Read Miss mode when combined with the -orm option. Sets GCM mode when combined with -gcm.

show — Shows detailed information about the devices in a composite group.

unbind — Unbinds device(s) from the thin pool.

uncompress — Starts data decompression on thin device(s). When combined with the -stop option, data decompression is stopped.

unhold — Releases the hold on all devices in the composite group. By default, all devices in the STD device list are acted upon.

unpin — Unset the device(s) from the user pinned state.

unset — Clears the persistent indicator for allocations on thin device(s) when combined with the -persistent option. Clears GCM mode when combined with -gcm.

write_disable — Disables writes for all devices in the composite group. By default, all devices in the STD device list are acted upon.

KEYWORDS

dev — Performs the action against a Symmetrix device given its Symmetrix device name.

devs — Performs the action against multiple Symmetrix devices given their Symmetrix device names.

dg — Performs the action against a device group given its device group name.

ld — Performs the action against a Symmetrix device given its logical device name.

name — Performs the action against the specified RDF group’s logical name.

pd — Performs the action against a Symmetrix device given its physical (host) device name.

OPTIONS

-add — Specifies that the devices will be added to the RDF Consistency enabled CG from the staging area.

-all — Used with the free operation in order to specify that all allocations associated with the indicated devices are to be freed, regardless of whether data has been written or not.

-apidb — Lists the composite groups that are in the SYMAPI database. On a create or import, overrides the options file setting and stores the RDF CG in the SYMAPI database only.

-bcv — Specifices all devices in the BCV device list of a composite group.

-brbcv — Specifies all devices in the BRBCV device list of a composite group.

-c — Specifies the number (count) of times to attempt to acquire an exclusive lock on the Symmetrix host database and (for control operations) on the local and/or remote Symmetrix arrays.

The time to wait between attempts to acquire a needed lock is specified by -i (interval).

If neither -c nor -i is specified, operations will fail if unable to acquire a requested lock.

If -c is not specified, and -i is specified, the program will loop continuously until the operation has acquired the locks it needs and can start.

-cap — Sets a minimum device size to the selection criteria of devices.

-captype <mb | cyl>
Specifies the units of capacity in megabytes or cylinders. If the unit of measurement is not specified, the default is mb.

-celerra — Allows controls on Celerra FBA devices.

-cg — Specifies the composite group name.

-cg_rdfg — Specifies the RDF group(s) within the CG to which devices will be added or from which devices will be removed. For a concurrent CG, two groups must be specified (separated by a comma). These RDF groups are associated (in order) with the RDF groups specified by the -stg_rdfg option.

-cg_r21_rdfg
: Specifies the RDF group connecting the R21 and R2 Symmetrix arrays of a cascaded CG. It is only valid for operations involving cascaded R1 devices. This RDF group is associated with the RDF group specified by the -stg_r21_rdfg option.

-orm
: Allows setting the Optimized Read Miss mode to system default, on, or off for the specified devices.

-devs
: Specifies the ranges of Symmetrix devices to add and remove.

-file
: For the import and export action, specifies the file to be used.

  For the modify action, specifies the filename containing the list of devices to be acted upon. Only the SymDevName specified on the first column of each line is used.

-force
: Forces the deletion of a composite group, with or without members. Also used with disable to perform the action when devices are in unexpected RDF modes and states. When used with remove, if the device is enabled, it will be disabled and removed.

  Also applies to the following actions: write_disable, rw_enable, ready, not_ready, hold, unhold, host_active and relabel.

-gcm
: Allows setting or clearing the device GCM mode.

-grpfile
: Specifies an alternate group database file for use in list, show, export and exportall actions.

-h
: Provides brief, online help information.

-hop2
: Indicates that the device is two hops away.

  If used for add/remove dev, addall, rmall, moveall, or copyall operations, then -rdfg, -remote_rdfg, and -tgt or -vdev must also be specified.

  When holding or readying devices, the hop2 flag must be used with -bcv, -vdev, or -tgt.

  If used for enable/disable operations, targets the operation at the group’s second-hop devices in a Cascaded RDF relationship. For example, given an RDF1 group, the R21->R2 pair of the R1->R21->R2 relationship will be enabled or disabled.

-i
: Specifies the interval, in seconds, to wait between attempts to acquire an exclusive lock on the Symmetrix host database and, for control operations, on the local and/or remote Symmetrix arrays.

The default interval is 30 seconds.
The minimum interval is 5 seconds.
For passive actions the minimum interval is 15 seconds. Passive actions are actions that do not acquire an exclusive lock.

-identity — Lists the external identity information for each device. The external identity will be displayed if the device has external identity set. Otherwise, the device native identity will be displayed. When used with reset command, the original identity of the device(s) is restored.

-inactive — Lists or shows inactive composite groups.

-lock — Displays whether the devices have any exclusive locks.

-name — Specifies the RDF group name(s). Reset it by setting it to the null string (for example: symcg -cg w1 set -rdfg 13:55).

-noprompt — Eliminates the prompt for user confirmation.

-noRDF — Adds, copies, moves, or removes non RDF devices only.

-novalidate — Lists groups without output of the Valid column.

-offline — Obtains information from the Symmetrix host configuration database.

-persistent — Used with set or unset it specifies operation on the persistent indicator. Used with allocate or reclaim it specifies the use of persistent storage.

-pool — Specifies a Thin Pool Name.

-R1 — Adds, copies, moves, or removes RDF1 (R1) devices only.

-R2 — Adds, copies, moves, or removes RDF2 (R2) devices only.

-R21 — Adds, copies, moves, or removes RDF21 (R21) devices only.

-rbcv — Chooses all devices in the RBCV device list of a composite group.

-rdf — Exports the remote Symmetrix ID(s) and remote Symmetrix device names. This allows the composite group to be imported on a host connected to the remote Symmetrix arrays.

When used in conjunction with -vdev, it indicates that the device is an RVDEV device. When used in conjunction with -tgt, it indicates that the device is an RTGT device.

-rdf_consistency

Creates or imports the CG, allowing it to be enabled for RDF consistency once devices have been added to the CG.

-rdfg — Adds, removes, or controls devices that belong to the specified RDF (RA) group(s). For an add operation, this parameter is only valid with -vdev or -tgt. For control operations (ready, not_ready, hold, unhold), this parameter must be used with a remote device type (-rbcv, -brbcv, -rrbcv, -rvdev, -rtgt).

-metro — When specified with ready and not_ready identifies the devices being controlled are part of an RDF/Metro configuration.

-recover — Recovers the failed modify add or modify remove operation and puts the CG into a known state.

-recovery_rdfg — Specifies a STAR recovery RDF (RA) group. This is a ones-based number. Reset it by setting it to zero.

-remote_rdfg — Specifies the RDF (RA) group to access a two-hop device from the first hop.

-remove — Specifies that the devices will be removed from the RDF Consistency enabled CG to the staging area.

-rename — Specifies a new name for the CG.

-rp — Indicates that the action is targeted for devices tagged for RecoverPoint.

-rrbcv — Specifies all devices in the RRBCV device list of a composite group.

-rtgt — Specifies all devices in the RTGT device list of a composite group.

-rvdev — Specifies all devices in the remote VDEV device list of a composite group.

-sel_rdfg — Indicates the Symmetrix RA (RDF) group number of the devices to be added via an addall operation.

-sid — Specifies the unique Symmetrix ID.

-star — Indicates that the action is targeted for devices in STAR mode.

-stg_r21_rdfg — The RDF group connecting the R21 and R2 Symmetrix arrays in the staging area that is used to add or remove cascaded devices from a cascaded CG. It is only valid for operations involving cascaded R1 devices. This RDF group is associated with the RDF groups specified by the -cg_r21_rdfg option.

-stg_rdfg — Specifies the RDF group(s) that comprise the staging area. For a concurrent CG, two groups must be specified (separated by a

comma). These RDF groups are associated (in order) with the RDF groups specified by the -cg_rdfg option.

-stop — Specifies that the compress, uncompress, allocate, free or reclaim operation will be stopped.

-symforce — Forces the operation (when used with remove or delete) if the device is enabled for consistency and it cannot be disabled.

Also applies to the following actions: unhold, set -gcm and unset -gcm.

-tgt — Targets the indicated action at the devices in TGT list of the composite group.

-type — Specifies the composite group type. Values are: REGULAR, RDF1, RDF2, RDF21, and ANY. The default type is REGULAR.

-v — Provides a more detailed, verbose listing.

-vdev — Chooses all devices in the VDEV device list of a composite group.

PARAMETERS

CgGrpNum — The RDF (RA) group number of an RDF Group within the enabled CG.

CgName — The composite group name assigned by the user. The name must be unique to this host.

CgR21GrpNum — The RDF (RA) group number connecting the R21 Symmetrix to the R2 Symmetrix of a Cascaded R1 CG.

DestCgName — The destination composite group name for copy/copyall or move/moveall operations.

DgName — The name of the device group to be contained by the composite group.

FileName — For the import or export action, the text file name to be used.

For the modify action, the file name that contains a list. Only the SymDevName specified on the first column of each line is used.

GrpDbFileName — Specifies an alternate group database file for use in list, show, export and exportall actions.

GrpNum — The RDF (RA) group number.

LdevName — The device logical name assigned by the user or automatically assigned when a device is added to a composite group.

NewCgName — The new composite group name.

OldCgName — The current composite group name.

PdevName — The host physical name for the device (for example: /dev/rhdiskpower61).

R21GrpNum — The RDF (RA) group number of the R1 Mirror of the R21 device.

RdfGroupName — The logical name associated with the RDF (RA) group(s).

RemoteGrpNum — The RDF (RA) group number from the first hop to the second hop.

SelRdfGrpNum — The Symmetrix RA (RDF) group number of the devices to be added via an addall operation.

SymDevEnd — The last Symmetrix device name in a sequence (for example: 00B6).

SymDevName — The Symmetrix device name, unique per Symmetrix (for example: 001C).

SymDevStart — The first Symmetrix device name in a sequence (for example: 001C).

SymmID — The 12-digit ID of the Symmetrix array.

RETURN CODES

Code #	Code Symbol
0	CLI_C_SUCCESS
1	CLI_C_FAIL
2	CLI_C_DB_FILE_IS_LOCKED
19	CLI_C_GK_IS_LOCKED All GateKeepers to the Symmetrix array are currently locked.

FILES

The output file created by the export argument and read by the import argument has the following format:

```
  <CG TYPE>
S <SYMMETRIX_ID> <SYMMETRIX_DEVICE_NAME>
B <SYMMETRIX_ID> <SYMMETRIX_DEVICE_NAME>
R <SYMMETRIX_ID> <SYMMETRIX_RA_GROUP_NUM>
<LOGICAL_DEVICE_NAME>
Z <SYMMETRIX_ID> <SYMMETRIX_RA_GROUP_NUM>
<LOGICAL_DEVICE_NAME>
Y <SYMMETRIX_ID> <SYMMETRIX_RA_GROUP_NUM>
<REMOTE_SYMMETRIX_RA_GROUP_NUM>
<LOGICAL_DEVICE_NAME>
D <SYMMETRIX_ID> <SYMMETRIX_RA_GROUP_NUM>
<HOP_2_SYMMETRIX_RA_GROUP_NUM>
<LOGICAL_DEVICE_NAME>
V <SYMMETRIX_ID> <LOGICAL_DEVICE_NAME>
W <SYMMETRIX_ID> <SYMMETRIX_RA_GROUP_NUM>
<LOGICAL_DEVICE_NAME>
E <SYMMETRIX_ID> <SYMMETRIX_RA_GROUP_NUM>
<HOP_2_SYMMETRIX_RA_GROUP_NUM>
<LOGICAL_DEVICE_NAME>
T <SYMMETRIX_ID> <LOGICAL_DEVICE_NAME>
```

```
X <SYMMETRIX_ID> <SYMMETRIX_RA_GROUP_NUM>
<LOGICAL_DEVICE_NAME>
F <SYMMETRIX_ID> <SYMMETRIX_RA_GROUP_NUM>
<HOP_2_SYMMETRIX_RA_GROUP_NUM>
<LOGICAL_DEVICE_NAME>
N <SYMMETRIX_ID> <SYMMETRIX_RA_GROUP_NUM>
<SYMMETRIX_RA_GROUP_NAME>
C <SYMMETRIX_ID> <SYMMETRIX_RA_GROUP_NUM>
<SYMMETRIX_RECOVERY_RA_GROUP_NUM>
G <DgName>
```

In the output described above, the <CG TYPE> values are:

Symbolic Name	Value
SYMAPI_C_CGTYPE_NA	0
SYMAPI_C_CGTYPE_RDF1	1
SYMAPI_C_CGTYPE_RDF2	2
SYMAPI_C_CGTYPE_REGULAR	3
SYMAPI_C_CGTYPE_RDF21	4
SYMAPI_C_CGTYPE_ANY	5

The single-character codes that represent device types are:

Code	Description
S	Identifies an STD device.
B	Identifies a local BCV device.
R	Identifies an RBCV device.
Z	Identifies a BRBCV device.
Y	Identifies an RRBCV device.
D	identifies a HOP 2 BCV device.
V	Identifies a VDEV.
W	Identifies an RVDEV device.
E	Identifies a HOP 2 VDEV device.
T	Identifies a TGT device.
X	Identifies an RTGT device.
F	Identifies a Hop 2 TGT device.
N	Identifies an RA Group Name.
G	Identifies a DG

The parameters used to describe the devices include:

Parameter	Description
<SYMMETRIX_ID>	The 12-digit identifier of the Symmetrix upon which the local STD devices resides.
<SYMMETRIX_RA_GROUP_NUM>	The RDFG (RA) number that can be used in combination with the local Symmetrix ID to get to the remote Symmetrix.
<REMOTE_SYMMETRIX_RA_GROUP_NUM>	The RDFG (RA) number that can be used in combination with the local RA group number and the local Symmetrix ID to get to the Symmetrix that is one hop away.
<HOP_2_SYMMETRIX_RA_GROUP_NUM>	The RDFG (RA) number that can be used in combination with the local and remote RA group numbers and the local Symmetrix ID to get

to the Symmetrix that is two hops
away.

<SYMMETRIX_DEVICE_NAME>
The Symmetrix device number.

<LOGICAL DEVICE NAME>
The name of the device in the
group.

<SYMMETRIX_RA_GROUP_NAME>
The RDFG (RA) group name.

<SYMMETRIX_RECOVERY_RA_GROUP_NUM>
The RDFG (RA) recovery RDFG (RA)
group number.

The file will have as many device description lines as
the composite group has members.

Lines in the file that are blank or have a pound sign
(#) in the first column will be ignored.

EXAMPLES

To create a REGULAR Symmetrix composite group called
mycg1, enter:

```
symcg create mycg1
```

To list all of the Symmetrix composite groups in a detailed
format, enter:

```
symcg -v list
```

To show information about composite group oracg, enter:

```
symcg show oracg
```

To export composite group oracg to a text file called
’oracg.txt’, enter:

```
symcg export oracg -file oracg.txt
```

To delete Symmetrix composite group oracg, regardless
of whether the group has devices in it, enter:

```
symcg -force delete oracg
```

To import composite group oracg from a text file,
called ’oracg.txt’, enter:

```
symcg import oracg -file oracg.txt
```

To add a device to Symmetrix composite group oracg,
enter:

```
symcg -cg oracg add pd /dev/rhdiskpower61
```

To remove a device from a Symmetrix composite group
oracg, enter:

```
symcg -cg oracg -sid 55 remove dev 00C
```

symchg

Marks areas of Symmetrix disk storage so that objects occupying those areas can be monitored for changes by the Change Tracker.

SYNOPSIS

```
symchg  [-h]

symchg -sid <SymmID> [-v]

   create dev <SymDevName> [-type <read | write | rw>]

   mark   dev <SymDevName>

   delete dev <SymDevName> [-force]

   remove dev <SymDevName> [-force]

symchg <-g <DgName> | -cg <CgName>> [-bcv] [-v]

   create [-type <read | write | rw>]

   mark

   delete [-force]

   remove [-force]

symchg -sg <SgName> -sid <SymmID> [-v]

   create [-type <read | write | rw>]

   mark

   delete [-force]

   remove [-force]

symchg -file <DevFile> -sid <SymmID> [-v]

   create [-type <read | write | rw>]

   mark

   delete [-force]

   remove [-force]

symchg [-sid <SymmID>] [-v]

   list

   list -session [dev]  [-type <read | write | rw>]

symchg [-v]
   [-i <Interval>] [-c <Count>] [-reset | -multi]

   view dev <SymDevName> -sid <SymmID> [-native]

   view <-g <DgName> | -cg <CgName>> [-bcv]
        [-native | -log <LogFile>]

   view <-g <DgName> | -cg <CgName>> ld <LdevName>
        [-native]
```

```
   view -sg <SgName> -sid <SymmID> [-native |
        -log <LogFile>]

   view -sid <SymmID> -file <DevFile> [-native |
        -log <LogFile>]

symchg -session [-v]

   view dev <SymDevName> -sid <SymmID>

   view <-g <DgName> | -cg <CgName>> [-bcv]

   view <-g <DgName> | -cg <CgName>> ld <LdevName>

   view -sg <SgName> -sid <SymmID>

   view -sid <SymmID> -file <DevFile>

symchg [-ra <NumRAs>] [-rate <KB/s>]
       [-resync [<mmmm|hh:mm>]]
   [-start <mmddyyyy hh:mm>] [-stop <mmddyyyy hh:mm>]
   [-backend] [-native] [-v]
   [-devs <<SymDevStart>:<SymDevEnd> | <SymDevName>
        [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]>]

   report -log <LogFile> <-g <DgName> | -cg <CgName>>
      [-bcv]

   report -log <LogFile> -sg <SgName> -sid <SymmID>

   report -file <DevFile> -log <LogFile> -sid <SymmID>
```

DESCRIPTION

The symchg command (also known as Change Tracker) timestamps and marks an area of Symmetrix disk storage occupied by a disk storage object using an SDDF bitmap. This allows the changes to a marked area to be viewed at a later time. You can mark storage objects such as Symmetrix devices, lists of Symmetrix devices in a device file or a storage group, and the standard or BCV devices of a Symmetrix device group or a composite group.

When viewing the changes for an object, the output can either be sent to stdout or saved to a user-defined log file. The information saved to the log file is stored in a comma-delimited format so the data to be imported into a spreadsheet for analysis. The results in the log file can be summarized by specifying the report argument to symchg.

The information saved to the log file can be the number of tracks (delta) changed between sample intervals (by specifying the -reset option), or the cumulative tracks changed over time (sum), or both the changed and cumulative tracks (by specifying -multi). The sum method is the default.

The symchg utility calculates certain RDF capacity planning values based on the amount of change to the monitored objects. The RDF values that can be calculated are re-synchronization time, the number of remote adapters (RA) and the RA transfer rates. Two of these values are specified on the command line, and the third value is calculated.

ARGUMENTS

create — Creates a Change Tracker session that monitors changes to logical objects. You can mark and examine more than one object per session. This command supports Symmetrix devices using the Symmetrix device name, logical device name, device group, or BCV devices of a device group.

delete — Removes the marked object from the symchg database and deletes the Change Tracker session.

list — Lists all the disk storage objects that were marked. Lists all Symmetrix devices that have an established Change Tracker session.

mark — Marks the area of disk storage to be monitored. A session must first be created for the device containing the object to be marked.

remove — Removes the marked object from the symchg database without terminating the Change Tracker session.

report — Generates a report on the amount of change for the specified object in the specified log file. Reports are generated for device groups only.

view — Processes the bitmap that corresponds to the disk storage object and displays the amount and the rate of change. Also shows whether a Change Tracker session exists for a specified disk device. The possible storage objects are specified by selecting a Symmetrix device name, logical device name, device group, or BCV devices of a device group.

OPTIONS

-backend — Displays the DA and back-end disk address of the device whose changes are being measured.

-bcv — Limits the action to the BCV devices of the device group.

-c — Specifies a count for the number of times to repeat execution of the view action. If -c is not specified and the -i option is specified, the change information for the object is displayed continuously at every interval.

-cg — Specifies the composite group name.

-devs — Specifies a set of Symmetrix device ranges and/or individual Symmetrix devices for the report action.

-file — Specifies the device list file.

-force — Works with delete and remove operations.

With the delete operation, the -force option

forces the cancellation of the Change Tracker session on each device and forces the removal of the devices from the symchg database. This action is performed even if one or more devices does not exist in the symchg database or if the object type does not match.

With the remove operation, the -force option forces the removal of the devices from the symchg database. This action is performed even if one or more devices does not exist in the symchg database or if the object type does not match.

-g — Specifies a Symmetrix device group name.

-h — Provides brief, online help information.

-i — Specifies the repeat interval (seconds) between the executions of the view action. The default interval value between counts is 60 seconds. The minimum value between counts is 30 seconds. If the -c option is not specified, the change information is displayed continuously at every interval.

-log — Specifies a log file (owned by the user) that is read (report action) from or written (view action) to. The data written in log file are based on 32k track size.

-multi — Reports changed tracks for both the Sum and the Delta methods. This option requires the -i or -c options.

-native — Reports or views data based on native track size.

-ra — Specifies the quantity of RA directors in your Symmetrix configuration for the report action.

-rate — Specifies the RA director’s transfer rate (in KB/sec) to the report action.

-reset — Causes the specified storage device or object to be marked unchanged. This option requires the -i or -c options.

-resync — Specifies the specified time window (in minutes or hours and minutes) to synchronize the SRDF pair to the report action.

-session — Causes the list argument to list all of the physical devices with created Change Tracker sessions. Causes the view argument to show if a Change Tracker session exists for the devices that the object spans.

-sg — Specifies the storage group name.

-sid — Specifies a unique Symmetrix ID.

-start — Specifies a start date/time to begin processing data in the log file. The default operation is to start at the

beginning of the log file.

-stop — Specifies a stop date/time to stop processing data in the log file. The default operation is to stop at the end of the log file.

-type — Specifies the Change Tracker session type to be created or listed. If -type is not specified the default session type is write.

-v — Provides a more detailed, verbose listing.

PARAMETERS

CgName — The composite group name.

DevFile — The device list file containing the list of devices to be acted upon. The file should have one SymDevName per line.

DgName — The device group name. The name must be unique to this host.

LogFile — The log file name.

LdevName — The logical device name either named by the user or automatically assigned when a logical device is added to a device group.

SgName — The storage group name.

SymDevEnd — The Symmetrix device name, ending the contiguous range of selected devices, such as 00C.

SymDevName — The Symmetrix device name, unique per Symmetrix, such as 01C.

SymDevStart — The Symmetrix device name, starting the range of selected devices, such as 002.

SymmID — The 12-digit ID of the Symmetrix array.

read — The change Tracker read session.

rw — The change Tracker read_write session.

write — The change Tracker write session.

RETURN CODES

Code #	Code Symbol
0	CLI_C_SUCCESS
1	CLI_C_FAIL

EXAMPLES

To create a Change Tracker write session for Symmetrix device 17C, enter:

```
symchg create dev 017C
```

To create a Change Tracker read session for each Symmetrix device in device group ProdDB, enter:

```
symchg create -g ProdDB -type read
```

To examine the amount and rate of change for logical device Ldev1 in device group ProdDB, enter:

```
symchg view -g ProdDB ld Ldev1
```

To save changed information in logfile /tmp/logfile for device group ProdDB, performing 10 samples and resetting the changed information after each sample, enter:

```
symchg view -g ProdDB -reset -log /tmp/logfile -c 10
```

To list all of the disk storage objects that are currently marked, enter:

```
symchg list
```

To list all of the devices with Change Tracker write session that are currently marked, enter:

```
symchg list -session
```

To list devices that are visible to the host with Change Tracker read sessions, enter:

```
symchg list -session dev -type read
```

To generate a summary report on each of the Symmetrix devices of device group ProdDB for /tmp/logfile, enter:

```
symchg report -g ProdDB -log /tmp/logfile -v
```

To generate a summary report (/tmp/logfile) on device group ProdDB with an RA transfer of 1000 KB/sec, and perform a resync of the changed tracks every 10 minutes, enter:

```
symchg report -g ProdDB -log /tmp/logfile -rate 1000
  -resync 10
```

To delete the marks and terminate the Change Tracker session for Symmetrix device 14D, enter:

```
symchg delete dev 14D
```

To unmark all the Symmetrix devices associated with device group ProdDB without deleting the Change Tracker session for these devices, enter:

```
symchg remove -g ProdDB
```

symchksum

Performs Symmetrix checksum operations on the RDBMS database devices, RDBMS tablespace devices, or Symmetrix devices.

SYNTAX

```
symchksum [-h]

symchksum [-type <RdbType>] [-sid <SymmID>]
[-range <SymDevStart>:<SymDevEnd>]

list [-v]

symchksum [-sid <SymmID>]

show dev <SymDevName> [-blocks|-kb|-mb]

symchksum -type generic -sid <SymmID>
[dev <SymDevName>] [-range <SymDevStart>:<SymDevEnd>]

enable

disable

symchksum -type generic [-g <DeviceGroup>]

enable

disable

symchksum -type oracle [-db <DbName>] [-tbs <TbsName>]
[-control] [-redo] [-h]

enable [-reject_io] [-fractured_reject_io] [-discard]
[-phone_home] [-v]
[-check_dba] [-check_all_blocks] [-straddle]
[-suppress_feature <Keyword>[,<Keyword>]]

disable [-v]

validate [-v]

verify [-v]

disable -force -sid <SymmID> dev <SymDevName>
```

DESCRIPTION

The symchksum command enables checksum checking on a set of Symmetrix devices. Relational databases or generic applications are selected using the -type option.

For the generic type, the feature is enabled or disabled on a device basis. Devices can be specified as a single device, a range of devices, or a group of devices. The generic type operations do not perform the checks or provide the options available with the relational type.

For relational types, the I/O of enabled extents on a specified device are checked by executing a specified checksum algorithm. When checksum errors are detected, an error is logged to the Symmetrix array. Users can decide whether to reject the I/O or have the Symmetrix phone home when a checksum error is detected.

By default, the symchksum command attempts to select the following tests:
- block checksum value (Checksum)
- blocksize verification (BlkSize)
- 3-bit magic number (MagicNumber) in Oracle data blocks (not available for redo logs)
- checks for non-zero data block addresses (NonZeroDba)

To suppress one or more of these operations, use the -suppress_feature option and supply the name of the feature. Multiple feature names can be separated by commas.

The following three operations can be manually enabled:

- Check all blocks in the I/O - This option causes magic number and non-zero data block address checks to be performed on all blocks in the write. Normally, these checks are performed on the first block of the write only.To select this feature, use the -check_all_blocks option.

- Check for I/O straddling of Oracle extents - If a single I/O spans beyond the bounds of defined EMC DoubleChecksum extents, the I/O is said to straddle. This options checks for straddle I/O. Do not use this feature if autoextend is used on the RDBMS datafiles. To select this feature, use the -straddle option.

- Compare data block addresses and target blocks of I/O (-check dba) - When this option is enabled, extended data is stored on each extent. This limits the ability of symchksum to collapse adjacent extents. To select this feature, use the -check_dba option.

If an I/O is not a multiple of the object blocksize, the user can choose to reject the I/O. This is called a fractured I/O and is selected with the -fractured_reject_io option. When using this option, the -reject_io option must also be used.

When extents are enabled with -discard, EMC Double Checksum writes blocks to disk until a failed block is detected. The -discard option divides a large I/O into smaller units of 32K bytes each. When a checksum failure is detected, all blocks in that unit and subsequent units are discarded.

When a database, control file, redo log, or tablespace name is specified, the device information that defines those objects is obtained using SYMAPI database mapping routines. The extents of those devices are used when defining the set of extents to be checked for checksum errors.

When the disable action is specified for a Symmetrix device, the -force option is required. Disabling extents in this way can cause a mapped tablespace or database to be only partially protected, so use the -force option with caution. All the extents being monitored for checksum errors on the specified Symmetrix device are disabled.

There is a maximum of 200 extents per device that can be monitored at one time for Enginuity 5x70 rev 50 and higher. The maximum is 31 extents for earlier Enginuity revisions.

When listing the physical devices that have checksum

checking enabled, the information reported is from the first extent encountered. If you want to see all the extent details for a particular device, use the show command.

The only checksum algorithm type supported is for Oracle RDBMS products. The Oracle instance must be configured to perform checksum checking when used with the Symmetrix checksum functionality.

The database user login information must be supplied with the SYMCLI_RDB_CONNECT environment variable. The user login information is specified in the following format: username/password@service. The username and password must be non-NULL.

To execute the symchksum utility, you must have the proper application software installed and the environment variables set.

You can specify the database type and database name with environment variables. The command line options take priority over the environment variables:

- SYMCLI_RDB_TYPE - database type, instead of -type
- SYMCLI_RDB_NAME - database name, instead of -db

Only one restrictor clause can be used at one time. Restrictors are -tbs, -redo, -control.

In Oracle, the database name does not need to be specified. The default will be taken from the instance represented by the connection arguments and environment variables. Client/server mode is not supported.

Note: For Oracle, the specified database user must have one of the following to run this utility:

- "select any table" privilege (Oracle 8i and earlier)
- SELECT_CATALOG_ROLE (Oracle 9i and later)
- DBA role
- SYSDBA system role

ARGUMENTS

disable Disables checksum checking on the extents of the specified devices.

enable Enables checksum checking on the extents of the specified devices.

list Lists all the devices that currently have checksum checking enabled.

show Shows the extents of a specified device that is having checksum checking performed.

validate Validates if a specified database of tablespace devices can have checksum checking enabled.

verify Verifies if a specified database or tablespace has checksum checking enabled on all their devices.

KEYWORDS

BlkSize A default operation that checks I/O blocksize to block header.

Checksum A default operation that checks the computed checksum of the block where the checksum value is stored.

MagicNumber A default operation that verifies the 3-bit magic number.

NonZeroDba A default operation that checks for non-zero data block addresses.

OPTIONS

-check_all_blocks
Checks all blocks in the I/O. Otherwise, only the first block is checked.

-check_dba Compares the data block address and target block of I/O.

-control Specifies a relational database control file.

-db Specifies a relational database name.

-discard Discards bytes in the current buffer and subsequent buffers when checksum failure is detected.

-force Forces checksum to disable.

-fractured_reject_io
Rejects I/O if it is not a multiple of the blocksize.

-g Specifies a device group.

-h Provides brief online help information.

-phone_home Phones home when a checksum error is detected.

-range Specifies a range of devices.

-redo Specifies the relational database redo logs.

-reject_io Rejects I/O when a checksum error is detected.

-sid Specifies the unique Symmetrix ID.

-straddle Checks for writes that straddle defined database extents.

-suppress_feature
Turns off a default operation.

-tbs Specifies a relational database tablespace name.

-type Specifies a relational database type.

-v Provides a more detailed, verbose listing.

PARAMETERS

DbName A relational database name.

RdbType An application type, such as Oracle or generic.

SymDevName A Symmetrix device name.

SymmID The 12-digit ID of the Symmetrix array.

TbsName A relational database tablespace name.

RETURN CODES

Code #	Code Symbol
0	CLI_C_SUCCESS
1	CLI_C_FAIL
18	CLI_C_ALREADY_IN_STATE The device(s) are already in the desired checksum state. Applicable only for checksum control actions.

Return codes for symchksum verify

14	CLI_C_NOT_ALL_VALID Only returned if some of the specified RDBMS extents are checksum enabled.
15	CLI_C_NONE_VALID Only returned if none of the specified RDBMS extents are checksum enabled.

EXAMPLES

To list the devices on the specified Symmetrix array that have extents being checked for checksum errors, enter:

```
symchksum -sid 3890 list
```

To show all the extents of the specified Symmetrix device that are being checked for checksum errors, enter:

```
symchksum show dev 0A1
```

To enable Oracle checksum type checking on the extents of all the devices that define the current Oracle database instance, and when an error is detected, phone home, enter:

```
symchksum -type Oracle -phone_home enable
```

To enable Oracle checksum type checking on the extents of all the devices that define the specified Oracle tablespace, and when an error is detected, log that error, enter:

```
symchksum -type Oracle -tbs SYSTEM enable
```

To enable Oracle checksum type checking on the extents of all the devices that house the Oracle control files, and when an error is detected, log that error, enter:

```
symchksum -type Oracle -control enable
```

To verify that the specified Oracle tablespace has checksum checking for all devices that define it, enter:

```
symchksum -type Oracle -tbs USER01 verify
```

To disable checksum checking on the current Oracle instance, enter:

```
symchksum -type Oracle disable
```

To disable checksum checking for all checksum extents on the specified Symmetrix device, enter:

```
symchksum -sid 3890 disable dev 0A1 -force
```

To validate the Oracle tablespace, enter:

```
symchksum validate -tbs ACCOUNTING_1
```

To enable Oracle checksum and add the optional operations of check dba and check all blocks, enter:

```
symchksum -type Oracle enable -check_dba
-check_all_blocks
```

symcli

Provides a brief description of all the commands included in the Solutions Enabler Symmetrix Command Line Interface (SYMCLI) and the environment variables.

SYNOPSIS

```
symcli [-env] [-def] [-h] [-v]
```

DESCRIPTION

The symcli command provides a brief explanation of the commands included in the Symmetrix Command Line Interface. Manual pages are available for each individual command listed using the symcli command.

ARGUMENTS

None

OPTIONS

-def	Displays the SYMCLI environmental variable values that are currently set.
-env	Displays the list of environmental variables that can be used with SYMCLI commands.
-h	Provides brief, online help information.
-v	Provides a brief description of all the SYMCLI commands.

PARAMETERS

None

RETURN CODES

Code #	Code Symbol
0	CLI_C_SUCCESS
1	CLI_C_FAIL

EXAMPLES

To obtain a list of all SYMCLI commands, enter:

```
symcli -v
```

To obtain a list of the environment variables that can be set, enter:

```
symcli -env
```

symclient - Description of client server installation, management, and execution.

The client would be a SYMCLI command running on a local host.

The server would be the SYMAPI Server running on a remote host.

INSTALLATION

The SYMCLI installation procedure installs the client server SYMAPI server in the binaries directory. The executable is storsrvd. It also installs two files, netcnfg and symapinlck, in the configuration directory. On both a client host or server host the netcnfg file lists a network service available from that local host or network services available from one or more remote hosts. The symapinlck file is used as a lock file on a server host to guarantee a single port listener for a network service. To execute a remote SYMCLI session, both the client and server host netcnfg file should have the identical network service entry in it, to generate a TCP/IP network connection between them.

The netcnfg file is a template. The system administrator can use a text editor to add one or more network service entries to it. The syntax of the network service entry is as follows, on one line:

service-name domain-name network-protocol server-node-name server-network-address port-number

The domain-name should unspecified and substituted by a hyphen (-). An unspecified server-node-name or server-network-address can be substituted by a hyphen (-). But at least one must be specified. Both service-name and port-number are mandatory. The currently supported network protocol is TCP/IP and its field is required.

Example:

```
SYMAPI_SERVER -  TCPIP  node001  12.345.670.89  7777
BACKUP_SERVER -  TCPIP  node002  -              6666
```

A comment line can be specified by a hatch (#) as the first character.

SERVER SYMAPI SERVER STARTUP

On the server host the storsrvd executable for a network service can be started either by a startup script or interactively, by command line:

stordaemon start storsrvd [-args -port Port]

SERVER SYMAPI SERVER SHUTDOWN

Either on a client host or the server host the storsrvd executable for a network service can be stopped either by a shutdown script or interactively, by command line:

```
stordaemon shutdown storsrvd

INVOKING CLIENT SERVER OPERATIONS

The following is a description of invoking a client
server session using the SYMCLI commands.

To invoke a client server session, first start the
SYMAPI Server executable, storsrvd, on the remote
host as already described.

Then set the environment variable SYMCLI_CONNECT to
the desired service name, as entered in the netcnfg
file, on the local host.

Then optionally set the environment variable
SYMCLI_CONNECT_TYPE to the desired connection
type: LOCAL, REMOTE.  If REMOTE, operations
will be executed on the remote server.
If LOCAL, they are executed locally and the
SYMCLI_CONNECT environment variable is ignored.

Run any SYMCLI executable.

EXAMPLE:

setenv SYMCLI_CONNECT SYMAPI_SERVER
/usr/symcli/bin/symcfg list
```

symclone

Performs TimeFinder/Clone control operations on
a device group, composite group, or devices in
a device file.

SYNOPSIS

```
symclone -h

symclone -g <DgName> [-v] [-noprompt] [-force]
     [-i <Interval>] [-c <Count>] [-star]
     [-preserveTGTLocks -lockid <lockNum>]
     [-tgt [-bcv] | -rdf [-bcv | -tgt] |
      -rbcv -tgt | -rrbcv | -hop2 [-tgt]]

   create [-opt | -exact] [-concurrent] [-skip]
     [[-nocopy | -vse] [-nodifferential]] |
     [[-copy | -precopy] [-differential | -nodifferential]]

   split [-skip]

   activate [-consistent [-both_sides]]
     [-preaction <ScriptFile>] [-postaction <ScriptFile>]
     [-not_ready] [-skip] [-concurrent]

   terminate [-symforce] [-skip] [-restored] [-not_ready]

   recreate [-skip] [-precopy]

   establish [-consistent [-both_sides]]
     [-preaction <ScriptFile>] [-postaction <ScriptFile>]
     [-not_ready] [-skip]
     [-full [-opt|-exact] [-vse]] [-concurrent]

   restore [-preaction <ScriptFile>][-postaction
     <ScriptFile>] [-full]

symclone -g <DgName> [-offline] [-i <Interval>][-c <Count>]
     [-bcv | -rdf [-bcv] | -rbcv | -rrbcv | -hop2]

   query [-multi] [-summary] [-mb | -gb | -tb]

   verify [-created | -copied | -copyinprog |
     -copyonaccess | -copyonwrite | -precopy [-cycled] |
     -recreated | -restored | -restinprog | -split]
     [-force] [-concurrent] [-summary]

symclone -g <DgName> [-force]
     [-tgt [-bcv] | -rdf [-bcv | -tgt] |
      -rbcv -tgt | -rrbcv | -hop2 [-tgt]]

   set mode <copy | nocopy | precopy>

symclone -sid <SymmID> <-file <DeviceFileName> [-noprompt] |
     -noprompt 'redirect stdin'>
     [-force] [-i <Interval>] [-c <Count>] [-star]
     [-preserveTGTLocks -lockid <LockNum>] [-v]

   create [-skip]
     [[-nocopy | -vse] [-nodifferential]] |
     [[-copy | -precopy] [-differential | -nodifferential]]

   split [-skip]

   activate [-consistent]
```

```
     [-preaction <ScriptFile>] [-postaction <ScriptFile>]
     [-not_ready] [-skip]

   terminate [-symforce] [-skip] [-restored] [-not_ready]

   recreate [-skip] [-precopy]

   establish [-consistent]
     [-preaction <ScriptFile>] [-postaction <ScriptFile>]
     [-not_ready] [-skip]
     [-full [-opt | -exact] [-vse]]

   restore [-preaction <ScriptFile>][-postaction
     <ScriptFile>] [-full]

symclone -sid <SymmID> <-file <DeviceFileName> [-noprompt] |
     -noprompt 'redirect stdin'> [-force]
     [-i <Interval>] [-c <Count>]

   set mode <copy | nocopy | precopy>

symclone -sid <SymmID> <-file <DeviceFileName> |
     'redirect stdin'>
     [-i <Interval>] [-c <Count>]

   query [-multi] [-summary] [-mb | -gb | -tb]

   verify [-created | -copied | -copyinprog |
     -copyonaccess | -copyonwrite | -precopy [-cycled] |
     -recreated | -restored | -restinprog | -split]
     [-force] [-concurrent] [-summary]

symclone [-sid <SymmID>] [-i <Interval>] [-c <Count>]
     [-offline] [-mb | -gb | -tb]

   list [-copy | -nocopy | -precopy | -vse]

symclone -cg <CgName> [-v] [-noprompt] [-force]
     [-i <Interval>] [-c <Count>] [-star]
     [-tgt [-bcv] | -rdf [-bcv | -tgt] |
      -rbcv -tgt | -rrbcv | -hop2 [-tgt]]
     [-sid <SymmID> |
     -rdfg <SymmID>:<GrpNum>[,<GrpNum>,...]|<all>[,...] |
      name:<RdfGroupName>[,<RdfGroupName>,...]>]

   create [-opt | -opt_rag | -exact] [-concurrent] [-skip]
     [[-nocopy | -vse] [-nodifferential]] |
     [[-copy | -precopy] [-differential | -nodifferential]]

   split [-skip]

   activate [-consistent]
     [-preaction <ScriptFile>] [-postaction <ScriptFile>]
     [-not_ready] [-skip] [-concurrent]

   terminate [-symforce] [-skip] [-restored] [-not_ready]

   recreate [-skip] [-precopy]

   establish [-consistent]
     [-preaction <ScriptFile>] [-postaction <ScriptFile>]
     [-not_ready] [-skip]
     [-full [[-opt | -opt_rag] [-vse | -exact]]
     [-concurrent]

   restore [-preaction <ScriptFile>][-postaction
     <ScriptFile>] [-full]
```

```
symclone -cg <CgName> [-offline] [-i <Interval>]
     [-bcv | -rdf [-bcv] | -rbcv | -rrbcv | -hop2]
     [-c <Count>] [-sid <SymmID> |
     -rdfg <SymmID>:<GrpNum>[,<GrpNum>,...]|<all>[,...] |
      name:<RdfGroupName>[,<RdfGroupName>,...]>]

   query [-multi] [-summary] [-mb | -gb | -tb]

   verify [-created | -copied | -copyinprog |
     -copyonaccess | -copyonwrite | -precopy [-cycled] |
     -recreated | -restored | -restinprog | -split]
     [-force] [-concurrent] [-summary]

symclone -cg <DgName> [-force]
     [-tgt [-bcv] | -rdf [-bcv | -tgt] |
      -rbcv -tgt | -rrbcv | -hop2 [-tgt]]

   set mode <copy | nocopy | precopy>
```

DESCRIPTION

The symclone command performs TimeFinder/Clone operations on a device or composite group or on pairs listed in a device file.

These operations include creating and activating a source device with a target device in a copy session, terminating the session, and querying the state of the device pair.

You can perform all clone operations on a group or a device file.

Before you can copy a source device to a target device, the target device must have been previously associated with the device group and the target device must be the same size as the source device.

ARGUMENTS

activate Activate an internal copy session with the devices in the device or composite group and one or more target devices associated with the group.

While the operation is in progress, the state of the device pair is either CopyInProgress or CopyOnAccess. When the operation completes, the state changes to Copied.

create Creates an internal copy session with the devices in the device or composite group and one or more target devices associated with the group.

While the operation is in progress, the state of the device pair is CreateInProgress. When the operation completes, the state changes to Created.

establish Creates and activates an internal copy session with the devices in the group and one or more target devices associated with the group. Specifying this argument without the -full option performs a recreate

followed by an activate operation.

list	Lists all copy sessions created on the Symmetrix array.
query	Returns clone state information about all device pairs in a group or device file.
recreate	Copies only the tracks that changed since the last activate action to the target device. The session must have been created with the -differential option. You must issue a subsequent activate action to establish a new point-in-time copy.
restore	Initiates a copy from the target device to a source device. When you specify this argument with the -full option, a full copy of the data currently on the target device will occur. The device must be in a Copied state for a restore to take place.
set mode	Changes a session in the Created state to Precopy, Copy, or NoCopy. Once a session is in the Precopy state, you cannot change it to one of the other states.
split	Splits a clone device pair that is in the Restored state. Once in the Split state, you can either recreate or restore the pair.
terminate	Stops the existing internal copy session between the specified source and target devices in a group.
verify	Verifies whether all device pairs in a group are in the Copied state.

KEYWORDS

Locks	Locks on the target devices. The target devices must already be locked by the same lock holder ID. Requires -lockid.
name	Indicates an RDF group’s logical name.
SRCDEVS	Specifies to use the path names from the standard devices being controlled.

OPTIONS

-bcv	When used with the -rdf option, executes the operation on the remote BCV devices (RBCVs).
-both_sides	Activates all locally and remotely associated clone pairs in an SRDF group.
-c	Specifies the number (count) of times to display or to acquire an exclusive lock on the Symmetrix host database. If you do not specify this option and specify an interval (-i), the program will loop continuously to display or start the mirroring operation.
-cg	Applies a composite group name to the command.

-concurrent	When used with the verify argument, this option verifies the standard device and multiple target devices. When used with the active commands, this option performs the operation on a pair with an additional target device.
-consistent	Consistently activates the source and target pairs.
-copied	Verifies that the copy session(s) are in the Copied state.
-copy	Performs the device copy in the background. Normally, tracks are not copied unless the source device is written to, or the target device is read from or written to. When the create -copy operation executes, the state of the device pair is CopyInProgress. If all tracks are eventually moved to the target device, the state changes to Copied. When used with the list command, lists only sessions with background copy active.
-copyinprog	Verifies that the copy session(s) are in the CopyInProg state.
-copyonaccess	Verifies that the copy session(s) are in the CopyOnAccess state.
-copyonwrite	Verifies that the copy session(s) are in the CopyOnWrite state.
-created	Verifies that the copy session(s) are in the Created state.
-cycled	Verifies that the copy session(s) have completed one precopy cycle. Requires the -precopy option.
-differential	Creates a differential session, which you can recreate at a later time.
-exact	Selects pairs according to the exact order in which the source and target devices were added to the specified group. This option overrides all other pairing algorithms and only applies to group operations.
-file	Applies a device file to the command. The device file contains device pairs (SymDevnames) listing a pair per each line (the source device first, a space, and the target device last within each line entry). Device files can include comment lines that begin with the pound sign (#). A Symmetrix ID is required for this option. -f is synonymous with -file.
-force	Attempts to force the operation even though one or more paired devices in the device group may not be in the normal, expected state(s) for the specified operation.
-full	Copies data from all tracks to the target device. This option is only valid with a

Restore or Establish command.

-g Applies a device group name to the command.

-gb Displays counts in gigabytes.

-h Provides brief online help information.

-hop2 Performs the specified action on the Symmetrix array two hops away.

-i Specifies the repeat interval, in seconds, to display or to acquire an exclusive lock on the Symmetrix host database.
The default interval is 30 seconds.
The minimum interval is 5 seconds.
For passive actions the minimum interval is 15 seconds. Passive actions are actions that do not acquire an exclusive lock.

When used with the verify action, the number of seconds specified indicates the interval of time (in seconds) to repeat the verify command before the verify action finds and reports the pairs fully synchronized.

-lockid Specifies the lock holder ID for preserving the target locks on the control operation. The lock number ID must be a hexadecimal number.

-mb Displays counts in megabytes.

-multi Applies to the query operation in a multi-copy session environment to show all targets that are paired with the source devices.

-nocopy Creates a session without a background copy. This option is not allowed with the -differential option. When used with the list command, lists only sessions with background copy inactive.

-nodifferential
Creates a non-differential session, which cannot be recreated without a full copy of the data.

-noprompt Requests to not return a prompt after you enter a command. The default is to prompt for confirmation.

-not_ready Following the clone operation, for activate and establish leaves the target device(s) Not Ready to the host and for terminate leaves the target(s) in their prior Ready or Not Ready state.

-offline Specifies that the Symmetrix data connection is offline from the host in-memory database.

-opt Applies the symclone create command to optimize the device pair selection (source and target devices) to achieve the highest

copy speed between them. For remote BCV operations, use the -opt_rag option.

-opt_rag — Applies to the full create or establish operations for remote device optimization to distribute the I/O load so that the remote adapters are not connected to the same devices of the selected pair. Requires that you also specify the -rdf or -rrbcv option.

-postaction — Executes the script argument after a copy session has been activated.

-preaction — Executes the script argument before a copy session has been activated.

-precopy — Performs the device copy in the background before the activate starts. Normally, the copying of tracks does not start until the activate occurs. The precopy process continuously runs in the background until the activate is called. Precopy implies -copy. When used with the list command, lists only sessions with precopy active.

-preserveTGT — Performs the action without taking out device locks on the target devices. The target devices must already be locked by the same lock holder ID. Requires the -lockid option.

-rdf — Performs the action on the remote Symmetrix array.

-rdfg — Performs the requested action on a subset of the composite group defined by one or more Symmetrix/RA Group combinations supplied as the argument to -rdfg.

-restored — With the verify command, verifies that the copy session(s) are in the Restored state. With the terminate command, terminates a restored VP Snap session.

-rrbcv — Performs the action on the multi-hop Symmetrix array.

-sid — Applies the command to the specified Symmetrix ID. Specify this option with the -file option to select the Symmetrix array on which to perform the operation, or specify this option with -cg option to restrict the operation to a single Symmetrix array.

-skip — Skips the source locks action. This option will not lock the source devices if all of the specified source devices are either locked or unlocked. Applicable only when the source device is a standard device (not a BCV).

-tb — Displays counts in terabytes.

-vse — Sets the copy mode to VP Snap nodifferential. When used with the list

command, lists only VP Snap sessions.

-split Verifies that the Copy session(s) are in the Split state.

-star Targets the action at devices in STAR mode.

-summary Shows device state summary.

-symforce Forces the operation to execute when normally it is rejected. Use extreme caution with this option.

-tgt Uses TGT devices as clone targets. When you use this option with the -rdf option, the operation will use RTGT devices.

-v Provides a more detailed, verbose listing.

PARAMETERS

CgName The composite group name.

DgName The device group name.

DeviceFileName The device file name. The device file contains device pairs (SymDevNames) listing a pair each line (the source device first, a space, followed by the target device name on each line).

GrpNum The RDF (RA) group number.

LockNum The hexadecimal value of the lock holder ID.

RdfGroupName The logical name associated with the RDF (RA) group(s).

ScriptFile The full pathname of a script file to execute.

RETURN CODES

Code #	Code Symbol
0	CLI_C_SUCCESS
1	CLI_C_FAIL
2	CLI_C_DB_FILE_IS_LOCKED
18	CLI_C_ALREADY_IN_STATE The device or device group is already in the desired Copy state.
19	CLI_C_GK_IS_LOCKED All gatekeepers to the Symmetrix array are currently locked.
22	CLI_C_NEED_FORCE_TO_PROCEED Requires the force flag to proceed.
23	CLI_C_NEED_SYMFORCE_TO_PROCEED Requires the symforce flag to proceed. CAUTION: Extreme caution should be exercised

when using this option.

Return codes for symclone verify

53	CLI_C_NOT_ALL_COPYINPROG Not all source devices are in the CopyInProg state.
54	CLI_C_NONE_COPYINPROG No source devices are in the CopyInProg state.
55	CLI_C_NOT_ALL_COPIED Not all source devices are in the Copied state.
56	CLI_C_NONE_COPIED No source devices are in the Copied state.
57	CLI_C_NOT_ALL_COPYONACCESS Not all source devices are in the CopyOnAccess state.
58	CLI_C_NONE_COPYONACCESS No source devices are in the CopyOnAccess state.
60	CLI_C_NOT_ALL_CREATED Not all source devices are in the Created state.
61	CLI_C_NONE_CREATED No source devices are in the Created state.
68	CLI_C_NOT_ALL_RECREATED Not all source devices are in the Recreated state.
69	CLI_C_NONE_RECREATED No source devices are in the Recreated state.
73	CLI_C_NOT_ALL_PRECOPY Not all source devices are in the Precopy state.
74	CLI_C_NONE_PRECOPY No source devices are in the Precopy state.
75	CLI_C_NOT_ALL_PRECOPY_CYCLED Not all source devices have completed one precopy cycle.
76	CLI_C_NONE_PRECOPY_CYCLED No source devices have completed a precopy cycle.

EXAMPLES

To create the device group ProdDB as a REGULAR device group, enter:

```
symdg create ProdDB
```

To define device group ProdDB as the default device group, enter:

```
setenv SYMCLI_DG ProdDB
```

To create a clone copy of the source devices in group ProdDB with target devices (associated with the group), enter:

```
symclone create -g ProdDB
symclone activate -g ProdDB
```

To wait until the BCV pairs are fully copied, polling every 30 seconds, enter:

```
symclone -i 30 verify -g ProdDB
```

To query information about all paired devices in device group ProdDB, enter:

```
symclone query
```

symconfigure

Allows you to create and delete Symmetrix devices.

Allows you to modify Symmetrix devices, ports, RDF characteristics, and host assignments.

Provides SAVE device (SAVEDEV) and DATA device pool management.

Provides management and support for device reservations.

Provides a query option to allow monitoring the progress of a Symmetrix configuration change.

Provides an abort option for situations where the host connection to the configuration server was dropped and left a configuration change session in progress.

Provides a verify option to determine if the Symmetrix configuration complies with the requirements for host-based configuration changes.

Provides a list option to display information about the available space on the physical disks in the Symmetrix array.

Provides support to create gatekeeper devices and set disk group names and reset them back to their default names.

Provides support to create external disk groups and add/remove external disks to/from external disk groups.

Provides support for setting parameters on Storage Resource Pools (SRPs).

SYNOPSIS

symconfigure -h

symconfigure -sid <SymmID> [-h] [-v]
[-file <CmdFile> | 'redirect stdin' | -cmd "Cmd"]
[-noprompt] [-noecho] [-i <Interval>] [-c <Count>]
[-reserve_id <ResvID>[,<ResvID>[,<ResvID>]]]
[-remote_reserve_id <ResvID>[,<ResvID>[,<ResvID>]]]

preview

prepare

commit

symconfigure -sid <SymmID>

abort -session_id <SessionID>

verify

symconfigure -sid <SymmID> [-i <Interval>] [-c <Count>]

```
          [-session_id <SessionID>] [-v]

       query

    symconfigure -sid <SymmID>
         <-v | -freespace [-units CYLINDERS | MB] >

       list

    symconfigure -version [-v] [-sid <SymmID>]

    symconfigure -sid <SymmID> [-expire <ExpirationDate>]
         [-f[ile] <CmdFile> | 'redirect stdin' | -cmd "Cmd"]
         [-noprompt] [-v]
         -owner <Owner> -comment "UserComment"
         [-enforce | -advise]

       reserve

    symconfigure -sid <SymmID>
         -reserve_id <ResvID>[,<ResvID>[,...]]
         [-noprompt]

       release

    symconfigure -sid <SymmID> -reserved

       list

    symconfigure -sid <SymmID> -reserve_id <ResvID>

       show

COMMAND FILE (or -cmd) SYNTAX

Adding a new device for 5876:

    create dev count=<n>,
         size = <n> [MB | GB | CYL],
         emulation=<EmulationType>,
         config=<DevConfig>
         [, data_member_count=<n>]
         [, remote_config=<DevConfig>, ra_group=<n>
           [, remote_data_member_count=<n>]
           [, remote_mvs_ssid=<n>]
           [, dynamic_capability =
             <DYN_RDF | DYN_RDF1_ONLY | DYN_RDF2_ONLY>]]
         [, mvs_ssid=<n>]
         [, attribute=<CKD_META | SAVEDEV | DATADEV>
           [, in pool=<PoolName>]
           [, member_state=<ENABLE | DISABLE>]]
         [, meta_member_size = <n> [MB | GB | CYL]]
         [, meta_config = <STRIPED | CONCATENATED>]
         [, disk_group=<<n>| name:<DskGrpName>>
           [, remote_disk_group=<<n> | name:<DskGrpName>>]]
         [, binding to pool=<PoolName> [, remote_pool=<PoolName>]
           [, preallocate size = <ALL>
             [, allocate_type = PERSISTENT]]]
         [, binding to <pool=<PoolName> | remote_pool=<PoolName>>
           [, preallocate size = <ALL>
             [, allocate_type = PERSISTENT]]]
         [, mapping to dir <director_num:port>
           [starting] target = <scsi_target>,
           lun=<scsi_lun>, vbus=<fibre_vbus>
           [starting] base_address = <cuu_address>[,...]]
         [, device_attr =
           <SCSI3_PERSIST_RESERV | ACLX | DIF1 |
```

```
              AS400_GK>[,...]]
          [, device_name=<DeviceName> [,number=<n | SYMDEV>]]
          [, sg=<SgName> [,remote_sg=<SgName>]];
```

Adding a new device for 5977 and above:

```
     create dev count=<n>,
          size = <n> [MB | GB | CYL],
          emulation=<EmulationType>,
          config=<DevConfig>
          [, preallocate size = <ALL>
            [, allocate_type = PERSISTENT]]
          [, sg=<SgName>]
          [, mapping to dir <director_num:port>
            [starting] target = <scsi_target>,
            lun=<scsi_lun>, vbus=<fibre_vbus>
            [starting] base_address = <cuu_address>[,...]]
          [, mapping to cu_image = <cu_image_num>,
            split_name=<split_name>,
           [starting] base_address=<base_address>
           [, mvs_ssid=<n>]]
          [, device_attr =
            <SCSI3_PERSIST_RESERV | DIF1 |
              AS400_GK>[,...]]
          [, device_name=<DeviceName> [,number=<n | SYMDEV> ]];
```

Adding a gatekeeper device for 5876
with emulation type = AS/400_D910_099:

```
     create gatekeeper count=<n>,
          emulation=<EmulationType>,
          type=thin,
          binding to pool=<PoolName>
          [, sg=<SgName>]
          [, mapping to dir <director_num:port>
             [starting] target = <scsi_target>,
             lun=<scsi_lun>, vbus=<fibre_vbus>
             [starting] base_address = <cuu_address>[,...]]
```

Adding a gatekeeper device:

```
     create gatekeeper count=<n>,
          emulation=<EmulationType>,
          [, type=thin
              [, binding to pool=<PoolName>]]
          [, sg=<SgName>]
          [, [mapping to dir <director_num:port>
             [starting] target = <scsi_target>,
             lun=<scsi_lun>, vbus=<fibre_vbus>
             [starting] base_address = <cuu_address>[,...]]
          [, mapping to cu_image = <cu_image_num>,
             split_name=<split_name>
             [starting] base_address = <base_address>
             [MVS_SSID = <n>]];
```

Adding a meta member:

```
     add dev <SymDevName>[:<SymDevName>]
          to meta <SymDevName>
          [, protect_data=[TRUE | FALSE],
             bcv_meta_head=<SymDevName>];
```

Adding an RDF mirror for 5876:

```
     add rdf mirror to dev <SymDevName>[:<SymDevName>]
          ra_group=<n>, mirror_type = [RDF1 | RDF2],
          remote_dev = <SymDevName>,
```

```
                invalidate = <invalidate_opt>,
                start_copy = [YES | NO]
                [,rdf_mode = [sync|semi|acp_wp|acp_disk|async]];

        Adding a director:
             add dir slot_num = <director_slot_num> type  = [FA | FE | FN | SE | RF | R
E];

        Assigning a PAV alias address range to a CU image:

             add pav alias_range addr <n:n>
                to mvs_ssid=<n>;

        Configuring a device by copying a similar device:

             configure [<n.n> [MB | GB] | <n> devices]
                copying dev <SymDevName>
                [mapping to cu_image = <cu_image_num>,
                   split_name=<split_name>
                   [starting] base_address=<base_address>]
                [mapping to dir <director_num>:<port_num>
                [masking hba [awwn=<awwn> | wwn=<wwn> |
                              iscsi=<iscsi> |
                              aiscsi=<aiscsi>]
                              [host_lun=<lun> |
                               dynamic_lun]]]
                [,device_name=<DeviceName>
                 [,number=<n | SYMDEV> ]]
                [overriding
                         [size=<n> [MB | GB | CYL]]
                         [emulation=<EmulationType>]
                         [config=<DevConfig>]
                         [data_member_count=<n>]
                         [mvs_ssid=<n>]
                         [disk_group=<n>| name:<DskGrpName>]];

        Converting a device’s configuration:

             convert dev <SymDevName>[:<SymDevName>] to
                <DevConfig>
                [emulation=CELERRA_FBA,]
                [ ra_group=<n>, remote_dev=<SymDevName>,
                  invalidate=<invalidate_opt>,
                  [remote_mvs_ssid=<n>],
                  start_copy=<YES | NO> ]
                [mvs_ssid=<n>] [raidset = [TRUE | FALSE]];

        Converting a metadevice’s configuration:

             convert meta <SymDevName>
                config=<meta_option>
                [, stripe_size=<meta_stripe_size>[cyl]],
                [, protect_data=[TRUE | FALSE],
                   bcv_meta_head=<SymDevName>];

        Converting an RDF device from static RDF to dynamic RDF:

             convert rdf dev <SymDevName>[:<SymDevName>]
                to dynamic;

        Converting a director’s type:

             convert dir <director_num> to type = <FA | RF>;

        Deleting a Symmetrix device:

             delete dev <SymDevName>[:<SymDevName>]
```

```
          [, raidset = [TRUE | FALSE]];
```

Dissolving a metadevice:

```
     dissolve meta dev <SymDevName>[:<SymDevName>]
        [[,<SymDevName>[:<SymDevName>]]...];
```

Forming a metadevice:

```
     form meta from dev <SymDevName>,
          config=<meta_option>
          [, stripe_size=<meta_stripe_size> [cyl] ]
          [, count=<member_count>];
```

Mapping a device to an address, or mapping a range of devices to consecutive addresses by specifying a starting address:

```
     map dev <SymDevName>[:<SymDevName>]
          to dir <director_num>:<port_number>,
          [starting] [target=<scsi_target>,] lun=<scsi_lun>
          [, vbus=<fibre_vbus>]
          [, awwn=<awwn> | wwn=<wwn> | iscsi=<iscsi> |
             aiscsi=<aiscsi> ]
          [, masking [host_lun=<lun> | dynamic_lun] ]
          [, emulation=CELERRA_FBA];
```

Mapping a range of devices to EF (mainframe) ports:

```
     map dev <SymDevName>[:<SymDevName>]
          to dir <director_num>:<port_number>,
          [starting] base_address=<cuu_address>
          [mvs_ssid=<n>];
```

Mapping a range of devices to EF (mainframe) ports in Enguinity 5977 and above:

```
     map dev <SymDevName>[:<SymDevName>]
          to cu_image = <cu_image_num>,
          split_name=<split_name>,
          [mvs_ssid=<n>,]
          [starting] base_address=<base_address>;
```

Removing a director:

```
     remove dir <director_num>;
```

Removing a metamember:

```
     remove dev <SymDevName>[:<SymDevName>]
          from meta <SymDevName>;
```

Removing PAV alias address range from a CU image:

```
     remove pav alias_range from mvs_ssid=<n>;
```

Removing an RDF mirror for 5876:

```
     remove rdf mirror from
          dev <SymDevName>[:<SymDevName>],
          ra_group=<n>;
```

Setting a device’s emulation type:

```
     set dev <SymDevName>[:<SymDevName>]
          emulation=<EmulationType>;
```

Setting a device attribute:

```
     set dev <SymDevName>[:<SymDevName>]
          attribute = [NO ]<DevAttr>;

     set dev <SymDevName>[:<SymDevName>]
          identity = NO identity;
```

Setting a device identifier:

```
     set dev <SymDevName>[:<SymDevName>]
          [device_name='<DevName>' | NO device_name]
          [[hp_identifier='<hp_id>' | NO hp_identifier] |
           [vms_identifier=<vms_id> | NO vms_identifier]];
```

Setting port characteristics:

```
     set port <director_num>:<port_number>
          [ <flag_name>=<ENABLE | DISABLE> [, ...] ]
          [[gige] [primary_ip_address=<IPaddress>],
           [primary_netmask=<IPaddress>],
           [default_gateway=<IPaddress>],
           [isns_ip_address=<IPaddress>],
           [primary_ipv6_address=<IPaddress>],
           [primary_ipv6_prefix=<0-128>],]
          [fa_loop_id=<integer>] [hostname=<HostName>];

     set port <director_num>:<port_number>,
          copying port <director_num>:<port_number>,
          [fa_loop_id=<integer>] [hostname=<HostName>];
```

Setting a Storage Resource Pool (SRP) parameter:

```
     set srp <SRP Name>,
          <[resv_cap = <n | NONE>]
          [,rdfa_dse = <ENABLE | DISABLE>]
          [,description = 'SRP Description']>;
```

Setting a name to a Service Level (SL):

```
     set sl <SLName>
<name=NewSLName | BASE NAME>;
```

Setting a Symmetrix configuration metric:

```
set symmetrix
<metric_name> = <metric_value>
[, <metric_name> = <metric_value>];
```

Unmapping a device:

```
unmap dev <SymDevName>[:<SymDevName>] from dir
< ALL:ALL | ALL:<port_number> |
<director_num>:ALL |
<director_num>:<port_num> >
[, emulation=CELERRA_FBA];
```

Unmapping a range of devices from EF (mainframe) ports:

```
unmap dev <SymDevName>:[<SymDevName>] from dir
< ALL:ALL | ALL:<port_num> |
<director_num>:ALL |
<director_num>:<port_num> >
[new_ssid=<n>];
```

Unmapping a range of devices from EF (mainframe) ports in Enguinity 5977 and above:

```
unmap dev <SymDevName>[:<SymDevName>]
from cu_image = <cu_image_num>,
split_name=<split_name>;

Reserving a device:

reserve dev <SymDevName>[:<SymDevName>];

Creating a snap, RDFA_DSE, or thin pool:

create pool <PoolName>,
type = <snap | rdfa_dse | thin>
[, max_subs_percent= <n>]
[, rebalance_variance = <n>]
[, max_dev_per_rebalance_scan = <n>]
[, pool_resv_cap = <n>]
[, vp_compression = <ENABLE>];

Deleting a snap, RDFA_DSE, or thin pool:

delete pool <PoolName>,
type = <snap | rdfa_dse | thin>;

Setting thin pool attributes:

set pool <PoolName>,
type = <thin>,
<[max_subs_percent= <n | NONE>]
[, rebalance_variance = <n>]
[, max_dev_per_rebalance_scan = <n>]>
[, pool_resv_cap = <n | NONE>]
[, vp_compression = <ENABLE | DISABLE>];

Renaming a snap, RDFA_DSE, or thin pool:

rename pool <PoolName> to <NewPoolName>
type = <snap | rdfa_dse | thin>;

Renaming a Storage Resource Pool (SRP):

rename SRP <SRP Name> to <New SRP Name>;

Adding a SAVEDEV to a snap or RDFA_DSE pool, or
adding a DATADEV to a thin pool:

add dev <SymDevName>[:<SymDevName>] to pool <PoolName>
type = <snap | rdfa_dse | thin>
[, member_state = <ENABLE | DISABLE> ];

Removing a SAVEDEV or DATADEV from a named pool:
(Once removed, the device is available and can be
added to any snap, RDFA_DSE, or thin pool.)

remove dev <SymDevName>:[<SymDevName>]
from pool <PoolName>,
type = <snap | rdfa_dse | thin>;

Enabling a SAVEDEV or DATADEV for use:

enable dev <SymDevName>:[<SymDevName>] in pool
<PoolName>, type = <snap | rdfa_dse | thin>;

Disabling a SAVEDEV or DATADEV from use:

disable dev <SymDevName>:[<SymDevName>] in pool
<PoolName>, type = <snap | rdfa_dse | thin>;
```

```
Activating a DATADEV for use:

activate dev <SymDevName>:[<SymDevName>] in pool
<PoolName>, type = <thin>;

Activating an edisk for use:

activate external_disk <wwn=<wwn> | spid=<SpindleID>>;

Deactivating a DATADEV for use:

deactivate dev <SymDevName>:[<SymDevName>] in pool
<PoolName>, type = <thin>;

Starting a drain operation on a DATADEV:

start drain on dev <SymDevName>[:<SymDevName>]
in pool <PoolName> type = <thin>;

Starting a drain operation on a external disk

start drain on external_disk <wwn=<wwn> | spid=<SpindleID>>;

Stoping drain operation on a external disk

stop drain on external_disk <wwn=<wwn> | spid=<SpindleID>>;

Stopping a drain operation on a DATADEV:

stop drain on dev <SymDevName>[:<SymDevName>]
in pool <PoolName> type = <thin>;

Starting a pool balancing operation on a thin pool:

start balancing on pool <PoolName>;

Stopping a pool balancing operation on a thin pool:

stop balancing on pool <PoolName>

Setting a disk group attribute:

set disk_group <DskGrpNum | name:DskGrpName>
group_attribute_name = value

Creating a disk group:

create disk_group DskGrpName
disk_location = <external>

Deleting a disk group:

delete disk_group <DskGrpNum | name:DskGrpName>

Adding an external device from the external array to a
disk group for 5876:

add external_disk wwn=<wwn>
[to disk_group=<DskGrpNum | name:DskGrpName>]
encapsulate_data=<NO | YES [pool=<PoolName>]>
[meta_member_size=<n> [MB | GB | CYL]]
[member_size_equal=<YES | NO>] [dir=<Director_num>];

Adding an external disk for 5977 and above:

add external_disk wwn=<wwn>
encapsulate_data=<YES | NO <SRP=<SRPName>>>
```

```
[dir=<Director_num>];

Adding an external disk for 5977 Q12016 SR and above:

add external_disk wwn=<wwn>
encapsulate_data=<YES | NO
<keep_data=<YES [SG=<sgname>] | NO [SRP=<SRPName>]>>>
[dir=<Director_num>];


Removing an external device from a disk group:

remove external_disk <wwn=<wwn> | spid=<SpindleID>>
[force_remove=<YES | NO>];

Associating a port to a director emulation:

associate port <port_num>[,<port_num>...] to
dir <director_num>;

Disassociating a port from a director emulation:

disassociate port <port_num>[,<port_num>...] from
dir <director_num>;

Creating an IP Interface:

create ip_interface dir <director_num> port <port_number>,
ip_address = <IPaddress>, ip_prefix = <ip_prefix>,
network_id = <network_id>, vlanid = <vlanid>
[, mtu = <mtu>];

Modifying an IP Interface:

modify ip_interface dir <director_num>,
ip_address = <IPaddress>, network_id = <network_id>
[, new_network_id = <network_id>]
[, new_ip_address = <IPaddress>]
[, ip_prefix = <ip_prefix>]
[, mtu = <mtu>];

Deleting an IP Interface:

delete ip_interface dir <director_num>, ip_address = <IPaddress>,
network_id = <network_id>;

Adding an IP Route:

add ip_route dir <director_num>,
ip_address = <IPaddress>, ip_prefix = <ip_prefix>,
gateway = <IPaddress>
[, network_id = <network_id>];

Removing an IP Route:

remove ip_route dir <director_num>,
ip_address = <IPaddress>
[, network_id = <network_id>];

Creating an iSCSI target:

create iscsi_tgt dir <director_num>,
network_id = <network_id>
[, iqn = <IQN>] [, ip_address = <IPaddress> [,...]]
[, set_default_flags = <ENABLE | DISABLE>]
[, flag_name = ENABLE [,...]]
[, tcp_port = <tcp_port>];
```

```
Valid iSCSI Target port flags are:
SOFT_RESET
ENVIRON_SET
DISABLE_Q_RESET_ON_UA
AVOID_RESET_BROADCAST
SCSI_3
SPC2_PROTOCOL_VERSION
SCSI_SUPPORT1
VOLUME_SET_ADDRESSING
OPENVMS
ISID_PROTECTED

Modifying an iSCSI target:

modify iscsi_tgt
<[iqn = <IQN>] | [iscsi_dirport = <director_num>:<port_ number>]>
[, flag_name = ENABLE | DISABLE [,...]][, tcp_port = <tcp_port>]
[, network_id = <network_id>];

Valid iSCSI Target port flags are same as in creating an iSCSI target.

Renaming an iSCSI target:

rename iscsi_tgt
<[iqn = <IQN>] | [iscsi_dirport = <director_num>:<port_number>]>
to new_iqn = <IQN>;

Deleting an iSCSI target:

delete iscsi_tgt
<[iqn = <IQN>] |
[iscsi_dirport = <director_num>:<port_number>]>;

Attaching an IP Interface to an iSCSI target:

attach ip_interface ip_address = <IPaddress>,
to iscsi_tgt
<[iqn = <IQN>] | [iscsi_dirport = <director_num>:<port_number>]>;

Detaching an IP Interface from an iSCSI target:

detach ip_interface ip_address = <IPaddress>
from iscsi_tgt
<[iqn = <IQN>] | [iscsi_dirport = <director_num>:<port_number>]>;

DESCRIPTION

The symconfigure command allows you to modify aspects of a
Symmetrix configuration. A lock may be taken out by the
specified Symmetrix configuration server while the
configuration change session is active. A session is
processed in a series of stages which may be monitored by
using the query option.

EMC has restricted what state the Symmetrix configuration
must be in before allowing changes to be applied from a
host system. To determine if a Symmetrix array can be
modified, use the verify command.

Not all stages in the change process must be executed.
The user indicates which stages should be completed to
allow command files to be debugged without reconfiguring
the Symmetrix array. The preview argument verifies the
syntax and correctness of each individual change defined,
and then aborts the session.
```

The prepare argument performs the preview checks and also verifies the appropriateness of the resulting configuration definition against the current state of the Symmetrix array and then aborts the session.

The commit argument completes all stages and activates the changes within the Symmetrix array.

The definitions of changes may be placed into a command file, or provided with the command line argument -cmd, which is read and processed by the utility. Alternatively, stdin redirection can be used with "here documents" in UNIX shell scripts. Each change in the file is terminated by a semi-colon (;). The parsing of the file is case insensitive.

In the same session, all different types of changes can be performed, except RDF changes and pool changes.

Earlier Enginuity versions allowed multiple changes to be made in one session, but all the changes had to be in the same class. An exception to this is that increasing and decreasing mirror protection could always be done in the same session.

Mapping and unmapping devices includes options for modifying the device masking associated with those devices. See symmask and symmaskdb for additional information. On Symmetrix arrays running Enginuity 5876 and later, the masking options for mapping and unmapping commands are not supported. See the symaccess command for additional information.

The classes of changes available are:

Creating:
- Creating new Symmetrix devices
- Configuring new Symmetrix devices

Mapping:
- Mapping/unmapping a Symmetrix device to a front-end port
- Mapping/unmapping a range of devices that form a mainframe CU image to a front-end port
- Copying device mappings from one EF port to another
- Adding or removing PAV aliases from mainframe devices

FBA metadevice handling:
- Forming/dissolving metadevices
- Adding/removing metamembers
- Converting meta types

CKD metadevice handling:
- Creating/deleting metadevices

Device configuring:
- Adding/removing BCV attributes
- Adding/removing DRV attributes

Increase mirroring:
- Adding mirror(s) to an existing device; for example, converting RDF1 to RDF1+Mir

Decrease mirroring:
- Removing mirror(s) from an existing device, resulting in the creation of a new device

SAVEDEV and DATADEV pool management:

- Adding or deleting pools
- Adding or removing pool members
- Enabling or disabling pool members
- Renaming pools

Setting device attributes:
- Marking devices as available for use as dynamic RDF devices, or VCM/ACLX database devices
- Enabling the SCSI3 Persistent Reservation option for clustered devices
- Restoring a device’s identity to its original value

Setting device emulation:
(Note: device emulation can be changed among FBA emulation types only.)

Setting front-end port attributes:
- Setting/resetting SCSI or fibre-port flags
- Setting fibre FA loop ID
- Setting port connection’s host name
- Setting Gig-E front-end port IP addresses and netmasks

Setting Symmetrix metrics, including the following:
- auto_meta
- auto_meta_config
- auto_meta_member_size
- dse_max_cap
- min_auto_meta_size
- pav_alias_limit
- pav_mode
- rdfa_cache_percent
- rdfa_host_throttle_time

Disk group management:
- Setting and resetting disk group names

External disk management:
- Adding and removing an eDisk to and from an external disk group.

Port Asssociation/Disassociation:
- Association of port to director emulation
- Diassociation of port from director emulation

Storage Resource Pool (SRP) management:
- Setting the reserve capacity for the SRP
- Setting the SRP to be used for RDFA DSE allocations

The overall processing time for the application of the changes can vary from seconds to over an hour, depending on the class of the changes and the complexity of the Symmetrix configuration. RDF changes will also be applied to the remote Symmetrix array. Depending on the state of the RDF pairs, additional processing time will be needed if the devices are to be synchronized.

Pool sessions are executed directly within the Symmetrix array and do not communicate with the configuration server. The query and abort arguments are not available for pool sessions.

Support for reserving devices and front-end mapping addresses for future, planned configuration changes is provided through the use of the reserve, release, list, and show commands. A reserve ID is assigned to a reservation for use with configuration change and masking operations. A reservation can be defined as requiring

enforcement or as an advisory. Management of advisories varies with the application. Management of enforced reservations is done by the SYMAPI and requires the user to specify the reserve ID to be able to use the devices.

Devices are reserved by using the RESERVE argument. All standard configuration change commands that deal with devices can be used or a reserve command can be used to reserve devices.

The configure command option allows you to create new devices by copying an existing device, optionally changing some of the attributes and/or mapping the devices to a specified port. This allows a series of commands (create, map, convert, form) to be specified in a single command file entry. The port addresses for the mapping will be generated. If the device being copied is a metadevice, the newly created devices will also be metadevices. Mapping to EF ports is not supported.
In addition to mapping the configure command also supports masking options. On Symmetrix arrays running Enginuity 5876 and later, the masking options are not supported. See the symaccess command for additional information.

The list argument currently has two options (-v and -freespace). The -v option lists out configuration information that is not stored in the SYMAPI database and that needs to be retrieved directly from the configuration server.

The -freespace option focuses solely on the free physical disk space within the Symmetrix array as it can be used to create new Symmetrix devices for different emulation modes. If a physical disk has been partially used to create a device, that disk is considered to be formatted, and the rest of the available space can only be used for devices of the same emulation mode. Free disk space on unformatted disks is shown as available for all emulation modes.

Freespace is shown in units of cylinders by default. To have it converted to megabytes, use the -units MB option.

ARGUMENTS

abort Attempts to gain control of an existing session to abort it and free the configuration lock. On Enginuity 5876 and higher, if there is more than one session running, a user can use the -session_id option to abort a particular session.

commit Activates the changes defined in the command file into the specified Symmetrix array.

list Lists the relevant details, depending on the option:

- When used with -freespace, shows the free physical disk space within the Symmetrix array as it can be used to create new Symmetrix devices for different emulation modes. Free disk space on unformatted disks is shown as available for all emulation modes. If a physical disk has been partially used to create a device, that device is considered formatted and the rest

of the available space can only be used for devices of the same emulation mode.

- When used with -v, displays configuration information that is not stored in the SYMAPI database and that needs to be retrieved directly from the configuration server.

prepare Verifies the validity of the command file changes and their appropriateness for the specified Symmetrix array. The prepare action has no function for pool sessions.

preview Verifies the validity of the command file changes.

query Returns information about the status of a configuration change session. If there is more than one session running, a user can use the -session_id option to query a particular session.

release Releases the specified device reservation.

reserve Processes the command file to reserve the indicated devices and displays the resulting reserve ID.

show Shows the details of the specified device reservation.

verify Verifies that the configuration currently running in the specified Symmetrix array complies with the requirements for host-based configuration changes.

OPTIONS

-advise Marks a device reservation as being an advisory to all users that the included devices have been scheduled for a pending configuration change or device masking change. Applications may allow the user to proceed with using these devices.

-c Specifies the number (count) of times to attempt the requested action. If this option is not specified and an interval (-i) is specified, the process will loop querying for status until the session for the specified Symmetrix array completes. When used with the preview/prepare/commit actions, the process will attempt -c number of times waiting for the database or configuration locks.

-cmd Specifies the list of commands containing the configuration change definitions. Each command must be terminated by a semicolon.

-comment Provides additional details about a device reservation request.

-enforce Marks a device reservation for enforcement, by requiring you to supply the device reservation ID to use the devices. This

is the default behavior when reserving devices. Older applications may not be able to process reservation IDs.

-expire Sets the date and time for a reservation to expire, which will cause it to be removed from the reservation database automatically.

-f[ile] Specifies the command file containing the configuration change definitions.

-h Provides brief, online help information.

-i Specifies the repeat interval for retrying the requested action. For a query, this option indicates how often to display the session's status. For a preview/prepare/commit, this option indicates how often to attempt to get the needed resources to start a new session. The default interval is 30 seconds. The minimum interval is 5 seconds. For passive actions the minimum interval is 15 seconds. Passive actions are actions that do not acquire an exclusive lock.

-noecho Blocks the printing of session status and progress messages during the configuration change session. For use with PREVIEW, PREPARE, and COMMIT actions. Cannot be used with the -v option.

-noprompt Bypasses the confirmation requests issued to the output terminal. Intended for use in script files.

-owner Specifies the individual or organization that owns the devices being reserved. This is a required field.

-remote_reserve_id
Specifies the identification number assigned to a reservation on a remote Symmetrix array.

-reserve_id Specifies the identification number assigned to a reservation.

-sid Identifies the Symmetrix (Symmetrix ID) configuration to change.

-v Echoes the contents of the command file to the output terminal. Cannot be used with the -noecho option.

-version Lists the SYMCLI, SYMAPI, and configuration server version information. Also connects to the configuration server managing the the Symmetrix configuration to request its version. If a host is connected to more than one Symmetrix array, the -sid option is required.

PARAMETERS

CmdFile The name of an ASCII text file containing the set of commands to process.

ExpirationDate
The optional date and time for a device reservation to be expired. The default is no expiration. The format is:

[mm/dd[/yy]:][hh:mm[:ss]]

If only the hh:mm is provided, the current day will be assumed. If only mm/dd is provided, the current year is assumed. A four-digit year can also be specified.

Owner The name of the owner of the reservation.

ResvID The device reservation ID.

SymmID The 12-character ID of the Symmetrix array.

UserComment The user comment detailing the device reservation.

COMMAND FILE (or -cmd) PARAMETERS

alias An alternative front-end mapping address, used by a mainframe host to access a device.

awwn The alias WWN which identifies the HBA by a user-specified alias.

base_address The base address to be assigned to the first device in the mapping request. It will be incremented by one for each device in the range of devices being mapped.

bcv_meta_head
When adding new members to an existing striped metadevice, or when reconfiguring the stripe format, if the data on the metadevice is to be protected, you must specify the name of a bcv_meta that matches the original metadevice in capacity, stripe count, and stripe size.

ckd_meta An option available when creating a device with an emulation type of CKD-3380 or CKD-3390. Indicates that the device should be a striped metadevice. CKD metadevices must be created in sets of four devices.

count A positive integer.

cuu_address A base or alias address for a device being mapped to an EF port. These are mainframe ports which expect devices to be mapped in groups to form CU images. The first digit in the address is the CU image number, which can range from 0 to 0xF. The remaining two digits can range from 00 to 0xFF.

cycle_time The minimum time to wait before attempting an RDF/A cycle switch. Possible values range from 1 to 60 seconds.

cylinders A positive integer. A cylinder for FBA emulation is 1920 512-byte blocks

Devices that are going to be used as BCV, RDF, or metamembers need to precisely match the corresponding devices in size. Use the symdev/sympd show command or the symdev/sympd list -cyl command to see relevant device sizes.

With the exception of the two variable size emulation types of AS/400_M2107_099 and AS/400_M2107_050, AS/400 devices have specific sizes that must be used:

Meta Device Type	Cyl	Size
AS/400_M2105_A01	17484	
AS/400_M2105_A81	17484	
AS/400_M2105_A02	35720*	8930
AS/400_M2105_A82	35720*	8930
AS/400_M2105_A84	143584*	8974
AS/400_M2105_A04	143584*	8974
AS/400_M2105_A05	71568*	8946
AS/400_M2105_A85	71568*	8946
AS/400_M2107_A06	143576*	17947
AS/400_M2107_A07	287152*	17947
AS/400_M2107_A86	143576*	17947
AS/400_M2107_A87	287152*	17947
AS/400_M2107_A02	17860*	4465
AS/400_M2107_A04	71792*	17948
AS/400_M2107_A05	35784*	8946
AS/400_M2107_A82	17860*	4465
AS/400_M2107_A84	71792*	17948
AS/400_M2107_A85	35784*	8946

* These device models cannot be created as a single volume, as they exceed the maximum size supported by the Symmetrix array. Users can directly create metadevices by enabling the auto meta feature.

datadev An option available when creating a device, which indicates that the device should become part of the thin device pool for use with thin devices.

data_member_count
When creating a RAID-5 or RAID-6 device on a Symmetrix array set the value to 3 or 7 for RAID-5 (3+1) and RAID-5 (7+1) or, 6 or 14 for RAID-6 (6+2) and RAID-6 (14+2).

default_gateway
The gateway or router address for a front-end Gig-E port.

DevAttr A device attribute that restricts how a device can be accessed. These include:

- Dyn_rdf (can be either R1 or R2)
- Dyn_rdf1_only
- Dyn_rdf2_only
- ACLX
- SCSI3_persist_reserv
- RCVRPNT_TAG
- DIF1
- AS400_GK

The use of the RAD attribute is no longer supported. EMC will be using this attribute in the mainframe environment.

DevConfig A valid SYMAPI device configuration. When used to apply or remove the BCV, DRV, or RDF attribute, if the device configuration would result in a change to the device's mirroring, the change will be denied.

The mirroring protection of a device can be increased, but this must be done in a separate session from BCV/DRV/RDF changes.

The mirroring protection of a device can also be decreased, but must be done in a separate session from BCV/DRV/RDF changes. As a result of removing mirrors from a device, a new device is created from the discarded mirrors, resulting in a new symdevice. If the original or new device is unprotected, it cannot be mapped to a host.

This cannot be used to convert RDF1 devices to RDF2.

Configurations include:

- Unprotected
- 2-Way-Mir
- RDF1
- RDF2
- RDF1+Mir or RDF1-Mir
- RDF2+Mir or RDF2-Mir
- BCV
- 2-Way-BCV-Mir
- 2-Way-DRV-MIR
- RDF1+TDEV or RDF1-TDEV
- RDF2+TDEV or RDF2-TDEV
- RDF1-BCV *
- RDF2-BCV *
- RDF1-BCV+MIR or RDF1-BCV-MIR *
- RDF2-BCV+MIR or RDF2-BCV-MIR *
- VDEV
- TDEV
- DLDEV
- RAID-5
- RAID-6
- BCV+R-5 or BCV-R-5
- BCV+R-6 or BCV-R-6
- RDF1+R-5 or RDF1-R-5
- RDF1+R-6 or RDF1-R-6
- RDF2+R-5 or RDF2-R-5
- RDF2+R-6 or RDF2-R-6
- RDF1-BCV+R-5 or RDF1-BCV-R-5 *
- RDF2-BCV+R-5 or RDF2-BCV-R-5 *
- RDF1-BCV+R-6 or RDF1-BCV-R-6 *
- RDF2-BCV+R-6 or RDF2-BCV-R-6 *
- RDF1+DLDEV or RDF1-DLDEV
- RDF2+DLDEV or RDF2-DLDEV
- BCV+TDEV or BCV-TDEV
- RDF1-BCV+TDEV or RDF1-BCV-TDEV
- RDF2-BCV+TDEV or RDF2-BCV-TDEV

* Not allowed for create device

DevName A user-defined device name of up to 64

characters in length.

DeviceName The <DeviceName> is the user specified name with a maximum of 64 characters including the suffix. The legal characters for the device name include all ASCII characters except any of these characters [\n\"\’;=,:\[\]].
The device name plus optional suffix can have a maximum of 64 characters. If using a numerical suffix, the device name will be limited to 50 characters (prefix) and the trailing numerical suffix number will be limited to 14 characters. If not using a numerical suffix, all 64 characters can be specified for the device name.

The maximum starting suffix is 1000000. The <n> in the number= option represents the user supplied number for the starting suffix while using SYMDEV will mean that the corresponding Symmetrix Device Number will be used as the suffix.

director_num The director number (for example, 16A).

DskGrpName A disk group name of up to 32 characters in length. If the name is reset, it goes back to its default name in the format DISK_GROUP_xxx, where xxx is the disk group number.

dynamic_lun Specifies to use the dynamic LUN addressing features, but does not require the user to give a LUN address for each device. The LUN addresses will be assigned based on what may already be in use for that host HBA.

encapsulate_data
Option available when adding an eDisk, which indicates that the data that is currently in the external device be preserved (encapsulating).

When setting this option to NO, to not preserve the data, any data that is currently on the external device will be deleted.

EmulationType
A valid SYMAPI emulation type, including:

- FBA
- CELERRA_FBA
- VME_512_FBA
- CKD-3380
- CKD-3390
- AS/400_M2107_A02
- AS/400_M2107_A82
- AS/400_M2107_A04
- AS/400_M2107_A84
- AS/400_M2107_A05
- AS/400_M2107_A85
- AS/400_M2107_099
- AS/400_M2107_050
- AS/400_D910_099

When changing a device’s emulation, changes can only be among FBA emulation types.

fibre_vbus The virtual bus address. This is only used when mapping to an FA port using volume set addressing.

FlagName A SCSI, fibre, or iSCSI port flag name. SCSI port flags can be set on SA, SE, and FA ports, unless otherwise noted. FA port flags can only be set on FA and SE ports.

SCSI port flags:
- Negotiate_Reset [N]
(SA ports only)
- Soft_Reset [S]
- Environ_Set [E]
- Common_Serial_Number [C]
- Disable_Q_Reset_on_UA [D]
- Avoid_Reset_Broadcast [ARB]
- Server_On_AS400 [A4S]
- SCSI_3 [SC3]
- SPC2_Protocol_Version [SPC2]
- SCSI_Support1 [OS2007]

Fibre port flags:
- Volume_Set_Addressing [V]
- Non_Participating+ [NP]
- Init_Point_to_Point+ [PP]
(Not Supported 5977+)
- Unique_WWN+ [UWN]
(Not Supported 5977+)
- Access_Logix [ACLX]
- OpenVMS [OVMS]
- AS400 [AS4]
- Auto_Negotiate [EAN]
- Show_ACLX_Device

+ Not available for GigE ports

force_remove Option available when removing an eDisk, set to YES for forced removal of eDisk when it has IVTOC TRACKS.

gige Indicates that one or more network address values are going to be specified for a front-end Gig-E director.

group_attribute_name
The disk group attribute to be set. Possible values are:

Attribute Name	Description
disk_group_name	The new disk group name. To reset a name to the default disk group name use: DEFAULT NAME

HostName The host name, 12 characters.

host_lun The LUN addresses to be used for each device that is to be added for the host HBA.

hp_id A user-defined HP device identifier less than or equal to 128 characters in length.

invalidate_opt
The RDF device to invalidate so that a full

copy can be initiated from the remote mirror.

Allowed values are R1 (invalidate the source),
or R2 (invalidate the target).

IPaddress A valid IPv4 or IPv6 address is expected.

ip_prefix IPv4 or IPv6 prefix length of IP Interface and IP Route.
IPv4 prefix length range: 1-32
IPv6 prefix length range: 1-128
Only the default gateway's IPv4/IPv6 route will have
ip_prefix = 0

iqn IQN (iSCSI Qualified Name). A user will specify an
iSCSI target as either the iSCSI target name (IQN)
or its iSCSI virtual port equivalent.

iscsi The iSCSI name.

isns_ip_address
The IP address for the iSNS name server
associated with a front-end Gig-E port.

max_subs_percent
The maximum limit (in %) for the pool
subscription. If there is no limit, set to
NONE.

max_dev_per_rebalance_scan
The maximum limit for the pool rebalance
scan device range. It can be set in the
range of 1 to 1024. The default value
is 256.

member_count The total number of devices to add to the new
metadevice, including the head. To be used
when the configuration server should select
the members from the pool of unmapped devices.
Only devices that match the specified head
in size, emulation, protection, and
attributes will be selected. Omit this option
if you are going to select the members
yourself using the add dev command.

member_size_equal
Indicates that the meta members created on an
eDisk will have the same size. Available when
adding an eDisk with encapsulation.

meta_option The meta type configuration. Possible values
are: CONCATENATED or STRIPED.

meta_stripe_size
The metadevice stripe size, specified in 512
byte blocks, or cylinders. If specifying
cylinders, the keyword cyl must follow the
size field (for example, 2cyl).

For Enginuity 5876, 960 blocks or 1 cylinder
will be set as the stripe size.

metric_name The Symmetrix metric to be set. Possible
values are:

Metric Name	Description
auto_meta	Enables the auto meta

feature. Can be enabled
only if the other auto
meta parameters
min_auto_meta_size,
auto_meta_config, and
auto_meta_member_size
are set to valid values.
Possible values are
ENABLE or DISABLE.
Not applicable for
CKD meta devices.

auto_meta_config Specifies the default
meta config when auto
meta feature is enabled.
Possible values are
CONCATENATED, STRIPED,
or NONE.

auto_meta_member_size
Specifies the default
meta member size in
cylinders when auto_meta
feature is enabled.
Possible values
range from 0 to the
max size specified
in the table below:

32K_compatibility	Max Cylinders
Disabled	262668
Enabled	525336

dse_max_cap The maximum capacity
for DSE in GB. The value
NOLIMIT is used to set
an unlimited capacity.

pav_alias_limit If PAV is enabled,
specifies the maximum
number of aliases that
can be assigned to a
device. Possible
values are: 1-127.

min_auto_meta_size Specifies the size
threshold that triggers
auto meta creation.
When users try to
create a device greater
than or equal to the
min_auto_meta_size, and
auto_meta is enabled, a
meta will be created.
Possible values are
between 0 and the maximum
value from the table
below (the default value
is the same as the
maximum value):

32K_compatibility	Max/Default cyl
Disabled	262669
Enabled	525338

PAV_mode Enables the use of PAV (Parallel Access Volumes).
Possible values are:

PAV_mode	Description
NONE	No PAV volumes can be configured in the Symmetrix array. (Not available for customer use.)
STANDARD	Standard PAV volumes with static aliasing can be configured.
DYNAMIC_STANDARD	Standard PAV volumes with dynamic aliasing can be configured.
SIEMENS	Standard PAV volumes with static aliasing modified for Fujitsu Siemens hosts. (Not available for customer use.)
DYNAMIC_SIEMENS	Standard PAV volumes with dynamic aliasing modified for Fujitsu Siemens hosts. (Not available for customer use.)

rdfa_cache_percent The percentage of write pending cache that can be used by RDF/A. Possible values are from 0 to 100 percent.

rdfa_host_throttle_time
The number of seconds to throttle host writes to RDF/A devices when cache is full, before dropping RDF/A sessions.
Throttling will delay a write from the host until a cache slot becomes free. Possible values are from 0 to 65535
metric_value
Either a numeric or string value to be assigned to the specified metric.

meta_config Used with the create dev command to override the default Symmetrix-wide auto_meta_config. User can only override the meta_config when the Symmetrix-wide auto meta is enabled and the total device size is greater than the Symmetrix-wide min_auto_meta_size.

meta_member_size
When it is used with the create dev command, it overrides the default Symmetrix-wide auto_meta_member_size. User can only override the meta_member_size when the Symmetrix-wide auto meta is enabled and the total device size is greater than the Symmetrix-wide min_auto_meta_size.

When it is used with the add external disk command, it allows user to chose the meta member size when encapsulating data on the external disk.

mtu The mtu field allows the user to specify MTU (maximum transmission unit) value in the range of 1200-9000 for the IP interface. The default value is 1500.

mvs_ssid See remote_mvs_ssid below.

network_id The network_id/namespace can be used for the purpose of isolating IP routing tables. It is used to support multiple IP addresses with same IP subnet on a SE director emulation. Valid values:
For 5977 Q32016 SR and above: 1 to 16383. (Please note, maximum number of unique network_id that could be associated with an iSCSI instance is 511.)
For prior 5977 Q32016 SR: 1 to 511.
Default value is 1.

NewPoolName The new pool name. See PoolName.

pav alias See alias.

pool Pools can either contain SAVE devices or DATA devices. A SAVEDEV pool can be a SNAP pool, which is used for snap sessions or a RDFA_DSE pool which is used for RDF/A spillover. A DATADEV pool is used for thin devices.

PoolName A sequence of 1 to 12 alphanumeric or '-' (hyphen) or '_' (underscore) characters.

pool_resv_cap
Pool reserve capacity (in %) for the thin pool. Valid pool_resv_cap percentage values range from 1 to 80.

port_num The port number.

primary_ip_address
The IPv4 address for a front-end Gig-E port.

primary_ipv6_address
The IPv6 address for a front-end Gig-E port.

primary_netmask
The IPv4 netmask for a front-end Gig-E port.

primary_ipv6_prefix
The IPv6 mask prefix for a front-end Gig-E
port. Can be 0 - 128, indicating the number
of initial bits in the subnet that are
identical.

protect_data When adding members to an existing striped
meta, you must specify whether the data on
the existing metadevice needs to be
protected. Valid settings are TRUE and
FALSE.

ra_group A positive integer specifying the RDF group.

raidset When requesting a change to a member of a
RAID-S group, this option indicates that all
members of the group should be processed. It
is not necessary to list the other members.
This option is available for converting
RAID-S groups to unprotected devices or
deleting all members of the group.

rdfa_dse When used with an SRP, enabling it will
allow that SRP to be used for RDFA DSE
spillover.

rebalance_variance
The value (in %) for the pool Rebalance
Variance. It can be set in the range of 1 to
50. Its default value is 1.

refresh_opt The RDF device to mark for refresh
from the remote mirror. Allowed values are:
R1 (mark the source device for refresh) or R2
(mark the target device for refresh).

remote_config
When creating an RDF device, indicates the
device configuration of the corresponding
remote device.

remote_data_member_count
When creating an RDF RAID-5 or RAID-6 device
on Symmetrix arrays the user should set the
value to 3 or 7 for RAID-5 (3+1) and RAID-5 (7+1)
respectively, or 6 or 14 for RAID-6 (6+2) and
RAID-6 (14+2), respectively.

remote_dev A hexadecimal value specifying the Symmetrix
device name.

remote_mvs_ssid
When creating a device in a Symmetrix array
that also contains CKD devices, an mvs_ssid
value must be provided so the new FBA
devices are not seen as part of an existing
SID group.

Only one mvs_ssid and remote_mvs_ssid can
be used in a session. They will be applied
to all devices created within that session.

When using the convert device command to

reduce mirroring, the removed mirror
becomes a new stand-alone symdev. An
attempt will be made to use the same
mvs_ssid for the new device. If the mvs_ssid
group is full, you will need to specify a
new mvs_ssid.

When mapping a set of devices to an EF port,
the current mvs_ssid assigned to the devices
may need to be changed. If the devices are
becoming part of an existing CU image, they
will be assigned the mvs_ssid of the devices
already mapped. If a new CU image is being
formed and mapped, a new mvs_ssid can be
assigned during the map request. It is not
valid to have some devices with a particular
SSID mapped and some unmapped.

remote_sg When creating an RDF device, indicates the
remote storage group in which the remote
device will be added upon creation.
(supported on Enginuity 5977 and higher).

resv_cap Percentage of the capacity of the SRP that
will be reserved for devices write IO
activities. Valid values for the percentage
are from 1 to 80, or NONE to disable it
(supported on Enginuity 5977 and higher).

savedev An option available when creating a device,
that indicates the device should become part
of the SAVE device pool for use with
virtual copy sessions or rdfa_dse.

scsi_lun The SCSI logical unit number (a hex value).

scsi_target The SCSI target ID (a hex value).

sess_priority
The priority used to determine which RDF/A
sessions to drop if cache becomes full.
Possible values range from 1 to 64, with 1
being the highest priority (that is, the last
to be dropped).

SgName User specified storage group name in which
the devices will be added upon creation.

spid Indicates the specific spindle ID of the
target eDisk.

SRP Name User specified storage resource pool (SRP).

SLName User specified service level (SL).

NewSLName New name for the user specified Service Level.

start_copy Indicates whether an RDF pair should be
synchronized after the configuration change
is committed.

tcp_port TCP port on which the iSCSI target will be listening
on for incoming connections. The default value is 3260.

vlanid Virtual LAN Identifier. Max value is 4094.

vms_id A user-defined VMS device identifier from

0 to 32766.

vp_compression
Enables or disables the compression feature
on the thin device pool. Possible values are
ENABLE or DISABLE.

wwn   The World Wide Name.

RETURN CODES

Code #	Code Symbol
0	CLI_C_SUCCESS
1	CLI_C_FAIL
2	CLI_C_DB_FILE_IS_LOCKED
19	CLI_C_GK_IS_LOCKED All Gatekeepers to the Symmetrix array are currently locked.
48	CLI_C_CONFIG_LOCKED A configuration change session cannot be started because another application has the Symmetrix configuration locked.

EXAMPLES

To add four new Symmetrix devices to Symmetrix array
12345 as two-way mirrored devices with a size of 1100
cylinders, using FBA emulation, and to create the hypers
to support the new devices in disk group 3, enter:

```
symconfigure commit -sid 12345 -file add_new_symdevs.cmd
```

Where add_new_symdevs.cmd contains:

```
create dev count=4 size=1100,
emulation=FBA, config=2-Way-Mir,
disk_group=3;
```

To create three new DATA devices as two-way-mirrored with
a size of 1200 cylinders, using FBA emulation, and to add
it in a thin pool tp_pool with member state enabled, enter:

```
symconfigure commit -sid 12345 -file addnewdatadevs.cmd
```

Where addnewdatadevs.cmd contains:

```
create dev count=3 size=1200,
emulation=FBA, config=2-Way-Mir,
attribute=datadev, in pool=tp_pool
member_state=ENABLE;
```

To create 10 2 GB thin devices, bind them to a thin pool
tp_pool, and preallocate 1 GB to each thin device, enter:

```
symconfigure commit -sid 12345 -file add_new_tdevs.cmd
```

Where add_new_tdevs.cmd contains:

```
create dev count=10, size=2 GB
emulation=FBA, config=TDEV,
binding to pool=tp_pool, preallocate size=1 GB;
```

```
NOTE: default unit for <size> and <preallocate size> are
in CYL. User should ensure that units for <size>
and <preallocate size> are either both defaulted
or match.

To create a CKD metadevice of 1200 cylinders by first
creating four Symmetrix devices of 300 cylinders and then
forming a metadevice, enter:

symconfigure commit -sid 12345 -file add_new_ckdmeta.cmd

Where add_new_ckdmeta.cmd contains:

create dev count=4 size=1200,
emulation=ckd-3390, config=2-Way-Mir,
attribute=ckd_meta;

To add 4 FBA thin devices of 1200 cylinders and map them to
director 7E:0 and 8H:0 on a Symmetrix array with Enginuity
5876 and higher, use the create device command with
the mapping clause. Note that the command can support more
than one mapping definition:

create dev count=4, size = 1200 cyl, emulation=FBA,
config=TDEV,
mapping to dir 7E:0, starting lun = F04,
mapping to dir 8H:0, starting lun = F14;

To create 4 devices of 1200 cylinders and set the
SCSI3_persist_reserv device attribute on a Symmetrix array
running Enginuity 5876 and higher, use the
create device command with the device_attr clause:

create dev count=2, size = 1200, emulation=FBA,
config=2-Way-Mir,
device_attr = SCSI3_PERSIST_RESERV;

Note: A create device command can also have both the
mapping and device_attr clause, as in the example below:

create dev count=2, size = 30, emulation=FBA,
config=unprotected,
device_attr = SCSI3_PERSIST_RESERV
mapping to dir 7E:0, starting lun = F02;

To create a striped metadevice in Symmetrix array 12345,
using device 030 as the metahead and 031-033 as members,
enter:

symconfigure commit -sid 12345 -file add_meta.cmd

Where add_meta.cmd contains:

form meta from dev 030 config=striped,
stripe_size = 2 cyl;
add dev 031:033 to meta 030;

To verify the mapping command file is correct and later
map the metahead to director 16A, port 0, SCSI
target/lun 0/7, enter:

symconfigure preview -sid 12345 -file meta_map.cmd

Where meta_map.cmd contains:

map dev 030 to dir 16A:0 target=0, lun=7;
```

To map a device to a fibre port that uses volume set addressing, enter:

```
symconfigure commit -sid 12345 -file map_vsa.cmd
```

Where map_vsa.cmd contains:

```
map dev 122 to dir 03A:0, vbus=0A, target=0F, lun=3;
```

To unmap devices 020-024 from all front-end directors, enter:

```
symconfigure commit -sid 12345 -file unmap_dev.cmd
```

Where unmap_dev.cmd contains:

```
unmap dev 020:024 from dir ALL:ALL;
```

To unmap half the devices in CU image 0x07 and assign them a new SSID, enter:

```
symconfigure commit -sid 12345 -file unmap_range.cmd
```

Where unmap_range.cmd contains:

```
unmap dev 040:54 from dir ALL:ALL, new_ssid=620;
```

To enable the use of VCM/ACLX for masking device visibility to host systems for director 03A, port 0, enter:

```
symconfigure commit -sid 3160 -file setup_fa_port.cmd
```

Where setup_fa_port.cmd contains:

```
set port 03A:0 Access_Logix=enable;
```

To setup the symmetrix auto meta parameters:

```
symconfigure commit -sid 3160 -file set_symm_metrics.cmd
```

Where set_symm_metrics.cmd contains:

```
set symmetrix auto_meta_config = concatenated,
auto_meta_member_size = 65521;
```

To change the emulation type of three devices from FBA to CELERRA FBA, enter:

```
symconfigure commit -sid 3160 -file device_set.cmd
```

Where device_set.cmd contains:

```
set dev 01A:01C emulation=CELERRA_FBA;
```

To remove the device masking database attribute from a device masking database device, enter:

```
symconfigure commit -sid 3160 -file device_reset.cmd
```

Where device_reset.cmd contains:

```
set dev 01A, attribute = NO ACLX;
```

To configure 6 GB of available free space into BCVs matching an existing standard device and mapping them to port 14B:0, enter:

```
symconfigure commit -sid 3420 -file cfg_bcv.cmd
```

Where cfg_bcv.cmd contains:

```
configure 6 gb copying dev 07A,
mapping to dir 14B:0, overriding
config=bcv;
```

To create a new pool and move some SAVE devices from an existing pool into it, enter:

```
<session 1>
disable dev 01D:01F in pool FINANCE, type=snap;

<session 2>
create pool HR, type=snap;
add dev 01D:01F to pool HR, type=snap,
member_state=ENABLE;
```

To rename an existing pool HR of type snap to newHR, enter:

```
symconfigure commit -sid 12345 -file renamepool.cmd
```

Where renamepool.cmd contains:

```
rename pool HR to newHR, type=snap;
```

To rename an existing SRP TestAndDev to NewTestAndDev, enter:

```
symconfigure commit -sid 12345 -file renamesrp.cmd
```

Where renamesrp.cmd contains:

```
rename SRP TestAndDev to NewTestAndDev;
```

To reserve a set of devices for later conversion to BCV devices, enter:

```
symconfigure reserve -sid 3241 -file bcv.cmd
-owner "LabMgr" -comment "Adding BCV devs to dept xxx"
```

Where bcv.cmd (the actual change definition) contains:

```
convert dev 030:03A to 2-way-bcv-mir;
```

or:

```
symconfigure reserve -sid 3241 -file reserve.cmd
-owner "LabMgr" -comment "Adding BCV devs to dept xxx"
```

Where reserve.cmd (strictly a reservation file) contains:

```
reserve dev 030:03A;
```

To set a disk group name for disk group 2, enter:

```
set disk_group 2, disk_group_name = MY_DISK_GROUP;
```

To reset the disk group name for disk group 2, enter:

```
set disk_group 2, disk_group_name = DEFAULT NAME;
```

To create a disk group for external disks and give it the name my_external_disk_group, enter:

```
symconfigure commit -sid 3241 -file cmdfile.cmd
```

```
Where cmdfile.cmd contains:

create disk_group my_external_disk_group
disk_location = external;

To delete a disk group with the name
my_external_disk_group, create and commit a command file
that contains:

delete disk_group name:my_external_disk_group;

To add an external disk to disk group
my_external_disk_group with the encapsulating option,
create and commit a command file that contains:

add external_disk wwn=60000970000194900306533030314341
to disk_group=name:my_external_disk_group,
encapsulate_data=YES;

To remove an eDisk, create and commit a command file that
contains:

remove external_disk
wwn=60000970000194900306533030314341;
```

symconnect

Allows the administrator to setup or modify
the Symmetrix security functionality,
including CHAP authentication and
Radius server.

SYNOPSIS

```
symconnect -h

symconnect -sid <SymmID> [-dir <#|all> -p <#|all>] [-v]
   list [chap | radius]
     [-iscsi <iSCSIName> | -initiator] [-dir_port]

symconnect -file <Filename> [-dir <#|all> -p <#|all>] [-v]
   list [-iscsi <iSCSIName> | -initiator] [-dir_port]

symconnect -sid <SymmID> -iscsi <iSCSIName>
     <-dir <#> -p <#> | -dir all -p all>

   set chap -cred <Credential> -secret <Secret>

   set radius

symconnect -sid <SymmID> -iscsi <iSCSIName>
     [-dir <#|all> -p <#|all>]

   enable <chap | radius>

   disable <chap | radius>

   delete <chap | radius>

symconnect -sid <SymmID> -dir <#> -p <#>

   set chap -cred <Credential> -secret <Secret>

   set radius
     -server <ServerName> -ip <Ip> -key <Key>
      [-port <PortNum>]
     -rank <Primary | Backup1 | Backup2>
     [-c <Count> -i <Interval>]

symconnect -sid <SymmID> [-dir <#|all> -p <#|all>]

   enable chap

   disable chap

   delete chap

   enable radius [-rank <Primary | Backup1 | Backup2>]

   disable radius [-rank <Primary | Backup1 | Backup2>]

   delete radius [-rank <Primary | Backup1 | Backup2>]

symconnect -sid <SymmID> -file <Filename> [-noprompt]

   backup <chap | radius>

   restore <chap | radius>

   init <chap | radius>
```

DESCRIPTION

The symconnect command lists the security information from the Symmetrix array or a backup file. In addition, this command allows the following:

- CHAP credentials to be set, enabled, disabled, or deleted from the database for either the director/port or a specified iSCSI initiator.

- The use of a Radius server to be set, enabled, disabled, or deleted from the database for a specified iSCSI initiator.

- The Radius server information to be set, enabled, disabled, or deleted from the database for a director/port.

- The CHAP credential or Radius server information to be initialized, as well as backed up or restored, from a backup file.

- Enginuity level 5874 and above, all support for CHAP credential or Radius server information to be set, enabled or disabled for a given iSCSI initator through symaccess.

ARGUMENTS

backup	Makes a copy of all the security data present in a backup file that can be used later for restore.
delete	Deletes security information that was previously set for either a director/port or an iSCSI initiator.
disable	Disables security information that was previously set for either a director/port or an iSCSI initiator.
enable	Enables security information that was previously set for either a director/port or an iSCSI initiator.
init	Initializes the security information.
list	Lists the security information.
restore	Restores the security information from a backup file.
set	Allows security information to be established for either a director/port or an iSCSI initiator.

KEYWORDS

chap	Specifies the iSCSI CHAP credential.
radius	Specifies the iSCSI Radius server.

OPTIONS

-c	Indicates the number of times to retry.
-cred	Specifies the credential name associated with the CHAP protocol’s authentication

data.

-dir — Confines the action to a director number.

-dir_port — Lists all the information for the director/port.

-file — Specifies a backup file name.

-h — Provides brief, online help.

-i — Repeats the interval in seconds.
The default interval is 30 seconds.
The minimum interval is 5 seconds.
For passive actions the minimum interval is 15 seconds. Passive actions are actions that do not acquire an exclusive lock.

-initiator — Lists all the information for initiators.

-ip — Indicates an IP address.

-iscsi — Specifies an iSCSI name.

-key — Specifies the key associated with the Radius server data.

-noprompt — Requests that no prompts are returned after the command is entered.

-p — Identifies a specific port.

-port — Identifies a Radius server port.

-rank — Specifies the Radius server rank.

-secret — Designates the secret associated with the CHAP protocol’s authentication data.

-server — Specifies the Radius server name.

-sid — Supplies the Symmetrix serial number or ID.

-v — Provides a more detailed, verbose listing.

PARAMETERS

— A specific director or port number.

All — All directors or ports.

Backup1 — The first backup for the Radius server.

Backup2 — The second backup for the Radius server.

Count — The number of times to retry the Radius server.

Credential — The CHAP protocol’s credential name.
For Microsoft users, the string should be between 8 and 256 characters.

Filename — The name of a backup file.

Interval — The time (in seconds) between retries of the Radius server.

Ip	The IP address of the Radius server.
iSCSIName	An iSCSI name.
Key	The Radius server key.
PortNum	A Radius server port, the default value is 1812.
Primary	The primary Radius server to be used.
Secret	The CHAP protocol’s secret value, a user-defined string of up to 32 ASCII characters, or 64 binary characters. Binary values should be prefixed with the string 0X. Microsoft users must specify between 12 and 16 characters.
ServerName	The Radius server name.
SymmID	The Symmetrix serial number or ID.

EXAMPLES

To set CHAP credentials for a host initiator so the array knows to challenge the host, enter:

```
symconnect -sid 847
  -iscsi iqn.1991-05.com.microsoft:laqa0227.lss.emc.com
  set chap -cred laqa0227credentials -secret mysecrets123
  -dir 15c -p 0
```

To set CHAP credentials for the array so that the array is able to respond to a challenge from the host, enter:

```
symconnect -sid 847 -dir 15C -p 0 set chap
 -cred MyPeerCredentials -secret MyPeerSecret1234
```

To list the CHAP records for a given director and port, enter:

```
symconnect -sid 847 -dir 15c -p 0 list chap
```

To set Radius server for a host initiator, enter:

```
symconnect -sid 847
 -iscsi iqn.1991-05.com.microsoft:laqa0227.lss.emc.com
 set radius -dir 15c -p 0
```

To list the Radius server records for a given director and port, enter:

```
symconnect -sid 847 -dir 15c -p 0 list radius
```

symdev

Performs operations on a specific Symmetrix device.

SYNOPSIS

```
symdev -h

symdev [-sid <SymmID>] [-offline] [-v] [-resv | -pgr]

   list [-FA <# | ALL> [-p <#>] | -SA <# | ALL> [-scsi] [-fibre] [-p <#>]]
     [-wwn | -wwn_encapsulated [-detail] | -wwn_non_native] [-all]
     [-devs <<SymDevStart>:<SymDevEnd> | <SymDevName>
         [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]>]
     [-mb | -gb | -tb]
     [-cap <#> [-captype <mb | gb | tb | cyl>]] [-N <#>]
     [-vcm | -aclx] [-held] [-gige] [-[no]gcm]
     [-ficon] [-escon] [-dldev] [-dd] [-dm]
     [-noport | -firstport | -multiport] [-[no]bcv | -drv]
     [-meta] [-nomember]
     [-spare] [-dynamic] [-vdev]
     [-savedev [-nonpooled]] [-raids]
     [-disk_group <DskGrpNum> | name:<DskGrpName>]
     [-disk_director <# | ALL>]
     [-ext_spid <SpindleID>]
     [-rg][-sec_raid] [-unprotected] [-sec_unprotected]
     [-raid1] [-sec_raid1] [-bcv_emulation]
     [-raid5 [-protection <3+1 | 7+1>]]
     [-raid6 [-protection <6+2 | 14+2>]]
     [-sec_raid5 [-sec_protection <3+1 | 7+1>]]
     [-sec_raid6 [-sec_protection <6+2 | 14+2>]]
     [-emulation <fba | ckd | ckd3390 | ckd3380 | as400 | celerra>]
     [-star_mode] [-star_sync_target] [-star_async_target]
     [-[no]reserved] [-cyl] [-geometry_set]
     [-service_state [not]degraded | [not]failed | [not]normal]
     [-tdev [-bound | -unbound]] [-datadev [-nonpooled]]
     [-migr_tgt] [-pinned] [-host_passive] [-identity_set]
     [-identity [-detail]] [-sg <SgName>] [-rp]
     [-host_cache] [-dif1] [-as400_gk] [-fast]
     [-orm system | on | off] [-vvol] [-pedev]
     [-technology <EFD | FC | SATA>]
     [-internal | -external | -encapsulated [-limited]]
     [-R1] [-R2] [-R21] [-notrdf] [-rdfg <RdfGrpNum>]
     [-rdfa]] [-half_pair] [-dup_pair] [-metro] [-metroDR]
     [-insg | -notinsg]
     [-ficon_split] [ -device_id <compatibility| mobility> ]
     [-no[eff_encrypt_capable] [-no[eff_encrypted [-no[key]] [-rekey]]]

   list pd [-FA <# | ALL> [-p <#>] |
     -SA <# | ALL> [-scsi] [-fibre] [-p <#>]]
     [-wwn | -wwn_encapsulated [-detail] | -wwn_non_native] [-all]
     [-devs <<SymDevStart>:<SymDevEnd> | <SymDevName>
         [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]>]
     [-mb | -gb | -tb]
     [-cap <#> [-captype <mb | gb | tb | cyl>]] [-N <#>]
     [-vcm | -aclx] [-held] [-gige] [-[no]gcm]
     [-ficon] [-escon]
     [-noport | -firstport | -multiport] [-[no]bcv | -drv]
     [-meta] [-nomember]
     [-spare] [-dynamic] [-vdev]
     [-savedev [-nonpooled]] [-raids]
     [-disk_group <DskGrpNum> | name:<DskGrpName>]
     [-disk_director <# | ALL>]
     [-ext_spid <SpindleID>]
     [-rg][-sec_raid] [-unprotected] [-sec_unprotected]
```

```
     [-raid1] [-sec_raid1] [-bcv_emulation]
     [-raid5 [-protection <3+1 | 7+1>]]
     [-raid6 [-protection <6+2 | 14+2>]]
     [-sec_raid5 [-sec_protection <3+1 | 7+1>]]
     [-sec_raid6 [-sec_protection <6+2 | 14+2>]]
     [-emulation <fba | ckd | ckd3390 | ckd3380 | as400 | celerra>]
     [-star_mode] [-star_sync_target] [-star_async_target]
     [-[no]reserved] [-cyl] [-geometry_set]
     [-service_state [not]degraded | [not]failed | [not]normal]
     [-tdev [-bound | -unbound]]
     [-migr_tgt] [-pinned] [-host_passive] [-identity_set]
     [-identity [-detail]] [-sg <SgName>] [-rp]
     [-dif1] [-as400_gk]
     [-technology <EFD | FC | SATA>]
     [-host_cache]
     [-orm system | on | off]
     [-internal | -external | -encapsulated [-limited]]
     [-R1] [-R2] [-R21] [-notrdf] [-rdfg <RdfGrpNum>]
     [-rdfa] [-half_pair] [-dup_pair]
     [-insg | -notinsg]
     [-ficon_split]
     [-no[eff_encrypt_capable] [-no[eff_encrypted [-no[key]] [-rekey]]]

symdev [-sid <SymmID>] [-offline]

   list -space <-DA <# | ALL> | -DX <# | ALL>> [-cyl] [-spindle]

   list -inventory

symdev [-sid <SymmID>] [-hyper <# | ALL>] [-spindle]
       [-firstport] [-host_cache]
       [-orm system | on | off] [-offline]

   list [-internal | [-DA <# | ALL>] [-interface <# | ALL>]
       [-disk <# | ALL>]]

   list [-external | [-DX <# | ALL>] [-encapsulated [-limited]]]

symdev [-sid <SymmID>] [-v]

   list -lock
     [-devs <<SymDevStart>:<SymDevEnd> | <SymDevName>
         [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]>]

   list -identifier <device_name | hp_id | vms_id>
     [-devs <<SymDevStart>:<SymDevEnd> | <SymDevName>
         [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]>]
     [-sg <SgName>]

symdev [-sid <SymmID>] [-offline] [-cyl] [-ficon_split]
       [-mb | -gb | -tb]

   show <SymDevName>

   show -wwn <WWN> | -wwn_non_native <WWN>

symdev -sid <SymmID> [-force]

   release [-lock <#>] [-noprompt]
     [-devs <<SymDevStart>:<SymDevEnd> | <SymDevName>
         [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]>]

symdev -sid <SymmID> [-noprompt] [-rp] [-star] [-celerra]

   rw_enable     <SymDevName>  [[-SA | -FA] <# | ALL>
     [-p <#>]]
```

```
   write_disable <SymDevName>  [[-SA | -FA] <# | ALL>
     [-p <#>]]

   ready         <SymDevName> [-metro]

   not_ready     <SymDevName> [-metro]

   hold          <SymDevName>

   unhold        <SymDevName>  [-symforce]

   set -geometry <SymDevName> -cyl <#> <-symm6 | -symm7 |
      -symm9 | -sec_trk <#> -trk_cyl <#>>
      [-host_capacity <NumBlocks>]

   set -geometry <SymDevName> -host_capacity <NumBlocks>

   set -geometry <SymDevName> -default

symdev -sid <SymmID> [-noprompt] [-star]

   relabel           <SymDevName>

   pin               <SymDevName>

   unpin             <SymDevName>

   host_active       <SymDevName> [-force]

   set -persistent   <SymDevName>

   unset -persistent <SymDevName>

   set -orm < system | on | off >  <SymDevName>

   set -gcm          <SymDevName>

   unset -gcm        <SymDevName>

   reset -identity  <SymDevName>

symdev -sid <SymmID> [-noprompt] [-v] [-sg <SgName>]

   create -tdev -cap <#> [-bcv]
      [-captype <cyl | mb | gb | tb>][-N <#>]
      [-emulation <ckd3380 | as400 | celerra>]
      [-device_name <DeviceName> [-number <n | SYMDEV>]]

   create -tdev -emulation <ckd3390> [-N <#>] [-bcv]
      <<-model <1 | 2 | 3 | 9 | 27 | 54>> |
       <-cap <#> [-captype <cyl | mb | gb | tb>]>>
      [-device_name <DeviceName> [-number <n | SYMDEV>]]

   create -tdev -emulation <fba | celerra> [-mobility] [-bcv]
       -cap <#> [-captype <cyl | mb | gb | tb>] [-N <#>] [-dif1]
       [-eff_encrypt_capable] [-device_name <DeviceName>
                                  [-number <n | SYMDEV>]]

symdev -sid <SymmID> [-noprompt] [-v]

   create  <-as400_gk |
            -gk [-mobility] [-emulation <fba | celerra>] [-dif1] |
            -pedev [-emulation <fba>] >
      [-N <#>]
      [-device_name <DeviceName> [-number <n | SYMDEV>]]

   delete  <SymDevName>
```

```
   modify  <SymDevName> -tdev -cap <#> [-rdfg <RdfGrpNum>]
      [-captype <cyl | mb | gb | tb>]

   set <SymDevName> <-as400_gk | -bcv | -dif1>

   set <SymDevName> -device_id <compatibility | mobility>

   set <SymDevName> -device_name <DeviceName> [-number <n | SYMDEV>]

   set <SymDevName> -emulation < fba | celerra >

   unset <SymDevName> <-as400_gk | -bcv | -dif1>

symdev -sid <SymmID> -file <FileName>
  [-noprompt] [-rp] [-star] [-celerra]

   rw_enable     [-SA <# | ALL> [-p <#>]]

   write_disable [-SA <# | ALL> [-p <#>]]

   ready [-metro]

   not_ready [-metro]

   hold

   unhold        [-symforce]

   set -geometry -cyl <#> <-symm6 | -symm7 | -symm9 |
       -sec_trk <#> -trk_cyl <#>>
       [-host_capacity <NumBlocks>]

   set -geometry -host_capacity <NumBlocks>

   set -geometry -default

symdev -sid <SymmID> -file <FileName>
  [-noprompt] [-star]

   relabel

   pin

   unpin

   host_active   [-force]

   compress [-stop]

   uncompress [-stop]

   bind -pool <PoolName>

   unbind

   rebind -pool <PoolName>

   allocate [-persistent]

   allocate -stop

   free [-all]

   free [-all] -stop

   reclaim [-persistent]
```

```
   reclaim [-stop]

   set -persistent

   unset -persistent

   set -orm < system | on | off >

   set -gcm

   unset -gcm

   reset -identity

symdev -sid <SymmID> [-noprompt] [-rp] [-star] [-celerra]
  -devs <<SymDevStart>:<SymDevEnd> | <SymDevName>
      [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]>

   rw_enable     [-SA <# | ALL> [-p <#>]]

   write_disable [-SA <# | ALL> [-p <#>]]

   ready [-metro]

   not_ready [-metro]

   hold

   unhold        [-symforce]

   set -geometry -cyl <#> <-symm6 | -symm7 | -symm9 |
       -sec_trk <#> -trk_cyl <#>>
       [-host_capacity <NumBlocks>]

   set -geometry -host_capacity <NumBlocks>

   set -geometry -default

symdev -sid <SymmID> [-noprompt] [-star]
  -devs <<SymDevStart>:<SymDevEnd> | <SymDevName>
      [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]>

   relabel

   pin

   unpin

   host_active   [-force]

   compress [-stop]

   uncompress [-stop]

   bind -pool <PoolName>

   unbind

   rebind -pool <PoolName>

   allocate [-persistent]

   allocate -stop

   free [-all]
```

```
      free [-all] -stop

      reclaim [-persistent]

      reclaim [-stop]

      set -persistent

      unset -persistent

      set -orm < system | on | off >

      set -gcm

      unset -gcm

      reset -identity

   symdev -sid <SymmID> [-noprompt] [-v]
     -devs <<SymDevStart>:<SymDevEnd> | <SymDevName>
        [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]>

      delete

      modify -tdev -cap <#> [-captype <cyl | mb | gb | tb>]
        [-rdfg <RdfGrpNum>]

      set <-as400_gk | -bcv | -dif1>

      set -emulation <fba | celerra>

      set -device_id <compatibility | mobility>

      set -device_name <DeviceName> [-number <n | SYMDEV>]

      unset <-as400_gk | -bcv | -dif1>

      unset -device_name

   symdev -sid <SymmID> -file <DevPairFile>
     [-noprompt]

      copy -geometry

   symdev [-sid <SymmID>]

      list -ppi [-oid [-v]]
        [-devs <<SymDevStart>:<SymDevEnd> | <SymDevName>
            [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]>]
```

DESCRIPTION

The symdev command displays information about all or selected Symmetrix devices regardless of whether they are visible to the local host. You can release a Device External Lock (DEL) on one or more specified Symmetrix devices.

The symdev command also performs the following control actions on all or selected Symmetrix devices: write_disable, rw_enable, ready, not_ready, hold, unhold, pin, unpin, compress, uncompress, relabel, set, unset, bind, unbind, rebind, allocate, free, reclaim, create, delete or modify.

ARGUMENTS

allocate	Allocates storage in the thin pool.
bind	Binds the thin device(s) to the pool.
create	Creates device(s).
compress	Starts data compression on thin device(s). When combined with the -stop option, data compression is stopped.
copy	Copies the source device geometry to the target device geometry when the -geometry option is specified.
delete	Deletes device(s).
free	Frees storage in the thin pool.
hold	Sets the hold bit on a device. The hold bit is automatically placed on a target device during a TimeFinder/Snap operation.
host_active	Sets the host active mode on device(s). The device(s) must be in a host passive mode for this operation to succeed.
list	Lists all nonprivate Symmetrix devices that are configured in one or more Symmetrix arrays connected to this host.
list pd	Lists all host visible Symmetrix devices that are configured in one or more Symmetrix arrays connected to this host.
modify	Expands device(s) capacity.
not_ready	Sets the device(s) to be Not Ready. The device must be in a User Ready state for this operation to succeed.
pin	Sets the device(s) to the user pinned state. User pinned device(s) will not be moved via FAST controller, but they can be moved via Optimizer or symmigrate.
ready	Sets the device(s) to be Ready. The device must be in a User Not Ready state for this operation to succeed.
rebind	Rebinds the device(s) to the thin pool.
reclaim	Reclaims storage from the thin pool.
relabel	Applies the defined label to the device. The device must be in a User Not Ready state for this operation to be accepted. Refer to the symlabel command to learn how to define a device label.
release	Releases the Device External Lock associated with one or more devices within a Symmetrix array. Caution: Use the release lock action only if it appears that a Symmetrix lock is currently hung and there are no other operations in progress on the Symmetrix

array (local or remote). Also, make sure that your application is authorized to use the specified lock number.

rw_enable — Sets the device(s) to be Read and Write Enabled to the local hosts on the specified front director port(s). If no ports are specified, then the device(s) will be Read and Write Enabled on all ports where the device is visible.

reset — Sets the device to its original identity when combined with -identity option.

set — Sets the device geometry when combined with -geometry option. Sets the persistent indicator for allocations on thin device(s) when combined with the -persistent option. Sets the Optimized Read Miss mode when combined with the -orm option. Sets GCM mode when combined with -gcm.

show — Shows detailed information about a specific Symmetrix device, such as 000C.

unbind — Unbinds device(s) from the thin pool.

uncompress — Starts data decompression on thin device(s). When combined with the -stop option, data decompression is stopped.

unhold — Resets the hold bit on a device. The hold bit is automatically removed from a target of a Snap device when the TimeFinder/Snap pair is stopped. However, the unhold argument can be used if there was a problem removing the hold bit.

unpin — Resets the device(s) from the user pinned state.

unset — Clears the persistent indicator for allocations on thin device(s) when combined with the -persistent option. Clears GCM mode when combined with -gcm.

write_disable — Sets the device(s) to be Write Disabled to the local hosts on the specified front-end director port(s).

OPTIONS

-aclx — Lists all device masking devices in Symmetrix arrays running Enginuity 5874 and higher.

-all — Lists all private devices (Vault devices, SAVE devices, DRV devices, SFS devices, and COVD devices) along with other Symmetrix devices.

When used with the free operation, specifies that all allocations associated with the indicated devices are to be freed, regardless of whether data has been written or not.

-as400_gk Creates or lists the Symmetrix devices that have the as400_gk attribute set.

-bcv Creates or lists the BCV device(s).

-bcv_emulation Lists clone emulated BCV devices.

-bound Lists devices bound to a Thin pool.

-cap Sets the device capacity to a specific value (in megabytes, gigabytes, terabytes or cylinders). See the '-captype' switch to set the units used.

-captype Sets the capacity units to a specific value (either 'mb', 'gb', 'tb' or 'cyl').

-celerra Allows controls on Celerra FBA devices.

-cyl Lists the device capacity in cylinders, or specifies the device capacity when setting the device geometry. The default is megabytes (MB).

-orm When combined with the set action, allows setting the Optimized Read Miss mode to system default, on, or off for the specified device(s). When used with the list action, lists devices with Optimized Read Miss mode set to system default, on, or off.

-DA Lists the Symmetrix devices that are mapped to a certain DA director number. The interface, disk, and hyper IDs can also be used to limit the list further, but the default is ALL unless otherwise specified.

-datadev Lists the data device(s).

-dd Lists the Symmetix devices that are identified as Data Domain devices.

-default Specifies the device's native geometry when setting the device geometry.

-detail Lists the full WWN of the device when displaying the external identity information. When used with the -wwn_encapsulated option, lists vendor information for the device.

-devs Specifies a set of Symmetrix device ranges and/or individual Symmetrix devices.

-device_id Lists the Symmetrix devices matching the corresponding type of device IDs.

-device_name Used to specify a user friendly name to the device(s).

-dif1 Lists the Symmetrix devices that have the dif1 attribute set. Data Integrity Field (DIF) was the name previously used for the current T10 Protection Information (PI) standard.

-disk Lists the Symmetrix devices that are mapped to a certain disk Target ID. The DA,

interface, and hyper IDs can also be used to limit the list further, but the default is ALL unless otherwise specified.

-disk_director Lists the Symmetrix devices with hypers residing on spindles configured to the specified back-end disk director and engine to which that director belongs.

-disk_group Lists the Symmetrix devices whose hypers are contained on disks within the specified disk group.

-dldev Lists the diskless devices.

-dm Lists the Symmetix devices that are in data migration sessions.

-drv Lists the DRV devices.

-dup_pair Lists the SRDF devices in duplicate pair configurations.

-DX Lists the Symmetrix devices that are mapped to a certain DX director number. The hyper IDs can also be used to limit the list further, but dfaults to ALL unless specified.

-dynamic Lists the dynamic SRDF devices capable of being formed into SRDF pairs. When used with -R1, lists RDF1-capable devices. When used with -R2, lists RDF2-capable devices. When used with both -R1 and -R2, lists RDF1-capable or RDF2-capable devices. When used without -R1 or -R2, lists RDF1-capable, RDF2-capable, and RDF1 or RDF2-capable devices.

-eff_encrypt_capable
Lists the Symmetix devices capable of being encrypted by encryption software on an application host.

Creates devices which are End-to-end Efficient Encryption capable.

-eff_encrypted Lists the Symmetix devices that are encrypted with encryption software on an application host.

-emulation Lists devices that match the corresponding emulation type. Used with set/create command, to set the emulation type of the device.

-encapsulated Lists devices that are created on external spindles with encapsulation.

-escon Lists devices mapped to front-end Escon directors.

-external Lists the Symmetrix devices that are created on external spindles.

-ext_spid Lists the Symmetrix devices with hypers residing on the specified external spindle.

-FA Lists devices mapped to a specific fibre or

gige front-end director number.

-fast — Lists the Symmetrix devices that are in FAST Managed Storage Groups.

-fibre — Lists devices mapped to front-end fibre directors.

-ficon — Lists devices mapped to front-end Ficon directors.

-ficon_split — Lists/Shows device(s) with Split and CU Image mapping information.

-file — Specifies a file name with a list of devices to be acted upon. In copy geometry operations, the file must contain two SymDevNamed per line. Otherwise, the file must contain one SymDevName per line.

-firstport — Limits the display to just the first port of information for devices that are mapped to more than one port.

-force — Causes a device lock to be released independent of other options currently controlling the use of device locks.

Also applies to the host_active option.

-gb — Lists or shows capacity in gigabytes.

-gcm — Allows setting or unsetting the device GCM attribute. Also lists devices with GCM attribute.

-geometry — Specifies the operation that will set or reset the device geometry.

-geometry_set — Lists all devices that have the device geometry set.

-gk — Creates the Symmetrix GK devices.

-gige — Lists devices mapped to front-end GigE directors.

-h — Provides brief, online help information.

-half_pair — Lists SRDF devices in a half-pair configuration.

-held — Lists Symmetrix devices in the device group that have device holds for a TimeFinder/Snap session.

-host_cache — Lists Symmetrix devices which are registered for control by host cache cards.

-host_capacity — Specifies the device host usable capacity in 512 bytes blocks. Supported only on Symmetrix arrays running Enginuity 5876 and above.

-host_passive — Lists devices set with host passive mode.

-hyper — Lists the Symmetrix devices that are mapped

to a certain hyper ID. The DA, interface, and disk IDs or the DX can also be used to limit the list further, but the default is ALL unless otherwise specified.

-identifier — Lists the Symmetrix device identifiers assigned to devices by the user or other applications. The user must choose one of the four identifier types currently supported to be displayed.

-identity — Lists the external identity information for each device. The external identity will be displayed if the device has external identity set. Otherwise, the device native identity will be displayed. When used with reset command, the original identity of the device(s) is restored.

-identity_set — Lists devices whose external identity has been changed as part of a federated live migration operation.

-insg — Indicates that only devices contained within a Storage Group are listed.

-interface — Lists the Symmetrix devices that are mapped to a certain DA director interface path. The DA, disk, and hyper IDs can also be used to limit the list further, but the default is ALL unless otherwise specified.

-internal — Lists the Symmetrix devices that are created on internal spindles.

-inventory — Returns a table listing the number of configured Symmetrix devices for each supported emulation type.

-key — Lists the Symmetrix devices that are encrypted with valid keys

-limited — Lists encapsulated devices that are geometry limited.

-lock — Lists devices that have a device external lock, or identifies a lock to release.

-mb — Lists or shows capacity in megabytes.

-meta — Lists meta-head devices.

-migr_tgt — Lists the Symmetrix devices that are usable as a migration target.

-mobility — Create FBA device with mobility safe ID.

-model — Indicates one of the CKD3390 models: CKD3390-1, CKD3390-2, CKD3390-3, CKD3390-9, CKD3390-27, CKD3390-54.

-multiport — Lists the Symmetrix devices that are mapped to multiple front-end director ports.

-N — Sets the number of devices to create or list. When used for list operation the # specifies the maximum number of devices to return.

The actual number returned may be less than the specified number if fewer devices exist. The default is to list all devices.

-nobcv
Lists standard devices only; excludes BCV devices.

-noeff_encrypt_capable
Lists the Symmetix devices not capable of being encrypted by encryption software on an application host.

-noeff_encrypted
Lists the Symmetix devices that are not encrypted.

-nogcm
Lists only devices that do not have GCM attribute.

-nokey
Lists the Symmetrix devices that are encrypted but with no valid keys

-nomember
Lists only devices that are not meta members. This includes non-meta devices as well as meta head devices.

-nonpooled
Lists devices that are ready to be assigned to a pool.

-noport
Lists the Symmetrix devices that are not mapped to any front-end director ports.

-noprompt
Requests that no prompts are returned after the command is entered. The default is to prompt the user for confirmation.

-noreserved
Lists all Symmetrix devices that are not reserved as part of a device reservation (see symconfigure).

-notrdf
Lists only devices that are not RDF devices.

-notinsg
Indicates that only devices not contained within a Storage Group are listed.

-number
It is a numerical suffix to be appended to the name specified with the -device_name option. It is limited to 14 characters.

-offline
Displays the Symmetrix devices from the Symmetrix configuration database without refreshing the data from the Symmetrix array.

-pedev
Creates or lists Protocol Endpoint (PE) device(s).

-persistent
Used with set or unset it specifies operation on the persistent indicator. Used with allocate or reclaim it specifies the use of persistent storage.

-pgr
Lists the WWN and reservation key of Symmetrix devices that have SCSI3 persistent group reservations.

-pinned
Lists the devices that were pinned.

-pool	Specifies a Thin Pool Name.
-protection	Applies to -raid5 or -raid6 only. Further filters raid devices based on protection types (3+1, 7+1, 6+2 or 14+2).
-p	Lists devices mapped to a specific SCSI or front-end fibre director port. By default, all ports are selected.
-R1	Lists RDF1 (R1) devices. When used with -dynamic, lists devices that are RDF1 capable (see -dynamic).
-R2	Lists RDF2 (R2) devices. When used with -dynamic, lists devices that are RDF2 capable. Also, see -dynamic.
-R21	Lists RDF21 (R21) devices.
-raid1	Lists RAID-1 devices.
-raid5	Lists RAID-5 devices.
-raid6	Lists RAID-6 devices.
-raids	Lists RAID-S devices by RAID group number. The -raid option is synonymous with -raids.
-rekey	Lists the Symmetrix devices that are being encrypted or re-encrypted.
-metro	Lists devices that are in an RDF/Metro configuration. When specified with ready and not_ready identifies the devices being controlled are part of an RDF/Metro configuration.
-metroDR	Lists the Symmetix devices that are identified as MetroDR devices.
-rdfa	Lists devices that are RDFA-backed.
-rdfg	Lists devices that belong to the specified SRDF group. When used with modify it specifies the SRDF group associated with the SRDF devices and indicates that both sides of the SRDF pair, which are associated with the SRDF group, should be expanded.
-resv	Lists all Symmetrix devices that have SCSI reservations.
-reserved	Lists all Symmetrix devices that have device reservations (see symconfigure).
-rg	Lists primary and secondary RAID group information for each device.
-rp	Allows controls on devices that have been tagged for RecoverPoint use. Lists the Symmetrix devices that are RecoverPoint devices.

-SA Lists devices mapped to a specific SCSI or fibre front-end director number.

-savedev Lists devices that are Symmetrix SAVE devices.

-scsi Lists devices mapped to SCSI front-end directors (SAs).

-sec_protection
Applies to -sec_raid5 or -sec_raid6 only. Further filters devices with secondary RAID-5 or RAID-6 RAID groups based on protection types (3+1, 7+1, 6+2 or 14+2).

-sec_raid Lists devices with a secondary RAID group.

-sec_raid1 Lists devices with a secondary RAID-1 RAID group.

-sec_raid5 Lists devices with a secondary RAID-5 RAID group.

-sec_raid6 Lists devices with a secondary RAID-6 RAID group.

-sec_trk Specifies the number of sectors per track.

-sec_unprotected
Lists devices with a secondary Unprotected RAID group.

-service_state Lists devices that match the corresponding service state.

-sg Lists the Symmetrix devices that belong to a specified storage group. For the create device command, it specifies the storage group to which the devices will be added.

-sid Specifies a unique Symmetrix ID.

-space Shows the available or unconfigured storage space for the specified list of disks.

Capacities refer to physical usage on-disk (spindle). Disks containing device hypers of different emulation types are listed with a Format of "Mixed." Disks containing no hypers (such as empty or spare) are not included.

Note that the summary Total and Unconfigured values are repeated for each emulation type. These values refer to cumulative capacities for the listed disk(s) and are not specific to each emulation type.

-spare Lists devices that have a spare disk invoked against them during dynamic sparing.

-spindle Displays spindle information instead of the standard disk address information.

-star Indicates that the action is targeted for devices in STAR mode.

-star_async_target
Lists devices in STAR ASYNC target mode.

-star_mode Lists devices in STAR mode.

-star_sync_target
List devices in STAR SYNC target mode.

-stop Specifies that the compress, uncompress, allocate, free or reclaim operation will be stopped.

-symforce Requests that the Symmetrix force the operation to be executed when normally it is rejected. Use extreme caution when using this option.

-symm6 Specifies the Symmetrix 6 device geometry. It indicates 64 sectors per track and 15 tracks per cylinder.

-symm7 Specifies the Symmetrix 7 device geometry. It indicates 128 sectors per track and 15 tracks per cylinder.

-symm9 Specifies the Symmetrix 9 device geometry. It indicates 256 sectors per track and 15 tracks per cylinder.

-tb Lists or shows capacity in terabytes.

-tdev Creates or lists the thin device(s).

-technology Specifies the drive technology type of the primary local back-end storage for the device.

-trk_cyl Specifies the number of tracks per cylinder.

-unbound Lists devices that are not bound to a Thin pool.

-unprotected Lists unprotected devices.

-v Provides a more detailed, verbose listing.

-vcm Lists all of the device masking (or VCM) devices in the Symmetrix array. This option is obsolete in Enginuity 5874 and higher, and is replaced by -aclx.

-vdev Lists devices that are Symmetrix virtual devices.

-vnx Specifies the VNX device geometry. It indicates 16 sectors per track and 32 tracks per cylinder.

-vvol Lists all VVol devices.

-wwn Lists the full WWN of all devices, or select a device by its WWN.

-wwn_encapsulated
Lists the WWN of the backing external spindle for the device, if the spindle is encapsulated. Devices which are not

backed by encapsulated external spindles will not be listed.

-wwn_non_native
List or show device(s) with External Device Identity set to non-native WWN(s).

-oid Display the Oracle instance name with the Powerpath mount status of Symmetrix devices. The standard display will trunctate the OID to 12 characters, but the full unabridged records can be displayed by adding the -v switch

-ppi Display the Powerpath mount status of Symmetrix devices.

PARAMETERS

DevPairFile The name of the file that contains pairs of source and target device. One pair per line with the format of source device first, a space and the target device.

For example:
10A 20A
10B 20B

DskGrpName The disk group name.

DskGrpNum The disk group number.

EFD The Enterprise Flash disk drives.

FC The Fibre Channel disk drives.

FileName The name of the file that contains a list (one SymDevName per line) of devices to be acted upon.

NumBlocks The number of 512 bytes blocks.

RdfGrpNum The SRDF Group number.

SATA The SATA disk drives.

SgName The storage group name.

SymDevName The Symmetrix device name, unique per Symmetrix, such as 01C.

SymDevStart The first Symmetrix device name in a sequence, such as 001C.

SymDevEnd The last Symmetrix device name in a sequence, such as 00B6.

SymmID 12-digit ID of the Symmetrix array.

RETURN CODES

Code #	Code Symbol
0	CLI_C_SUCCESS
1	CLI_C_FAIL

```
  19        CLI_C_GK_IS_LOCKED
            All GateKeepers to the Symmetrix
            array are currently locked.
```

EXAMPLES

To list all nonprivate Symmetrix devices that are configured in Symmetrix arrays connected to this host, enter:

```
symdev list
```

To show detailed information about Symmetrix device 00C in a Symmetrix array with the specified ID, enter:

```
symdev -sid 870 show 00C
```

To list the first 20 BCV devices starting at Symmetrix device 01F that are configured in a Symmetrix array enter:

```
symdev list -bcv -devs 01F -N 20
```

To list details about devices in the range from Symmetrix device (0000 to 000A) with device external lock of 9 locked, enter:

```
symdev list -sid 870 -lock 9 -devs 0000:000A -v
```

To release all Symmetrix devices in Symmetrix 870 that have a device external lock of 9, enter:

```
symdev -sid 3009 release -lock 9
```

symdg

Performs operations on a Symmetrix device group.

SYNOPSIS

```
symdg -h

symdg [-i <Interval>] [-c <Count>] [-v]

   create <DgName>
     [-type REGULAR | RDF1 | RDF2 | RDF21 | ANY]

   delete <DgName> [-force]

   rename <DgName> <NewDgName>

   export <DgName> [-delete] [-file <FileName>]
     [[-rdf [-rdfg <GrpNum>]] | [-sid <SymmID>]]
     [-grpfile <GrpDbFileName>]

   exportall [-delete] [-file <FileName>]
     [[-rdf [-rdfg <GrpNum>]] | [-sid <SymmID>]]
     [-grpfile <GrpDbFileName>]

   import <DgName> [-file <FileName>]

   importall [-file <FileName>]

   list [-sid <SymmID>] [-offline] [-v [-mb | -gb | -tb]]
     [-grpfile <GrpDbFileName>]

   list [-inactive]

   show <DgName> [-inactive] [-offline | -lock ]
     [-mb | -gb | -tb] [-grpfile <GrpDbFileName>]

   activate <DgName> [-noprompt]

   activateall [-noprompt]

   dg2file <DgName> [-file <FileName>]
     [-ftype STD | R1BCV | STD_BCV |
              STD_R1BCV | STD_VDEV | BCV_VDEV]

   file2dg <DgName> [-file <FileName>]
     [-type REGULAR | RDF1 | RDF2 | RDF21]

   dg2cg <DgName> <CgName> [-rename] [-force]
     [-bcv [-hop2] | -nobcv | -rbcv | -rrbcv | -brbcv |
      -vdev [-hop2] | -rvdev | -tgt [-hop2] | -rtgt]
     [-apidb | -rdf_consistency]

   dg2sg <DgName> <SgName> [-bcv | -vdev | -tgt ]


symdg list -novalidate
     [-sid <SymmID>]
     [-grpfile <GrpDbFileName>]


symdg -g <DgName> [-i <Interval>] [-c <Count>] [-v]
      [-offline]

   add pd  <PdevName> [LdevName]

   addall [-sid <SymmID>] [pd | -host <HostName>]
```

```
     [-SA <# | ALL>] [-p <#>] [-N <#>]
     [-cap <#> [-captype <mb> | <cyl>]]
     [<-rdf | -hop2> <-vdev | -tgt>
     [-rdfg <GrpNum> [-remote_rdfg <RemoteGrpNum>]]]
     [-sel_rdfg <SelRdfgNum>]
     [-devs <<SymDevStart>:<SymDevEnd> | <SymDevName>
         [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]>]

   list ld [-v] [-cyl] [-held] [-offline | -resv]

   show ld <LdevName> [-geometry]

   rename ld <LdevName> <NewLdevName>

   remove ld <LdevName> [-force]

   move ld <LdevName> <DestDgName> [-force] [-rename]

   copy ld <LdevName> <DestDgName> [-force] [-rename]


symdg -g <DgName> [-i <Interval>] [-c <Count>] [-v]
     [-offline] [-sid <SymmID>]
     [<-rdf | -hop2> <-vdev | -tgt>
     [-rdfg <GrpNum> [-remote_rdfg <RemoteGrpNum>]]]

   add dev <SymDevName> [LdevName]

   addall devs
     [-SA <# | ALL>] [-p <#>] [-N <#>]
     [-cap <#> [-captype <mb> | <cyl>]]
     [-sel_rdfg <SelRdfgNum>]
     [-devs <<SymDevStart>:<SymDevEnd> | <SymDevName>
         [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]>]

symdg -g <DgName> [-i <Interval>] [-c <Count>] [-v]
     [-offline] [-sid <SymmID>]
     [-SA <# | ALL>] [-p <#>] [-N <#>]
     [-cap <#> [-captype <mb> | <cyl>]]
     [-sel_rdfg <SelRdfgNum>]
     [-devs <<SymDevStart>:<SymDevEnd> | <SymDevName>
         [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]>]
     [-R1 | -R2 | -R21 | -noRDF]

   copyall <DestDgName> [-force] [-rename]
     [<-vdev | -tgt> [-hop2] | -rvdev | -rtgt]

   moveall <DestDgName> [-force] [-rename]
     [<-vdev | -tgt> [-hop2] | -rvdev | -rtgt]

   rmall [-force]
     [<-vdev | -tgt> -rdf [-rdfg <GrpNum>] | -hop2]

symdg -g <DgName> [-noprompt] [-i <Interval>] [-c <Count>]
     [-bcv | -vdev | -tgt] [-rp] [-star] [-celerra]

   rw_enable [-p <#>] [-SA <#|ALL>]

   write_disable [-p <#>] [-SA <#|ALL>]

symdg -g <DgName> [-noprompt] [-i <Interval>] [-c <Count>]
     [-bcv | -vdev | -tgt] [-star]

   relabel [-force]

   pin
```

```
      unpin

      host_active [-force]

   symdg -g <DgName> [-noprompt] [-i <Interval>] [-c <Count>]
         [-bcv [-hop2] | -rbcv  | -brbcv | -rrbcv |
         -vdev [-hop2] | -rvdev | -tgt [-hop2] |
         -rtgt] [-rp] [-star] [-celerra]

      ready [-metro]

      not_ready [-metro]

      hold

      unhold [-symforce]

      compress [-stop]

      uncompress [-stop]

      bind -pool <PoolName>

      unbind

      rebind -pool <PoolName>

      allocate [-persistent]

      allocate -stop

      free [-all]

      free [-all] -stop

      reclaim [-persistent]

      reclaim -stop

      set -persistent

      unset -persistent

      set -orm < system | on | off >

      set -gcm [-symforce]

      unset -gcm [-symforce]

      reset  -identity

   symdg [-i <Interval>] [-c <Count>]

      release <DgName> [-force] [-lock #] [-noprompt]

      break -g <DgName> [-noprompt] [-vdev]
```

DESCRIPTION

The symdg command performs the following operations specific to device groups: creating new device groups, importing ASCII group files, exporting groups to files, deleting groups, translating groups to/from Symmetrix Manager files, renaming groups, and listing and showing information about a device group.

The symdg command also performs the following operations specific to a device in a device group: adding a device to a device group, adding all available devices to a device group, listing all devices in a device group, removing a device from a device group, removing/moving a device or all devices from a device group, renaming a device in a device group, and showing detailed information about a device in a device group.

The symdg command also performs the following control actions on some or all of the devices in a device group: write_disable, rw_enable, ready, not_ready, hold, unhold, pin, unpin, compress, uncompress, relabel, set, unset, bind, unbind, rebind, allocate, free and reclaim. By default the actions will only be applied to the standard devices in the group. The -bcv and -tgt switches must be specified to operate on those types of devices.

ARGUMENTS

activate	Activates a specified device group (imports to GNS). If GNS is enabled on the host, this command allows device groups to be imported from the host database into the GNS repository. If GNS is not enabled on the host, this command has little or no use. For example, if GNS was previously enabled, and the GNS groups were copied to the host's configuration database, then the command would import those GNS device groups to the host's device group list.
activateall	Activates all the inactive device groups. If GNS is enabled on the host, this command allows device groups to be imported from the host database into the GNS repository. If GNS is not enabled on the host, this command has little or no use. For example, if GNS was previously enabled, and the GNS groups were copied to the host's configuration database, then the command would import those GNS device groups to the host's device group list.
add dev	Adds any ungrouped Symmetrix device (given its Symmetrix device name) to an existing device group.
add pd	Adds to an existing device group an ungrouped device by specifying its physical (host) device name.
addall devs	Adds all ungrouped Symmetrix devices from a specified Symmetrix array to an existing device group.
addall	Adds all ungrouped physical devices from a specified Symmetrix array to an existing device group.
allocate	Allocates storage in the thin pool.
bind	Binds the thin device(s) to the thin pool.
break	Breaks SCSI device reservations on one or all devices in the device group.

compress	Starts data compression on thin device(s). When combined with the -stop option, data compression is stopped.
copy ld	Copies one standard device from one existing device group to another existing device group. The source and destination device groups must have compatible types.
copyall	Copies all standard devices from one existing device group to another existing device group. The source and destination device groups must have compatible types.
create	Creates an empty device group of type REGULAR, RDF1, RDF2, or RDF21. Only RDF devices can belong to an RDF device group. Only non-RDF devices can belong to the REGULAR group. All devices added to a group must belong to the same Symmetrix array. If you do not specify a type, the device group will be created using type REGULAR.
delete	Deletes an existing device group. If the device group has member or gatekeeper devices that are associated with it, the command will fail unless the -force option is used. If the -force option is specified, the devices that are members of the group are removed from the group and become ungrouped devices.
dg2cg	Adds selected members of a device group to a target composite group.
dg2file	Creates a file in the same format used by the EMC Symmetrix Manager (SM-CLI) from a DG. This action should not be used to make a backup copy of a device group; use export for that purpose.
dg2sg	Adds selected members of a device group to a target storage group.
export	Creates a text file that details the members of an existing device group. The device group can later be recreated from this file using the import command.
exportall	Creates a text file that details the members of the existing device groups. The device groups can later be recreated from this file using the importall command.
file2dg	Creates a device group from an EMC Symmetrix Manager (SM-CLI) format device file.
free	Frees storage in the thin pool.
hold	Creates a hold on all available devices from an existing device group. When a hold is placed on a device, TimeFinder and Snap operations will be blocked.

host_active	Sets the host active mode on device(s). The device(s) must be in a host passive mode for this operation to succeed.
import	Creates a device group from data contained in a text file previously created using the export command.
importall	Creates device groups from data contained in a text file previously created using the exportall command.
list	Lists all of the device groups that have been created for this host. If -inactive is specified, it lists all of the device groups from the inactive group list.
list ld	Lists all of the available devices from an existing device group.
move ld	Moves one standard device from one existing device group to another existing device group. The source and destination device groups must have compatible types.
moveall	Moves all standard devices from one existing device group to another existing device group. The source and destination device groups must have compatible types.
not_ready	Sets the device(s) to be Not Ready. The device must be in a User Ready status for this operation to succeed.
pin	Set the device(s) to the user pinned state. User pinned device(s) will not be moved via FAST controller, but they can be moved via Optimizer or symmigrate.
ready	Sets the device(s) to be Ready. The device must be in a User Not Ready status for this operation to succeed.
rebind	Rebinds the device(s) to the thin pool.
reclaim	Reclaims storage from the thin pool.
relabel	Applies the defined label to the device. The device must be in a User Not Ready status for this operation to be accepted. Refer to the symlabel command for details about how to define a device label.
release	Releases a device external lock associated with all devices within a device group.
remove ld	Removes a standard device from an existing device group.
rename	Renames an existing device group. Use a device group name that is unique to this host.
rename ld	Renames a device within a device group.

reset — Sets the device to its original identity when combined with -identity option.

rmall — Removes all standard devices from an existing device group.

rw_enable — Sets the devices to be Read and Write Enabled to their local hosts on the specified front director ports. If no ports are specified, then the devices will be Read and Write Enabled on all ports on which the devices are visible.

set — Sets the persistent indicator for allocations on thin device(s) when combined with the -persistent option. Sets the Optimized Read Miss mode when combined with the -orm option. Sets GCM mode when combined with -gcm.

show — Shows information about a device group including: group type, Symmetrix ID, creation time, number of devices that are members, a list of associated gatekeeper devices, and a list of associated Business Continuance Volume (BCV) devices.

show ld — Shows status information about a device in the device group.

unbind — Unbinds device(s) from the thin pool.

uncompress — Starts data decompression on thin device(s). When combined with the -stop option, data decompression is stopped.

unhold — Releases devices that were previously set to the hold state.

unpin — Unsets the device(s) from the user pinned state.

unset — Clears the persistent indicator for allocations on thin device(s) when combined with the -persistent option. Clears GCM mode when combined with -gcm.

write_disable — Sets the devices to be Write Disabled to their local hosts on the specified front director ports. If no ports are specified, then the devices will be Write Disabled on all ports on which the devices are visible.

OPTIONS

-all — Used with the free operation in order to specify that all allocations associated with the indicated devices are to be freed, regardless of whether data has been written or not.

-apidb — Creates the device group in the SYMAPI configuration database only.

-bcv
This flag may be used for one of the following reasons:

- Adds only BCV devices to the target device or storage group.

- Targets the operation to the specified BCV device(s) that are locally associated with the device group.

-brbcv
This flag may be used for one of the following reasons:

- Adds only the BRBCV devices to the target device group.

- Targets the action at the specified remotely associated RDF BCV device(s) in the device group.

-c
Specifies the number (count) of times to attempt to acquire an exclusive lock on the Symmetrix host database and (for RDF control operations) on the local and/or remote Symmetrix arrays.

The time to wait between attempts to acquire a needed lock is specified by -i (interval).

If neither -c nor -i is specified, operations will fail if unable to acquire a requested lock.

If -c is not specified, and -i is specified, the program will loop continuously until the operation has acquired the locks it needs and can start.

-cap
Sets a minimum device size to the selection criteria of devices.

-captype <mb | cyl>
Specifies the units of capacity in megabytes or cylinders. The default unit of measure is mb.

-celerra
Allows controls on Celerra FBA devices.

-orm
Allows setting the Optimized Read Miss mode to system default, on, or off for the specified devices.

-cyl
Displays the device capacity in cylinders. The default unit of measure is megabytes (MB).

-delete
Deletes the device group after the group is exported to a file (when used with the export argument). The default is to export the device group to the file without deleting the device group.

-devs
Specifies the ranges of Symmetrix devices to add, remove, and move.

-file Specifies a file to write to or read from.

-force This flag may be used for one of the following reasons:

- Forces a deletion of a device group, with or without members, or forces a partial device group conversion (dg2cg) of the devices to a consistency group (even though some devices cannot be converted).

- Applies force with the specified action on a device group that would otherwise be rejected. Forces standard devices to be removed or moved from a device group without querying the Symmetrix array for the device's BCV pair states (if any).

-ftype Specifies the device type to create a Symmetrix Manager file containing a list of the specified type of devices. Only one device type can be specified. If no type is specified, a list of standard devices will be used to create the file.

Possible values are:

-ftype	Description
STD	Creates a file containing a list of all of the standard devices in the group.
R1BCV	Creates a file containing a list of all of the R1 BCV devices in a group.
STD_BCV	Creates a file containing a list of all of the standard/BCV pairs that are in the group. This includes both established and split pairs (from the point of view of the standard device).
STD_R1BCV	Creates a file containing a list of all of the standard/R1 BCV pairs that are in the group. This is a subset of the list provided by the STD_BCV option, but it does not include devices that have never been paired.
STD_VDEV	Creates a file containing a list of all of the standard/virtual device pairs that are in the group.
BCV_VDEV	Creates a file containing a list of all of the BCV/virtual device pairs that are in the group.

-g Specifies a device group name.

-gcm
Allows setting or clearing the device GCM mode.

-geometry
Shows device geometry.

-grpfile
Specifies an alternate group database file for use in list, show, export and exportall actions.

-h
Provides brief, online help information.

-held
Lists devices in the device group that have device holds.

-hop2
Indicates that the device is two hops away. If used, -vdev, -tgt, or -bcv must also be specified.

When adding, removing, moving, or copying devices, -rdfg, -remote_rdfg, and -tgt or -vdev must also be specified.

When holding or readying devices, the hop2 flag must be used with -bcv, -vdev, or -tgt.

-host
Limits the devices added to only those mapped to the host's front-end directors.

-i
Specifies the interval, in seconds, to wait between attempts to acquire an exclusive lock on the Symmetrix host database and (for RDF control operations) on the local and/or remote Symmetrix arrays.
The default interval is 30 seconds.
The minimum interval is 5 seconds.
For passive actions the minimum interval is 15 seconds. Passive actions are actions that do not acquire an exclusive lock.

-identity
Lists the external identity information for each device. The external identity will be displayed if the device has external identity set. Otherwise, the device native identity will be displayed. When used with reset command, the original identity of the device(s) is restored.

-inactive
Lists or shows inactive device groups. When GNS is enabled on the host, the inactive groups are those that were previously defined in the host's configuration database file. When GNS is disabled on the host, this may show group definitions present the last time GNS was enabled on the host (if they were captured in the host database file).

-lock
Displays existing device external locks on devices within the group. Use with the show argument.

-N
Sets the number of devices to add, remove, or move.

-nobcv
Adds only STD devices to the target device group.

-noprompt
Disables the prompt for confirmation feature. The default is to prompt the user for confirmation before executing the operation.

-noRDF
Adds, copies, moves, or removes non RDF devices only.

-novalidate
Lists groups without output of the V(alid DG) flags column.

-offline
Obtains information from the Symmetrix host configuration database.

-p
Specifies the front-end (SCSI or Fibre) director port number to only select devices that are primarily visible through this director port. By default, all ports are selected.

-persistent
Used with set or unset it specifies operation on the persistent indicator. Used with allocate or reclaim it specifies the use of persistent storage.

-pool
Specifies a Thin Pool Name.

-R1
Adds, copies, moves, or removes RDF1 (R1) devices only.

-R2
Adds, copies, moves, or removes RDF2 (R2) devices only.

-R21
Adds, copies, moves, or removes RDF21 (R21) devices only.

-rbcv
This flag may be used for one of the following actions:

- Adds only RBCV devices to the target device group.

- Targets the action at the device group’s locally associated RDF BCV devices that can be BCV paired with the remote mirrors of the standard RDF devices.

-rdf
This flag may be used for one of the following actions:

- Allows the group to be imported on the remote Symmetrix array from the file which is created. When exporting an RDF group, this will use the remote Symmetrix ID and device names and toggle the RDF group type from R1 to R2 or vice versa.

- Indicates that remote VDEVs or TGTs are being added to the group.

-rdf_consistency
Creates a CG and allows it to be enabled for RDF consistency after adding devices to the CG.

-rdfg
This flag may be used for one of the following actions:

- Supplies the RA (RDF) group number to only add devices that belong to this RA group number.

- Indicates the Symmetrix RA (RDF) group number to reach the remotely associated BCV device.

-metro
When specified with ready and not_ready identifies the devices being controlled are part of an RDF/Metro configuration.

-remote_rdfg
Specifies the RDF (RA) group to access a two-hop device from the first hop.

-rename
This flag may be used for one of the following actions:

- Assigns new logical device names to all added devices.

- Renames the standard device(s) to the default names as they are moved from their current device group to the destination device group.

-resv
Lists devices in the device group that have SCSI reservations.

-rp
Indicates that the action is targeted for devices tagged for RecoverPoint.

-rrbcv
This flag may be used for one of the following actions:

- Adds only the RRBCV devices to the target device group.

- Targets the action at the specified remotely associated remote BCV device(s) in the device group.

-rtgt
This flag may be used for one of the following actions:

- Adds only RTGT devices to the target device group.

- Targets the indicated action at the devices in RTGT list of the device group.

-rvdev
This flag may be used for one of the following actions:

- Adds only devices that are Symmetrix remote virtual devices to the target device group.

- Targets the indicated action at the specified remote VDEV device(s) that are associated with the device group.

-SA
Specifies the front-end (SCSI or Fibre)

director number to only select devices that are primarily visible through this director. Alternatively, if ALL (the default) is specified, all devices satisfying any other selection criterion will be selected.

-sel_rdfg — Indicates the Symmetrix RA (RDF) group number of the devices to be added via an addall operation.

-sid — This flag may be used for one of the following actions:

- Lists the device group information for a specified Symmetrix ID.
- Supplies the Symmetrix ID to add only devices belonging to the specified Symmetrix array.

-star — Indicates that the action is targeted for devices in STAR mode.

-stop — Specifies that the compress, uncompress, allocate, free or reclaim operation will be stopped.

-symforce — Forces the operation to be executed when normally it would be rejected.

Caution: Extreme caution should be exercised when using this option.

-tgt — This flag may be used for one of the following reasons:

* Adds only TGT devices to the target device or storage group.
* Targets the indicated action at the devices in TGT list of the device group.

-type — Identifies the type of device group, either REGULAR, RDF1, RDF2, R21 or ANY. The default type is REGULAR.

-v — Provides a more detailed, verbose listing.

-vdev — This flag may be used for one of the following actions:

- Adds only devices that are Symmetrix virtual devices to the target device or storage group.
- Targets the indicated action at the specified VDEV device(s) that are associated with the device group.

PARAMETERS

CgName — The target composite group name.

DestDgName — The destination device group to which the standard devices are moved.

DgName	The device group name assigned by the user. The name must be unique to this host.
FileName	The data file used to export or import a device list, or used in the translation of a Symmetrix Manager device list.
GrpDbFileName	Specifies an alternate group database file for use in list, show, export and exportall actions.
GrpNum	The RDF (RA) group number.
HostName	The name of the host system.
LdevName	The logical device name that is named by the user or automatically assigned when a device is added to a device group.
NewDgName	The renamed device group name.
SelRdfGrpNum	The Symmetrix RA (RDF) group number of the devices to be added via an addall operation.
SgName	The target storage group name.
SymDevEnd	The last Symmetrix device name in a sequence, such as 00B6.
SymDevName	The Symmetrix device name, unique per Symmetrix, such as 01C.
SymDevStart	The first Symmetrix device name in a sequence, such as 001C.
SymmID	The 12-digit ID of the Symmetrix array.

FILES

The export argument creates a group file (ASCII text) and the import argument reads the file to import a device group. The file will contain as many device description lines as devices and gatekeepers that are being defined in the group list. Any lines that are blank or have a pound sign (#) in the first column are ignored.

Group files contain device parameters in the following format:

```
<GroupType> <SymmID>
<DeviceType> <DeviceParameters>
<DeviceType> <DeviceParameters>
. . .
```

The following list describes the various parameters within the file:

Parameter	Description
<GroupType>	Specifies an integer value that defines the type of group for this group list. Possible values include:

Symbolic Name	Value
SYMAPI_C_DGTYPE_REGULAR	0

SYMAPI_C_DGTYPE_RDF1	1
SYMAPI_C_DGTYPE_RDF2	2
SYMAPI_C_DGTYPE_RDF21	3
SYMAPI_C_DGTYPE_ANY	4

<SymmID> Specifies the 12-digit ID of the Symmetrix array associated with the group.

<DeviceType> Defines the device type for the group member being exported or imported:

Type	Meaning
S	Standard Device (STD)
B	Local BCV Device (BCV)
R	Remote BCV Device (RBCV)
Z	BCV Remote BCV Device (BRBCV)
Y	Remote Remote BCV Device (RRBCV)
D	Hop 2 BCV Device (2BCV)
G	Gatekeeper Device
V	VDEV Device
W	Remote VDEV Device (RVDEV)
E	Hop 2 VDEV Device (2VDEV)
T	Target Device
X	Remote Target Device (RTGT)
F	Hop 2 Target Device (2TGT)

<DeviceParameters> Defines the device parameters:

Type	Parameters
S	<SymmID> <LdevName>
B	<SymmID> <LdevName>
V	<SymmID> <LdevName>
R	<SymmID> <LdevName> <RDFGrpNum>
W	<SymmID> <LdevName> <RDFGrpNum>
E	<SymmID> <RDFGrpNum> <HOP2RDFGrpNum> <LdevName>
Z	<SymmID> <LdevName> <RDFGrpNum>
Y	<SymmID> <RDFGrpNum> <RemoteRDFGrpNum><LdevName>
D	<SymmID> <RDFGrpNum> <HOP2RDFGrpNum> <LdevName>
G	<PdevName>
T	<SymmID> <LdevName>
X	<SymmID> <LdevName> <RDFGrpNum>
F	<SymmID> <RDFGrpNum> <HOP2RDFGrpNum> <LdevName>

The parameters are:

Parameter	Description
<HOP2RDFGrpNum>	The RDF group number of the device on an array two hops from the standard.
<LdevName>	A logical device name (for example: DEV002).
<PdevName>	The gatekeepers physical device name (for example: /dev/rdsk/c2t0d2s2).

Appears for a gatekeeper (G) type only.

<RDFGrpNum> The RDF group number of the standard device with which the BCV is paired.

<RemoteRDFGrpNum> The RDF group number of the device on a remote array.

<SymmID> The 12-digit ID of the Symmetrix array associated with the group.

RETURN CODES

Code #	Code Symbol
0	CLI_C_SUCCESS
1	CLI_C_FAIL
2	CLI_C_DB_FILE_IS_LOCKED
18	CLI_C_ALREADY_IN_STATE The device or device group is already in the desired state. Applicable only for the rw_enable and write_disable actions.
19	CLI_C_GK_IS_LOCKED All GateKeepers to the Symmetrix unit are currently locked.

EXAMPLES

To create a Symmetrix device group mydg_r1 of type RDF (R1), enter:

```
symdg -type RDF1 create mydg_r1
```

To list all Symmetrix device groups in detailed format, enter:

```
symdg -v list
```

To rename Symmetrix device group mydg_r1 to oradg_rdf1, enter:

```
symdg rename mydg_r1 oradg_rdf1
```

To show information about device group oradg_rdf1, enter:

```
symdg show oradg_rdf1
```

To export the device group to a file named oradg_rdf1.txt and then delete the device group, enter:

```
symdg -file oradg_rdf1.txt -delete export oradg_rdf1
```

To recreate the device group from the file oradg_rdf1.txt, enter:

```
symdg -file oradg_rdf1.txt import oradg_rdf1
```

To translate the device group into a device file named devices.txt, enter:

```
symdg -file devices.txt -ftype STD_BCV dg2file oradg_rdf1
```

To delete Symmetrix device group oradg_rdf1, regardless of whether the group has members or associated gatekeeper or BCV devices, enter:

```
symdg -force delete oradg_rdf1
```

To recreate the device group from the device file devices.txt, enter:

```
symdg -file devs.txt -type REGULAR  file2dg oradg_rdf1
```

To add the device group's BCV devices to a composite group named oracg, enter:

```
symdg -bcv dg2cg oradg_rdf1 oracg
```

To add a Symmetrix host device to group ProdDB, and assign a device logical name temp1, enter:

```
symdg -g ProdDB add pd /dev/rdsk/c2t0d2s2 temp1
```

To add a Symmetrix device to group ProdDB and assign a device logical name temp2, enter:

```
symdg -g ProdDB add dev 01C temp2
```

To add to group ProdDB all devices that are primarily visible from this host on Port 0 (top port) of Symmetrix SCSI director 1, enter:

```
symdg -g ProdDB -SA 1 -p 0 addall pd
```

To add to group ProdDB all devices that are primarily visible from this host and fall in the following device range, enter:

```
symdg -g ProdDB -devs 000:00F addall pd
```

To list all devices in device group ProdDB, enter:

```
symdg -g ProdDB list ld
```

To rename device DEV001 to log1 in group ProdDB, enter:

```
symdg -g ProdDB rename ld DEV001 log1
```

To remove device log1 from device group ProdDB, enter:

```
symdg -g ProdDB remove ld log1
```

To show detailed information about device log1, enter:

```
symdg -g ProdDB show ld log1
```

To write disable device DEV001 on Symmetrix director 16A and Port 0, enter:

```
   symdg -g ProdDB -SA 16A -p 0 write_disable DEV001
```

To relabel all BCV devices that are locally-associated with device group ProdDB, enter:

```
   symdg -g ProdDB -bcv -v relabel
```

To add to group ProdDB, all VDEVs that are primarily visible from this host and fall in the following device range, enter:

```
   symdg -g ProdDB -devs 000:00F addall pd -vdev
```

symdisk

Reports on the configuration and status of disks (spindles) and their hypers for Symmetrix arrays.

SYNOPSIS

```
symdisk [-h]

symdisk [-sid <SymmID>] [-offline] [-cyl | -mb | -gb | -tb]
        [-disk_group <DskGrpNum | name:<DskGrpName> | ALL>
         [-all]] [-failed]

   list [-spindle [-internal]] [-isspare] [-nospares]
        [-v [-hypers] [-spare_info] [-gaps]]
        [-DA <# | ALL>] [-interface <# | ALL>]
        [-tid <# | ALL>]

   list [-spindle
          [-external [-detail] [-encapsulated [-free]]]]
        [-v [-hypers] [-gaps]]
        [-DX <# | ALL>]

symdisk [-sid <SymmID>] -external -spindle -state
        [-spid <SpindleID>]

   list

symdisk [-sid <SymmID>] -external -spindle -paths
        [-spid <SpindleID> |-DX <# | ALL> [-port <# | ALL>]]

    list -detail

    list [-offline]

symdisk -sid <SymmID> -external
        [-wwn <ExternalWWN> | -spid <SpindleID>]
        <-draining | -drained | -active | -disabled>
        [-i <Interval>] [-c <Count>]

   verify

symdisk [-sid <SymmID>] [-offline] [-cyl | -mb | -gb | -tb]

   list -dskgrp_summary [-v]
        [-disk_group <DskGrpNum | name:<DskGrpName> | ALL>]
         | -internal | -external]

   list -dskgrp_summary -by_engine [-v | -detail]
        [-disk_group <DskGrpNum | name:<DskGrpName> | ALL>]
         | -internal | -external]

   show <DiskAddress> [-gaps_only]

symdisk -sid <SymmID> [-offline] [-cyl | -mb | -gb | -tb]

   show -spid <SpindleID> [-gaps_only]

   show -wwn <ExternalWWN> [-gaps_only]
```

DESCRIPTION

The symdisk command allows access to the configuration information of the disks (spindles) that make up a Symmetrix array. It can be used to list all of the disks for a Symmetrix array or only those that match certain criteria.

The selection criteria allows the user to return only data about the disks on a certain disk director (DA), disk interface (INT), or disk Target ID (TID). In addition, the -isspare flag may be used to select only those disks that are configured as spare disks.

Using the -v option will provide more detailed information. The -hypers flag can be used with -v to display additional information about each of the logical hypers on a given disk (including which Symmetrix devices they make up).

ARGUMENTS

list	Lists all disks. With the use of options, the list can be restricted. The -v and -hypers options will return additional information about each disk.
show	Displays detailed information about the disk(s) that match the given DA, INT, and TID.
verify	Verifies whether one or all disks are in a given state.

OPTIONS

-active	The specified disk is ready to accept new writes.
-all	Includes spare disks in the -disk_group disk listing. This option is specific to -disk_group and is not compatible with -dskgrp_summary.
-c	Specifies the number (count) of times to execute the verify operation. If this option is not specified but an interval (-i) is specified, the program will loop continuously. The looping may be terminated during verification if all devices enter the requested state.
-cyl	Displays the disk capacities in terms of cylinders. The default is megabytes (MB).
-DA	Indicates the disk director number. A value of ALL returns all disk directors.
-detail	Displays detailed path information for external spindles. Alternatively, displays external array id and external device name information for external spindles.
-disabled	The specified disk will not accept new writes.
-disk_group	Lists the disks that are members of the specified disk group name or number. When used with the ALL qualifier, all disk groups are listed. When used with the -all option, any spare disks are also included in the disk group listing.
-by_engine	When used with the -dskgrp_summary option, displays aggregate information about

spindles configured to specific engines within a disk group. The default output is in table format with additional information provided when used with the -v (verbose) or -detail listing option. All engines supporting spindles are included in the listing unless otherwise restricted by additional filter options.

-draining — Drain operation is in progress for the specified disk.

-drained — Drain operation is complete for the specified disk.

-dskgrp_summary — Displays summary information for disk groups. The default output is in table format. When used with the -v option, the output is in expanded format (one field per line).

-DX — Indicates the external disk director number. A value of ALL returns all disk directors.

-encapsulated — Lists only encapsulated spindles.

-external — Lists only external spindles.

-failed — Lists only those spindles that have been flagged as failed.

-free — Lists only those external encapsulated spindles that have no allocations on all of their datadevs or have no RAID groups existing on the spindle.

-gaps — Lists the size and location of freespace gaps as they occur within the list of hyper information.

-gaps_only — Shows only the gap information.

-gb — Displays the disk capacities in gigabytes.

-h — Provides brief, online help information.

-hypers — Shows hyper information when used with -v.

-i — Specifies the repeat interval in seconds to execute the verification operation. The default interval is 30 seconds. The minimum interval is 5 seconds.

-isspare — Indicates that only spare disks should be displayed.

-interface — Indicates the disk interface number. A value of ALL returns all interfaces.

-internal — Lists only internal spindles.

-mb — Displays the disk capacities in megabytes.

-nospares — Indicates that only internal disks (spindles) that are capable of being covered by a spare, but currently are not, should be

displayed. This option is only supported on Enginuity version 5876 and above. No disks (spindles) will be displayed if this option is issued for a Symmetrix running a prior Enginuity version. In addition, this option is not applicable for external spindles (external spares do not exist) and no external spindles will be returned when this option is specified.

-paths — Displays path information for external spindles. The default display provides a DX port summary, with counts representing DX ports which have connections. When combined with the -detail option, all paths are displayed, including multiple paths per DX port (if configured).

-sid — Indicates a unique ID of the Symmetrix array.

-spare_info — If the disk is a spare and it has been invoked against a failed disk during dynamic sparing, this flag will return information about the failed disk. The -v option must also be specified. Note that this display is not applicable for permanent sparing.

-spid — Indicates the specific spindle ID of the target disk.

-spindle — Displays spindle information instead of the standard disk address information.

-state — Reports the current state of the disk(s).

-tb — Displays the disk capacities in terabytes.

-tid — Indicates a target ID. A value of ALL returns all targets.

-v — Provides a more detailed, verbose listing.

-wwn — Display detailed information for the external spindle specified by ExternalWWN.

PARAMETERS

— The number of the disk group, director, disk interface, or target ID.

ALL — All of the directors, disk interfaces, target IDs, or disk groups.

DiskAddress — The disk adapter, interface, and TID values in the format <XXX:YZ>, where XXX is the disk director, Y is interface, and Z is the TID. It is also acceptable to supply those same values in the format <XXX,Y,Z>.

DskGrpName — The disk group name.

DskGrpNum — The disk group number.

ExternalWWN — The WWN of the target external spindle.

SpindleID — The spindle ID of the disk in hex format.

SymmID The 12-digit ID of the Symmetrix array.

RETURN CODES

Code #	Code Symbol
0	CLI_C_SUCCESS
1	CLI_C_FAIL

EXAMPLES

To list all of the disks for Symmetrix 012345678901, enter:

```
symdisk list -sid 012345678901
```

To display additional verbose information about disk 01A:C3 and its hypers, enter:

```
symdisk list -sid 012345678901 -da 01A -interface C
  -tid 3 -v -hypers
```

To display detailed information about the disk whose DA is 16B, INT is D, and TID is 5, enter:

```
symdisk show 16B:D5
```

To display detailed information about the disk whose DA is 16B, INT is D, and TID is 5 as shown in the previous example using an alternative input format, enter:

```
symdisk show 16B,D,5
```

To list all disks organized by disk group number, including spare disks, enter:

```
symdisk list -disk_group ALL -all
```

To list all disks with spindle information, organized by disk group number, including spare disks, enter:

```
symdisk list -disk_group ALL -all -spindle
```

To list all disks for Symmetrix 012345678901 with their spindle ID information, enter:

```
symdisk list -spindle -sid 012345678901
```

To display detailed information about the disk whose spindle ID is 0x11D0, enter:

```
symdisk show -spid 11D0
```

Performs a Data Migration of an Application to a PowerMax/VMAX3. Migrations from one PowerMax/ VMAX3/VMAX to another PowerMax/VMAX3 preserve application availability throughout the process, except when the -offline option is used with the create argument.

SYNOPSIS

```
symdm -h

symdm -src_sid <SymmID> -tgt_sid <SymmID>
     [-i <Interval>] [-c <Count>] [-noprompt]

   environment <-setup | -remove | -validate>

symdm -src_sid <SymmID> -tgt_sid <SymmID> -sg <SgName>
     [-i <Interval>] [-c <Count>] [-noprompt]
     [-tgt_srp <SRPName>] [-tgt_pg <PgName>]
     [-nocompression] [-validate]

   create [-precopy]

   create -offline [-move_identity] [-precopy]

symdm -sid <SymmID> -sg <SgName>
     [-i <Interval>] [-c <Count>] [-noprompt]

   cancel [-revert]

   commit

   cutover

   readytgt

   recover [-force | -validate]

   sync <-start | -stop>


symdm [-sid <SymmID>]

   list  [-sg <SgName>]
     [-i <Interval>] [-c <Count>]
     [-v [-detail [-sg_info] [-pg_info] [-ig_info]
     [-view_info] [-pairs_info]]]

   list -environment [-offline]
```

DESCRIPTION

The symdm command performs a Data Migration of an application’s resources to a target PowerMax/VMAX3 array.

The command is used to orchestrate the migration of an application from the source to the target array using the command set provided:

- symdm environment -setup
- symdm create
- symdm cutover/readytgt
- symdm commit
- symdm environment -remove

While multiple migrations can be in progress concurrently, a single environment -setup action is required to provide the required migration infrastructure before any migrations can start, and a single environment -remove action is required when all have completed, to remove that infrastructure.

Additional actions are available to monitor the state of a migration, to back out of a migration, to recover from a migration failure or to pause/resume data synchronization between the arrays of a Data Migration session.

Additional options are available to specify the type of Data Migration and specific create behaviors.

Upon successful completion of a migration, the application will run using resources on the target PowerMax/VMAX3 rather than the resources it originally used on the source array.

When migrating data from one PowerMax/VMAX3/VMAX to another, the application remains available for use throughout the migration process, except for a short application downtime when the -offline argument is used.

ARGUMENTS

cancel — Cancel a Data Migration that is in progress. Resources allocated by the migration for the application on the target array will be removed.

commit — Commit the application to permanently run on the target array by removing the replication session between the arrays and the accessibility of the source devices to the application host. This operation can only be used if the migration session is in either the CutoverSync or Synchronized state.

create — Duplicate the source application on the target array, creating equivalently provisioned devices visible to the host, and begin replicating application data to the target array.

With the -validate option, makes no changes to the target array but validates that the target array is capable of supporting the migration.

cutover — Configures the application over to run only on the target array. Subsequent updates to data on the target array will be replicated back to the source array. This operation can only be used if the migration session is in the CutoverReady state.

environment — Setup, remove, or validate the environment required to perform a Data Migration. The required environment includes connectivity between the two arrays, and between both arrays and the management host, and between both arrays and the application host.

Once a migration environment has been set

up, it can be used to migrate multiple applications from the source to the target array.

list — List all migration sessions or migration environments depending on options. When used without the -environment option, list sessions that are running on a specified PowerMax, VMAX3, or VMAX array, or filter the list by SG name. When used with the -environment option, list environments configured for data migration.

readytgt — Will reconfigure the replication so the IOs are replicated to the other array and will make the Target devices visible to the host. This operation can only be used if the migration session is in the Precopy state.

recover — A recover operation should be used only after a step in the migration completes with a "failed" state, and is not normally required as part of a migration.

Assuming that the administrator has corrected the issue(s) that caused a migration step to fail, a recover will allow the failed step (create, cutover, commit, cancel) to complete.

A recover with the -validate flag is used when the data migration is in an "Invalid" state to run validation checks and report errors found to the symapi log file.

sync — Controls target-to-source replication after a cutover.

OPTIONS

-c — Used with list, specifies the number (count) of times to display information.

Used with control operations, specifies the number (count) of times to attempt to acquire an exclusive lock on the PowerMax/ VMAX3/VMAX host database.

If you do not specify this option and specify an interval (-i), the program will loop continuously to list or start the control operation.

-detail — Provide detailed information on the state of a single migration session, to help identify the cause of a failed migration or a blocked operation.

-environment — Report summary information for configured migration environments. By default, all local and remote arrays will be queried.

-force — Attempts to force the operation even though the devices in the migration session might not be in the normal, expected state for the operation. The -force option should be used cautiously and only when the user understands the implications of the action.

-i	Specifies the repeat interval, in seconds, to wait, either between successive iterations of a list operation or between control operation attempts to acquire an exclusive lock on the PowerMax/VMAX3/VMAX host database. The default interval is 30 seconds. The minimum interval is 5 seconds. For passive actions the minimum interval is 15 seconds. Passive actions are actions that do not acquire an exclusive lock.
-ig_info	Only report the detailed information of IGs that were part of the migration.
-move_identity	Only used with the -offline Data Migrations, the same device identities as used to access the devices on the source array will be used by the target array devices.
-nocompression	When given all FAST managed storage groups on the target array will have compression disabled.
-noprompt	Requests that prompts are not displayed after the command is entered. The default is to prompt the user for confirmation.
-offline	When used with the create, it specifies the migration will be minimally disruptive and require a short application downtime. When used with the list -environment, obtains data only from the configuration database on the host. No connections are made to any Symmetrix.
-pairs_info	Only report the detailed information of the device pairs that were part of the migration.
-precopy	When used with the create, it configures the replication to copy the data to target and leaves the target devices not visible to the host.
-pg_info	Only report the detailed information of PGs that were part of the migration.
-remove	Used with the environment action to remove the migration infrastructure created by the -setup option after all necessary application migrations have been completed.
-revert	Move the application back from the target to the source array, when the session is in a CutoverSync state, This option is only valid for migration sessions from arrays running Enginuity 5876 that were not created with the -offline option.
-setup	Used with the environment action to create the infrastructure required for migrations, on both the source and target arrays. Once the environment has been set up, it can be used to perform multiple migrations.

-sg The name of the Storage Group that represents the application being migrated.

-sg_info Only report the detailed information of SGs that were part of the migration.

-sid For the cutover, cancel, recover, commit and list operations, the PowerMax/ VMAX3/VMAX ID of the array participating in the migration session.

-src_sid For the environment and create operations, the PowerMax/VMAX3/VMAX ID of the array from which data is being migrated (source array).

-start Used with the sync action to start or restart replication from the target array to the source array.

-stop Used with the sync action to stop replication from the target array to the source array.

-tgt_pg Specifies the name of the port group used on the target array.

-tgt_sid For the environment and create operations, the PowerMax/VMAX3 ID of the array to which data is being migrated (target array).

-tgt_srp The name of the Storage Resource Pool (SRP) from which to create storage on the target array. If not specified, the default SRP will be used.

-v Provides a more detailed, verbose listing.

-validate Can be used with the create action to validate that: the source-array devices are suitable for migration; and that the target array has sufficient available storage to accept the migrated data; and that the required migration infrastructure exists on both arrays.

Validation is normally performed by the create action before the command is started; this option allows for validation without actually performing the action.

Can be used with the environment action to validate that the environment meets requirements for the migration to be performed.

Can be used with the recover action to report changes in the data migration session that have made the session state "Invalid".

-view_info Only report on the detailed information of the masking views that were part of the migration.

PARAMETERS

Count	The number of times (count) to repeat.
Interval	The interval between repetitions, in seconds.
PgName	The port group name.
SgName	The storage group name.
SRPName	The SRP name.
SymmID	The 12-digit ID of a PowerMax, VMAX3, or VMAX array.

symdrv

Displays information for selected Dynamic Reallocation Volume (DRV) devices.

SYNOPSIS

```
symdrv [-sid <SymmID>] [-h] [-offline] [-v]

   list [-cap <#>] [-i <Interval>] [-c <Count>]
```

DESCRIPTION

The symdrv command lists all the DRV devices that are configured on Symmetrix arrays attached to this host.

ARGUMENTS

list	Lists all the DRV devices (SymDevNames) that are configured on Symmetrix arrays attached to this host.

OPTIONS

-c	Specifies the number (count) of times to display DRV devices. If this option is not specified and an interval (-i) is specified, the list of statistics will be displayed continuously.
-cap	Specifies to display only those devices whose capacities match (in megabytes) that value.
-h	Provides brief, online help information.
-i	Specifies the repeat interval in seconds. The default interval is 30 seconds. The minimum interval is 5 seconds. For passive actions the minimum interval is 15 seconds. Passive actions are actions that do not acquire an exclusive lock.
-offline	Displays Symmetrix devices from the Symmetrix configuration database without refreshing the data from the Symmetrix array.
-sid	Supplies the Symmetrix ID to limit the scope of the operation to the specified Symmetrix array.
-v	Provides a more detailed, verbose listing.

PARAMETERS

Count	Number of iterations to execute before exiting.
Interval	Interval between polls, in seconds.
SymmID	The 12-digit ID of the Symmetrix array.

RETURN CODES

Code #	Code Symbol
0	CLI_C_SUCCESS
1	CLI_C_FAIL
19	CLI_C_GK_IS_LOCKED All GateKeepers to the Symmetrix array are currently locked.

EXAMPLES

To list all DRV devices that are configured on Symmetrix arrays attached to this host, enter:

```
symdrv list
```

symevent

Enables the monitoring and tracking of events
on Symmetrix arrays.

SYNOPSIS

```
symevent -h

symevent [-sid <SymmID>] [-v] [-warn | -error | -fatal]

   monitor [-i <Interval>] [-c <Count>]

   list [-start <Date:Time>] [-end <Date:Time>] [-dir]
```

DESCRIPTION

The symevent command allows an administrator to monitor events within a Symmetrix array that may affect its operation. In most cases, a reported event represents a condition that has already been repaired. This tool allows an administrator to track those events to understand the events that have occurred, or are occurring, on your Symmetrix array.

The monitor action sets the command to run in the foreground where it polls the Symmetrix array for new events every interval, defined in seconds, until the iteration count is satisfied or the program is stopped.

The list action reports on the history of events, which is stored on the Symmetrix array. Specifying a start and end time allows you to retrieve events that occurred between the specified time bounds.

In addition, you can restrict the query to a specific Symmetrix array and restrict the events reported to a minimum severity level (warnings, errors, or fatal events).

When run against a Symmetrix at Enginuity 5671 or higher, event timestamps are calculated relative to the host (i.e., the same time zone) from which the command is issued. In client/server mode, this is the server host. For a Symmetrix array with an earlier Enginuity level, timestamps are relative to the time on the Symmetrix array, usually GMT/UTC.

Note: Beginning with Enginuity 5761, the Error Number is reported using a new numbering scheme. This allows the error numbers to remain constant across all future Enginuity releases.

ARGUMENTS

list	Lists events which have occurred over time on the Symmetrix array.
monitor	Monitors the Symmetrix array in real time for new events.

OPTIONS

-c	Specifies the number (count) of times to poll for events. If this option is not specified, symevent will continuously poll

for events.

-dir — Displays events sorted on the basis of reporting director.

-end — Specifies a date and time before which to report on events. Used with the list argument.

-error — Displays only events with a severity of Error or greater.

-fatal — Displays only events with a severity of Fatal.

-h — Provides brief online help information.

-i — Defines the time interval, in seconds, between polls.
The default interval is 30 seconds.
The minimum interval is 5 seconds.
For passive actions the minimum interval is 15 seconds. Passive actions are actions that do not acquire an exclusive lock.

-sid — Specifies a unique Symmetrix ID.

-start — Specifies a date and time after which to report on events. Used with the list argument.

-v — Provides a more detailed, verbose listing.

-warn — Displays only events with a severity of Warning or greater.

PARAMETERS

Count — The number of iterations to execute before exiting.

Date:Time — The date and time specification of the form [mm/dd/yyyy]:[hh:mm[:ss]].
The current date and time will be substituted for omitted fields.

Interval — The interval between polls, in seconds.

SymmID — The 12-digit ID of the Symmetrix array.

RETURN CODES

Code #	Code Symbol
0	CLI_C_SUCCESS
1	CLI_C_FAIL

REPORTED EVENTS

The following list contains the reported events using the following format:

Event Code Symbol
Severity: <severity-level>

```
  Event Code Text

DIAG_TRACE_TRIG
  Severity: Informational
  A Symmetrix diagnostic event-trace was triggered

DIAG_TRACE_TRIG_REMOTE
  Severity: Informational
  A diagnostic event-trace was triggered for a
  Symmetrix remotely attached via RDF links

TOO_MANY_SUSPHALT_CHAINS
  Severity: Informational
  Too many suspend/halt chains switching to Adaptive
  Copy Write Pending Mode

MEM_DISABLE_INVOKED
  Severity: Informational
  One or more memory banks were disabled due to cache
  errors

SPARE_INVOKED
  Severity: Informational
  A Spare Disk was invoked against a Symmetrix Disk

M2_RESYNC_WITH_M1
  Severity: Informational
  An M2 mirror of a Symmetrix Device is resynchronizing
  with the M1 mirror

M1_RESYNC_WITH_M2
  Severity: Informational
  An M1 mirror of a Symmetrix Device is resynchronizing
  with the M2 mirror

DISK_ADAPTER_DEAD
  Severity: Fatal
  A Symmetrix Disk Director is not responding

ALL_DEVICES_MIGRATED
  Severity: Informational
  All Symmetrix migration devices have completed the
  data migration

DEVICE_RESYNC_STARTED
  Severity: Informational
  A Symmetrix device resynchronization process has
  started

SPARE_INVOKED_REMOTE
  Severity: Informational
  A Spare Disk was invoked against the Disk of the R2
  mirror in Symmetrix remotely attached via RDF links

RDF_SIM_MESSAGE
  Severity: Informational
  The RDF subsystem initiated a SIM message to a
  Symmetrix remotely attached via RDF links

RDF_ERROR
  Severity: Error
  The RDF subsystem has experienced an error

FC_OPTICAL_MOD_ERROR
  Severity: Warning
  A Fibre Channel optical module has experienced
  a problem
```

```
ALL_RDF_LINKS_DOWN
  Severity: Warning
  No RDF links in an RDF group are operational

ALL_RDF_LINKS_NOW_UP
  Severity: Informational
  All RDF links in an RDF group are now operational

BUS_PROBLEM
  Severity: Informational
  Bus Arbiter problem: primary arbiter has
  experienced a problem

TEMPERATURE_PROBLEMS
  Severity: Warning
  The Symmetrix is experiencing temperature problems

ALARM_SIGNAL
  Severity: Warning
  An alarm signal was set but no alarm was found

ALARM_SIGNAL_POWER
  Severity: Warning
  An alarm signal was set indicating a power subsystem
  error

MIRROR_NR
  Severity: Warning
  A device has a mirror that is Not Ready

MIRROR_WD
  Severity: Warning
  A Symmetrix device has a member or a mirror that is
  Write Disabled

RDF2_DEVICE_NR
  Severity: Informational
  One of the RDF2 devices was found to be Not Ready

SP_NOT_RESPONDING
  Severity: Warning
  The Symmetrix Service Processor is not communicating
  with the Symmetrix

SP_PHONEHOME_FAIL
  Severity: Warning
  The Symmetrix Service Processor could not complete a
  Call Home for service

12_VOLTS_ON
  Severity: Warning
  One of the Symmetrix subsystems is running in the
  abnormal 12-Volts mode

SENSE_CABLE_MISSING
  Severity: Warning
  A Symmetrix power subsystem Environment sense cable
  is missing

AC_LINE_INTERRUPTED
  Severity: Warning
  A Symmetrix power subsystem AC line interruption was
  detected

HIGH_CHARGE_MISSING
  Severity: Warning
```

```
  The Symmetrix battery system is not fully charged

LATCHED_ALARMS
  Severity: Warning
  A Symmetrix power subsystem discovered latched
  alarms

ONE_RDF_LINK_DOWN
  Severity: Warning
  A single RDF link in an RDF group is not operational

ONE_RDF_LINK_NOW_UP
  Severity: Warning
  A single RDF link in an RDF group is now operational
  after a 'Single Link Down' event

SP_PHONEHOME_SUCCESS
  Severity: Informational
  The Symmetrix Service Processor completed a Call Home
  for service

UNABLE_TO_SET_REGISTER
  Severity: Warning
  A Symmetrix communication subsystem was unable to set
  a shared register

DISABLED_MEMORY_BANK
  Severity: Warning
  A Symmetrix Director reported Disabled Memory Bank
  to a host

INVALID_ENVIR_BITS
  Severity: Warning
  A validity problem was detected during an
  environmental test

ENABLED_ENVIR_TESTING
  Severity: Informational
  An event was detected to enable environmental testing
  in diagnosis mode

COMM_BOARD_MISMATCH
  Severity: Error
  The Symmetrix communication board software data has
  a mismatch

OLD_BOARD_MISMATCH
  Severity: Error
  The Symmetrix communication board old information
  does not match current information

THERMAL_DET_FAILED_TEST
  Severity: Error
  The Symmetrix thermal tests detected a failure

THERMAL_EVENT
  Severity: Error
  A thermal event was detected in the Symmetrix

POWER_ON_TIME_FAILED_TEST
  Severity: Error
  The Symmetrix environment tests found
  inconsistencies in Power-on-Time

SP_CONNECT_TIME_NOT_FOUND
  Severity: Error
  The Symmetrix has no records of the last Service
```

```
  Processor connection time

SP_CONNECT_VIA_SERIAL_LINE
  Severity: Informational
  The Service Processor is currently communicating
  via a serial line

SYMREMOTE_CONNECTED
  Severity: Informational
  A SymmRemote session is currently connected to the
  Service Processor

SYMREMOTE_REJECTED
  Severity: Informational
  A SymmRemote session to the Service Processor was
  denied access

SYMREMOTE_DISCONNECTED
  Severity: Informational
  A SymmRemote session to the Service Processor was
  disconnected

SP_EXCESS_MEMORY_USAGE
  Severity: Warning
  The Service Processor software detected excessive
  memory usage

BATTERY_FAILED_TEST
  Severity: Warning
  Automatic battery tests detected failures

NO_COMM_TO_MII_DIR
  Severity: Warning
  The Service Processor could not communicate to
  a director

CANT_QUERY_MII_DIR
  Severity: Warning
  The Service Processor could not query a director

SP_COMM_TO_MII_DIR
  Severity: Informational
  The Service Processor is communicating via a
  local director

CANT_READ_ENVIR_SENSOR
  Severity: Warning
  The Service Processor failed to read an
  environmental sensor

UNRECOGNIZED_EPO_CARD
  Severity: Warning
  The Service Processor has detected a failed or
  unrecognized communication card

ENVIR_READING_OUT_OF_LIMIT
  Severity: Warning
  The Service Processor found environmental readings
  to be out of limits

HIGH_TEMP_DETECTED
  Severity: Warning
  The Service Processor detected high temperature

EXCESS_TEMP_DETECTED
  Severity: Warning
  The Service Processor detected excessive temperature
```

```
SP_DISK_FULL
  Severity: Warning
  The Service Processor disk is full

SMOKE_DETECT_MALFUNCTION
  Severity: Warning
  The Service Processor detected a malfunction in the
  smoke detector

SMOKE_DETECT_ALERT
  Severity: Warning
  The Service Processor detected a smoke detector
  alert

PHONEHOME_TRIGGER
  Severity: Informational
  A certain event triggered a Call Home for service

DB_CHECKSUM_TRIGGER
  Severity: Informational
  A Database Double Checksum detection event was
  triggered

RDF_CG_TRIGGER
  Severity: Informational
  An RDF CG trip event was triggered

SP_REBOOT_SUCCESS
  Severity: Informational
  The Service Processor has successfully rebooted

SAVEDEVS_FULL
  Severity: Error
  The save or data device pool is full

SRDFA_INACTIVE
  Severity: Warning
  SRDF/A is now inactive

SRDFA_ACTIVE
  Severity: Informational
  SRDF/A is now active

ACCESS_TO_NR_DEVICE
  Severity: Warning
  Access was attempted to a Not Ready device

SAVEDEVS_NEAR_FULL
  Severity: Warning
  The save or data device pool is almost full

SAVEDEV_NOT_RDY
  Severity: Error
  An active device in the Timefinder/Snap save device
  pool has gone Not Ready

DIRECTOR_DEAD
  Severity: Fatal
  A Symmetrix Director is not responding

TIMEOUT_R2_WP_LIMIT
  Severity: Error
  Timeout writing to an R2 device. Maximum writes
  pending is reached

RDFA_SESS_DROP_WPL_DSBL
```

```
  Severity: Error
  SRDF/A Session dropped, write pending limit reached.
  Host throttling disabled

RDFA_SESS_DROP_WPL_ENBL
  Severity: Error
  SRDF/A Session dropped, write pending limit reached.
  Host throttling enabled

RDFA_SESS_DROP_DEV_NR_OFF
  Severity: Error
  SRDF/A Session dropped, device not ready. Tolerance
  mode is off

RDFA_SESS_DROP_DEV_NR_CG
  Severity: Error
  SRDF/A Session dropped, device not ready through
  consistency group

RDFA_SESS_DROP_NO_RDF_LNK
  Severity: Error
  SRDF/A Session dropped, no RDF links operational

RDFA_SESS_DROP_TIMEOUT_MSC
  Severity: Error
  SRDF/A Session dropped, time out in MSC mode

RDFA_SESS_DROP_TIMEOUT_HA
  Severity: Error
  SRDF/A Session dropped, time out on an HA

RDFA_SESS_DROP_TIMEOUT_RA
  Severity: Error
  SRDF/A Session dropped, time out on an RA

GEN_CHECKSUM_TRIGGER
  Severity: Informational
  A Generic Double Checksum detection event was
  triggered

POWER_ZONE_COUNTDOWN_STARTED
  Severity: Error
  Power zone count down started. One of the power
  zones is down, count down (20 hours) for
  Vault-Shutdown is started

POWER_ZONE_5_HOURS_BEFORE_SHUTDOWN
  Severity: Error
  Power zone 5 hours before shutdown, one of the power
  zones is down. 5 hours before Vault shutdown

POWER_ZONE_ILLEGAL_STATUS
  Severity: Error
  Power zone down, illegal status, found file with
  old information

POWER_ZONE_DOWN_FAILED
  Severity: Error
  Power zone down failed

POWER_ZONE_WAIT_TIME_CHANGED_ON_GUI
  Severity: Error
  Power zone down - wait time changed on the GUI

RDFA_DROP_ISSUED_FROM_HOST
  Severity: Warning
  SRDF/A session drop requested [host software
```

```
  initiated]

RDFA_DEACTIVATE_ISSUED_FROM_HOST
  Severity: Warning
  SRDF/A session transition out of Asynchronous mode
  requested [host software initiated]

RDFA_PENDING_DROP_ISSUED_FROM_HOST
  Severity: Warning
  SRDF/A session drop at cycle boundary requested
  [host software initiated]

RDFA_CONSISTENT_DEACTIVATE_ISSUED_FROM_HOST
  Severity: Warning
  SRDF/A session transition from Asynchronous to
  Synchronous mode requested [host software initiated]

RDFA_DROP_ISSUED
  Severity: Warning
  SRDF/A session drop requested

RDFA_DEACTIVATE_ISSUED
  Severity: Warning
  SRDF/A session transition out of Asynchronous mode
  requested

RDFA_PENDING_DROP_ISSUED
  Severity: Warning
  SRDF/A session drop at cycle boundary requested

RDFA_CONSISTENT_DEACTIVATE_ISSUED
  Severity: Warning
  SRDF/A session transition from Asynchronous to
  Synchronous mode requested

DAE_PSA_M_FAN_FAULT
  Severity: Error
  Environmental Error: Power Supply A multiple Fan fault

DAE_PSA_S_FAN_FAULT
  Severity: Error
  Environmental Error: Power Supply A single Fan fault

DAE_PSA_FAULTED
  Severity: Error
  Environmental Error: Power Supply A faulted

DAE_PSA_SHUTDOWN
  Severity: Error
  Environmental Error: Power Supply A shutdown

DAE_PSB_M_FAN_FAULT
  Severity: Error
  Environmental Error: Power Supply B multiple fan fault

DAE_PSB_S_FAN_FAULT
  Severity: Error
  Environmental Error: Power Supply B single fan fault

DAE_PSB_FAULTED
  Severity: Error
  Environmental Error: Power Supply B faulted

DAE_PSB_SHUTDOWN
  Severity: Error
  Environmental Error: Power Supply B shutdown
```

```
DAE_LCC_A_TEMP_HIGH
  Severity: Error
  Environmental Error: Link Card Controller A
  temperature high

DAE_LCC_B_TEMP_HIGH
  Severity: Error
  Environmental Error: Link Card Controller B
  temperature high

DAE_SPS_INTRN_FAULT
  Severity: Error
  Environmental Error: Supplemental Power Supply
  internal fault

DAE_SPS_BAT_ENDLINE
  Severity: Error
  Environmental Error: Supplemental Power Supply battery
  end of line

DAE_SPS_LOW_VOLTAGE
  Severity: Error
  Environmental Error: Supplemental Power Supply low
  input AC Voltage

RDFA_SE_TRANS_IDLE
  Severity: Error
  SRDF/A Session entering transmit idle state

RDFA_SR_TRANS_IDLE
  Severity: Warning
  SRDF/A Session recovered from a transmit idle state

RDFA_TO_TRANS_IDLE
  Severity: Error
  SRDF/A Session dropped, transmit idle state timeout

RDFA_SD_NO_ONL_RA
  Severity: Error
  SRDF/A Session dropped, no online RAs

AUDIT_HIGH_ACTIVITY:
  Severity: Warning
  Entries are being written to the audit log at an
  unusually high rate

AUDIT_SFS_MIRR_OFF:
  Severity: Error
  Audit log has lost its redundancy due to an SFS
  mirror being offline

AUDIT_LOG_WRAPPED:
  Severity: Warning
  Audit log entries have been overwritten in an
  unusually short time period

RDFA_SE_DROP_WPL_CP:
  Severity: Error
  SRDF/A Session dropped, write pending limit reached
  on a cache partition

RDFA_NO_CYCLE_SWITCH:
  Severity: Warning
  There has been no SRDF/A cycle switching within the
  past hour

SRDF_JFC_STATE_CHANGE:
```

```
          Severity: Warning
          The job flow control setting has changed on RDF group
```

EXAMPLES

To list the events for a specific Symmetrix array, enter:

```
symevent -sid 54 list
```

To report on all events, on all locally connected Symmetrix arrays, every 10 seconds, forever, enter:

```
symevent monitor
```

To poll for and display events of severity Warning or greater on Symmetrix 012345678901 every 10 minutes for a 24-hour period, enter:

```
symevent monitor -sid 012345678901 -i 600 -c 144 -warn
```

To retrieve a verbose list of the events that have occurred on the given Symmetrix array between 9 a.m. and 5 p.m. today, enter:

```
symevent list -sid 012345678901 -v -start 9:00
  -end 17:00
```

NAME

Performs FAST operations on tiers, storage groups, policies, and the FAST controller.

SYNOPSIS

```
symfast -h

symfast -fp -sid <SymmID> [-i <Interval>] [-c <Count>]

    create -name <FastPolicyName> [-tier_name <TierName>
       [-max_sg_percent <MaxSgPercent>]]

    delete -fp_name <FastPolicyName> [-force]

symfast -fp -sid <SymmID> [-i <Interval>] [-c <Count>]
        -fp_name <FastPolicyName>

    add -tier_name <TierName>
        [-max_sg_percent <MaxSgPercent>]

    remove -tier_name <TierName>

    modify -tier_name <TierName>
           -max_sg_percent <MaxSgPercent>

    rename -name <NewFastPolicyName>

symfast -sid <SymmID> [-i <Interval>] [-c <Count>]
        -fp_name <FastPolicyName>

    associate -sg <SgName> [-priority <PriorityValue>]
        [-rdf_coordination <ENABLE | DISABLE>]

    disassociate -sg <SgName>

    modify -sg <SgName> [-priority <PriorityValue>
        [-rdf_coordination <ENABLE | DISABLE>]

symfast -sid <SymmID> [-i <Interval>] [-c <Count>]
        -sg <SgName>

    reassociate -fp_name <FastPolicyName>

symfast -sid <SymmID> [-i <Interval>] [-c <Count>]
        [-noprompt]

    enable [-dp | -vp]

    disable [-dp | -vp]

    set -control_parms
      [-approval_mode <AUTO_APPROVE | USER_APPROVE>]
      [-vp_data_move_mode <AUTO | NONE>]
      [-min_perf_period <PerfTime>]
      [-workload_period <WorkTime>]
      [-max_simult_devs <MaxSimultDevs>]
      [-max_devs <MaxDevs>]
      [-vp_reloc_rate <VPRate>]
      [-swap_notvisible_devs <ENABLE | DISABLE>]
      [-allow_only_swap <ENABLE | DISABLE>]
      [-pool_resv_cap <ResvPct>]
      [-vp_allocation_by_fp <ENABLE | DISABLE>]
      [-time_to_compress <NumDays | never>]
      [-fast_compression_rate <FastCompRate>]
```

```
    symfast -plan -sid <SymmID> [-i <Interval>] [-c <Count>]
          [-noprompt]

        approve -id <PlanID> [-begin_at <TimeVal>]

        decline -id <PlanID>

    symfast [-sid <SymmID>] [-offline]

        list -fp [-v] [-dp | -vp [-ckd | -fba]]

        list -association [-demand [-sg <SgName> |
             -fp_name <PolicyName>] [-mb]]


    symfast -sid <SymmID> [-offline]

        list -demand [-v]
           <[-technology <EFD | FC | SATA | ALL>]
            [-external | -internal]>
           [-dp | -vp] [-allocated]

        list -control_parms

    symfast -sid <SymmID>

        list -history [-v]
           [-start_date <TimeVal>] [-end_date <TimeVal>]

    symfast -sid <SymmID> [-i <Interval>] [-c <Count>]

        list -plan [-v]

        list -state [-dp | -vp]

    symfast [-sid <SymmID>] [-offline]

        show -association -sg <SgName> [-all] [-v]

        show -fp_name <FastPolicyName>

DESCRIPTION

    The symfast command provides the ability to perform the
    following actions:

     - Create a FAST policy.

     - Delete a FAST policy.

     - Add a tier to a FAST policy.

     - Remove tiers from a FAST policy.

     - Modify a FAST policy.

     - Rename a FAST policy.

     - Associate a FAST policy to a storage group.

     - Dissassociate a FAST policy from a storage group.

     - Modify a FAST policy and also its association.

     - Set FAST control parameters.

     - Enable the FAST controller.
```

- Disable the FAST controller.

- Query the FAST controller.

- Display FAST policies, associations, plans, history and reports. Plans and history are not displayed for FAST VP.

ARGUMENTS

add	Adds a tier to the specified FAST policy.
approve	Approves the FAST plan based on plan ID (not applicable to FAST VP).
associate	Associates a storage group with a FAST policy.
create	Creates a FAST policy.
decline	Declines the FAST plan based on plan ID (not applicable to FAST VP).
delete	Deletes the FAST policy.
disable	Disables FAST controller.
disassociate	Disassociates a storage group from a FAST policy.
enable	Enables the FAST controller.
list	Lists the FAST policies, associations, plans, history, and reports. Plans and history are not listed for FAST VP.
modify	Modifies the properties of a FAST policy or of an association between a policy and a storage group.
reassociate	Re-associates the storage group to a different FAST policy.
remove	Removes a tier from the FAST policy.
rename	Renames the FAST policy.
set	Sets FAST control parameters.
show	Shows FAST policies or associations between policies and storage groups.

OPTIONS

-all	Displays all of the devices in a storage group. Includes even those devices which are not managed by FAST.
-allocated	Displays Max SG Demand in the FAST VP demand report based on the allocated capacity of thin devices under FAST control. By default, Max SG Demand displays based on the configured capacity.
-allow_only_swap	Indicates that the FAST controller can only perform a full swap of devices and

cannot move devices to unconfigured space. Valid values are ENABLE and DISABLE.

-approval_mode — Specifies the mode of the FAST controller either in automatic(AUTO_APPROVE) or user approval mode (USER_APPROVE).

-association — Displays FAST associations that exist between storage groups and FAST policies.

-begin_at — Schedules the plan to run at a specific time.

-c — Indicates the number (count) of times to attempt the action. If this option is not specified, and an interval (-i) is specified, the process will attempt -c number of times waiting for the database lock.

-ckd — Indicates devices with CKD emulation.

-control_parms — Specifies the control parameters for the FAST controller.

-demand — Displays demand reports for FAST.

-dp — Specifies that the operation be performed on FAST policies containing disk group-provisioned tiers.

-end_date — Identifies the end date and time for reading the history entries.

-external — Indicates the devices that are provisioned on external disk groups.

-fast_compression_rate — Indicates how aggressively FAST VP should compress the data. Valid range is between 1 and 10. Default value is 5.

-fba — Indicates devices with FBA emulation.

-force — Allows a non-empty FAST policy to be deleted.

-fp — Indicates that the action is related to a FAST policy.

-fp_name — Specifies a FAST policy name.

-h — Provides brief, online help information.

-history — Displays all data movement history including both FAST and Optimizer command history.

-i — Specifies the repeat interval for retrying the requested action. This option indicates how often to attempt to get the needed resources to start a new session. The default interval is 30 seconds. The minimum interval is 5 seconds. For passive actions the minimum interval is 15 seconds. Passive actions are actions that do not acquire an exclusive lock.

-id Specifies the plan ID number.

-internal Indicates the devices that are provisioned on internal disk groups.

-max_devs Specifies the maximum number of devices that can be moved in a 24-hour period.

-max_sg_percent Specifies an upper limit of space allowed for a tier in a policy as a percentage of total storage group capacity.

-max_simult_devs Specifies the maximum number of devices that can be moved simultaneously.

-mb Lists capacity in megabytes.

-min_perf_period Specifies the minimum amount of workload sampling that the FAST controller should complete before analyzing the samples for the first time.

-name Specifies the name of the FAST policy being created or the new name if a policy is being renamed.

-noprompt Requests no confirmation prompt before performing an action.

-offline Displays information about FAST policies and associations from the Symmetrix configuration database without refreshing the data from the Symmetrix array.

-plan Specifies that the action is related to FAST plans.

-pool_resv_cap Specifies the capacity from each DATA device pool that will be reserved for non-FAST activities. Possible values are 1 to 80.

-priority Specifies the priority of the association between storage group and a policy. The priority values can be 1 (highest), 2, or 3 (lowest).

-rdf_coordination
Specifies that RDF coordination is required on RDF devices in the associated storage group.

-sg Specifies a storage group name.

-sid Specifies the Symmetrix ID.

-start_date Identifies the start date and time for reading the history entries.

-state Displays the state of the FAST controller.

-swap_notvisible_devs
Enables or disables the FAST controller’s functionality to use devices that are not visible to the host to do a full swap with devices in storage groups.

Valid values are ENABLE and DISABLE.

-technology Specifies the drive type. The currently supported types are EFD, FC, or SATA. If the user specifies ALL, it includes EFD, FC, and SATA.

-time_to_compress
Indicates how many days FAST VP will wait before start compressing data in the tier. Valid values are between 40 days to 400 days or never. Default value is never.

-vp Specifies that the operation be performed on FAST policies containing virtual provisioned tiers.

-vp_allocation_by_fp
Indicates whether VP allocation comes from a bound pool or from any pool within the policy.

-vp_data_move_mode
Specifies the Virtual Provisioning data movement mode of the FAST controller either in AUTO or NONE mode. In AUTO mode, the FAST controller performs data movement for thin devices without user intervention within the data movement window. In NONE mode, the FAST controller will not perform any data movement for thin devices.

-vp_reloc_rate
Specifies the aggressiveness of the data movements for thin devices. The lower the value, the more aggressive FAST will be. Possible values are 1 to 10.

-tier_name Specifies the storage tier name.

-v Provides a more detailed, verbose listing.

-workload_period Specifies the amount of workload sampling that the FAST controller should maintain for the sample analysis. It is specified in units of time (hours).

PARAMETERS

ALL All the drive types (EFD,FC,SATA).

AUTO_APPROVE The automatic mode of the FAST controller.

Count The number of iterations to execute before exiting.

DISABLE Disables a FAST controller setting.

EFD The enterprise Flash disk drives.

ENABLE Enables FAST controller setting.

FastPolicyName The FAST policy name.

FC The Fibre Channel disk drives.

Interval	The interval between polls, in seconds.
MaxDevs	The number of devices that can be moved or swapped in a 24-hour period. Possible values are 2 to 200 devices.
MaxSgPercent	The percentage of total logical device capacity in the storage group. Valid values are 1 to 100.
MaxSimultDevs	The number of devices that can be moved or swapped simultaneously. Possible values are 2 to 32 devices.
NewFastPolicyName	The new FAST policy name.
PerfTime	The minimum length of the time (in hours) to be used for performance analysis if the user does not want to wait until the entire workload period has elapsed. Allowed values are: Minimum: 2 hours Maximum: Current value of the workload period parameters.
PlanID	The plan ID number. The format is: mmddyyyy:hhmmss.
PriorityValue	The priority of the storage group associated with the policy.
SATA	The SATA disk drives.
SgName	The storage group name.
SymmID	The 12-digit ID of the Symmetrix array.
TierName	The storage tier name.
TimeVal	A specific date and time (MMDDYYYY:HHMMSS format).
USER_APPROVE	The user approval mode of the FAST controller.
WorkTime	A time in hours that the FAST controller should maintain for sample analysis. Valid values are 2 to 672 hours.

RETURN CODES

Code #	Code Symbol
0	CLI_C_SUCCESS
1	CLI_C_FAIL
2	CLI_C_DB_FILE_IS_LOCKED
19	CLI_C_GK_IS_LOCKED All Gatekeepers to the Symmetrix array are currently locked.

EXAMPLES

```
To create a FAST policy, enter:

   symfast -sid 207 -fp create -name DBPolicy

To create a FAST policy and add a tier to it, enter:

   symfast -sid 207 -fp create -name DBPolicy
      -tier_name PrimeDBTier -max_sg_percent 30

To delete a FAST policy, enter:

   symfast -sid 207 -fp delete -fp_name DBPolicy

To add a tier to a policy, enter:

   symfast -sid 207 -fp add -tier_name ArchiveDBTier
      -max_sg_percent 10 -fp_name DBPolicy

To remove a tier from a policy, enter:

   symfast -sid 207 -fp remove -tier_name ArchiveTier
      -fp_name FinanceData

To modify policy tier capacity, enter:

   symfast -sid 207 -fp modify -fp_name DBPolicy
      -tier_name PrimeDBTier -max_sg_percent 70

To rename a policy, enter:

   symfast -sid 207 -fp rename -fp_name DBPolicy
      -name OraDBPolicy

To list policies containing DP tiers, enter:

   symfast -sid 207 list -fp -dp

To list policies containing VP tiers, enter:

   symfast -sid 207 list -fp -vp

To list all policies in a Symmetrix array, enter:

   symfast -sid 207 list -fp

To show a policy in a Symmetrix array, enter:

   symfast -sid 207 show -fp_name DBPolicy

To associate a storage group to a policy, enter:

   symfast -sid 207 associate -sg OraSales
      -fp_name DBPolicy -priority 1

To list associations, enter:

   symfast -sid 207 list -association

To show a association, enter:

   symfast -sid 207 show -association -sg
      Finance2009

To enable the FAST controller, enter:

   symfast -sid 207 enable
```

To set the control parameters of the FAST controller, enter:

```
symfast -sid 207 set -control_parms
   -approval_mode AUTO_APPROVE -min_perf_period 100
   -workload_period 200
```

To approve a plan, enter:

```
symfast -plan -sid 207 approve -id 04142009:130114
   -begin_at 07132009:9:45
```

To decline an approved or a running plan, enter:

```
symfast -plan -sid 207 decline -id 04142009:130114
```

To query the FAST controller state, enter:

```
symfast -sid 207 list -state
```

To list the FAST controller settings, enter:

```
symfast -sid 207 list -control_parms
```

To list the FAST controller plan, enter:

```
symfast -sid 207 list -plan
```

To display FAST data movement history, enter:

```
symfast -sid 207 list -history
```

To list the compliance report, enter:

```
symfast -sid 207 list -association -demand
```

To list the FAST tech demand report, enter:

```
symfast -sid 207 list -technology ALL -demand -dp
```

To list the FAST VP tech demand report, enter:

```
symfast -sid 207 list -tech ALL -demand -vp
```

symhost

Displays host configuration information
and performance statistics.

SYNOPSIS

```
symhost show -config [-h]

symhost stats [-h] [-i <Interval>] [-c <Count>]
   [-type CPU | MEMORY | DISK | ALL]
```

DESCRIPTION

The symhost command displays host configuration information and performance statistics. The performance statistics are displayed for CPU, memory, and host devices. Note that not all statistics are available for all hosts.

ARGUMENTS

show	Shows detailed configuration information.
stats	Shows performance statistics.

OPTIONS

-c	Indicates the number (count) of times to display statistics. If this option is not specified, and an interval (-i) is specified, stats will be displayed continuously.
-config	Shows detailed configuration information.
-h	Provides brief online help information.
-i	Repeats the interval in seconds. The default interval is 30 seconds. The minimum interval is 5 seconds. For passive actions the minimum interval is 15 seconds. Passive actions are actions that do not acquire an exclusive lock.
-type	Specifies the type of performance information to display. The default is to display ALL statistics. Individual CPU, memory, and disk statistics can be selected by specifying one of the following:

-type	Description
CPU	Reports user, system, wait I/O, idle CPU time, interrupts, system calls, and context switch statistics for each processor and overall.
MEMORY	Reports system-wide page in-page out and swap in-swap out statistics.
DISK	Reports read, write, busy, and idle time statistics for each host disk.

PARAMETERS

None.

RETURN CODES

Code #	Code Symbol
0	CLI_C_SUCCESS
1	CLI_C_FAIL

EXAMPLES

To display configuration information for the local host, enter:

```
symhost show -config
```

To display statistics about all processors, memory, and disk every 60 seconds, enter:

```
symhost stats -i 60
```

To display statistics about all host processors, every 30 seconds for one hour, enter:

```
symhost stats -i 30 -c 120 -type CPU
```

```
H:M:S CPU  %User %Sys  %WIO  %Idle Int/s Calls/s  CtxSw/s
  A    B     C     D     E      F    G     H        I
```

A Time of day
B CPU number/id
C 100 * (CPU busy time in user mode / elapsed time)
D 100 * (CPU busy time in system mode / elapsed time)
E 100 * (CPU idle time for wait I/O / elapsed time)
F 100 * (CPU idle time / elapsed time)
G Interrupts per second
H System calls per second
I Process context switches per second

To display statistics about host memory, every 30 seconds for one hour, enter:

```
symhost stats -i 30 -c 120 -type MEMORY
```

```
H:M:S      Pi/s Ppi/s Po/s Ppo/s Si/s Psi/s So/s Pso/s
  A         B     C    D     E    F    G     H     I
```

A Time of day
B Page in requests per second
C Number of pages paged in per second
D Page out requests per second
E Number of pages paged out per second
F Swap in requests per second
G Number of pages swapped in per second
H Swap out requests per second
I Number of pages swapped out per second

To display statistics about all host disks every 30 seconds for one hour, enter:

```
symhost stats -i 30 -c 120 -type DISK
```

```
H:M:S  Disk  RW/s  R/s  W/s  KbRW/s KbR/s KbW/s %Busy %Wait
```

```
A     B     C     D    E     F     G     H      I     J
```

```
    A  Time of day.
    B  Disk name
    C  Read and write requests per second
    D  Read requests per second
    E  Write requests per second
    F  KB read and written per second
    G  KB read per second
    H  KB written per second
    I  100 * (disk active time / elapsed time)
    J  100 * (non-empty wait queue time / elapsed time)
```

symhostfs

Displays information about file systems, directories, and regular files that are defined on the host system.

SYNOPSIS

```
symhostfs [-h] [-meta | -data] [-kb | -blocks | -mb]

   list [ObjName] [-v] [-R]

   list [-file <ObjName> | -dir <ObjName> | -nfs] [-v] [-R]

   show <ObjName>
        [-no_extents | -expand | -collapse |
         -phys_collapse]
```

DESCRIPTION

The symhostfs command displays mapping information specific to the file systems that are defined on the host system.

A list of file systems, files, or directories can be obtained with symhostfs. The default is to list the mounted local file systems. The list of files or subdirectories of a given parent directory can be obtained by specifying the -dir or -file option.

Detailed information can be retrieved for file systems, files, or directories. The attributes will be shown for each object type. For file systems that are mounted on Symmetrix devices, you can obtain logical to physical information of where the file extents are mapped on these Symmetrix devices.

Note that with network based file systems, the file system device is represented with the remote host and remote path. For Unix and Linux, the format is "remote host name (or IP address) : directory exported from the remote host". For Windows, it is represented as a UNC path. The remote host field is either host name or IP address depending upon what is used while mounting NFS.
Network based file systems are represented as NFS independent of the protocol (NFS or CIFS) the file system is based on. NFS based on NFS protocol is supported on Linux, Solaris, HP_UX and AIX. NFS based on CIFS protocol is supported on Linux and Windows.

ARGUMENTS

list	Lists file systems, files or directories on the current host system. Note that the default is to show only local file systems. Network based file systems are not shown by default.
show	Shows detailed mapping information about a file system, file or directory on the current host system.

OPTIONS

-blocks	Displays size information in 512-byte blocks.

```
     -collapse      Collapses the extents of a file or file
                    system, if possible. This is the default.

     -data          Displays a file's data extents only.

     -dir           Displays directory information.

     -expand        Expands the extents of a file or file
                    system, if possible.

     -file          Displays regular file information. This
                    is the default for list when an object
                    name is specified.

     -h             Provides brief, online help information.

     -kb            Displays size information in Kilobytes.

     -mb            Displays size information in Megabytes.
                    This is the default.

     -meta          Displays a file's meta data extents only.
                    Default is to show both meta data and
                    data extents.

     -nfs           Displays information on mounted network
                    based file systems.

     -no_extents    Excludes extent information from the
                    display.

     -phys_collapse
                    Physical collapse of the extents of a
                    file or file system.

     -R             Displays directory or file information
                    recursively.

     -v             Provides a more detailed, verbose listing.

PARAMETERS

     ObjName        A directory or file name.

RETURN CODES

     Code #     Code Symbol
     ------     -----------

       0        CLI_C_SUCCESS

       1        CLI_C_FAIL

EXAMPLES

     To list all the mounted file systems, enter:

        symhostfs list

     To recursively list all the subdirectories of
     directory "/usr", enter:

        symhostfs list "/usr" -dir -R

     To display detailed information about directory
     /usr/guest, enter:
```

```
   symhostfs show "/usr/guest"
```

To display detailed information about the file system /mountp1, with the extent information collapsed, and all sizes reported in megabytes, enter:

```
   symhostfs -collapse -mb show "/mountp1"
```

syminq

Issues a SCSI INQUIRY command, and optionally a SCSI READ CAPACITY, on one or all devices. In addition, it can be used to obtain a list of the local host's HBAs.

SYNOPSIS

```
syminq [-h]

syminq [-sym] [-bcv] [-powerpath] [PdevName] [-mobility]
   [-symmids] [-la | -nocapacity]
    [-pdevfile] [-cache | -nocache]
    [-v [-nocapacity]]
    [-copa] [-wwn [-colons]]
    [-winvol] [-native] [-host_cache]

syminq [-clariion] [-powerpath] [PdevName]
    [-cids] [-la | -nocapacity]
    [-v [-nocapacity]]
    [-wwn [-colons]]

syminq [-mapinfo] [PdevName] [-mobility]
    [-sym[-powerpath]|-clariion[-powerpath]]
    [-cache | -nocache] [-colons] [-winvol]

syminq hba [-fibre | -scsi | -iscsi | -snia]

syminq -identifier <device_name|nice_name|hp_id|vms_id>
    [PdevName] [-sym [-bcv]| -clariion] [-mobility]
```

DESCRIPTION

The syminq command can issue a SCSI INQUIRY, and optionally a SCSI READ CAPACITY, on one or all devices. By default, the scope of the command is for all disk devices. You can limit the scope to Symmetrix or CLARiiON devices.

The syminq command also lists the HBAs in the local host, for fibre, SCSI, or both.

The option -cache attempts to recover the results of a previous SCSI interrogation from a running base daemon for increased speed.

ARGUMENTS

None.

OPTIONS

-bcv	Displays Symmetrix BCV devices only.
-cache	Attempts to recover the results of a previous SCSI interrogation from a running base daemon for increased speed.
-cids	Displays CLARiiON IDs.
-clariion	Displays CLARiiON devices only.
-colons	Indicates to use a colon separator between bytes of WWN data.

-copa	Lists physical device names only in a format to input into EMC's COPA tool.
-fibre	Modifies the request for listing HBAs to include Fibre HBAs only.
-h	Provides brief, online help information.
-host_cache	Displays devices registered for control by host cache cards only.
-identifier	Lists the Symmetrix device identifiers assigned to devices by the user or other applications. The user must choose one of the four identifier types currently supported to be displayed. If nice_name is specified, nice names for CLARiiON devices also display.
-iscsi	Modifies the request for listing HBAs to include iSCSI HBAs only.
-la	Lists physical device names, Symmetrix IDs, or CLARiiON IDs, only in a left-aligned format.
-mapinfo	Displays target mapping information for devices mapped through Fibre HBAs.
-native	Displays the native inquiry data if the inquiry data is altered.
-nocache	Bypasses the cache and rescans the devices.
-nocapacity	Skips issuing a SCSI READ CAPACITY to the device(s).
-pdevfile	Lists physical device names in a format for use as pdevfile entries.
-powerpath	Displays EMC PowerPath devices only.
-scsi	Modifies the request for listing HBAs to include SCSI HBAs only.
-snia	Indicates the use of only the SNIA API to gather HBA data. This implies -fibre.
-sym	Displays Symmetrix devices only.
-symmids	Displays Symmetrix IDs.
-v	Provides a more detailed, verbose listing. Cannot be used with -copa or -la options.
-winvol	Displays Windows volumes only.
-wwn	Displays the device WWN.
[-mobility]	Displays only Mobility Devices

PARAMETERS

hba	The host bus adapter.
PdevName	The host name for the device, such as

/dev/rdsk/c2t0d2s3.

EXAMPLES

To issue a SCSI INQUIRY to all Symmetrix devices that are visible to this host, enter:

```
syminq -sym -nocap
```

To issue a SCSI INQUIRY and READ CAPACITY to a device, enter:

```
syminq /dev/rdsk/c2t0d2s3
```

To issue a SCSI INQUIRY and READ CAPACITY to a device and display with more detailed, verbose information, enter:

```
syminq -v /dev/rdsk/c2t0d2s3
```

To list the SCSI HBAs in the local host, enter:

```
syminq hba -scsi
```

symioctl

Sends I/O control commands to a specified application.

SYNOPSIS

```
symioctl -type <DbType> [-h] [-noprompt]

   begin backup [<object> [<object>...]] [-checkpoint]

   freeze [<object> [<object>...]] [-checkpoint]

   checkpoint [<object> [<object>...]]

   end backup [<object> [<object>...]]

   thaw [<object> [<object>...]]

   archive log

   begin snapshot <object> SAVEFILE <SaveFile>
     [-checkpoint] [-overwrite]

   restore snapshot <object> SAVEFILE <SaveFile>
     [-norecovery | -standby]

   end snapshot <object>

   abort snapshot <object>
```

DESCRIPTION

The symioctl command allows control actions to be sent to a specified application. This utility is intended to be used in conjunction with a split operation. The symioctl freeze command suspends updates from writing to disk. Once the freeze action completes, you can perform a TimeFinder or SRDF split. After the split is complete, use the command symioctl thaw to resume normal application activity.

Additionally, for Oracle, Hot Backup control of all tablespaces to be backed up must be performed before and after a freeze/thaw command. The steps to split a group of BCV devices are:

1. symioctl begin backup
2. symioctl freeze
3. Split standard and BCV pairs. This may involve several steps depending on your environment.
4. symioctl thaw
5. symioctl end backup

For SQLServer 2000 or higher, the snapshot commands support the SQLServer BACKUP and RESTORE database with snapshot operations using the Virtual Device Interface (VDI). The database can be restored in recovery, norecovery, or standby mode.

The database user login information must be supplied using the SYMCLI_RDB_CONNECT environment variable. The user login information is specified in the following format: username/password@service. The username and password must be non-NULL. If the NT trusted authentication is used, the user login information is specified in the following

format: @sqlservername.

The object list is not always required, as shown in the following table:

Action	RDBMS	Objects	List
Freeze/ Thaw	Oracle	database server	No
	SQL Server	database name	Yes
	Informix	database server	No
	IBM DB2/UDB	database name	Yes
	Sybase	database name	No
Checkpoint	Oracle	database server	No
	SQL Server	database name	Yes
	Informix	database server	No
Begin/End Hot Backup	Oracle	tablespace	Yes
Archive Log	Oracle	database server	No
Begin/End Abort/Restore Snapshot	SQLServer	database name	No

Important Note: The user of the symioctl command must have database administrator privileges.

To execute the symioctl utility, you must have the appropriate application software installed and the environment variables set.

You can specify the database type information from an environment variable. The command line option takes priority over the follwoing environment variable:

Environment Variable	Can be Used Instead of	To Specify
SYMCLI_RDB_TYPE	-type	Application type to perform requested action.

If an argument specified on the command line contains special shell characters (e.g., $, ’, \\, etc.), those characters must be escaped with a backslash (\\).

If symioctl is being run in client/server mode, and the required RDBMS environment variables are set in the client’s environment, they are sent to the server to be used.

For IBM UDB/DB2, if the user tries to freeze all user databases’ I/O while one or more databases’ I/O have already been frozen, the freeze operation will fail.

ARGUMENTS

abort snapshot
: For SQLServer 2000 and higher. The BACKUP DATABASE SQL command for the specified database will terminate and database writes will resume.

archive log
: Archives the current log. This is an Oracle-specific command.

begin backup — Places the specified tablespace objects into Hot Backup mode. This is an Oracle-specific command.

begin snapshot — For SQLServer 2000 and higher. A BACKUP DATABASE TO VIRTUAL_DEVICE WITH SNAPSHOT SQL command is sent to SQLServer which will begin the snapshot backup and suspend writes for the specified database. After the BCV mirrors are split, the end snapshot command should be issued to save the snapshot meta-data to a file.

checkpoint — Issues a checkpoint to the RDBMS.

end backup — Ends the Hot Backup for the specified tablespace objects. This is an Oracle-specific command.

end snapshot — For SQLServer 2000 and higher. The "BACKUP DATABASE" SQL command for the specified database will complete, database writes will resume and the snapshot meta-data will be saved to the save-file (which is needed for a subsequent snapshot restore).

freeze — Suspends I/O at the application layer. Each application has a slightly different behavior, but they all provide a mechanism to halt modifications while a split operation occurs.

restore snapshot — For SQLServer 2000 and higher. A "RESTORE DATABASE FROM VIRTUAL_DEVICE WITH SNAPSHOT" SQL command for the specified database is sent to SQLServer. The previously saved snapshot meta-data is used by SQLserver to logically restore the database. The -norecovery and -standby options allow the RESTORE to operate in NORECOVERY or STANDBY mode. The undo file for -standby option is automatically generated. It is in the same location as the savefile with the file name undo_(database name).ldf.

thaw — Resumes I/O at the application layer.

OPTIONS

-checkpoint — Requests a checkpoint prior to the specified action.

-h — Provides brief, online help information.

-noprompt — Turns off the prompt for user confirmation.

-norecovery — Restores an SQLServer database with the NORECOVERY option.

-overwrite — Allows the backup process to overwrite an existing save file. By default,

existing save files are protected.

-standby Restores the SQLServer database with the STANDBY option.

-type Identifies the database type on which to perform the requested action. Types include: Informix, Oracle, SQLServer, IBMUDB, and Sybase.

PARAMETERS

object Database or tablespace name(s). If no objects are specified, the action defaults to all objects of the specified type database.

SaveFile For SQLServer snapshot only. Name of the save file used by begin snapshot and restore snapshot. For client/server mode, the meta-data file is saved by the server. It is recommended that an absolute path (e.g. C:\\TEMP\\PUBS.SAV) be specified (especially for client/server) to ensure that the file can be located for protection or subsequent restores.

RETURN CODES

Code #	Code Symbol
0	CLI_C_SUCCESS
1	CLI_C_FAIL

EXAMPLES

The following example will backup and then restore the SQLServer pubs database on a group of BCV devices. The snapshot meta data will be saved in C:\\TEMP\\PUBS.SAV. The database login parameters will be set via the environment to be user = sa, passwd = pass and service = sqlserv. Do the following:

```
setenv SYMCLI_RDB_CONNECT "sa/pass@sqlserv"
setenv SYMCLI_RDB_TYPE SQLServer
symioctl begin snapshot pubs SAVEFILE C:\\TEMP\\PUBS.SAV
Split standard and BCV pairs.
symioctl end snapshot pubs
Protect the PUBS.SAV save file.
- - - - -
Restore the PUBS.SAV save file.
Shut down SQLServer.
Restore the standard devices from BCVs.
Restart SQLServer.
symioctl restore snapshot pubs SAVEFILE
  C:\\TEMP\\PUBS.SAV
```

To freeze all the I/O in the Oracle instance represented by the connection information without prompting for confirmation, enter:

```
setenv SYMCLI_RDB_CONNECT "scott/tiger"
  symioctl freeze -type Oracle -noprompt -checkpoint
```

The database will perform a checkpoint prior to executing

the freeze command. The database login parameters will be set via the environment to be user = scott and passwd = tiger.

To thaw all the I/O in the Oracle instance represented by connection information, enter:

```
setenv SYMCLI_RDB_CONNECT "scott/tiger@acme"
  symioctl thaw -type Oracle
```

The database login parameters will be set via the environment to be user = scott, passwd = tiger and service = acme.

To place all tablespaces in the Oracle instance represented by connection information into Hot Backup mode, enter:

```
setenv SYMCLI_RDB_TYPE oracle
  symioctl backup begin -noprompt
```

The database type parameter will be set via the environment.

To archive the current log of the Oracle instance represented by connection information, enter:

```
setenv SYMCLI_RDB_TYPE oracle
  symioctl archive log -noprompt
```

The database type parameter will be set via the environment.

To freeze I/O for the Informix database without prompting for confirmation, enter:

```
setenv SYMCLI_RDB_CONNECT "infadm/pass@infserv"
  symioctl freeze -type Informix -noprompt
```

The database login parameters will be set via the environment to be user = infadm, passwd = pass, and service = infserv.

To perform a checkpoint for the Informix database, enter:

```
setenv SYMCLI_RDB_TYPE informix symioctl checkpoint
```

The database type parameter will be set via the environment.

symipsec

Displays or sets parameters that control the behavior of IPSec encryption on Gigabit Ethernet connections.

SYNOPSIS

```
symipsec -h

symipsec -sid <SymmID> -dir <#>|ALL [-port <#>|ALL]

   list -priority <<Level#> | -all>

   list -stats -type <StatsType> [-local_addr <IPendPt>]
      [-priority <Level#>]

   list -spi

   show -priority <Level#> | -all

symipsec -sid <SymmID> -dir <#> [-port <#>]
   set spi <on <SpiStart> [ length <SpiLen> ] | off>

symipsec -sid <SymmID> -file <FileName> -dir <#> [-port <#>]

   preview

   commit
```

DESCRIPTION

The symipsec command allows you to display and set the values of control parameters for Symmetrix IPSec encryption and authentication support.

Commands for listing or retrieving policies, or retrieving statistics may be executed directly from the command line. The results will be returned to the screen for viewing.

Commands for setting, modifying, or clearing policies may be placed in a command file, which will then be processed by this utility. Alternatively, stdin redirection can be used with "here documents" in UNIX shell scripts. Each command in the file has to be terminated by a semi-colon (;). There is no limit on the number of commands or the type of commands that can be placed in a command file.

Prior to making any changes, the preview argument can be used to verify that the command file is syntactically correct without applying the changes to the Symmetrix array.

When using the commit argument, commands are executed sequentially, and do not execute within the context of a session. Therefore, if there are three or more commands in the file and the second one fails, processing will abort, and the effects of the first command will remain.

The commands in the command file are not case sensitive however, the parameters entered are case sensitive.

ARGUMENTS

list	Displays the priority number(s) for one or more policies. Optionally, can retrieve and display statistical information about IPSec processors or list the SPI range reserved for manual IPSec policies.
show	Shows detailed information about one or more policies.
preview	Verifies the syntax of the changes specified in the command file.
commit	Updates the Symmetrix with the changes defined in the command file.
set	Reserves a range of SPI values for manual IPSec policies.

OPTIONS

-all	Targets all policy level numbers.
-dir	Targets a specific director.
-file	Specifies the command file that holds the policy definitions.
-port	Targets a specific Port. Currently, only port 0 is a valid value and the default.
-priority	Targets a specific policy priority level number.
-local_addr	Specifies a local endpoint IP address.
-stats	Retrieves and displays statistical information about IPSec processors within the Symmetrix array.
-sid	Targets a specific Symmetrix array ID.
-spi	Lists reserved SPI (Security Parameter Indexing) range for the specified director.
-length	Specifies the number of SPI values in the range to be reserved for manual IPSec policies.
-type	Selects the type of statistics to retrieve.
-h	Provides brief, online help information.

PARAMETERS

#	Specific director or port number. Optionally ALL may be supplied, during a policy list or show to select all applicable directors on the Symmetrix array.
FileName	The target command file name that holds the policies.
IPendPt	The local endpoint or IP address to obtain

IKE errors from.

Level# — The selected IPSec policy priority number (0-110). The value "ALL" may be supplied, during a policy list or show to select all policy level numbers.

SpiLen — The number of values in the range to be reserved for manual IPSec policies. Default value is 1.

SpiStart — The starting SPI value to be reserved for manual IPSec policies. Must be 0 to remove reservations or > 255 to set reservations.

StatsType — The statistical type of report to return. Possible values are:

- ike_errors
- ipsec_details

When retrieving IKE errors, the local address parameter must be supplied. Likewise, the priority level number must be provided when retrieving IPSec details.

SymmID — The ID of the Symmetrix array (up to 12-digits).

COMMAND FILE SYNTAX

The following shows how to define and modify policies using command file entries.

When executing a command that changes the array configuration, the preview operation will syntax-check the command file for errors, and the commit operation will send the policy changes to the array.

Note that adding and modifying a policy are almost identical. The former requires that the policy not exist, and the latter requires that the policy already exists and will be overwritten.

Note: Currently, you can only define one proposal and one transform per policy declaration.

Remove an existing policy:

```
policy delete -priority Level#;
```

Add or modify a policy:

```
policy add|modify
  -priority Level#
  -action discard|secure|bypass
  [-assoc_ike_policy Level#]
    #(only if proposal_type is IPSEC)
  -local_addr IPaddr
      [-ipproto IPprotocol#|all][-ipport IPport#|all]
      [-mask IPaddr]
  -remote_addr IPaddr
      [-ipproto IPprotocol#|all][-ipport IPport#|all]
      [-mask IPaddr]
  [-remote_tunnel_addr IPaddr]
    #(only if esp_mode is 'tunnel')
```

```
        [-selectivity {destip|destport|srcport} SPECIFICITY]
          #(for policies with wildcarded ipaddr/port/proto only)
        -proposal_set
        -proposal_set_type auto|manual|ike
        [-key_format hex|ascii]
        [-presharedkey Keystring]
          #(if proposal_set_type is IKE only)
        [-inenc_key Keystring -outenc_key Keystring]
          #(if transform_type is ESP or IKE, and
          #propset_type is MANUAL)
        [-inauth_key Keystring -outauth_key Keystring]
          #(only if proposal_set_type is MANUAL
          # and auth_alg isn't NULL)
        [-in_spi SPI# -out_spi SPI#]
          #(only if proposal_set_type is MANUAL)
        [-in_nonce NONCE -out_nonce NONCE]
          #(only if proposal_set_type is MANUAL
          #and -enc_alg is one of the aes-cm modes)
        [-ike_mode main|aggressive]
          #(only if proposal_set_type is IKE)
        [-pfs on|off]
          #(only if proposal_set_type is IKE)
        -proposal
        -proposal_type ike|ipsec
        -transform
        -transform_type ike|esp
        [-auth_alg null|sha1|md5|xcbc]
        [-enc_alg null|des|3des|aes_128|aes_256|
          aes_cm_128|aes_cm_256]
          #(only if transform_type is ESP or IKE)
        [-dhgroup 1|2|3|4]
          #(only if transform_type is IKE)
        [-esp_mode tunnel|transport]
          #(only if transform_type is ESP)
        [-lifetime [LifeParam1][,][LifeParam2]]
        [-auth_method preshared_key|dsa|rsa]
          #(only if transform_type is IKE)
```

COMMAND FILE OPTIONS

-priority	Specifies the index number of the policy to be retrieved or modified. When packets arrive, policies with lower numbered priorities are examined first. Also, any IKE policies must have a lower priority index number (higher priority) than the corresponding IPSec policy.
-action	Specifies the kind of action to run: discard, secure, or bypass.
-assoc_ike_policy	For IPSec policies, specifies the IKE policy that will set up and maintain session details for this IPSec policy.
-local_addr	Specifies the local IP address.
-remote_addr	Specifies the remote IP address.
-ipproto	Specifies the IP protocol number.
-ipport	Specifies the IP port number.
-mask	Specifies the IP address to mask.
-remote_tunnel_addr	

	Specifies the remote tunnel IP address.
-selectivity	Specifies to use selectivity lists confined to destination or source points and optional specificity types/protocols for wildcarded proposals only. When a endpoint field's properties has been wildcarded, determines whether new connections will share an existing security association (selectivity POLICY), or if new connections will cause a new security association to be created (selectivity PACKET). Selecting PACKET results in a more secure configuration, since encryption keys won't be shared between connections, but consumes more resources. Selecting POLICY conserves security associations, when this is desired. Properties that may be wildcarded include IP address, IP port number, and IP protocol number.
-proposal_set	Starts a proposal set declaration.
-proposal_set_type	Specifies the type of proposal to set for key management: auto, manual, or ike
-key_format	Specifies the format in which the keys provided in the policy are presented. Default value is hex. Note that ASCII strings will be half the length of hex strings, but security is slightly diminished, since 1/8 of the available hex key space is not available to ASCII strings.
-presharedkey	Specifies the preshared key string. (Same secret string shared between security points.)
-inenc_key	Specifies an encryption key string used for encrypting/decrypting incoming traffic. Must match the corresponding field on the remote endpoint. For manual IPSec proposals with a non-NULL ESP mode only.
-outenc_key	Specifies an encryption key string used for encrypting/decrypting outgoing traffic. Must match the corresponding field on the remote endpoint. For manual IPSec proposals with a non-NULL ESP mode only.
-inauth_key	Specifies a hash key string used for authenticating incoming traffic. Must match the corresponding field on the remote endpoint. For manual IPSec proposals with a non-NULL auth mode only.
-outauth_key	Specifies a hash key string used for authenticating outgoing traffic. Must match the corresponding field on the

remote endpoint. For manual IPSec proposals with a non-NULL authenticate mode only.

-in_spi — Specifies the Security Parameter Indexing (SPI) number for incoming traffic decode security associations. Applies to a manual proposal only.

-out_spi — Specifies the Security Parameter Indexing (SPI) number for outgoing traffic encode security associations. Applies to a manual proposal only.

-in_nonce — Specifies a random nonce value on incoming traffic to counter replay attacks.

-out_nonce — Specifies a random nonce value on outgoing traffic to counter replay attacks.

-ike_mode — For IKE phase 1 negotiations specifies the intensity of examination. Main mode is more intense and secure, but time consuming. Aggressive mode provides faster negotiations but exposes identies of the peers to eavesdropping.

-pfs — Turns on, or off, Perfect Forward Secrecy (PFS) mode for IKE policies. (Typically, this should be left on, unless you have a special environment.)

-transform — Specifies the start of a transform declaration.

-transform_type — Specifies the type of transform to apply to the policy. Possible values are:

- ike -- IP Key Exchange
- esp -- Encapsulation Security Payload

Note that value ah for authentication header is not currently supported.

-auth_alg — Specifies the authentication algorithm for IKE or ESP policy transform hash functions. Possible values are:

- null
- sha1
- md5
- xcbc

-enc_alg — Specifies the encryption algorithm for IKE or ESP policy transforms. Possible values are:

- null
- des
- 3des
- aes_128
- aes_256
- aes_cm_128
- aes_cm_256

-dhgroup — Specifies which Diffie-Hellman (dh) group to use for the symmetrical key generation. Groups 1 through 4 are supported.

-esp_mode — Specifies the Ecapsulating Security Payload (ESP) transform mode:

- tunnel
- transport

-lifetime — Specifies the life of a policy with time and/or data size parameters.

-auth_method — Specifies the authentication method for IKE transforms. Possible values are:

- preshared_key
- dsa
- rsa

COMMAND FILE PARAMETERS

Level# — An unsigned 32-bit integer that specifies the priority level (policy index number 0-110).

IPaddr — The local, remote, or remote-tunnel endpoint IP address.
In addition to the IP address, the mask and IP protocol options can also be specified here:

```
[-mask IPaddr]
#(mask not valid when using "ipaddr all")
[-ipproto IPprotocol#|all]
#(defaults to all)
[-ipport IPport#|all]
#(defaults to all)
```

Where:

- IPprotocol# is a protocol number. For example, 6 for TCP, or 1 for IPv4-ICMP.
- IPport# is an IP port number. For example, 3260 for iSCSI, or 1748 for RDF.

The CLI will determine if this is an IPv4 or v6 address by looking for the presence of dots (’.’,v4) or colons (’:’,v6) in the address. If both are present in ’mixed-mode’ form (e.g., ::FFFF:a.b.c.d), only the v4 section will be used.

When specifying the remote tunnel address, -mask, -ipproto, and -ipport are not supported.

SPI# — Security parameter index number for Security Associations (SA’s). An unsigned 32-bit integer greater than 255.

NONCE — An unsigned 32-bit integer. Required when using AES counter mode. A nonce is a random value used to prevent replay attacks. It makes sure the sender is really participating in the conversation.

Keystring — A string of concatenated hexadecimal digit pairs, without per-byte delimeters, that represent an encryption or authentication key.

The following key length restrictions must be adhered to:

- MD5: exactly 16 bytes
- SHA1: exactly 20 bytes
- AES_CBC: exactly 16 bytes
- DES: exactly 8 bytes
- 3DES: exactly 24 bytes
- AES_128: exactly 16 bytes
- AES_256: exactly 32 bytes
- Preshared: between 1 and 64 bytes

SPECIFICITY
A string that defines what happens when a new packet matches this policy, and the policy contains a wildcard in the corresponding field (i.e., src/dest ip address, src/dest port, protocol). The following possible string values may only be specified when the corresponding object is wildcarded:

- packet: A new SA will be created to handle this connection. (fine-grained)

- policy: A single SA will be created that will handle all connections that match this policy (coarse-grained).

LifeParam Lifetime parameters concerning time and data size. Just one, or both values in any order, can be specified:

- a time value in minutes or hours (e.g., 90m or 5h)

- a data size value in megabytes or gigabytes (e.g., 50mb or 3gb)

If both are specified, apply a comma between parameters with no intervening space. The first limit reached will end life.

RETURN CODES

Code #	Code Symbol
0	CLI_C_SUCCESS
1	CLI_C_FAIL
2	CLI_C_DB_FILE_IS_LOCKED
19	CLI_C_GK_IS_LOCKED All gatekeepers to the Symmetrix array are currently locked.

EXAMPLES

To list the policy priorities on all directors within a specific Symmetrix, enter:

```
symipsec -sid 123456789012 -dir ALL -port ALL
  list -all
```

To show the policy details for a specific policy, enter:

```
symipsec -sid 123456789012 -dir 1A -port 0
```

```
     show -priority 20

To check the syntax of a command file, enter:

   symipsec -sid 123456789012 -dir 1A -port 0
     -file /tmp/commandfile preview

To display IPSec SA details for a specific priority, enter:

   symipsec -sid 123456789012 -dir 1A -port 0
     list -stats -type ipsec_details -priority 20

To add an IPSec iSCSI policy to the policy database, enter:

   symipsec -sid 0039 -dir 1A -port 0 -file /tmp/ap commit

Where /tmp/ap contains:

  policy add
    -priority 50 -assoc_ike_policy 40 -action secure
    -local_addr 172.23.195.20 -ipport 3260 -ipproto 6
    -remote_addr 50.60.70.80 -ipport 3260 -ipproto 6
    -selectivity destip packet -selectivity destport packet
    -proposal_set -proposal_set_type auto -proposal
    -proposal_type ipsec -transform -transform_type esp
    -encalg aes_cm_256  -lifetime 90m,5gb
  ;

To remove an IPSec policy to the policy database, enter:

  symipsec -sid 0039 -dir 1A -port 0 -file /tmp/dp commit

Where /tmp/dp contains:

  policy delete -priority 50;
```

symlabel

Performs device label operations on one or more devices.

SYNOPSIS

```
symlabel -h

symlabel -g <DgName> [-noprompt]

   define <LdevName> [label <Label>]

   undefine <LdevName> -type WNT

symlabel -g <DgName> [-type WNT] [-offline]

   list [-bcv | -vdev]

   show <LdevName>
```

DESCRIPTION

The symlabel command performs device label (signature) operations on the device(s) of a device group. A device label (or signature) is initially assigned to each Symmetrix device by the host operating system. These labels must be relabeled during TimeFinder operations using a symdg relabel command. For SYMCLI usage, you can define labels of devices in a device group in the SYMAPI configuration database. For WNT labels, you can list or show defined or actual labels for these devices and undefine device labels.

ARGUMENTS

define	Defines the device labels in the Symmetrix configuration database for the specified device in a device group.
list	For Windows only, lists the defined or actual labels of the devices in a device group. If the offline option is used, it lists only the defined labels in the Symmetrix configuration database.
show	For Windows only, shows the device label and information about a specified device in the device group.
undefine	Removes the device labels that were previously defined in the Symmetrix configuration database. Only labels of type WNT can be undefined.

KEYWORDS

label	Applies an 8-digit hexadecimal label for Windows.
WNT	Writes Windows type label on the device.

OPTIONS

-bcv	Targets the indicated action at the specified BCV device(s) that are locally associated with the device

group.

-g Specifies a device group name.

-h Provides brief, online help information.

-noprompt Disables the prompt flag. The default is to prompt the user for confirmation before executing the indicated operation.

-offline Obtains information from the Symmetrix host configuration database. Only defined labels will be displayed. If this option is omitted, the actual device label(s) will also be read from the device(s) and then displayed.

-range Applies the labeling action to a number of Symmetrix devices within a contiguous range.

-sid Supplies the Symmetrix ID to limit the scope of the operation to the specified Symmetrix array.

-type Specifies a device label type. Currently supports WNT.

-v Provides a more detailed, verbose listing.

-vdev Performs the action on VDEVs that are locally associated with a device group.

PARAMETERS

DgName The device group name.

Label An 8-digit hexadecimal label for Windows.

LdevName The device logical name, either named by the user, or automatically assigned when a device is added to a device group.

SymmID The 12-digit ID of the Symmetrix array.

RETURN CODES

Code #	Code Symbol
0	CLI_C_SUCCESS
1	CLI_C_FAIL
2	CLI_C_DB_FILE_IS_LOCKED
19	CLI_C_GK_IS_LOCKED All GateKeepers to the Symmetrix array are currently locked.

EXAMPLES

To define a device label for BCV device BCV001 in device group ProdDB, enter:

```
symlabel -g ProdDB define BCV001 label ABCDEF
```

To list the defined device labels for all BCV devices in device group ProdDB, enter:

```
symlabel -g ProdDB -bcv -offline list
```

To list both the defined and the actual device labels, for all BCV devices in device group ProdDB, enter:

```
symlabel -g ProdDB -bcv list
```

To undefine the device label for standard device DEV005 in device group ProdDB, enter:

```
symlabel -g ProdDB undefine DEV005
```

symlmf

This command is used to manage licenses
with Solutions Enabler.

SYNOPSIS

```
symlmf  <LicenseKey>

symlmf  add    -type se -lic <LicenseKey>

        delete -type se -license <LicenseKey>

        list   -type se [-v] [-summary]

        add    -type emclm -sid <SymmID> -file <FileName>

        list   -type emclm -sid <SymmID>

        query  -type emclm -sid <SymmID>

        show   -type emclm -sid <SymmID>

        list   -type sym -sid <SymmID>
```

DESCRIPTION

symlmf is used to manage licenses with Solutions Enabler.
There are two types of licenses that are supported:

Traditional Solutions Enabler licenses - indicated by
a type of "se".

New style EMCLM licenses - indicated by a type of
"emclm".

For the traditional SE licenses:
The ’add’ action can be used to register a license.

```
add -type se -lic <LicenseKey>
```

The existing format, without the ’-type se add -lic’
arguments, continues to be supported. The use of new
syntax is encouraged since the older format may be
deprecated in the future.

The ’delete’ action can be used to delete a license.

```
delete -type se -license <LicenseKey>
```

The ’list’ action can be used to display installed
licenses.

```
list -type se [-v] [-summary]
```

For the new EMCLM licenses:
The ’add’ action can be used to register licenses from a
file on disk.

```
add  -type emclm -sid <SymmID> -file <FileName>
```

The ’list’ action can be used to display installed
licenses.

```
list -type emclm -sid <SymmID>
```

The ’query’ action can be used to display information

about the current capacity of licensed features.

query -type emclm -sid <SymmID>

The 'show' action can be used to display the current licensing file of a particular Symmetrix.

show -type emclm -sid <SymmID>

For the Symmetrix features:
The 'list' action can be used to display only licenses that apply to a particular Symmetrix.

list -type sym -sid <SymmID>

If symlmf is invoked with no arguments, it prompts you to supply traditional style Solutions Enabler licenses. The use of this method is discouraged since it may be deprecated in the future.

The SYMCLI_CONNECT environment variable can be set to connect to a remote Solutions Enabler server.

ARGUMENTS

add	Register one or more licenses. SE licenses are supplied on the command line via the '-lic <LicenseKey>' option. EMCLM licenses are supplied by supplying the pathname to a license file on disk ('-file <FileName>').
delete	Delete one or more SE licenses. SE licenses are supplied on the command line via the '-license <LicenseKey>' option. EMCLM licenses may not be deleted.
list	Displays information about SE or EMCLM licenses that are currently installed.
query	Displays information about Symmetrix features, whether they have EMCLM licenses or are enabled by other means and the capacities these features are currently using.
show	Displays the current licensing file of a particular Symmetrix.

OPTIONS

-file	Specifies the path to a license file.
-license	Specifies a traditional SE license key.
-sid	Specifies Symmetrix serial number of the license.
-summary	Provides a list of all license types enabled by the installed SE licenses.
-type	Indicates the type of license to be operated upon. This takes a parameter of either "emclm" or "se".

```
    -v            Provides a more detailed, verbose listing.


PARAMETERS

    FileName      The path to a file.

    LicenseKey    A Traditional license key, with the
                  following syntax:
                      1234-5678-9ABC-DEF0.

    SymmID        Symmetrix serial number of the license.


RETURN CODES

    Code #    Meaning
    ------    -----------

       0      The operation succeeded.

       1      There was an error.  Perhaps an invalid
              key was supplied.

EXAMPLES

    To add EMCLM Symmetrix licenses from a license file,
    enter:

        symlmf add -type emclm -sid 000111222333
                   -file /tmp/hk111222333.lic

    To add an SE license from the command line, enter:

        symlmf add -type se -license 1234-5678-9ABC-EDF0

        or

        symlmf 1234-5678-9ABC-DEF0

    Note: The use of the above example should be
          discontinued as this method is being depreciated.


    To delete an SE license from the command line, enter:

        symlmf delete -type se -license 1234-5678-9ABC-EDF0

    To display the currently installed SE licenses, enter:

        symlmf list -type se

    To display the currently installed SE licenses with more
    information, enter:

        symlmf list -type se -v

    To display all the license types enabled by the currently
    installed SE licenses, enter:

        symlmf list -type se -summary

    To display the current licenses installed on a Symmetrix,
    enter:

        symlmf list -type emclm -sid <SymmID>
```

To display the current capacity usage of licensed and non-licensed features on a Symmetrix, enter:

```
symlmf query -type emclm -sid <SymmID>
```

To display the current license file of a particular Symmetrix, enter:

```
symlmf show -type emclm -sid <SymmID>
```

To display the current Symmetrix based licenses of a particular Symmetrix, enter:

```
symlmf list -type sym -sid <SymmID>
```

Displays performance statistics and detailed mapping information about one or more logical volumes that are defined in a logical volume group and performs control operations on logical volumes.

SYNOPSIS

```
symlv -g <VgName> [-type <VgType>] [-h] [-kb|-blocks|-mb]

   list  [-v]

   show  <LVolName> [-stripe_column]
     [-no_extents|-expand|-collapse|-pdev_extents]

   create <LVolName> -size Size [-nmirror <Mirrors>]
     [-striped|-RAID5] [-ncols <Columns>]
     [-strsize <StripeSize>] [-pd <Pdevname...>]

   delete <LVolName>

   add <LVolName> -nmirror <Mirrors>
     [-striped|-RAID5]
     [-ncols <Columns>] [-strsize <StripeSize>]
     [-pd <Pdevname...>]

   remove <LVolName> [-nmirror <Mirrors>] [-mir <MirName>]

   extend <LVolName> -size <Size> [-pd <Pdevname...>]

   reduce <LVolName> -size <Size> [-pd <Pdevname...>]

symlv stats [-type <VgType> [-g <VgName> [-lv <LVolName>]]]
     [-i <Interval>] [-c <Count>] [-h]
```

DESCRIPTION

The symlv command displays detailed logical-to-physical mapping information specific to a volume in a logical volume group. It also supports control operations, such as, create, delete, extend, reduce, and remove on logical volumes.

In the lists of mirror physical extents and mirror physical devices for the logical volume, CLARiiON devices are distinguished from other device types by a (C) indicator.

Volume group name length restriction:
The volume group name field is limited to 63 characters. This length restriction is inclusive of the name and absolute path of the volume group even if the user only specified the volume group name itself. The behavior is undefined if the length is exceeded.

Logical volume name length restriction:
The logical volume name field is limited to 63 characters. The behavior is undefined if the length is exceeded.

When mapping objects in the ASM volume manager, three environment variables are required in order to contact the ASM instance:

Environment Variable	Description
SYMAPI_ASM_HOME	Oracle Home of ASM instance
SYMAPI_ASM_SID	Oracle Sid of ASM instance
SYMCLI_ASM_CONNECT	username/password of ASM instance

ARGUMENTS

add — Adds mirror images to a logical volume of the specified type. The default VgType is assumed if a VgType is not specified. Not all VgTypes are supported.

create — Creates a logical volume of the specified type. The default VgType is assumed if a VgType is not specified. Not all VgTypes are supported.

delete — Deletes a logical volume of the specified type. The default VgType is assumed if a VgType is not specified. Not all VgTypes are supported.

extend — Extends/grows a logical volume of the specified type to given size. The default VgType is assumed if a VgType is not specified. Not all VgTypes are supported. Size parameter should be greater than the current size of the logical volume.

list — Lists all defined volumes in an existing logical volume group. The behavior is undefined if the volume group and volume name is more than 63 characters.

show — Shows detailed logical-to-physical mapping information about a volume in the logical volume group.

stats — Shows performance statistics about logical volumes. The default VgType is assumed if a VgType is not specified. Default VgType for AIX and SunOS is Veritas volume manager. For Linux it is LVM2 and for HPUX it is Native logical vol manager. For windows default VgType for statistics is same as other operations.
Note that, on the Windows platform, you may need to first run the command "diskperf -yv" in order to obtain performance statistics. On SunOS, metadevices need to be mounted before trying to get I/O statistics.

reduce — Reduces/shrinks a logical volume of the specified type to given size. The default VgType is assumed if a VgType is not specified. Not all VgTypes are supported. Size parameter should be lesser than the current size of the logical volume.

remove — Removes mirrors of a logical volume of the specified type. The default VgType is assumed if a VgType is not specified. Not all VgTypes are supported.

OPTIONS

-blocks	Displays size information in 512-byte blocks.
-c	Specifies the number of times to poll for data.
-collapse	Collapses the extents of a logical volume, if possible.
-expand	Expands the extents of a logical volume, if possible.
-g	Specifies a logical volume group name.
-h	Provides brief, Online help information.
-i	Specifies the repeat interval in seconds. The default interval is 30 seconds. The minimum interval is 5 seconds. For passive actions the minimum interval is 15 seconds. Passive actions are actions that do not acquire an exclusive lock.
-kb	Displays the size information in Kilobytes.
-lv	Specifies a logical volume name.
-mb	Displays the size information in Megabytes. This is the default.
-mir	Specifies the name of the mirror to remove.
-ncols	Specifies the number of stripe columns.
-no_extents	Specifies to not display extents information.
-nmirror	Specifies the number of mirrors in the logical volume. For the create action, if this option specifies a number less than or equal to 0, then 1 is assumed by default.
-pd	Specifies a list of the device names used for the operation. On the Solaris platform for SVM, the device pathname must be specified to create a metadevice (volume).
-pdev_extents	Specifies physical device-level extents only. (Does not expand extents to reflect an underlying meta device configuration).
-RAID5	Defines a RAID5 logical volume type.
-size	Defines the size of the logical volume in 512 byte blocks. An optional suffix can be specified to indicate the unit of size measurement. The optional suffixes supported are: - b = size in Blocks. - k = size in Kilobytes - m = size in Megabytes On the Windows platform if the size specified is less than 1 Megabyte, a

volume of size 1 Megabyte will be created. On the Solaris platform for SVM, the size option should not be specified for creating a volume. The volume size will be the same as the size of the device partition, on which the volume is created.

-stripe_column Displays the extent's stripe column number for striped volumes.

-striped Defines a striped logical volume type.

-strsize Specifies the size of each stripe column in 512 byte blocks. An optional suffix can be specified to indicate the unit of size measurement. The optional suffixes supported are:

- b = size in Blocks.
- k = size in Kilobytes
- m = size in Megabytes

-type Specifies the volume group type.

-v Provides a more detailed, verbose listing.

PARAMETERS

Columns The number of stripe columns.

Count A positive integer.

Interval The interval between polls, in seconds.

LVolName A logical volume name. On the Solaris platform for SVM, the volume name should follow the SVM guideline of naming metadevices (e.g., d11,d12,etc.).

MirName The name of the mirror.

Mirrors The number of mirrors in logical volume.

PdevName A fully qualified host or physical device name.

Size The logical volume size in 512 byte blocks or size with the appropriate suffix specified.

StripeSize The logical volume stripe size in 512 byte blocks or size with the appropriate suffix specified.

VgName A specific logical volume manager's volume group name.

VgType The volume group type. Possible values are:

Group Type	
DEFAULT	LINUX_LVM
AIX_LVM	LINUX_POOL
AIX_VXVM	LINUX_VXVM
AS400_LVM	ORACLE_ASM
DYNIX_SVM	SUN_SOLSTICE
EMC_PVM	SUN_VXVM

```
                  HP_LVM                WIN_LDM
                  HP_VXVM               WIN_VXVM

                Note: Statistics are not available for
                AIX_LVM, DYNIX_SVM, AS400_LVM and
                ORACLE_ASM.
```

RETURN CODES

Code #	Code Symbol
0	CLI_C_SUCCESS
1	CLI_C_FAIL

EXAMPLES

To list all volumes in the default volume group ProdVG, enter:

```
symlv -g ProdVG list
```

To display detailed logical-to-physical mapping information for the logical volume vol1 in the volume group ProdVG, enter:

```
symlv -g ProdVG show vol1
```

To create a simple volume called testlv of size 500m in the volume group named testvg, enter:

```
symlv -g testvg create testlv -size 500m
```

To list all volumes in the HP-UX VXVM volume group vg00, enter:

```
symlv -g vg00 -type HP_VXVM list
```

To display statistics about all logical volumes of VXVM on SunOS every 30 seconds for one hour, enter:

```
symlv stats -i 30 -c 120 -type SUN_VXVM
```

The output key follows:

```
Volume Group Type : SunOS VxVM
H:M:S VgName LvName RW/s R/s W/s KbRW/s KbR/s KbW/s %Busy %Wait
  A     B       C      D   E   F   G      H     I     J      K
```

A Time of day
B Volume group name
C Logical volume name
D Read and write requests per second
E Read requests per second
F Write requests per second
G KB read and written per second
H KB read per second
I KB written per second
J 100 * (logical volume active time / elapsed time)
K 100 * (non-empty wait queue time / elapsed time)

symmask

Allows the administrator to set up and modify
Symmetrix device masking functionality.

SYNOPSIS

```
symmask -h

symmask discover hba [-rename] [-v]

symmask list hba [-v]

symmask -sid <SymmID> <-wwn <wwn> | -awwn <awwn>> |
      <-iscsi <iscsi> | -aiscsi <aiscsi> -name <name>> |
      -host <hostname> -dir <#|all> -p <#|all>
      [-celerra] [-rp]
      [-reserve_id <ResvID>[,<ResvID>[,<ResvID>]]]

   add devs <SymDevStart:SymDevEnd> |
      <SymDevName> | <<SymDevName>,<SymDevName>...>
      [-dynamic_lun | -lun <Addr> | <<Addr>,<Addr>...>]
      [-noprompt] [-remove_unmapped] [-map]

   add -g <DgName> [-std] [-bcv] [-vdev] [-tgt] [-noprompt]
      [-dynamic_lun | -lun <Addr>] [-remove_unmapped]

   add -file <DeviceFileName> [src] [tgt] [-noprompt]
      [-dynamic_lun | -lun <Addr>] [-remove_unmapped]

   remove devs <SymDevStart:SymDevEnd> | <SymDevName> |
     <SymDevName,SymDevName,SymDevName...> [-unmap] [-force]

   remove -g <DgName> [-std] [-bcv] [-vdev] [-tgt] [-force]

   remove -file <DeviceFileName> [src] [tgt] [-force]

symmask -sid <SymmID> -wwn <wwn> | -awwn <awwn> |
      -iscsi <iscsi> | -aiscsi <aiscsi> | -host <hostname>

   set heterogeneous
      <on <host> | off> -dir <#|all> -p <#|all>

   set hba_flags
      <on <<flag>,<flag>...> <-enable | -disable> |
       off [<flag>,<flag>...] >
      -dir <#|all> -p <#|all>

symmask -sid <SymmID> -wwn <wwn> | -awwn <awwn> |
      -iscsi <iscsi> | -aiscsi <aiscsi>

   list logins [-dir all [-p all] | -dir # [-p <#|all>]]
     [-pdev <PdevName>] [-v]

   set lunoffset <on <offset> <base>|off> -dir <#> -p <#>

   set visibility <on|off> -dir <#> -p <#>

   replace <wwn | iscsi> [-noprompt]

symmask -sid <SymmID> <-wwn <wwn> | -awwn <awwn>> |
     <-iscsi <iscsi> | -aiscsi <aiscsi> [-name <name>]>

   delete [-dir all -p all | -dir # -p #] [-login]
      [-reserve_id <ResvID>[,<ResvID>[,<ResvID>]]]

symmask -sid <SymmID> -wwn <wwn> | -awwn <awwn>
```

```
   set lockdown <on <fcid> | off> -dir <#> -p <#>

symmask -sid <SymmID> -wwn <wwn> | -iscsi <iscsi>

   rename <awwn | aisci>

symmask -sid <SymmID> [-noprompt]

   refresh [-verify]
```

DESCRIPTION

The symmask command discovers the HBAs on the host and assigns ASCII or alias names to the login history table entries for those initiators that are not set.

This command also provides the following functions:

- Lists the hosts' HBA information.
- Adds devices to and removes devices from the device masking database (VCMDB).
- Displays the login history table.
- Sets a LUN offset. This feature was added to address the restrictions of LUN values allowed to be assigned to HBAs.
- Sets a heterogeneous host. This feature was added to allow the user to change some attributes for a different host type.
- Sets the HBA port flags. This feature is similar to the hetergeneous host feature, but the user is able to specify which settings are enabled and disabled.
- Sets volume visibility. This feature allows HP hosts to find all assigned devices, even if they are assigned non-continuously.
- Replaces the host HBA without losing established permissions.
- Removes an initiator from the device masking database, along with all of the devices associated with that initiator.
- Associates the Fibre Channel ID (FCID) of a switch in a fabric to the path from a host HBA to a Symmetrix array. This further restricts the path by which a host can connect to a Symmetrix array.
- Associates an ASCII name with the initiator as a convenience. To NULL the alias, use a slash (/) as input.
- Refreshes Fibre Channel and Gige directors with the latest copy of the data in the device masking VCMDB.

ARGUMENTS

add	Adds devices to the record in the database with the matching WWN, iSCSI name or the hostname part of the alias.

delete — Deletes the record(s) matching the WWN or iSCSI name from the database.

discover — Discovers the WWN or iSCSI names of the HBAs on the host and writes the ASCII names to the login history table, if empty.

list — Lists the requested data.

refresh — Updates the fiber and gige adapters with changes to the database.

remove — Removes devices from the record in the database with the matching WWN or iSCSI name.

rename — Changes the ASCII name or alias in the database and the login history table.

replace — Changes the WWN or iSCSI name in the database.

set — Allows certain device masking features to be enabled or disabled.

KEYWORDS

devs — Specifies devices to be added or removed.

hba — Specifies the WWN or iSCSI name of the HBA on the host.

heterogeneous — Sets the record in the database to hold information on the host type that may differ than the current setting on the corresponding FA.

hba_flags — Sets the record in the database to hold information on the HBA port setting that may differ than the current setting on the corresponding FA.

lockdown — Sets the FCID value in the database to correlate that entry with a specific path.

logins — Specifies the entries in the login history table.

lunoffset — Sets the record in the database to hold information of a LUN offset and base or starting values.

src — Acts on only the source devices when used with a device file.

tgt — Acts on only the target devices when used with a device file.

visibility — Sets information in the device masking database to note that the host should find all devices, even if they are not contiguous.

OPTIONS

-aiscsi — Specifies a user-given name, or alias

iSCSI name.

-awwn — Specifies a user-given name, or alias WWN.

-bcv — Acts on only the BCV devices when used with a device group.

-celerra — Allows controls on Celerra FBA devices.

-disable — Disables the overridden HBA port flags on a per initiator basis.

-dynamic_lun — Specifies to use the dynamic LUN addressing features but does not require the user to give a LUN address for each device. The LUN addresses will be assigned based on what may already be in use for that host HBA.

-enable — Enables the overided HBA port flags on a per initiator basis.

-file — Applies a device file to the command. The device file contains device pairs (SymDevnames) listing a pair per each line (the source device first, a space, and the target device last within each line entry). Device files can include comment lines that begin with the pound sign (#). A Symmetrix ID is required for this option. -f is synonymous with -file.

-force — Forces the SYMAPI server to allow actions that ordinarily would fail. The force flag should be used when removing devices that should not be masked, such as metamembers, or when a range of devices covers devices that may not be masked.

-g — Applies a device group name to the command.

-h — Provides brief, online help.

-host — Specifies the host name.

-iscsi — Specifies the iSCSI name.

-login — Deletes the entries in the login history table and the entries in the masking database.

-lun — Specifies the starting LUN addresses to be used for the devices being added to the host HBA. The user may supply a single starting LUN address for all devices being added, or a list of starting LUN addresses equal to the number of device ranges in the list.

-map — Allows the user to map the devices while adding them to the masking database. This option is supported for Enginuity 5773 only.

-name — Specifies the access logix record name. This option to be used along with -iscsi or -aiscsi option for Symmetrix arrays running Enginuity 5874.

-noprompt — Requests that no prompts are returned

after the command is entered. The default is to prompt the user for confirmation.

-pdev — Applies a physical device name (host path) to the list login action, which allows you to determine if an HBA is logged on to this device.

-p — Applies a port number designation.

-remove_unmapped — Assists the user to convert an existing device masking record to use the new dynamic feature in a single step. Records that currently contain unmapped devices are not able to be converted until those devices are removed. This option will allow the user to remove those unmapped devices at the same time as adding devices with dynamic addressing.

-rename — Forces the hostname/adapter or hostname/IP to be written out to both the login history table and device masking VCMDB, even if one is present. Overwrites any existing alias in the record.

-reserve_id — Specifies the device reservation IDs for the devices in the operation. For Enginuity 5874, device reservation IDs must be specified when a record is being deleted.

-rp — Allows controls on devices tagged for RecoverPoint use.

-sid — Specifies the unique Symmetrix ID.

-std — Acts on only the standard devices when used with a device group.

-tgt — Acts on only the target devices when used with a device group.

-unmap — Allows the user to unmap the devices while removing them from the masking database. This option is supported for Enginuity 5773 only.

-v — Provides a more detailed, verbose listing.

-vdev — Acts on only the vdevs when used with a device group.

-verify — Allows the user to compare the information in the database to what’s currently on the local directors to all changes before the refreshed.

-wwn — Specifies a World Wide Name.

PARAMETERS

— A specific director or port number.

Addr — The LUN address to be used for the corresponding device.

aiscsi — A user-given name, in two parts separated

by a slash (/).

all — All directors or ports.

awwn — A user-given name, in two parts separated by a slash (/).

base — The base value of the offset value in hexadecimal.

DeviceFileName — The name of the file where device pairings are listed.

DgName — The device group name.

fcid — A Fibre Channel ID associated with the switch.

flag — The overridden HBA port flags from the following values in []:

Port Flag	Value Indicator
Common_Serial_Number	[C]
Disable_Q_Reset_on_UA	[D]
Environ_Set	[E]
Siemens	[S]
Volume_Set_Addressing	[V]
Avoid_Reset_Broadcast	[ARB]
AS400	[AS4]
OpenVMS	[OVMS]
SCSI_3	[SC3]
Sunapee	[SCL]
Sequent	[SEQ]
SPC2_Protocol_Version	[SPC2]
SCSI_Support1	[OS2007]

host — The host type from the following:

Valid Host Types	
AS400	AS400_LSE
BULL_AIX	BULL_AIX_PP15
CELERRA	DEC_OVMS
DEC_UNIX	DG_AViiON
HP-UX	IBM_AIX
IBM_AIX_PP15	IBM_AIX_DMP
IBM_AIX_DMP_PP15	IBM_EMC
IBM_EMC_PP15	ICL_OPEN
FSC_BS2000	LINUX
LINUX_DMP	NCR
NCR_MP	NCR_NT
NCR_NT_MP	NOVELL
NOVELL_CLUSTER	PRIMEPOWER
PRIMEPOWER_DMP	PRIMEPOWER_PP15
RELIANT	SEQUENT
SEQUENT_FCSW	SOLARIS
SOLARIS_DMP	SOLARIS_PP15
SUN_CLUSTER	SUN_CLUSTER30
VERITAS	VERITAS20
VERITAS_DMP	VMWARE
WINDOWS	WINDOWS_DMP
WINDOWS_DMP_PP15	WINDOWS_HP
WINDOWS_HP_DMP	WINDOWS_HP_DMP_PP15
WINDOWS_HP_PP15	WINDOWS_PP15
LINUX_DMP_VCS	

hostname	The host name.
iscsi	An iSCSI name.
on	Turn feature on.
off	Turn feature off.
offset	The hex value of the offset in LUN addressing.
PdevName	A physical device name for the specified action.
ResvID	A device reservation ID.
SymDevEnd	The end of a range of logical devices.
SymDevName	A Symmetrix device to be added or removed.
SymDevStart	The start of a range of logical devices.
SymmID	The 12-digit ID of the Symmetrix array.
wwn	The World Wide Name.

symmaskdb

Allows the administrator to back up, restore, initialize, and show the contents of the device masking VCMDB. Also provides limited conversion and attribute options.

SYNOPSIS

```
symmaskdb -h

symmaskdb -sid <SymmID> | -file <FileName> [-v]

   list database [-dir all [-p all] | -dir <#>
     [-p <#|all>]][-wwn <wwn> | -awwn <awwn> |
      -iscsi <iscsi> | -aiscsi <aiscsi>]

symmaskdb -sid <SymmID> | -file <FileName>
   list devs [-wwn <wwn> | -awwn <awwn> |
     -iscsi <iscsi> | -aiscsi <aiscsi>]

symmaskdb -sid <SymmID>

   list assignment [-v] -devs <<SymDevStart>:<SymDevEnd> |
     <SymDevName> | <<SymDevName>,<SymDevName>...>>

   list no_assignment [-dir all [-p all] |
      -dir <#> [-p <#|all>]]

   list capacity -host <HostName>

symmaskdb -sid <SymmID> [-noprompt] -file <FileName>

   restore [-skip_authentication]

   backup

symmaskdb -sid <SymmID> -file <FileName>

   init

symmaskdb -sid <SymmID> [-noprompt]

   remove -meta_member
```

DESCRIPTION

The symmaskdb command provides the following options:

- Lists the device masking VCMDB.
- Lists the devices assigned to an HBA in the device masking VCMDB.
- Lists which HBAs have been assigned to the given devices.
- Lists which devices mapped to a given director and port have not yet been assigned.
- Lists the capacity of devices assigned to a particular host.
- Restores the device masking database from a backup file stored on the host.

- Backs up the device masking database to a user-named file on the host.

- Initializes the device masking database and also requires a user-named file on the host for an initial backup.

- Removes metamembers from the device masking database, while keeping the metaheads in place.

ARGUMENTS

backup	Specifies a backup of the database to be copied to a given file.
init	Initializes the database.
list	Lists various records in the database.
remove	Removes the metamember devices.
restore	Restores the database from a given file.

KEYWORDS

assignment	Names the HBAs that are assigned in the device masking VCMDB.
capacity	Specifies the size of the device.
database	Lists records within the device masking VCMDB.
devs	Lists devices assigned by records in the device masking VCMDB.
no_assignment	Lists devices that are mapped, but not yet assigned, in the device masking VCMDB.

OPTIONS

-aiscsi	Specifies a user-given name, or alias iSCSI name.
-awwn	Specifies a user-given name, or alias WWN.
-devs	Specifies a set of Symmetrix device ranges and/or individual Symmetrix devices.
-dir	Applies a director number designation.
-file	Applies a backup file to the specified action.
-h	Provides brief, online help information.
-host	Specifies the host name.
-iscsi	Specifies the iSCSI name.
-meta_members	Specifies the metamembers, other than the metaheads.
-noprompt	Requests that no prompts are returned after the command is entered. The default is to prompt the user for confirmation.

-p Applies a port number designation.

-sid Specifies the unique Symmetrix ID.

-skip_authentication
Skips over the authentication information in a backup file and does not restore it.

-v Provides a more detailed, verbose listing.

-wwn Applies a World Wide Name (WWN).

PARAMETERS

A specific director or port number.

aiscsi A user-given name in two parts, separated by a slash (/).

all All fibre directors or ports.

awwn A user-given name, in two parts separated by a slash (/).

FileName The name of the device masking backup file.

HostName The host name.

iscsi The iSCSI name.

SymDevEnd The end of a range of logical devices.

SymDevName A Symmetrix device to be added or removed.

SymDevStart The start of a range of logical devices.

SymmID The 12-digit ID of the Symmetrix array.

wwn The system-generated World Wide Name.

```
symmdr

        Integrates SRDF/Metro and SRDF/Async to allow
        highly-available Disaster Recovery (DR) for an
        SRDF/Metro environment.

        Control operations are targeted at one of the
        following:
        The entire SRDF/Metro Smart DR (MetroDR)
        environment
        The SRDF/Metro session of the environment, by
        providing the -metro option
        The DR session of the environment, by
        providing the -dr option

        SYNOPSIS

             symmdr -h

        List/Show/Query

             symmdr -sid <ArrayID>
                  [-i <Interval>] [-c <Count>]

                list [-tb]

             symmdr -sid <ArrayID> -name <EnvironmentName>
                  [-i <Interval>] [-c <Count>]

                show [-detail]

             symmdr -sid <ArrayID> -name <EnvironmentName>
                  [-i <Interval>] [-c <Count>]

                query [-tb]

        Operations specific to the SRDF/MetroDR environment

             symmdr -sid <ArrayID> -name <EnvironmentName>
                    -metro_rdfg <RDFGroup> -dr_rdfg <RDFGroup>
                    [-noecho] [-noprompt]
                    [-i <Interval>] [-c <Count>]

                environment -setup

             symmdr -sid <ArrayID> -name <EnvironmentName>
                  -dr_rdfg <RDFGroup>
                  [-noecho] [-noprompt]
                  [-i <Interval>] [-c <Count>]
                  [-force] [-symforce]

                environment -remove

          Recover

             symmdr -sid <ArrayID> -name <EnvironmentName>
                  [-noecho] [-noprompt]
                  [-i <Interval>] [-c <Count>]

                recover

        Operations specific to the SRDF/Metro session

             symmdr -sid <ArrayID> -name <EnvironmentName>
                  [-noecho] [-noprompt]
                  [-i <Interval>] [-c <Count>]
                  [-force] [-symforce]
```

```
          -metro

        establish

        restore

        suspend  [-keep <R1 | R2>]

Operations specific to the DR session

     symmdr -sid <ArrayID> -name <EnvironmentName>
          [-noecho] [-noprompt]
          [-i <Interval>] [-c <Count>]
          [-force] [-symforce]
          -dr

        establish

        failback

        failover

        restore

        split

        suspend

        update

        set mode <acp_disk | async>


DESCRIPTION

     The symmdr command provides the ability to setup, remove,
     control, list, show and query an SRDF/MetroDR environment.

     Running DR in Async mode provides highly-available DR for
     the SRDF/Metro session.

     The SRDF/Metro session must be configured to use Witness
     protection, and an appropriate Witness must be available
     when the environment is initially set up and while it
     remains in use.

ARGUMENTS

     establish      Resumes I/O traffic on the SRDF links and
                    initiates a data copy from the source
                    side to the target side.

                    Can be targeted at the SRDF/Metro or the DR
                    session.

     environment    Setup or remove an SRDF/MetroDR environment.

                    Environment Requirements:
                    All 3 arrays must be discovered and online.
                    All SRDF groups supporting the environment
                    must exist and be online.

     failback       Adjusts the SRDF/MetroDR environment after a
                    failover to allow the application to be
                    re-started from the SRDF/Metro site.

                    Resumes the I/O traffic on the SRDF links
```

and initiates a data copy from the DR side to the SRDF/Metro side.

Targeted at the DR session.

failover — Adjusts the SRDF/MetroDR environment to allow the application to be started from the DR site.

Targeted at the DR session and must be initiated from the DR site by specifying the array (sid) that represents the DR site.

list — Lists all SRDF/MetroDR environments on an array.

query — Displays the details of a specified SRDF/MetroDR environment.

recover — Attempts to recover an SRDF/MetroDR environment from an Invalid or Unknown state.

restore — Resumes I/O traffic on the SRDF links and initiates a data copy from the target side to the source side.

Can be targeted at the SRDF/Metro or the DR session.

set — Modifies the setting to the specified value.

Targeted at the DR session.

show — Displays the configuration of a specified SRDF/MetroDR environment.

split — Suspends I/O traffic on the SRDF links and sets the target side to be read and write-enabled to the host.

Targeted at the DR session.

suspend — Suspends I/O traffic on the SRDF links.

Can be targeted at the SRDF/Metro or the DR session.

update — Starts an update of the source side after a failover.

Targeted at the DR session.

KEYWORDS

mode — Sets the SRDF mode for the DR session to asynchronous (async) mode or adaptive copy disk (acp_disk) mode.

OPTIONS

-c — Used with the list, show and query operations, specifies the number (count) of times to repeat the operation, displaying results appropriate to the operation at each iteration.

Used with the control, set, environment setup and remove operations, specifies the number (count) of times to attempt to acquire an exclusive lock on the host database, or on the local and/or remote arrays.

The time to wait between operation iterations or between attempts to acquire a needed lock is specified by -i (interval). If neither -c nor -i is specified, these operations fail if unable to acquire a requested lock.

If -c is not specified and -i is specified, the command loops continuously, repeating the specified list, or show or query operation, or until the specified control operation has acquired the locks it needs and can start.

-detail — Displays detailed information for an SRDF/MetroDR environment, including the devices in the SRDF groups.

-dr — Indicates that the command is targeted at the DR session.

-dr_rdfg — Specifies a DR SRDF group number.

For the environment -setup operation, represents the DR SRDF group that should be used to pair SRDF/Metro devices with DR devices.

For the environment -remove operation, represents the DR SRDF group, on the SRDF/Metro array specified by -sid, from which the SRDF pairs between SRDF/Metro devices and DR devices will be deleted.

-force — Attempts to force the operation even though one or more devices may not be in the normal expected state or mode for that operation.

Use caution when using this option.

-h — Provides brief online help information.

-i — Specifies the interval, in seconds, to wait, either between successive iterations of a list, show or query operation or between attempts to acquire an exclusive lock on the host database or on the local and/or remote arrays for control operations.

The default interval is 30 seconds.
The minimum interval is 5 seconds.
For passive actions the minimum interval is 15 seconds. Passive actions are actions that do not acquire an exclusive lock.

-keep — Sets the winner side of the SRDF/Metro group to the R1 or the R2 side, as specified.

When the SRDF link becomes Not Ready (NR), devices on the winner side will be made accessible to the host and devices on the loser (non-winner) side will be made inaccessible to the host.

-metro — Indicates that the command is targeted at the SRDF/Metro session.

-metro_rdfg — Specifies an SRDF group number.

For the environment -setup operation, represents the Metro R2 SRDF group, on the array specified by -sid, that will participate in the SRDF/MetroDR environment.

-name — Specifies the name that uniquely identifies the SRDF/MetroDR environment on all three arrays.

-noecho — Does not echo the progress status of operations to stdout.

-noprompt — Requests that prompts are not displayed after the command is entered. The default is to prompt the user for confirmation.

-sid — Specifies a unique Array ID.

For the environment -setup operation, -sid must represent the MetroR2 array.

For the environment -remove operation, -sid must represent the array from which the specified dr_rdfg originates.

-symforce — Requests the operation be executed when normally it is rejected.

Use extreme caution when using this option.

CAUTION: Use care when applying -symforce, as data could be lost or corrupted. Use of this option is not recommended, except in an emergency.

-tb — Used with list or query to display capacity and invalids in terabytes.

PARAMETERS

acp_disk — Sets the DR session to adaptive copy disk mode.

ArrayID — 12-character ID that specifies the array.

async — Sets the DR session to asynchronous mode.

count — The number of times (count) to repeat.

EnvironmentName — Name that uniquely identifies the SRDF/MetroDR environment on all 3 arrays.

This name can be up to 16 characters in length.

interval — The interval between repetitions, in seconds.

RDFGroup — The SRDF group number.

R1 — Operation is targeted at the source (MetroR1) side.

When used with suspend -keep, indicates that the R1 side will remain accessible to the host when the suspend completes.

R2 — Operation is targeted at the target (MetroR2) side.

When used with suspend -keep, indicates that the current R2 side will be the R1 when the suspend completes and it will remain accessible to the host; the current R1 will be the R2 side when the suspend completes, and it will become inaccessible to the host.

symmigrate

Allows the physical disk space associated with a Symmetrix device to be changed to a different data protection scheme, or to be relocated to disks with different performance characteristics. The device can be migrated to use the storage of existing devices that do not contain live data or to unconfigured disk space.
Alternatively, a thin device may be migrated between thin pools or the data in a thin pool for a specified set of thin devices may be migrated to another thin pool.

SYNOPSIS

```
symmigrate -h
```

For configured space

Using device file input:

```
symmigrate [-v]
      [-i <Interval>] [-c <Count>] [-noprompt]
      -name <SessionName>
      -file <DevFile> -sid <SymmID>

    validate

    establish
```

Using target criteria:

```
symmigrate [-v]
      [-i <Interval>] [-c <Count>] [-noprompt]
      -name <SessionName>
      <-file <SrcFile> -sid <SymmID> |
       -sg <SgName>    -sid <SymmID> |
       -g <DgName>     [-bcv | -tgt]>
      -tgt_config -tgt_dsk_grp <DskGrp | name:<DskGrpName>>
      <-tgt_unprotected                  |
       -tgt_raid1                        |
       -tgt_raid5 -tgt_prot <3+1 | 7+1> |
       -tgt_raid6 -tgt_prot <6+2 | 14+2>>

    validate [-outfile <OutputFile>]

    establish
```

For unconfigured space:

```
symmigrate [-v]
    [-i <Interval>] [-c <Count>] [-noprompt]
    -name <SessionName>
    <-file <SrcFile> -sid <SymmID> |
     -sg <SgName>    -sid <SymmID> |
     -g <DgName>     [-bcv | -tgt]>
    -tgt_unconfig -tgt_dsk_grp <DskGrp | name:<DskGrpName>>
    <-tgt_unprotected                 |
     -tgt_raid1                       |
     -tgt_raid5 -tgt_prot <3+1 | 7+1> |
     -tgt_raid6 -tgt_prot <6+2 | 14+2>>

  validate

  establish
```

Using a target thin pool:

```
symmigrate [-v]
    [-i <Interval>] [-c <Count>] [-noprompt] [-force]
    -name <SessionName>
    <-file <SrcFile> -sid <SymmID> |
     -sg <SgName>    -sid <SymmID> |
     -g <DgName>     [-bcv | -tgt]>
     [-src_pool <PoolName>]
     -tgt_pool -pool <PoolName>

  validate

  establish
```

General operations:

```
symmigrate [-v] [-i <Interval>] [-c <Count>]
     -name <SessionName> -sid <SymmID>

  query [-detail | -summary]

  terminate [-noprompt]

  verify [-createinprog | -syncinprog    |
          -synchronized | -migrateinprog |
          -migrated | -failed | -invalid]
         [-summary]

symmigrate [-v] [-i <Interval>] [-c <Count>]

  list [-sid <SymmID>] [-names] [-detail]
```

DESCRIPTION

The symmigrate command performs LUN migration operations on a set of source devices. The target disk spaces can be either configured disk spaces or unconfigured disk spaces.

Both source devices and target devices can be specified by using a device file (containing pairs of devices). If a device file is not used, the source devices can be specified by using a device group, a storage group, or a source file (containing a list of source devices), however, the target disk spaces must then be specified by using a disk group number.

The operations include validating the migration request and establishing a migration session for a set of source devices and target disk spaces.

ARGUMENTS

establish	Starts a new session and begins the syncing process.
list	Lists all the sessions for a given Symmetrix array, or for all Symmetrix arrays.
query	Queries for the status of sessions.
terminate	Removes a migrated session from the Symmetrix array. When using -sg, -dg, or -file to specify devices, those devices must exactly match the devices in the session with the supplied

session name or an error will return.

validate — Verifies that the information provided is currently allowed. No changes will be made to the device(s) and the command can optionally output a file containing the device pairs (using configured space).

verify — Verifies that a session is in a specified state.

OPTIONS

-bcv — Indicates that the control operation is targeted at the BCV device in the device group.

-c — Specifies the number (count) of times to display or to acquire an exclusive lock on the Symmetrix host database. If this option is not specified but an interval (-i) is specified, the program will loop continuously to display or to start the control operation.

-createinprog — Verifies that the migration device pairs are in the CreateInProg state.

-detail — Lists detailed information for all migration sessions with information specific to the source devices and target devices or disks in each session.

-failed — Verifies that the migration device pairs are in the Failed state.

-file — Applies a DevFile or SrcFile to the command. A Symmetrix ID is required for this option. -f is synonymous with -file.

-force — Allows a migration to occur in a thin migration when the source devices include different emulation types and not all of the emulations match the emulation of the target pool. In this case, only the devices with emulations that match the pool’s emulation will be migrated.

-g — Applies a device group name to the command to specify the source devices.

-h — Provides brief, online help information.

-i — Specifies the repeat interval in seconds to display or to acquire an exclusive lock on the Symmetrix host database.
The default interval is 30 seconds.
The minimum interval is 5 seconds.
For passive actions the minimum interval is 15 seconds. Passive actions are actions that do not acquire an exclusive lock.

-invalid — Verifies that the migration device pairs are in the Invalid state.

-migrated — Verifies that the migration device pairs are in the Migrated state.

-migrateinprog	Verifies that the migration device pairs are in the MigrateInProg state.
-name	Indicates a session name when establishing or validating a migration session.
-names	Indicates that the list command should only show the session names for active migration sessions.
-noprompt	Requests that no prompts are returned after the command is entered. The default is to prompt the user for confirmation.
-outfile	Specifies an output file for a command.
-pool	Specifies the name of the pool to be used as target space.
-sg	Applies a storage group name to the command to specify the source devices.
-sid	Identifies the Symmetrix ID for the operation.
-src_pool	Specifies the name of the source pool to be used for a Virtual Provisioning migration when moving allocations from a source pool to a target pool.
-summary	Displays a summary of the session information.
-synchronized	Verifies that the migration device pairs are in the Synchronized state.
-syncinprog	Verifies that the migration device pairs are in the SyncInProg state.
-tgt	Indicates that the control operation is targeted at the TGT devices in the device group.
-tgt_config	Indicates that the configured disk space will be used as the target disk space.
-tgt_dsk_grp	Specifies the target disk group number or name (when preceeded by 'name:').
-tgt_pool	Indicates that this is a Virtual Provisioning migration.
-tgt_prot	Specifies the number of disks used to provide RAID-5 or RAID-6 protection.
-tgt_raid1	Indicates that the target protection type is RAID-1.
-tgt_raid5	Indicates that the target protection type is RAID-5.
-tgt_raid6	Indicates that the target protection type is RAID-6.
-tgt_range	Applies a range of Symmetrix devices to a command to specify the target devices.

-tgt_unconfig Indicates that the unconfigured disk space will be used as the target disk space.

-tgt_unprotected
Indicates that the target protection type is unprotected.

-v Provides a more detailed, verbose listing.

PARAMETERS

Count The number of iterations to execute before exiting.

DgName The device group name.

DskGrp The disk group number.

DskGrpName The disk group name.

Interval Interval between polls, in seconds.

DevFile A text file that contains pairs of source and target devices. One pair per line in the file. If a line is too long, the line continuation character '\\' can be used.

Examples:

```
10     20
12     22
14     27
```

Or for a meta SRC device with non meta TGT devices:

```
15     28, 29, 30, 31, \\
       32, 33, 34, 39
```

SessionName A name for the migration session.

SgName The storage group name.

SrcFile A text file that contains only source devices. One device per line in the file.

Example:

```
9A
9E
9F
```

SymmID The 12-digit ID of the Symmetrix array.

RETURN CODES

Code #	Code Symbol
0	CLI_C_SUCCESS
1	CLI_C_FAIL
2	CLI_C_DB_FILE_IS_LOCKED
19	CLI_C_GK_IS_LOCKED

```
                All GateKeepers to the Symmetrix array are
                currently locked.

       23       CLI_C_NEED_SYMFORCE_TO_PROCEED
                Requires the symforce flag to proceed.
                WARNING: Extreme caution should be exercised
                when using this option.
```

EXAMPLES

To migrate data by specifying source and target device pairs in a file, first define a pair file, such as input.txt with the device pair information. Pair file input.txt contains the following information:

```
10      20
11      21
12      22
13      23
```

To migrate data in the input.txt file defined above, validate the pairs defined in the file prior to establishing the data. In this example, the target protection type is RAID-5 (3+1), and the target disk group number is 1. Enter the following:

```
symmigrate validate -sid 123 -f input.txt
  -tgt_raid5 -tgt_prot 3+1 -tgt_dsk_grp 1
  -name DevFile

symmigrate establish -sid 123 -f input.txt
  -tgt_raid5 -tgt_prot 3+1 -tgt_dsk_grp 1
  -name DevFile
```

To migrate data on source devices that are in a file (input.txt) to target devices that are configured, create the input.txt file, as follows:

Source file input.txt contains:

```
10
11
12
13
```

To migrate data in the input.txt file defined above, with the target protection type of RAID-5 (3+1) and the target disk group number 3, enter:

```
symmigrate establish -sid 123 -f input.txt
  -tgt_raid5 -tgt_prot 3+1 -tgt_dsk_grp 3
  -tgt_config -name DGTargets
```

To list all migration sessions established on one Symmetrix array, enter:

```
symmigrate list -sid 123 -detail
```

To monitor a migration session named ’mysession’, enter:

```
symmigrate query -name mysession -sid 123 -i 5 -c 2
```

```
symmir

        Performs Symmetrix BCV control operations on a
        device group, composite group, or on devices
        within a file.

        SYNOPSIS

             symmir -h

             symmir -g <DgName> [-v] [-force] [-symforce] [-reverse]
                  [-noprompt] [[-rdf] [-bcv] | [-rrbcv] | [-hop2]]
                  [-i <Interval>] [-c <Count>]
                  [-preserveTGTLocks -lockid <LockNum>] [-star]
                  [-preaction <ScriptFile>] [-postaction <ScriptFile>]

                establish [-full [-opt | -opt_rag | -exact]]
                  [-protbcvest] [-skip] [-concurrent]

                restore [-full [-exact] [-remote] [-bypass]]
                  [-not_ready] [-protect]

                split [-remote] [-bypass] [-not_ready] [-diff]
                  [-protect] [-std_protect] [-skip]
                  [-consistent [-both_sides]]

             symmir -g <DgName> [[-rdf] [-bcv] | [-rrbcv] | [-hop2]]
                  [-offline] [-i <Interval>] [-c <Count>]

                query [[-attach] [-multi] [-protect] [-protbcvest]
                  [-bg [-percent]]] | [-summary] [-mb | -gb | tb]

                verify [-synchronized | -restored [-protect] |
                  -split [-bg] | -syncinprog | -restinprog [-protect] ]
                  [-concurrent] [-force] [-summary]

                verify -bcv_mirrors [-ready | -syncinprog | -restinprog]
                  [-concurrent]

             symmir -g <DgName> [-v] [[-rdf] [-bcv] | [-rrbcv] |
                  [-hop2]] [-i <Interval>] [-c <Count>] [-noprompt]

                attach

                cancel [-force] [-skip] [-star]

                detach

             symmir -cg <CgName> [-v] [-force] [-symforce] [-reverse]
                  [-noprompt] [[-rdf] [-bcv] | [-rrbcv] | [-hop2]]
                  [-i <Interval>] [-c <Count>] [-star]
                  [-preaction <ScriptFile>] [-postaction <ScriptFile>]
                  [-sid <SymmID> |
                   -rdfg <SymmID>:<GrpNum>[,<GrpNum>,...]|<all>[,...] |
                   name:<RDFGroupName>[,<RDFGroupName>,...]]

                establish [-full [-opt | -opt_rag | -exact]]
                  [-protbcvest] [-skip] [-concurrent]

                restore [-full [-exact] [-remote] [-bypass]]
                  [-not_ready] [-protect]

                split [-remote] [-bypass] [-not_ready] [-diff]
                  [-protect] [-std_protect] [-skip]
                  [-consistent [-both_sides]]

             symmir -cg <CgName> [[-rdf] [-bcv] | [-rrbcv] | [-hop2]]
```

```
      [-offline] [-i <Interval>] [-c <Count>]
      [-sid <SymmID> |
      -rdfg <SymmID>:<GrpNum>[,<GrpNum>,...]|<all>[,...] |
       name:<RDFGroupName>[,<RDFGroupName>,...]]

    query [[-attach] [-multi] [-protect] [-protbcvest]
      [-bg [-percent]]] | [-summary] [-mb | -gb | tb]

    verify [-synchronized | -restored [-protect] |
      -split [-bg]| -syncinprog | -restinprog [-protect]]
      [-concurrent] [-force] [-summary]

    verify -bcv_mirrors [-ready | -syncinprog |
       -restinprog]
      [-concurrent]

 symmir -cg <CgName> [-v] [[-rdf] [-bcv] | [-rrbcv] |
      [-hop2]] [-i <Interval>] [-c <Count>] [-noprompt]
      [-sid <SymmID> |
       -rdfg <SymmID>:<GrpNum>[,<GrpNum>,...]|<all>[,...] |
       name:<RDFGroupName>[,<RDFGroupName>,...]]

    attach

    cancel [-force] [-skip] [-star]

    detach

 symmir -sid <SymmID>
      <-file <DeviceFileName> [-noprompt] |
       -noprompt 'redirect stdin'>
      [-v] [-force] [-symforce] [-reverse]
      [-i <Interval>] [-c <Count>] [-star]
      [-preaction <ScriptFile>] [-postaction <ScriptFile>]
      [-preserveTGTLocks -lockid <LockNum>]

    establish [-full] [-protbcvest] [-skip]

    restore [-full] [-bypass] [-not_ready] [-protect]

    split [-diff] [-bypass] [-not_ready] [-skip]
       [-protect] [-std_protect] [-consistent]

symmir -sid <SymmID>
       <-file <DeviceFileName> | 'redirect stdin'>
       [-i <Interval>] [-c <Count>]

    query [[-attach] [-multi] [-protect] [-protbcvest]
       [-bg [-percent]]] | [-summary] [-mb | -gb | tb]

    verify [-synchronized | -restored [-protect] |
       -split [-bg]| -syncinprog | -restinprog [-protect] ]
       [-concurrent] [-summary] [-force]

    verify -bcv_mirrors [-ready | -syncinprog |
       -restinprog] [-concurrent] [-force]

symmir [-sid <SymmID>] [-i <Interval>] [-c <Count>]
       [-offline] [-mb | -gb | tb]

    list

symmir -sid <SymmID>
        <-file <DeviceFileName> [-noprompt] |
         -noprompt 'redirect stdin'> [-v]
        [-i <Interval>] [-c <Count>]
```

attach

detach

cancel [-skip] [-star]

DESCRIPTION

The symmir command performs mirroring operations on a device group, composite group, or on devices in a file.

These operations include establishing (mirroring) the devices with BCV devices, splitting the device pairs, restoring the devices from the BCV devices, and querying the state of the device pairs.

You can perform both the establish and restore operations fully (entire copy) or incrementally (only changed tracks are synchronized). By default, if you do not specify the -full option, the system will attempt an incremental establish or restore. Note that you cannot perform an incremental establish or restore if the BCV pair state is Never Established.

You can perform all of these operations on a group or a device file.

Before you can establish a BCV device with a standard device, the BCV device must have been previously associated with the device group and the BCV device must be the same size as the standard device.

ARGUMENTS

attach — Attaches a BCV device to a standard device as the preferred BCV device to be paired with the standard device when a full establish or full restore action is issued.

cancel — Cancels the existing internal SDDF session between the specified standard and BCV device(s). Once the SDDF session is cancelled, the corresponding BCV device goes into the SplitNoInc state, and the BCV pair can no longer be incrementally established or restored.

detach — Detaches a BCV device from the standard device and disassociates (unmarks) the pair as the preferred pair when full establish or restore operations occur.

establish — Establishes (mirrors) all standard devices in a device group with one or more BCV devices associated with the group. Depending on whether the establish operation is full or incremental, all or only the changed tracks are internally copied to the BCV device.

While the operation is in progress, the state of the device pair is SyncInProg. When the operation completes, the state changes to Synchronized.

list — Lists all the BCV sessions created on the Symmetrix array.

query — Returns mirror state information about all device pairs in a group or device file.

restore — Restores all standard devices in a device group from one or more BCV devices associated with the group. Depending on whether the restore operation is full or incremental, all or only the changed tracks are internally copied to the standard device.

While the operation is in progress, the state of the device pair is RestInProg. When the operation completes, the state changes to Restored.

split — Splits all BCV devices from the mirror pair(s) in a device group. While the operation is in progress, the state of the device pair is SplitInProg. When the operation completes, the state changes to Split.

verify — Verifies whether all device pairs in a device group are in the Synchronized or Restored states.

KEYWORDS

name — Specifies to perform the action against the specified RDF group’s logical name.

OPTIONS

-attach — Alters the query to display BCV attachment information for the standard device(s) in the device group.

-bcv — Indicates that the BCV control operation is targeted at the remote mirror of a locally attached BCV RDF device and the remotely attached BCV device that is associated with the device group. You can only use this option with the -rdf option.

-bcv_mirrors — Verifies that the mirrors of the BCV device(s) are in the indicated state. The default is to verify that the mirrors of the BCV device(s) are in the synchronized state. Alternatively, if you also specify the -syncinprog flag or the -restinprog flag, then the system will verify the mirrors of the BCV device(s) against the state that corresponds to the specified flag. If you specify the -ready option, the system will verify that the mirrors are ready to the host.

-bg — Applies to query and verify operations. With query, shows the BCV pairs that are still in the background split mode. With verify, verifies that the BCV pair(s) are in the Split state, and that they have completed splitting in the background. With verify, you can only use this option with the -split option.

-both_sides	Splits all locally and remotely associated BCV pairs in an RDF group.
-bypass	Bypasses device reservations by other hosts.
-c	Specifies the number (count) of times to display or to acquire an exclusive lock on the Symmetrix host database. If you do not specify this option and specify an interval (-i), the program will loop continuously to display or to start the mirroring operation.
-cg	Applies a composite group name to the command.
-concurrent	When used with the verify argument, this option verifies the STD device and the two most recent BCVs. When used with the establish argument, this option establishes a second available BCV device.
-consistent	Consistently splits the managed standard devices.
-diff	Indicates that the split operation should initiate a differential data copy from the first (moving) mirror of the BCV device to the rest of the BCV mirrors when the split operation completes.
-exact	Specifies to pair devices in the exact order that the standard and BCV devices have been added to the device group. This option applies to full establish or full restore operations.
-file	Applies a device file to the command. The device file contains device pairs (by device number) listing a pair per each line (the source device first, a space, and the VDEV target device last within each line entry). A Symmetrix ID is required for this option. -f is synonymous with -file.
-force	Attempts to force the operation even though one or more paired devices in the device group may not be in the normal, expected state(s) for the specified operation.
-full	Requests a full establish or restore operation. The default is incremental.
-g	Applies a device group name to the command.
-gb	Displays counts in gigabytes.
-h	Provides brief online help information.
-hop2	Performs the specified action on the Symmetrix array two hops away.
-i	Specifies the repeat interval in seconds to display or to acquire an exclusive lock on the Symmetrix host database. The default interval is 30 seconds.

The minimum interval is 5 seconds. For passive actions the minimum interval is 15 seconds. Passive actions are actions that do not acquire an exclusive lock.

-lockid — Specifies the lock holder ID for preserving the target locks on the control operation.

-mb — Displays counts in megabytes.

-multi — Shows all BCVs that can be incrementally established/restored to/from the standard device. This option only applies to a query command. This feature is available with Enginuity 5x66 or later.

-noprompt — Requests to not return a prompt after you enter a command. The default is to prompt for confirmation.

-not_ready — Performs the BCV control operation but leaves the target device(s) Not Ready. That is, each BCV device will be set Not Ready when the split completes. And each standard device will be set Not Ready on the initiation of the restore operation. This option only applies to a restore or split command.

-offline — Specifies that the Symmetrix array data connection is offline from the host in-memory database.

-opt — Applies to the full establish operation that optimizes the standard/BCV pair selection to achieve the highest copy speed between them. (Basically, the device pair selection is such that they are not connected to the same disk adapters to distribute the I/O.) This option overrides all other pairing algorithms, which allows devices to be assigned as BCV pairs regardless of the previous pair assignment, since previously paired standard devices are normally paired with the same BCV devices. For remote BCV operations, use the -opt_rag option.

-opt_rag — Applies to the full establish operation for remote device optimization to distribute the I/O load in that the remote adapters are not connected to the same devices of the selected pair. Requires that you also specify the -rdf option.

-percent — When used with the -bg option in a query, this option specifies to show the percentage of progress in a background split operation.

-postaction — Specifies to execute the script argument after an establish, restore, or split operation. You must specify the full pathname of the script.

-preaction — Specifies to execute the script argument before an establish, restore, or split operation. You must specify the full

pathname of the script.

-preserveTGTLocks
Specifies to not take out device locks on the target devices. The target devices must already be locked by the same lock holder ID. Requires the -lockid option.

-protbcvest Specifies to move all mirrors of the BCV device to join the mirrors of the standard device. Applies only with an establish or query command. Applies to two-way mirrored BCV devices. For query actions, displays whether the BCV devices were established with the -protbcvest flag.

-protect Specifies to write protect the BCV before initiating the restore operation. Must be used with the split command when splitting devices that were restored protected. For query actions, displays whether the restored BCVs were restored with the protect option. When used with verify (with -restored or -restinprog), verifies that the protected restore operation is completed.

-rdf Indicates that the BCV control operation is targeted at the remote mirror and the remotely attached BCV device is associated with the device group.

-rdfg Performs the requested action on a subset of the composite group defined by one or more Symmetrix/RA group combinations supplied as the argument to -rdfg.

-ready Verifies that all the BCV mirrors are ready to the host. You can only specify this option with -bcv_mirrors. This option is useful after a reverse split to indicate the data on the BCV is available.

-remote Specifies to propagate the data to the remote mirror of the RDF device and to resume the RDF link if necessary. Applies only to a split of a BCV RDF1 device, or to a restore from a BCV to a STD RDF1 device. If you do not specify this flag, the default is to not propagate the data to the remote mirror of the RDF device.

-restinprog Verifies that the BCV pair(s) are in the RestInProg state.

-restored Verifies that the BCV pair(s) are in the Restored state.

-reverse Specifying this option with a split operation, indicates that the split operation should initiate a reverse data copy from the rest of the BCV mirrors to the first (moving) mirror of the BCV when the split operation completes.

Specifying this option with an establish or restore operation requests to verify

that the BCV's non-moving mirror has valid data so that the next BCV split can be a reverse split.

-rrbcv — Indicates that the BCV control operation is targeted at both the remote mirror of the remotely attached BCV device (RBCV) and the remotely attached remote BCV (RRBCV) device associated with the device group.

-sid — Applies the command to the specified Symmetrix ID. Specify this option with the -file option to select the Symmetrix array on which to perform the operation or specify this option with -cg option to restrict the operation to a single Symmetrix array.

-skip — Skips the source locks action. This option will not lock the source devices if all of the specified source devices are either locked or are unlocked.

-split — Verifies that the BCV pair(s) are in the Split state.

-star — Targets the action at devices in STAR mode.

-std_protect — Specifying this option with the Split command checks that the STD device has either mirror protection (aside from the BCV) or RAID-5 protection prior to starting the Split operation.

-summary — Shows device state summary.

-synchronized — Verifies that the BCV pair(s) are in the Synchronized state.

-syncinprog — Verifies that the BCV pair(s) are in the SyncInProg state.

-symforce — Forces the operation to execute when normally it is rejected. On a Split, it causes the Symmetrix array to split a synchronizing BCV pair. On an Establish or Restore, it inhibits SYMCLI from verifying whether there are invalid tracks at the source device. You should use extreme caution with this option.

-tb — Displays counts in terabytes.

-v — Provides a more detailed, verbose listing.

PARAMETERS

CgName — The composite group name.

DgName — The device group name.

DeviceFileName — The device Filename. The device file contains device pairs (SymDevNames) listing a pair each line (the source device first, a space, followed by the target device name on each line).

GrpNum — The RDF (RA) group number.

LockNum — The hexadecimal value of the lock holder ID.

Name — The logical name associated with the RDF (RA) group(s).

ScriptFile — The full pathname of a script file to execute.

RETURN CODES

Code #	Code Symbol
0	CLI_C_SUCCESS
1	CLI_C_FAIL
2	CLI_C_DB_FILE_IS_LOCKED
18	CLI_C_ALREADY_IN_STATE The device or device group is already in the desired BCV state. Applies only to BCV control actions.
19	CLI_C_GK_IS_LOCKED All GateKeepers to the Symmetrix array are currently locked.
22	CLI_C_NEED_FORCE_TO_PROCEED Requires the force flag to proceed.
23	CLI_C_NEED_SYMFORCE_TO_PROCEED Requires the symforce flag to proceed. WARNING: Extreme caution should be exercised when using this option.
77	CLI_C_CONSISTENCY_TIMEOUT The operation failed because of a consistency window timeout.

Return codes for symmir establish

47	CLI_C_WONT_REVERSE_SPLIT There are either Write Pending I/O in the Symmetrix cache for a BCV device, which will prevent the establish action from starting or the mirrors of the BCV are not synchronized. You can automatically repeat the action using the -i and/or -c flags.

Return codes for symmir restore

20	CLI_C_WP_TRACKS_IN_CACHE There are write pending I/O in the Symmetrix cache for a standard device, which will prevent the restore action from starting. You can automatically repeat the action using the -i and/or -c flags

47 CLI_C_WONT_REVERSE_SPLIT
There is either write pending I/O in the Symmetrix cache for a BCV device, which will prevent the restore action from starting or the mirrors of the BCV are not synchronized. You can automatically repeat the action using the -i and/or -c flags.

Return codes for symmir verify

4 CLI_C_NOT_ALL_SYNCHRONIZED
Not all standard devices are in the Synchronized or the Restored states.

5 CLI_C_NONE_SYNCHRONIZED
No standard devices are in the Synchronized or the Restored states.

10 CLI_C_NOT_ALL_SYNCHED
Not all standard devices are in the Synchronized state.

11 CLI_C_NONE_SYNCHED
No standard devices are in the Synchronized state.

12 CLI_C_NOT_ALL_RESTORED
Not all standard devices are in the Restored state.

13 CLI_C_NONE_RESTORED
No standard devices are in the Restored state.

25 CLI_C_NOT_ALL_SPLIT
Not all standard devices are in the Split state.

26 CLI_C_NONE_SPLIT
No standard devices are in the Split state.

27 CLI_C_NOT_ALL_SYNCINPROG
Not all standard devices are in the SyncInProg state.

28 CLI_C_NONE_SYNCINPROG
No standard devices are in the SyncInProg state.

29 CLI_C_NOT_ALL_RESTINPROG
Not all standard devices are in the RestInProg state.

30 CLI_C_NONE_RESTINPROG
No standard devices are in the RestInProg state.

EXAMPLES

To create the device group ProdDB as a REGULAR device

group, enter:

```
symdg create ProdDB
```

To define the device group ProdDB as the default device group, enter:

```
setenv SYMCLI_DG ProdDB
```

To establish standard devices in group ProdDB with BCV devices (associated with the group), enter:

```
symmir -full establish -g ProdDB
```

To wait until the BCV pairs are fully synchronized, polling every 30 seconds, enter:

```
symmir -i 30 verify -g ProdDB
```

To split all established devices in group ProdDB and skip any devices already split, enter:

```
symmir split
```

To perform an incremental restore onto standard devices in group ProdDB from its paired BCV devices, enter:

```
symmir restore -g ProdDB
```

To query information about all paired devices in device group ProdDB, enter:

```
symmir query
```

symntctl

Implements the functionality available in the Symmetrix Integration Utility (SIU) on a Windows platform.

SIU consists of a number of software components that provide tools to augment the storage functionality of the Windows operating environment.

SYNOPSIS

```
symntctl -h

symntctl [-output xml_attribute | xml_element]

   list -datastore

   list -disk [-v]

   list -volume [-v]

   list -signature

   show -datastore DatastoreName

   show -pd diskN [-signature]

   show -vol VolName [-g VolGroup]

   show -guid VolGuid

   show -sid SymId -symdev SymDev [-signature]

   show -drive DriveLetter

   show -path MountPnt

symntctl

   flush -all

   flush -drive DriveLetter

   flush -path MountPnt

   flush -vol VolName [-g VolGroup]

   flush -guid VolGuid

symntctl [-drive DriveLetter] | [-path MountPnt]

   mount -vol VolName [-g VolGroup]

   mount -guid VolGuid

   mount -sid SymId -symdev SymDev [-part PartitionNum]

   mount -pd Pdev [-part PartitionNum]

symntctl <force>

   umount -drive Driveletter

   umount -path MountPnt
```

```
      umount -vol VolName [-g VolGroup]

      umount -guid VolGuid

      umount -sid SymId -symdev SymDev [-part PartitionNum]

      umount -pd Pdev [-part PartitionNum]

   symntctl [-sig Signature] | [-erase] | [-initialize]

      signature -pd diskN

      signature -sid SymId -symdev SymDev

      signature -initialize

   symntctl rescan

   symntctl

      update -all

      update -sid SymId -symdev SymDev

      update -pd Pdev

   symntctl

      mask -sid <SymmID> -symdev <SymDev> |
        <SymDevName>,<SymDev> |
        <SymDevStart>:<SymDevEnd>
        [-wwn <wwn> |-iscsi <iscsi>] [-no_refresh |
        -no_rescan | -no_discover] [-force]

      mask -pd <PdevName> [-wwn <wwn> |-iscsi <iscsi>]
        [-no_refresh | -no_rescan | -no_discover]
        [-force]

   symntctl

      unmask -sid <SymmID> -symdev <SymDev> |
        <SymDevName>,<SymDev> |
        <SymDevStart>:<SymDevEnd>
        [-wwn <wwn> |-iscsi <iscsi>]
        [-no_refresh | -no_rescan | -no_discover]

   symntctl [-set Flag] | [-clear Flag <-all>]

      flag -vol VolName [-g VolGroup]

      flag -guid VolGuid

      flag -sid SymId -symdev SymDev [-part PartitionNum]

      flag -pd Pdev [-part PartitionNum]

      flag -drive DriveLetter

      flag -path MountPnt
```

DESCRIPTION

This command integrates and extends the Windows resource management functionality so that it operates more effectively with and on the Symmetrix Business Continuance storage devices.

ARGUMENTS

flag	Sets and clears volume flags.
flush	Flushes to disk all pending unwritten file system entries in cache.
list	Lists all visible physical devices or volumes.
mask	Removes access to the specified Symmetrix device on all HBAs of the host where the specified device is visible.
mount	Mounts the specified volume to the specified drive letter or mount point.
rescan	Scans the drive connections and discovers any new disks available to the system.
show	Displays single disk or volume details.
signature	Manipulates disk signatures.
umount	Unmounts a volume from the drive letter and all mount points. The unmount process flushes any pending writes to the volume and takes the volume offline.
unmask	Adds access to the specified Symmetrix device on all HBAs of the host that is connected to the specified Symmetrix array.
update	Updates the partition table on a disk.

OPTIONS

-all	Performs the action on all disks or volumes.
-clear	Clears a volume flag (READONLY, HIDDEN, or NO_DEFAULT_DRIVE_LETTER).
-datastore	Confines the list action to just displaying VMware datastores in the environment. Specifies a datastore name as the target of the show action.
-disk	Confines the list action to just displaying disk configuration across the system.
-drive	Specifies a drive as the target for the specified action.
-erase	Erases the signature of the specified disk.
-force	Requests that a volume’s open handles be ignored during the umount. These open handles will be broken.
-g	Specifies a volume group as the target for the specified action.
-guid	Specifies a volume’s global unique identifier as the target for the specified action.

-initialize	Assigns a signature to any disks found without a signature.
-iscsi	Specifies the iSCSI initiator name.
-no_discover	Prevents disk discovery during masking/ unmasking actions.
-no_refresh	Prevents VCM database update, disk rescan, and discovery during masking/unmasking actions.
-no_rescan	Prevents disk rescan and discovery during masking/unmasking actions.
-path	Specifies a device mount point as the target to the specified action. This directory path (of the form device:\dir) must be empty.
-pd	Specifies a physical disk as the target for the specified action. Note that the host-visible disk identifiers of the form diskN can be viewed with the list -physical command.
-part	Specifies the partition number of a disk.
-set	Sets a volume flag (READONLY, HIDDEN, or NO_DEFAULT_DRIVE_LETTER).
-sid	Specifies the ID of the Symmetrix array containing the device or volume on which the action is to be performed.
-sig	Writes the signature passed in -sig to the specified disk.
-signature	Specifies a disk signature.
-symdev	Specifies the ID of the Symmetrix device on which the action is to be performed.
-vol	Specifies a volume name as the target for the specified action.
-volume	Confines the list action to just displaying volume configuration across the system.
-wwn	Specifies the World Wide Name of the HBA.

PARAMETERS

diskN	The number of the physical disk in the system of the form diskN.
DriveLetter	The drive letter designation as viewed in the Windows directories (such as E:).
iSCSI	The iSCSI initiator.
MountPnt	A device mount point or directory path (for example, device:\dir).
PartitionNum	The partition number (greater than, or equal to one) on the Symmetrix device that is to be mounted. If omitted, this value defaults to one.

Signature	An 8-digit hex number for each disk.
SymmID	A 2 to 12-digit Symmetrix ID.
SymDev	A 3 to 5-digit Symmetrix device ID.
VolGroup	The name of the volume group.
VolName	The volume name of the device.
VolGuid	The global unique identifier (128-bit integer) of a volume.
WWN	The HBA world wide name.

RETURN CODES

Code #	Description
0	Command is successful.
1	Command is not successful.
2	Applies to the openhandle command and indicates there is at least one open handle on the volume.

EXAMPLES

To flush buffers on Drive E, enter:

```
symntctl flush -drive E:
```

To list device information and configuration of disk2, enter:

```
symntctl show -pd disk2
```

To change a disk’s signature, enter:

```
symntctl signature -pd disk2 -sig 1234ABCD
```

To mount a volume, enter:

```
symntctl mount -drive W: -vol volume2
```

or:

```
symntctl mount -path C:\\mt -vol volume2 -dg Api172Dg0
```

To unmount a volume, enter:

```
symntctl umount -drive W:
```

To remove a Symmetrix device from the host, enter:

```
symntctl mask -pd disk4
```

or:

```
symntctl mask -sid 000190300186 -symdev 3ED
```

To present a Symmetrix device to the host, enter:

```
symntctl unmask -sid 000190300186 -symdev 3ED
```

symoptmz

Displays or sets parameters that control the behavior of Symmetrix Optimizer and displays the current status of Symmetrix Optimizer.

SYNOPSIS

```
symoptmz -h

symoptmz -sid <SymmID>

   enable

   disable

   clear_stats

   query [-v]

   sync -version

symoptmz -sid <SymmID>

   list [-range <[SymDevStart]:[SymDevEnd]>]
     [-n <NumDevs>]

   show [-v] -swap_list | -activity_list
     [-manual | -generated ]

   show -parms [-dp | -vp] [-offline]

   show -composite [-dp | -vp]

   show [-v]  -swap_hist | -rollback_list

   read -log_type <RUNTIME | ERROR>
     [-start <DateTime>] [-stop <DateTime>]

   release

symoptmz -sid <SymmID>
   <-file <CommandFile> |'redirect stdin'> [-v |-noecho]

   preview

   prepare

   commit
```

DESCRIPTION

The symoptmz command can be used to display and set the values of control parameters for Symmetrix Optimizer.

For every invocation, symoptmz implicitly starts an Optimizer API session. It closes the session just before exiting. For actions that update Optimizer, symoptmz will implicitly acquire an Optimizer API Lock. It releases the lock just before exiting.

Stdin redirection can be used with "here documents" in UNIX shell scripts. Each command in the file has to be terminated by a semi-colon (;). There is no limit on the number of commands or the type of commands that can be placed in a command file. All the commands in a command file are executed in a single Optimizer session.

The commands in the command file are not case sensitive, however, the parameters entered are case sensitive.

Prior to making any changes, the preview argument can be used to verify that the command file is syntactically correct.

The prepare argument will perform some range checks, in addition to the syntax checks of preview. Note that these changes are NOT submitted to the Symmetrix Optimizer.

In addition to performing the steps of preview and prepare, the commit argument will update the Symmetrix Optimizer with the modified parameters.

Users are highly encouraged to first run PREVIEW on their command files and ensure that there are no syntax errors. After the syntax errors are corrected, run the command files through PREPARE. This stage will perform some range checks on the supplied values. After the command file passes PREPARE, they may run COMMIT.

Symmetrix arrays running Enginuity 5874 only require the COMMIT action.

ARGUMENTS

clear_stats	Clears disk statistics maintained by the Optimizer.
commit	Updates Optimizer with the changes defined in the command file.
disable	Disables the Optimizer algorithm processing.
enable	Enables the Optimizer algorithm processing.
list	Displays Optimizer specific attributes of Symmetrix devices.
prepare	Performs extra range checks on changes specified in the command file.
preview	Verifies the syntax of the changes specified in the command file.
query	Queries Symmetrix Optimizer and displays the current state and version information of the Optimizer. If -v is specified, displays additional information about Optimizer version and open Optimizer API sessions.
read	Reads from Optimizer’s log file.
release	Attempts to gain control of an existing Optimizer API session to abort it and release the Optimizer API lock. Not supported in Optimizer Rev 8.1.
rollback	Rolls back a Symmetrix configuration to that of a previous point-in-time.
show	Shows information about the current Optimizer parameters. By default, displays

control parameters.

sync — Acquires version information from Optimizer. This argument is supported ONLY in conjunction with -version.

OPTIONS

-activity_list — Displays all the swap lists and migrations currently known to Optimizer (an alias for -swap_list).

-composite — Displays composite time windows. Composite time windows are generated by Optimizer by combining all known user-defined time windows.

-dp — Specifies the data movement time windows for standard devices.

-file — Specifies the name of the command file containing changes to Optimizer.

-generated — Specifies the Optimizer generated device swap plan.

-h — Provides brief online help.

-log_type — Defines the type of log file to be read. Types of log files supported are: RUNTIME and ERROR.

-manual — Specifies the user defined device swap plan.

-n — Specifies the number of devices to list or set.

-noecho — Blocks the printing of session status and progress messages during the Optimizer change session. For use with PREVIEW, PREPARE, and COMMIT actions. Cannot be used with the -v option.

-offline — Displays information from Symmetrix configuration database without refreshing the data from the Symmetrix array.

-parms — Displays information about the control parameters of Symmetrix Optimizer. This is the default option for show.

-range — Specifies the start and end Symmetrix device names.

-rollback_list — Displays a list of possible rollback points.

-sid — Specifies the unique Symmetrix ID.

-start — Identifies the log file entries whose time stamp is after this date and time that will be read.

-stop — Identifies the log file entries whose time stamp is before this date and time that will be read.

-swap_hist Displays the history of the swaps known to Optimizer.

-swap_list Displays all the swap lists and migrations currently known to Optimizer.

-v Provides a more detailed, verbose listing.

-version Displays version information of Symmetrix Optimizer. The symoptmz command queries Optimizer for version information and caches it in the SYMAPI database. Subsequent symoptmz calls with -version would read the version information from the SYMAPI database for efficiency. If sync is specified with -version, any cached information from SYMAPI database is cleared and symoptmz re-queries Optimizer.

-vp Specifies the data movement time windows for thin devices.

PARAMETERS

CommandFile The name of an ASCII text file containing the set of commands to process.

DateTime A specific date and time (MMDDYYYY:HHMMSS format).

ERROR The Optimizer error log.

NumDevs The number of devices to display or set.

RUNTIME The Optimizer activity log.

SymmID The 12-digit ID of the Symmetrix array.

SymDevEnd The last Symmetrix device name in a range.

SymDevStart The first Symmetrix device name in a range.

COMMAND FILE SYNTAX

Syntax of commands allowed in <command_file> are:

For setting control parameters:

```
set control_parms
                [swap_mode=<AUTO | USER_OK>,]
                [min_perf_period=<min_perf>,]
                [workload_period=<workload>,]
                [max_simult_swaps=<max_simult>,]
                [swap_rate=<max_swaps>];
```

For setting time windows:

```
set time_window id=<tw_id>,
                type=<SWAP [, provisioning=<VP | DP>]
                | PERF>,
                flag=<INCLUDE | EXCLUDE>,
                period=<ONCE | WEEKLY | WEEKLY_BY_DAY>,
                starting=<date_time>,
                ending=<date_time>,
                [days=<day_list>,
                 start_time=<hh:mm>, end_time=<hh:mm>];
```

Where <date_time> is in the form of MMDDYYYY:HHMMSS, and <day_list> is any comma-separated combination of MON, TUE, WED, THU, FRI, SAT, or SUN. For the case of WEEKLY, <day_list> should also include one of the following: MON_START, TUE_START, WED_START, THU_START, FRI_START, SAT_START, or SUN_START. Each of "*_START" represents the corresponding day of the week on which the time window starts. Refer to the example at the end of this man page.

Provisioning type can be specified only for swap time windows with Enginuity Version 5875 and higher.

For clearing time windows:

```
    clear time_window;
```

For setting manual swap lists with Enginuity version 5773 and earlier:

```
    set swap_list {Hyper1} with {Hyper2}
                    [, {Hyper3} with {Hyper4},... ]
                    [begin_at=<time_val>];
```

{HyperN} is of the form {DDD,I,T,HH} where:

- DDD is the director Identifier,
- I is the Director Interface,
- T is the Target ID, and
- HH is the Hyper Number.

time_val is in the form of MMDDYYYY:HHMMSS.

For setting manual swap lists with Enginuity version 5874 and higher:

```
    set dev_swap <SymDevName1> with <SymDevName2>
            [, <SymDevName3> with <SymDevName4>, ...]
            [begin_at=<time_val>];
```

time_val is in the form of MMDDYYYY:HHMMSS.

For managing swap lists:

```
    set swap <APPROVE | DECLINE>
                     [begin_at=<time_val>,]
                     TIMESTAMP=<time_val>
                     [,ROLLBACK];
```

Where time_val is in the form of MMDDYYYY:HHMMSS.

For setting swap priority:

```
    set swap_priority TO <NO_SWAP | NORMAL | HIGH> for
                       dev <SymDevStart>[:<SymDevEnd>];
```

For setting advanced parameters:

```
    set advanced_parms [max_rollback = <max_days>,]
                       [hot_spot = <TRUE | FALSE>];
```

The migration feature is not available with Enginuity 5874 and higher. Use symmigrate to migrate devices.

For initiating a migration with Enginuity 5773 and earlier:

```
    migrate
```

```
          dev[s] <SymDevStart1>[:<SymDevEnd1>]
            [,<SymDevStart2>[:<SymDevStart2>],...]
          TO disk[s] {disk1} [,{disk2},...]
          [unmapped=TRUE] [unmasked=TRUE]
          [begin_at=<time_val>];

        migrate
          device_group <DgName>
          TO disk_group_num <disk_group_num>
          [unmapped=TRUE] [unmasked=TRUE]
          [begin_at=<time_val>];

        migrate
          device_group <DgName>
          TO disk[s] {disk1} [,{disk2},...]
          [unmapped=TRUE] [unmasked=TRUE]
          [begin_at=<time_val>];

        migrate
          dev[s] <SymDevStart1>[:<SymDevStart1>]
            [,<SymDevStart2>[:<SymDevStart2>],...]
          TO disk_group_num <disk_grou_num>
          [unmapped=TRUE] [unmasked=TRUE]
          [begin_at=<time_val>];

        {diskN} is of the form {DDD,I,T} where
            DDD is the director Identifier,
            I   is the Director Interface, and
            T   is the Target ID

    time_val is in the form of MMDDYYYY:HHMMSS.

    See the Symmetrix Optimizer documentation for a
    description of the parameters and their legal values.

    While setting time windows, note that the time windows
    defined in a single command file are treated as a set and
    replace the current time window definitions in the
    Optimizer. They are not treated as additions to the
    current definitions. Swap time windows and performance
    time windows are treated as one set.

RETURN CODES

     Code #     Code Symbol
     ------     -----------

       0        CLI_C_SUCCESS

       1        CLI_C_FAIL

       2        CLI_C_DB_FILE_IS_LOCKED

      19        CLI_C_GK_IS_LOCKED
                All gatekeepers to the Symmetrix
                array are currently locked.

EXAMPLES

     To display the current Optimizer control parameters for
     a specific Symmetrix array, enter:

        symoptmz -sid 123456789012 show -parms

     To list the swap priorities of all Symmetrix devices
     that are configured in Symmetrix arrays connected to
     this host, enter:
```

```
   symoptmz list -sid 123456789012
```

To display the current Optimizer swap lists for a specific Symmetrix array, enter:

```
   symoptmz show -sid 123456789012 -swap_list
```

To define a sets of time windows, enter:

```
   symoptmz commit -sid 123456789012
     -file define_timewindow.cmd
```

Where define_timewindow.cmd contains:

```
    set time_window id =MySwapTimeWondow type=swap
      flag=include period=once
      starting=04152009:00000
      ending=12302009:000000
      days=Sun start_time=04:00 end_time=12:00;
    set time_window id = MyPerfTimeWindow type=perf
      flag=exclude period=WEEKLY_BY_DAY
      starting=01142009:000000
      ending=12302009:000000
      days=SAT start_time=04:00 end_time=12:00;
```

To set the control parameters, enter:

```
   symoptmz commit -sid 123456789012
     -file set_cntrlparms.cmd
```

Where set_cntrlparms.cmd contains:

```
   set control_parms swap_mode=auto
     swap_rate=30 max_simult_swaps=32
     min_perf_period=200 workload_period=300;
```

To set the swap priority of a range of Symmetrix devices, enter:

```
   symoptmz commit -sid 123456789012
     -file set_swap_priority.cmd
```

Where set_swap_priority contains:

```
   set swap_priority to high for dev 0078:0092;
```

To swap twp devices with the same size, enter:

```
   symoptmz commit -sid 123456789012
     -file swap_device.cmd
```

Where swap_device.cmd contains:

```
   set dev_swap 00C1 with 00C2;
```

sympart

Displays partition information about a host device.

SYNOPSIS

```
sympart [-h] [-v] [-kb|-blocks|-mb]

        show <HostDevName>

        list [-count FirstNMatches]
             [-device_type DeviceType]
             [-label PartLabel] [-type PartType]
             [-exclude]
```

DESCRIPTION

The sympart command displays detailed information about the geometry and layout of a host device. This includes information specific to the partitions of the device such as the partition name, type, attributes, offset into the full device, and the size of the partition.

ARGUMENTS

list	Lists partition information of all devices connected to the host. List can be filtered using filter options.
show	Shows detailed partition information for the specified host device.

OPTIONS

-blocks	Displays the partition offset and size information in 512-byte blocks.
-count	Lists only first N matches found.
-device_type	Lists partition information for only the specified device type.
-exclude	Excludes partitions meeting filter criteria and lists others.
-h	Provides brief, online help information.
-kb	Displays the offset and size information in kilobytes.
-label	Lists information of devices with the specified partition label.
-mb	Displays offset and size information in megabytes. This is the default.
-type	Lists information of partitions with the specified partition type.
-v	Displays partition data in an expanded (verbose) format.

PARAMETERS

DeviceType — The device type.
[SYMMETRIX | CLARIION]

FirstNMatches — The number of partitions, such as 10.

HostDevName — The physical (host) device name, such as c2t0d2s2.

PartLabel — The partition label of the physical device.
[MBR | GPT | VTOC | EFI]

PartType — The partition type:

[PART_UNASSIGNED	PART_BOOT
PART_ROOT	PART_SWAP
PART_USR	PART_FULL_DISK
PART_STAND	PART_VAR
PART_HOME	PART_ALT_SECTOR
PART_CACHE	PART_RESERVED
PART_VX_PUBLIC	PART_VX_PRIVAT
PART_EXTENDED	PART_FAT12
PART_FAT16	PART_FAT32
PART_FAT32_X13	PART_HUGE
PART_IFS	PART_LDM
PART_NTFT	PART_OS2BOOT
PART_PREP	PART_UNIX
PART_XENIX_1	PART_XINT13
PART_XINT13_EX	PART_VALID_NTFT
PART_UNUSED	PART_XENIX_2
PART_VERSION6	PART_SYSTEM_V
PART_VERSION8	PART_VERSION7
PART_BSD_4_1	PART_BSD_4_2
PART_ADVFS	PART_LSMPUBLIC
PART_LSMPRIVAT	PART_LSMSIMPLE
PART_LSMNOPRIV	PART_DATABASE
PART_RAWDATA	PART_DRD
PART_CNX	PART_CDFS
PART_LINUX	PART_LINUX_LVM]

RETURN CODES

Code #	Code Symbol
0	CLI_C_SUCCESS
1	CLI_C_FAIL

EXAMPLES

To display partition information in kilobytes about a physical device on a SunOS system, enter:

```
sympart -kb show c2t0d2s2
```

To display partition information in blocks about a physical device on a Windows system, enter:

```
sympart -blocks show PHYSICALDRIVE9
```

To list all of the partitions of the devices connected to the host, enter:

```
sympart list
```

To list all of the partitions of the Symmetrix devices connected to the host, with the size and offset displayed in kilobytes, enter:

```
   sympart list -kb -device_type symmetrix
```

To list all of the partitions of the devices with the partition style GPT on a Windows system, enter:

```
   sympart list -label GPT
```

To list all of the partitions of the devices with the partition style GPT or MBR on Windows systems, enter:

```
   sympart list -label "GPT|MBR"
```

To list all of the partitions of the devices with the partition style EFI on SunOS system, enter:

```
   sympart list -label EFI
```

sympd

Performs operations on a device given the device’s physical (host) name.

SYNOPSIS

```
sympd  -h

sympd [-offline] [-sid <SymmID>] [-v]

   list [-resv]

   list [-SA <#|ALL>] [-p <#>] [-fibre]
     [-escon] [-ficon] [-gige [ -iscsi_port <#>] ]
     [-powerpath] [-vcm | -aclx] [-pdevfile] [-cyl]

sympd [-offline] [-sid <SymmID>] [-v]

   list [-DA <#|ALL>] [-interface <#|ALL>]
     [-disk <#|ALL>] [-hyper <#|ALL>]
     [-spindle]

   list [-DX <#|ALL] [-spindle]

   list [-vm]

sympd show <PdevName> [-geometry]

sympd

   export -file <FileName>

   verify -file <FileName> [-v]
```

DESCRIPTION

The sympd command performs operations on a device given the device’s physical name, such as listing devices by Symmetrix ID, and showing device information given the device’s physical name, such as /dev/rdsk/c3t0d3s2.

The export option writes the current list of physical device names visible to the host to a file. It stores other information about the physical device, such as its Symmetrix ID, device number, and director/port.

The verify option compares the current list of physical device names to the ones stored in the specified file for any differences.

ARGUMENTS

export	Stores the physical device information to a specified file.
list	Lists all Symmetrix devices visible to this host.
show	Shows the status information about a Symmetrix device that is visible to this host.
verify	Compares the current physical device information to the information stored in a specified file.

OPTIONS

-aclx	Lists the device masking (ACLX) devices. This option is supported in Enginuity 5874 and higher.
-cyl	Lists the device capacity in cylinders. The default is in megabytes (MBs).
-DA	Lists the host-visible Symmetrix devices that match DA, interface, disk, and hyper values. Interface, disk, and hyper values default to ALL if not specified.
-disk	Lists the host-visible Symmetrix devices that match disk, DA, interface, and hyper values.
-DX	Lists the host-visible Symmetrix devices that match the DX director number. Default to ALL if not specified.
-escon	Lists the devices mapped to the ESCON front-end directors.
-fibre	Lists the devices mapped to the Fibre front-end directors.
-ficon	Lists the devices mapped to the FICON front-end directors.
-file	Names the file to store or compare physical device information.
-geometry	Shows device geometry feature.
-gige	Lists the devices mapped to the Gig-E front-end directors.
-h	Provides brief online help information.
-hyper	Lists the host-visible Symmetrix devices that match hyper, DA, interface, and disk values.
-interface	Lists the host-visible Symmetrix devices that match interface, DA, disk, and hyper values.
-iscsi_port	Specifies the iSCSI target port number.
-offline	Obtains information from the Symmetrix host configuration database.
-p	Specifies a front-end (Fibre) director port number.
-pdevfile	Lists the device names in a format for use as pdevfile entries.
-powerpath	Lists the host-visible EMC PowerPath devices, their Symmetrix configurations, and their alternate paths.
-resv	Lists all Symmetrix devices that are visible to this host and have SCSI reservations.

-SA Lists the front-end (Fibre) director number.

-sid Filters the display of visible Symmetrix devices by the specified Symmetrix array.

-spindle Displays spindle information instead of the standard disk address information.

-v Provides a more detailed, verbose listing.

-vcm Lists the device masking (VCM) devices. This option is obsolete in Enginuity 5874 and higher and replaced by the -aclx option.

-vm Displays the virtual machine device name. This option only displays valid virtual machine names on VMWare ESX server environments.

PARAMETERS

FileName The name of the file containing the list of devices.

PdevName The host name for the device, such as /dev/rdsk/c2t0d2s3.

RETURN CODES

Code #	Code Symbol
0	CLI_C_SUCCESS
1	CLI_C_FAIL
19	CLI_C_GK_IS_LOCKED All GateKeepers to the Symmetrix array are currently locked.

EXAMPLES

To list all Symmetrix devices visible to this host, enter:

```
sympd list
```

To show detailed information about a Symmetrix device, enter:

```
sympd show /dev/rdsk/c2t0d2s3
```

To list only Symmetrix devices visible to this host that are mapped to all Fibre Channel directors on port 0, enter:

```
sympd -sa all -p 0 -fibre list
```

symppath
- Displays or updates Symmetrix configuration information related to PowerPath.

It can be used to:

- Enable or disable PowerPath attributes, such as initiator or host registration.
- Display PowerPath host registration information.
- Trigger a PowerPath device/path scan on a host, even if the host is not running Solutions Enabler.

SYNOPSIS

```
symppath -h

symppath [-sid <SymmID>] [-v]

   list -ppreg [-host <Hostname> | -wwn <WWN>]

symppath -sid <SymmID>

   enable   -initiator_registration
   disable  -initiator_registration

   enable   -host_registration
   disable  -host_registration

   scan -host <Hostname[,Hostname,...]>
```

DESCRIPTION

The symppath command sets or displays the PowerPath configuration for a Symmetrix.

ARGUMENTS

disable	Disables PowerPath initiator and host registration on the array.
enable	Enables PowerPath initiator and host registration on the array.
list	Lists brief or detailed PowerPath host registration information.
scan	Instructs the array to tell PowerPath running on the specified hosts to perform a new device/path scan.

OPTIONS

-h	Provides brief online help information.
-host	Lists only application information for the specified host or specifies a host name during authorization actions. When used with "scan," specifies a list of one or more hosts on which PowerPath will be instructed to scan for new devices/paths.

-host_registration
Specifies that the "host registration" array attribute be enabled or disabled.

-initiator_registration
Specifies that the "initiator registration" array attribute be enabled or disabled.

-ppreg Displays host registration information.

-sid Specifies the unique Symmetrix ID.

-v Provides a more detailed, verbose listing.

-wwn Displays host registration information for the specified WWN.

PARAMETERS

HostName The host name.

SymmID The 12-digit ID of the Symmetrix array.

wwn The World Wide Name used to filter PowerPath host registration records.

RETURN CODES

Code #	Code Symbol
0	CLI_C_SUCCESS
1	CLI_C_FAIL

EXAMPLES

symqos

Provides Quality of Service controls on specified devices.

The maximum Open Replication bandwidth percentages for selected ports can also be viewed or changed with symqos.

SYNOPSIS

```
symqos -h

symqos -g <DgName>

   set <RDF|MIR|CLONE|VLUN>
     <pace|priority <URGENT|STOP|<Value>>>

   query

   list

symqos <-sg <SgName> |
        -devs <<SymDevStart>:<SymDevEnd> | <SymDevName>
             [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>
              . . .]>>

   set <RDF|MIR|CLONE|VLUN>
       <pace|priority <URGENT|STOP|<Value>>>
       -sid <SymmID>

   list [-sid <SymmID>]

symqos [-sid <SymmID>]

   list
     [-devs <<SymDevStart>:<SymDevEnd> | <SymDevName>
        [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]>]
     [-pace_range <[MinPace]:[MaxPace]>]

symqos -pst -sid <SymmID>

   enable

   disable

symqos -pst -sid <SymmID>
  -devs <<SymDevStart>:<SymDevEnd> | <SymDevName>
     [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]>

   set hostio priority <Value>

symqos -pst -g <DgName> [-std]

   set hostio priority <Value>

symqos -pst

   list -sid <SymmID>
        [-all | -hostio_priority <<Value>|ALL>]
        [-devs <<SymDevStart>:<SymDevEnd> | <SymDevName>
           [,<<SymDevStart>:<SymDevEnd> |
             <SymDevName>>...]>]

   list -g <DgName> [-std]
        [-hostio_priority <<Value>|ALL>]
```

```
symqos -cp -sid <SymmID>

   enable

   disable

   analyze

   set empty <preserve | remove>

symqos -cp -name <Name> -sid <SymmID>

   create -target <TargetPercent>
          -min <MinPercent>
          -max <MaxPercent>
          -wp <WritePendingLimit>
          -time <DonationTime>

   modify [-target <TargetPercent>]
          [-min <MinPercent>]
          [-max <MaxPercent>]
          [-wp <WritePendingLimit>]
          [-time <DonationTime>]

   rename -new_name <NewCPName>

   delete

symqos -cp -name <CPName> -sid <SymmID>

   add dev <SymDevName>

   remove dev <SymDevName>

symqos -cp -name <CPName> -sid <SymmID> [-force]
   [-devs <<SymDevStart>:<SymDevEnd> | <SymDevName>
      [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]> |
    -rdfg <GrpNum> |
    -pool <PoolName> <-snap | -rdfa_dse | -thin>]

   addall

   rmall

symqos -cp -name <CPName> -sg <SgName>

   addall

   rmall

symqos -cp -name <CPName> -g <DgName>

   addall [-std] [-vdev]

   rmall [-std] [-vdev]

symqos -cp [-name <CPName>] [-offline]

   list -sid <SymmID>
     [-devs <<SymDevStart>:<SymDevEnd> | <SymDevName>
        [,<<SymDevStart>:<SymDevEnd> |
          <SymDevName>>...]> |
      -rdfg <GrpNum>]

   list [-sid <SymmID>] -sg <SgName>

   list -g <DgName> [-std] [-vdev]
```

```
   list [-sid <SymmID>] -settings [-usage] [-v]

symqos -cp -sid <SymmID> [-offline]

   list [-all]
     [-devs <<SymDevStart>:<SymDevEnd> | <SymDevName>
        [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]>]

symqos -RA -sid <SymmID>

   enable -io

   disable -io

symqos -RA -sid <SymmID>

   set IO -default
        -sync <SyncPercent>
        -async <AsyncPercent>
        -copy <CopyPercent>

   set IO -dir <# | ALL>
        -sync <SyncPercent>
        -async <AsyncPercent>
        -copy <CopyPercent>

   reset IO -dir <# | ALL>

symqos -RA [-sid <SymmID>]

   list -io

symqos -rcopy -sid <SymmID>
   -dir <# | ALL> [-p <# | ALL>]

   set ceiling <<CeilingPercent> | DISABLE | NONE>

   list ceiling
```

DESCRIPTION

The symqos command is used to view or change copy priorities for a range of devices, or selected members of a device group.

The symqos is also used to view or change the priority service time of devices.
The default device priority is 1 - the highest priority.

Cache partitions can be created or modified for different device groupings in addition to the default cache partition to which all devices belong.

ARGUMENTS

add	Adds a device to a particular cache partition.
addall	Adds a range of devices to a particular cache partition.
analyze	Enables analysis for cache partitioning. This mode allows the user to see if the target cache allocation of their partitions is reasonable. The min must be set to 0%, the max must be set to 100%, and the

	donation time must be 0 for all partitions.
create	Creates a new cache partition if there is one available.
delete	Deletes a cache partition that is no longer being used.
disable	Disables the cache partition settings or disables the device priority QoS feature for a Symmetrix array. When used with -io, disables the workload percentage settings for Synchronous, Asynchronous and Copy I/Os for all the RDF directors on the Symmetrix array.
enable	Enables the cache partition settings or enables the device priority QoS feature for a Symmetrix array. When used with -io, enables the workload percentage settings for Synchronous, Asynchronous and Copy I/Os for all the RDF directors on the Symmetrix array.
list	Displays QoS copy priorities for a range of devices. Displays device priority for a range of devices or a device group. Displays cache partition settings for one or all partitions. Displays cache partition devices for a range of devices, an RDF group, or a device group. When used with -io, displays the defined workload percentages for Synchronous, Asynchronous and Copy I/Os on the Symmetrix array and all the RDF directors. When used with -rcopy, displays the maximum bandwidth percentage allowed for Open Replicator usage for a given director, port, director and port, or all directors and ports.
modify	Modifies one or more of the attributes for a given cache partition.
query	Displays QoS copy priorities for specified members of a device group. Defaults to viewing STD devices.
remove	Removes a device from a particular cache partition.
rename	Changes the name of a cache partition.
reset	Resets the RDF director I/O settings for Synchronous, Asynchronous and Copy I/O workloads to the Symmetrix default.
rmall	Removes a range of devices from a particular cache partition.

set — Allows the setting of the specified keyword.

When used with IO, the percentages given for -sync, -async and -copy must each be greater than 1 and must add up to 100.

When used with -rcopy, sets the maximum allowed bandwidth percentage for Open Replicator usage for a given director, port, director and port or all directors and ports.

KEYWORDS

ceiling — Used with -rcopy set and -rcopy list to set or display the percentage of director/port bandwidth allowed for Open Replicator usage.

dev — Specifies the action to a device by Symmetrix device name.

empty — Specifies the action to all empty partitions. By default, empty partitions, or those without device assignments, will be removed after 2 hours.

hostio priority — Assigns host I/O priority to a range of devices or devices in a device group. For device group, if no device type flag is specified, the priority will be set for the STD devices only.

IO — Sets or resets the percentage of RDF director CPU resources assigned to Synchronous, Asynchronous, and Copy I/Os on the Symmetrix array or the specified RDF Director(s).

ld — Specifies the action to a device by logical device name.

pace — Sets the QoS copy pace. Default is to set parameters on STD devices.
Pace setting: integers between 0 (fastest pace) to 16 (slowest pace) are valid.

priority — Sets the QoS copy priority. Default is to set parameters on STD devices.
Host I/O priority: integers between 1 (highest priority) to 16 (lowest priority) are valid.

OPTIONS

-all — Lists all Symmetrix devices, including internal devices (VAULT devices, etc.)

-async — Specifies the percentage of RDF director CPU resources assigned to the Asynchronous workload on the Symmetrix array or RDF director(s).

-copy — Specifies the percentage of RDF director CPU resources assigned to the Copy workload on

the Symmetrix array or RDF director(s).

-cp — Specifies a cache partition operation.

-default — Specifies the default setting of the specified keyword.

When used with IO, specifies the percentage of CPU resources assigned to Synchronous, Asynchronous, and Copy I/Os for all the RDF directors on the Symmetrix array.

-devs — Specifies one or more ranges of Symmetrix devices to add, remove, move, or on which to set priority.

-dir — Specifies the director ID.

-force — Forces the operation to proceed without verifying destination cache availability when adding or removing devices from a cache partition.

-g — Specifies a device group.

-h — Provides brief, online help.

-hostio_priority — Specifies a host I/O priority value range from 1 to 16. 1 is the highest, 16 is the lowest.

-io — Specifies an RDF Director I/O workload operation.

When used with list, lists the workload percentage settings for Synchronous, Asynchronous, and Copy I/Os on the Symmetrix array.

When used with enable or disable, enables or disables the workload percentage settings for Synchronous, Asynchronous, and Copy I/Os for all the RDF directors on the Symmetrix array.

-max — Specifies the maximum cache percentage for a cache partition.

-min — Specifies the minimum cache percentage for a cache partition.

-name — Indicates a cache partition name.

-new_name — Indicates a new cache partition name for a rename action.

-offline — Displays information from Symmetrix configuration database without refreshing the data from the Symmetrix array.

-p — Sets the port number on which to list or set the ceiling. Defaults to ALL.

-pace_range — Sets the minimum/maximum pace values for display.

-pst Specifies a priority service time operation.

-RA Specifies operation for an RDF Adaptor (RA), also referred to as an RDF director.

-rcopy Used with set or list ceiling to set or display the percentage of director/port bandwidth allowed for Open Replicator usage.

-rdfg Specifies devices that belong to the specified RDF (RA) group.

-settings Lists the general cache partition configuration for the Symmetrix array.

-sg Specifies the name of the storage group.

-sid Specifies the unique Symmetrix ID.

-std Limits the devices from the device group to standard devices.

-sync Specifies the percentage of RDF director CPU resources assigned to the Synchronous workload on the Symmetrix array or RDF director(s).

-target Specifies the target cache percentage for a cache partition.

-time Specifies the donation time in seconds - the time for when idle cache is made available to other cache partitions.

-usage Specifies the cache partition usage.

-v Provides a more detailed, verbose listing of the cache partition configuration.

-vdev Limits the devices from the device group to VDEV devices.

-wp Specifies the write pending limit percentage for a cache partition [40-80].

PARAMETERS

AsyncPercent The percentage of RDF director CPU resources assigned to Asynchronous I/Os.

CeilingPercent The percentage of director/port bandwidth allowed for open replicator usage. Acceptable values are 0-100, DISABLE or NONE. If the value DISABLE is used, this will block Open Replicator I/O on the director/port specified. If the value NONE is used, this will turn off the ceiling function on this director/port, effectively enabling Open Replicator pace for sessions using this director/port.

CopyPercent The percentage of RDF director CPU resources assigned to Copy I/Os.

CPName A name assigned to a cache partition. The name must be unique and up to 31 characters.

DgName — The device group name.

DonationTime — The donation time in seconds.

GrpNum — The RDF (RA) group number.

MinPace — The minimum pace value in a range.

MinPercent — The minimum cache percentage for a cache partition.

MaxPace — The maximum pace value in a range.

MaxPercent — The maximum cache percentage for a cache partition.

NewCPName — A new name that can be assigned to an existing cache partition.

SgName — The storage group name.

SymDevEnd — The last Symmetrix device name in a range.

SymDevName — The Symmetrix device name, unique per Symmetrix array, such as 01C.

SymDevStart — The first Symmetrix device name in a range.

SymmID — The 12-digit ID of the Symmetrix array.

SyncPercent — The percentage of RDF director CPU resources assigned to Synchronous I/Os.

TargetPercent — The target cache percentage value for a cache partition. The target cache percentage should be more than or equal to the minimum cache percentage and less than or equal to the maximum cache percentage.

Value — An integer value.

Pace setting: integers between 0 (fastest pace) to 16 (slowest pace) are valid.
If the value URGENT is used, the copy pace is classified as urgent, and may be faster than the default. If the value STOP is used, the background initiated copy will stop.

Host I/O priority: integers between 1 (highest priority) to 16 (lowest priority) are valid.

WritePendingLimit — The write pending limit percentage for a cache partition [40-80].

RETURN CODES

Code #	Code Symbol
0	CLI_C_SUCCESS
1	CLI_C_FAIL

EXAMPLES

To view the copy priorities for devices 10 through 20 on Symmetrix 1234, enter:

```
symqos -sid 1234 -devs 10:20 list
```

To view the copy priorities for all STD devices in device group DeviceGroup, enter:

```
symqos -g DeviceGroup query
```

To enable host I/O device priority feature for Symmetrix 1234, enter:

```
symqos -pst -sid 1234 enable
```

To assign all STD devices of device group DeviceGroup to host I/O priority 1, enter:

```
symqos -pst -g DeviceGroup set hostio priority 1 -std
```

To view device priorities for device 10 through 20 on Symmetrix 1234, enter:

```
symqos -pst -sid 1234 -devs 10:20 list
```

To enable the cache-partition feature for Symmetrix 1234, enter:

```
symqos -cp -sid 1234 enable
```

To create a cache partition on Symmetrix 1234, enter:

```
symqos -cp -name TestPartition -sid 1234 \\
-target 10 -min 5 -max 40 -wp 50 -time 10 create
```

To add device 00C to TestPartition on Symmetrix 1234, enter:

```
symqos -cp -name TestPartition -sid 1234 add dev 00C
```

To view the setting of cache partition TestPartition, enter:

```
symqos -cp -name TestPartition list -settings
```

To enable the workload percentage settings for Synchronous, Asynchronous and Copy I/Os on Symmetrix 1234, enter:

```
symqos -RA -sid 1234 enable -io
```

To set the default settings of the workload percentages on Symmetrix 1234 to 60% for Synchronous I/Os, 30% for Asynchronous I/Os and 10% for Copy I/Os, enter:

```
symqos -RA -sid 1234 set IO -default -sync 60
-async 30 -copy 10
```

To change the settings of the workload percentages on director 8G of Symmetrix 1234 to 50% for Synchronous I/Os, 30% for Asynchronous I/Os and 20% for Copy I/Os, enter:

```
symqos -RA -sid 1234 -dir 8G set IO -sync 50
-async 30 -copy 20
```

To reset the settings of the workload percentages on

director 8G of Symmetrix 1234, enter:

```
symqos -RA -sid 1234 -dir 8G reset IO
```

symrcopy

Performs remote copy control operations between storage arrays on a collection of device pairs.

SYNOPSIS

```
symrcopy [-h]

symrcopy <-file <FileName> [-noprompt] |
     -noprompt 'redirect stdin'> [-star]
     [-i <Interval>] [-c <Count>]

   create <-push <-hot | -cold>
     [-differential | -nodifferential] |
     -pull <-hot [-donor_update] | -cold>
     [-frontend_zero] [-force]>
     [-name <SessionName>] [-copy | -precopy | -nocopy]
     [-force_copy] [-pace <Pace>]

   create -pull -migrate -host_type <OsType>
     [-hba_type <HbaType>] [-mp_type <MpType>]
     [-name <SessionName>] [-pace <Pace>] [-force]

   remove

symrcopy <-file <FileName> [-noprompt] |
     -session_name <SessionName> [-noprompt] |
     -noprompt 'redirect stdin'>
     [-v] [-force] [-star] [-i <Interval>] [-c <Count>]

   activate [-consistent | -migrate]

   failback -migrate

   recreate [-name <SessionName>] [-pace <Pace>] [-precopy]

   restore [-name <SessionName>] [-pace <Pace>]
     [-donor_update]

   rename -name <SessionName>

   set mode <copy | nocopy | precopy>

   set pace <Pace> [-migrate]

   set donor_update off [-consistent] [-force]

   set frontend_zero off [-migrate]

   terminate [-symforce] [-all_sessions | -migrate]

   terminate -symforce -rp

symrcopy [-sid SymmID] [-i <Interval>] [-c <Count>]

   list [-offline] [-detail | -wwn]
        [-type <migrate | standard | recoverpoint>]

symrcopy <-file <FileName> | -session_name <SessionName> |
     'redirect stdin'>
     [-offline] [-i <Interval>] [-c <Count>]

   query [-detail | -wwn]

   query -summary
```

```
   verify [-summary]
     [-createinprog | -created | -recreateinprog |
     -recreated | -copyinprog | -copyonaccess |
     -copyonwrite | -copied | -terminateinprog |
     -failed | -verifyinprog | -restored | -restinprog |
     -precopy [-cycled] | -syncinprog | -synchronized |
     -failedback]

symrcopy [-sid <SymmID>]

   export -file <FileName> -session_name <SessionName>
```

DESCRIPTION

The symrcopy command controls Symmetrix Remote Copy sessions which can copy data from various types of arrays within the Storage Area Network (SAN). Copy operations are controlled from a host attached to a Symmetrix array (referred to as the control side). Data can be copied to (push operation) or from (pull operation) the remote side array. Copy operations are either online (hot) or offline (cold). Copying data from a Symmetrix to devices on remote storage arrays allows for data to be copied fully or incrementally.

ARGUMENTS

activate	Starts the copying process on an existing session.
create	Defines a new session.
export	Creates the specified file with all of the session information for sessions matching session_name.
failback	Stops a federated live migration session and returns the control device(s) to host active mode and the remote device(s) to host passive mode. The control device(s) will retain the WWN and geometry of the remote device(s).
list	Lists all the sessions for a given Symmetrix array, or for all Symmetrix arrays.
query	Queries for the status of sessions.
recreate	Creates an incremental session on an existing session. Only valid for sessions created with the -differential flag.
remove	Removes remote devices from a differential session in the Copied state.
restore	Restores copied data from the remote device of an incremental push to the control device. Restore can only be run on an incremental push session in the Copied state.
rename	Changes the name of a session. Can only be executed on a differential session in the Copied state.

set donor_update off
: Allows the donor_update portion of a session to be cancelled.

set frontend_zero off
: Turns off the feature that improves performance of pull sessions to thin devices by detecting incoming zero patterns.

set mode
: Allows the session mode to be changed to nocopy (while in the CopyInProg state), causing the session state to change to CopyOnAccess if the session was created with the pull flag, or to CopyOnWrite if the session was created with the push flag.

  Allows the session mode to be changed to copy (while in the CopyOnAccess or CopyOnWrite state), causing the session state to change to CopyInProg.

  Allows the session mode to be changed to precopy (while in the Created or Recreated state), causing the session state to change to Precopy.

set pace
: Allows the session pace to be changed while in the CopyInProg state. Valid values are from 0 to 9, where 0 is the fastest pace, and 9 is the slowest pace.

terminate
: Terminates a session and removes it from the Symmetrix array.

verify
: Verifies that a session is in a specified state.

OPTIONS

-all_sessions
: Terminates all sessions associated with the control device. Remote devices in the control file are ignored.

-c
: Specifies a count to perform a specific action.

-cold
: Indicates that any directors that are mapped to the local device and can reach the targets may be used. Therefore, the source devices must be set to User Not Ready.

  With a push operation, there may be up to 16 targets.

-consistent
: Causes the source and target pairs to be consistently copied.

-copy
: Causes the device copy to take place in the background. If -nocopy is specified, the copying of tracks is not completed unless all tracks have been accessed during a pull, or all tracks have been written during a push. Only used with create.

  After the operation is executed, the state

of the device pair is Copy in Progress. If all the tracks are eventually moved to the Target device, the state changes to Copied.

-cycled — Used with verify command and precopy flag to verify first cycle of precopy is completed.

-detail — Specifies a more detailed list or query. Note that output text may not fit in some displays.

-differential — Allows a session to be recreated for incremental copy. A session created with -nodifferential cannot be recreated for incremental copy.

-donor_update — Writes data to the control device and the remote device during a hot pull.

-file — Specifies a file name to be used for the action, which contains device pairs. A device pair is a control device and a remote device.

Note that data flows from the control to the remotes on a -push, but on a -pull, data flows from the remote to the control.

Control devices and remote devices may be specified as LUN World Wide Names (WWNs) or Symmetrix device names.

Valid tags are symdev and wwn.

Source devices and remote devices are space delimited.

Lines that start with a pound sign (#) are ignored. Control devices are always in the left hand column, and remote devices are always in the right.

The usual file format will be:

```
symdev=<SymmID>:<DevName> wwn=<LUN WWN>
 ...
 ...
```

To specify more than one target per source device, add subsequent records, using the same source device and new remote device. Valid only for -push -cold.

-force — Forces the SYMAPI server to allow an action that would ordinarily fail.

-force_copy — Forces a create operation even though one or more paired devices in the device file may not be large enough to contain the whole extents of the control device on a push, or the whole extents of the remote device on a pull.

If the operation is a pull, and the control device is too small, the session

will be created so that it will only copy the total number of blocks that will fit into the control device.

If the operation is a push, and the remote device is too small, the session will be created so that it will only copy the total number of blocks that will fit into the remote device, if it is visible to the API host.

-frontend_zero Feature that provides improved performance of pull sessions to thin devices through the detection of incoming zero patterns.

-h Provides brief, online help information.

-hba_type The application host's HBA driver type.

-host_type Specifies the application host operating system in a federated live migration operation.

-hot Indicates that all directors that are mapped to the local device must be able to reach the targets. The source devices may be Read Write.

With a push operation, there may be only one remote device.

-i Specifies the interval in seconds to repeat a specific action.
The default interval is 30 seconds if -c is used.
The minimum interval is 5 seconds.
For passive actions the minimum interval is 15 seconds. Passive actions are actions that do not acquire an exclusive lock.

-migrate Indicates a session performing a federated live migration operation.

-mp_type The application host's MP vendor and version.

-name Indicates a session name when creating, recreating, restoring, or renaming a session.

-nocopy Causes the copying of tracks only to happen for tracks accessed during a pull, or written during a push. The copying of tracks is not completed unless all tracks have been accessed or written. Normally, the device copy takes place in the background. Only used with create.

After the operation is executed, the state of the device pair is Copy on Access or Copy on Write. If all the tracks are eventually moved to the Target device, the state changes to Copied.

-nodifferential
Prevents a session from being recreated for incremental copy. A session must be created

with -differential to be recreated for incremental copy.

-offline — Relies on information from the database for updated session information; does not query the Symmetrix array. Used with list or query.

-pace — Sets the pace value used for the session Used for create and recreate.

-precopy — Begins copying data immediately in the background at create or recreate time, but without taking a point in time image of the device. Only available for hot push sessions.

-pull — Pulls data from the remote to the control device when the session is created with the -pull flag.

-push — Pushes data from the control device to the remote device(s) when the session is created with the -push flag.

-rp — Indicates a RecoverPoint session. Used with terminate and -symforce to clean up a RecoverPoint session.

-session_name — Specifies the name of the session (or group of sessions) that you want to control.

-star — Targets the action for devices that are in Star mode.

-summary — Lists the number of sessions in each state.

-symforce — Terminates a session that has not yet finished copying. Also used with -rp to terminate a RecoverPoint session. Used only with terminate.

-type — Specifies which type of sessions to include in the list command output. If -type is not specified, all sessions will be included.

Type	Description
migrate	Indicates federated live migration sessions
recoverpoint	Indicates sessions used by the RecoverPoint appliance
standard	Indicates sessions that are neither recoverpoint nor federated live migration

-v — Provides more detailed, verbose information.

-wwn — Specifies a World Wide Name (WWN). Used for list or query to display remote devices as WWNs.

PARAMETERS

FileName — The name of the file that contains the source and target pairs.

HbaType — The Host Bus Adapter type. Possible values are:

- Emulex
- Qlogic

MpType — The multi-path IO type. Possible values are:

- DMP (Dynamic multi-path)
- PPath (PowerPath 4.6+)
- PPath_45 (PowerPath 4.5 only)
- NATIVE (Native multi-path)

OsType — The application host operating system. Possible values are:

- AIX
- HPUX
- LINUX
- SunOS_Sparc
- VMWARE
- Windows

Pace — The pace value assigned to the session during create, recreate, or set pace commands.

SessionName — The optional assigned name used to control a session or group of sessions.

SymmID — The 12-digit Symmetrix ID number.

RETURN CODES

Code #	Code Symbol
0	CLI_C_SUCCESS
1	CLI_C_FAIL
2	CLI_C_DB_FILE_IS_LOCKED
18	CLI_C_ALREADY_IN_STATE The device or session is already in the desired state.
19	CLI_C_GK_IS_LOCKED All GateKeepers to the Symmetrix array are currently locked.

Return codes for symrcopy verify

209	CLI_C_NOT_ALL_VERIFIED Some but not all of the devices are in the specified state.
210	CLI_C_NONE_VERIFIED None of the devices are in the specified state.

EXAMPLES

To create a hot push session from an input file, enter:

```
symrcopy create -f input.txt -hot -push
```

To activate a session from an input file, enter:

```
symrcopy activate -f input.txt
```

To query a session from an input file, enter:

```
symrcopy query -f input.txt
```

To terminate a session from an input file, enter:

```
symrcopy terminate -f input.txt
```

Input file examples:

The following file will control a single session with control device 10 and remote device 34:

```
symdev=000000001234:10 symdev=000000005678:34
```

The following file will control two separate sessions with different control and remote devices:

```
symdev=000000001234:01 symdev=000000005678:42
symdev=000000001234:02 symdev=000000005678:43
```

The following file shows a mix of symdev and wwn usage:

```
symdev=000000001234:01 symdev=000000005678:42
symdev=000000001234:02 symdev=000000005678:43
symdev=000000001234:03 wwn=60060480000000000567853594D303434
```

The following file will control a session with one control device (01) and multiple remotes. (41 and 42) (Note: this can only be used with -cold and -push):

```
symdev=000000001234:01 symdev=000000005678:41
symdev=000000001234:01 symdev=000000005678:42
```

symrdb

Displays detailed mapping information about one or more schemas, tables, files, tablespaces, or segments that are defined in a specified database instance.

SYNOPSIS

```
symrdb [-h] -type <DbType> [-db <DbName>] [-kb|-blocks|-mb]

   list [TBS | FILE [-datalog] | SCHEMA | TABLE]
     [-v] [-unalloc]

   show [-no_extents | -expand | -collapse]
     [TBS <TbsName>|FILE <FileName>|SCHEMA <SchemaName> |
      TABLE <TableName>]

symrdb [-h] [-v] -type <DbType> [-db <DbName>]
     [-kb|-blocks|-mb]

   list [FILE | SEG | TABLE] -tbs <TbsName>

   list [FILE | SEG | TABLE] -schema <SchemaName>

symrdb [-h] -type <DbType> [-db <DbName>] [-kb|-blocks|-mb]
     [-no_extents | -expand | -collapse]

   show -tbs <TbsName>
     [FILE <FileName>|SEG <SegmentName>|TABLE <TableName>]

   show -schema <SchemaName>
     [FILE <FileName>|SEG <SegmentName>|TABLE <TableName>]

symrdb [-h] [-v] -type <DbType> [-db <DbName>]
     [-sid <SymmID>] [-rdfg <GrpNum>] [-R1|-R2]
     [-bcv | -nobcv | -vdev] [-data] [-log] [-control]
     [-force]

   rdb2dg <DgName> [-dgtype REGULAR | RDF1 | RDF2]

   rdb2cg <CgName> [-cgtype REGULAR | RDF1 | RDF2]
     [-apidb | -rdf_consistency]

symrdb [-h] [-v] -type <DbType> [-db <DbName>]
     -tbs <TbsName> [-sid <SymmID>] [-rdfg <GrpNum>]
     [-R1|-R2] [-bcv | -nobcv | -vdev] [-force]

     tbs2dg <DgName> [-dgtype REGULAR | RDF1 | RDF2]

     tbs2cg <CgName> [-cgtype REGULAR | RDF1 | RDF2]
     [-apidb | -rdf_consistency]

symrdb [-h] [-v] -type <DbType> [-db <DbName>]

   show -config [-all]

symrdb [-h] [-v] [-i <Interval>] [-c <Count>]

   stats stats_option

symrdb [-h] startup db_startup_options

symrdb [-h] shutdown db_shutdown_options
```

DESCRIPTION

The symrdb command displays mapping information specific to a schema, table, file, tablespace, or segment in a database instance. Detailed information can be obtained for tablespaces, schemas, files and tables by specifying the -v option with the list command, or issuing the show command.

The symrdb rdb2cg and rdb2dg commands are used to convert a specified database into a composite group or device group. The symrdb tbs2cg and tbs2dg commands are used to convert a specified tablespace into a composite group or device group.

For database objects that are defined on Symmetrix devices, you can obtain logical-to-physical information of where the file, table, schema, or tablespace extents are mapped on Symmetrix devices.

The database user login information must be supplied with the SYMCLI_RDB_CONNECT environment variable. The user login information is specified in the following format, "username/password@service". The username and password must be non NULL.

Note: For Oracle, the database user specified must have one of the following in order to run this utility:

- "select any table" privilege (Oracle8i and earlier)
- SELECT_CATALOG_ROLE (Oracle9i and later)
- DBA role

If list is specified without a key word, and neither -tbs or -schema are present on the command line, the default is to list the database names. If -db, -tbs, or -schema are present on the command line, the default is to list the tables of the specified database, tablespace, or schema.

Given a database name, you can obtain a list of:

- files defined to make up the database
- schemas defined within the database
- tablespaces defined within the database
- tables defined within the database

Given a database name and tablespace or schema name, you can obtain a list of:

- files that make up the tablespace or schema
- segments defined within the tablespace or schema
- tables defined within the tablespace or schema

To execute the symrdb utility, you must have the proper application software installed and environment variables set.

You can specify the database type, database, schema, and tablespace name from environment variables. The command line options take priority over the environment variables as follows:

Environment Variable	Use to Specify	Instead of
SYMCLI_RDB_TYPE	database type	-type
SYMCLI_RDB_NAME	database name	-db

```
SYMCLI_SCHEMA_NAME     schema name      -schema

SYMCLI_TBS_NAME        tablespace name  -tbs

In Oracle, the database name does not need to be
specified. The default will be taken from the instance
represented by the connection arguments.

If a name specified on the command line contains special
shell characters, those characters must be escaped with a
\\ back slash. (e.g. $, ', \\, etc.)

If symrdb is being run in client/server mode and the
required RDBMS environment variables are set in the
client's environment, they will be sent to the server
to be used.

On DB2, for additional information about what some of
the configuration values mean, please refer to
"Administration Guide: Performance" or Configuration
parameters in the DB2 Information Center.

ARGUMENTS

list           Lists the defined databases (or object type
when a keyword is specificed) for the
database instance. For Oracle, the name
of the database instance is returned.

rdb2cg         Translates the specified database into a
composite group.

rdb2dg         Translates the specified database into a
device group.

show           Shows database names, states pertaining to
the connected server, status and usage of
each of the databases. When a keyword is
specified, detailed logical-to-physical
mapping information about that object
is returned.

stats          Shows performance statistics about
a specified database (type). The current
supported databases are Oracle, SQL Server,
Sybase, and IBMUDB.

tbs2cg         Translates the specified database table
space into a composite group. Only
data database files are translated.

tbs2dg         Translates the specified database
tablespace into a device group. Only data
database files are translated.

KEYWORDS

FILE           Specifies the action on defined files for a
specified database instance.

REGULAR        Specifies a device or composite group type
of regular.

RDF1           Specifies a device or composite group type
of RDF1.
```

RDF2 Specifies a device or composite group type of RDF2.

SCHEMA Specifies the action on defined schemas for a specified database instance, tablespace, or schema.

SEG Specifies the action on defined segments for a specified tablespace or schema.

TABLE Specifies the action on defined tables for a specified database instance, tablespace, or schema.

TBS Specifies the action on defined tablespaces for a specified database instance.

OPTIONS

-all Displays all the possible database configuration values (use with -config).

-apidb Creates the CG in the SYMAPI DB only.

-bcv Associates only BCV devices to the target group.

-blocks Displays size information in 512-byte blocks.

-c Indicates the number (count) of times to display statistics. If this option is not specified and an interval (-i) is specified, stats will be displayed continuously.

-cgtype Specifies a composite group type.

-collapse Collapses the extents of a database object, if possible. This is the default.

-config Displays the database configuration value.

-control Specifies to only operate on control database files. Default is all database files.

-data Specifies to only operate on data database files.

-datalog Displays "data and log" as a separate file type item. The default is to treat data and log as file type data. Used for Sybase only.

-db Specifies a relational database name.

-dgtype Specifies a device group type.

-expand Expands the extents of a database object, if possible.

-force Attempts to force the operation even though one or more devices in the database or tablespace may already be part of another DG or CG group.

-h Provides brief, online help information.

-i Repeats the interval in seconds.
The default interval is 30 seconds.
The minimum interval is 5 seconds.
For passive actions the minimum interval is
15 seconds. Passive actions are actions that
do not acquire an exclusive lock.

-kb Displays size information in Kilobytes.

-log Operates only on log database files.

-mb Displays size information in Megabytes.
This is the default.

-nobcv Adds only standard devices to the target
group. The default behavior is to add
both standard and BCV devices.

-no_extents Excludes extents information from the
display.

-R1 Adds R1 devices to the target device
group.

-R2 Adds R2 devices to the target device
group.

-rdf_consistency
Creates a CG, allowing it to be enabled
for RDF consistency once devices have
been added to the CG.

-rdfg Selects RDF devices that belong to the
specified Symmetrix RA (RDF) group number.

-schema Specifies a database schema name.

-sid Specifies the unique Symmetrix ID.

-target Identifies the database statistics output
options, refers to the stats_option on
ora_options, sybase_options,
sqlserver_options, and ibmudb_options.

-tbs Specifies a database tablespace name.

-type Specifies a database type (DbType).

-unalloc Displays unallocated space, for Sybase
database file only.

-v Provides a more detailed, verbose listing.

-vdev Adds VDEVs to the target group.

PARAMETERS

CgName The composite group name.

DbName A specific database name.

db_startup_options
See below for specific database options.

db_shutdown_options
See below for specific database options.

DbTarget The metrics available for the specified database statistic.

DbType The database type. Supported values are:
- Oracle
- SQLServer
- Sybase
- MVSDB2
- IBMDB2
- IBMUDB
- Informix
- Exchange
- Exchange2007
- SharePoint

DgName The device group name.

FileName A specific database filename.

GrpNum The RDF (RA) group number.

SchemaName A specific database schema name.

SegmentName A specific database segment name.

stats_option See below for specific database options.

SymmID The 12-digit ID of the Symmetrix array.

TableName A specific database table name.

TbsName A specific database tablespace name.

Oracle Startup/Shutdown/Stats Options:

Startup...

```
symrdb [-h] startup -type Oracle
[ [FORCE] [RESTRICT] [PFILE=filename] [QUIET]
[MOUNT [-db dbname] |
[ [OPEN | OPEN_READ_ONLY | OPEN_READ_WRITE |
OPEN_READ_WRITE_RECOVER | OPEN_RECOVER]
[-db dbname] ] | NOMOUNT]] |
[ [PFILE=filename] MIGRATE [QUIET]]
```

where:

Option or Keyword	Description
dbname	Identifies the database name to mount or open. Refer to Oracle documentation for the definition.
filename	Specifies a filename to be used while starting up the instance.
FORCE	Shuts down the current Oracle instance (if it is running) with the shutdown option ABORT, before restarting it.
MIGRATE	Starts the database in OPEN MIGRATE mode and sets system initialization parameters to specific values required to enable the database upgrade or

downgrade scripts to run.

MOUNT Mounts the database; does not open it.

NOMOUNT Causes the database not to be mounted upon instance startup.

OPEN Mounts and opens the database.

OPEN_READ_ONLY
Specifies READ ONLY to restrict users to read-only transactions, and prevent them from generating redo logs.

OPEN_READ_WRITE
Specifies READ WRITE to open the database in read/write mode, allowing users to generate redo logs. This is the default.

OPEN_READ_WRITE_RECOVER
Specifies READ WRITE to open the database in read/write mode, and Specifies that media recovery should be performed, if necessary, before starting the instance.

OPEN_RECOVER
Specifies to open the database, and that media recovery should be performed, if necessary, before starting the instance.

QUIET Suppresses the display of System Global Area information for the starting instance.

RESTRICT Allows only Oracle users with the RESTRICTED SESSION system privilege to connect to the database.

Shutdown...

symrdb [-h] shutdown -type Oracle
[ABORT | IMMEDIATE | NORMAL |TRANSACTIONAL [LOCAL]]

where:

Keyword	Description
ABORT	Proceeds with the fastest possible shutdown. Does not wait for calls to complete or users to disconnect.
IMMEDIATE	Does not wait for current calls to complete, prohibits further connects, and closes and dismounts the database. Finally, shuts down the instance. Does not wait for connected users to disconnect. Does not require instance recovery on next startup.
LOCAL	Specifies a transactional shutdown only on the local instance.
NORMAL	Waits for currently connected users to disconnect from the database, prohibits further connects, and closes and

```
dismounts the database. Finally, shuts
down the instance. Does not require
instance recovery on the next startup.
NORMAL is the default option.

TRANSACTIONAL
Shuts down an instance while
minimizing interruption to clients.
No client can start a new
transaction on the instance.

Stats...

symrdb stats -type Oracle -target
< INSTANCE | SESSION | FILE | ALL >

where:

Keyword    Description
---------  ----------------------
INSTANCE   Specifies instance stats.
SESSION    Specifies session stats.
FILE       Specifies file stats.

Sybase Startup/Shutdown/Stats Options:

Startup...

symrdb startup -type Sybase
-f runserver_file [-m] [-t delay_time]

where:

Keyword    Description
---------  ----------------------
delay_time Estimated time to startup the
Sybase server in seconds.

-m         Starts database in a single user
mode.

runserver_file
The absolute path name of a
runserver file used as a reference
each time you restart a Sybase
server.

Shutdown...

symrdb shutdown -type Sybase
[-f srvname] [-w {wait | nowait}]

where:

Keyword    Description
---------  ----------------------
nowait     Shuts down the server immediately.

srvname    Specifies a logical name by which
the backup server is known in the
server's sysservers system table.

wait       Brings the server down gracefully.

Stats...

symrdb stats -type Sybase -target
```

```
< SERVER | OBJECT | DEVICE | ALL >

where:

Keyword    Description
---------  ---------------------
OBJECT     Specifies object stats.
DEVICE     Specifies device stats.
SERVER     Specifies server stats.


SqlServer Startup/Shutdown/Stats Options:

Startup...

symrdb startup -type SqlServer
-s instance [-c] [-f] [-m] [-n] [-x]
[-p master_file_path] [-e error_log_path]
[-l master_log_path] [-g virtual_addr_space]
[-t trace_number]

where:

Keyword    Description
---------  ---------------------
-c         Shortens startup time.

-f         Starts an instance with minimal
configuration.

error_log_path
Names the error log file.

instance   Specifies the instance name to start.

-m         Starts an instance in single-user mode.

master_file_path
Names the master database file.

master_log_path
Names the master database log file.

-n         Does not use the Windows application log
to record an SQL Server event.

trace_number
Specifies a trace number.

virtual_addr_space
Displays the amount of virtual address
space in megabytes.

-n         Does not use the Windows application log
to record an SQL Server event.

-x         Disables the keeping of CPU time and
cache-hit ratio statistics to allow
maximum performance.

Shutdown...

symrdb shutdown -type SqlServer -s instance

where:

Keyword    Description
```

```
---------  ----------------------
instance   Instance name to be shut down.

Stats...

symrdb stats -type SqlServer -target
< INSTANCE | DATABASE | OBJECT | FILE | ALL >

where:

Keyword    Description
---------  ----------------------
INSTANCE   Specifies instance stats.
DATABASE   Specifies database stats.
FILE       Specifies file stats.
OBJECT     Specifies object stats.

IBMUDB Startup/Shutdown/Stats Options:

Startup...

symrdb startup -type IBMUDB
[ [-n node
[ADDNODE -u hostname -p port
[-c computer] [-nt netname]
[NODE | CATALOG -tsn tablespace_node] ] |
[RESTART [-u hostname] [-p port] [-nt netname]] |
[STANDALONE]] ]

where:

Keyword or
Option     Description
---------  ----------------------
ADDNODE    Issues the ADD NODE command.

computer   Specifies the computer name.

CATALOG    Indicates that the containers for
the temporary tablespaces should be
the same as those for the catalog
node of each database.

hostname   Specifies the system name.

netname    Specifies the net name.

NODE       Indicates that the containers for the
temporary tablespaces should be the
same as those for the specified node.

node       Specifies the node number.

port       Specifies the port number.

profile    Specifies the name of the profile.

RESTART    Issues the RESTART DATABASE command.

STANDALONE Starts the node in STANDALONE mode.

tablespace_node
Specifies the node number from which
the temporary tablespace definitions
should be obtained.

username/password
```

```
Specifies how it is set in the
environment variable SYMCLI_RDB_CONNECT.
The parameters are mandatory with
option ADDNODE on Windows, but is
ignored on other operating systems.

Shutdown...

symrdb shutdown -type IBMUDB
[ [FORCE | DROP] [DROP_ACT | CONTINUE | TERMINATE]
[-f profile] [-n node] ]

where:

Keyword or
Option     Description
---------  ----------------------
CONTINUE   Subsequent call. Continue processing
after a prompt.

DROP       Drops the node from the db2nodes.cfg
file.

DROP_ACT   Initial call.

FORCE      Issues the FORCE APPLICATION (ALL)
command.

node       Specifies the node number.

profile    Specifies the name of the profile.

TERMINATE  Subsequent call. Terminate processing
after a prompt.

Stats...

symrdb stats -type Ibmudb -target
< DATABASE | TABLE | TABLESPACE | ALL >

where:

Keyword    Description
---------  ----------------------
DATABASE   Specifies database stats.
TABLE      Specifies table stats.
TABLESPACE Specifies tablespace stats.

RETURN CODES

Code #         Code Symbol
------         -----------

0           CLI_C_SUCCESS

1           CLI_C_FAIL

EXAMPLES

To define the database login parameters to be
user = scott, passwd = tiger, and service = acme, enter:

setenv SYMCLI_RDB_CONNECT scott/tiger@acme

To list all tables that reside in tablespace
tbl_space1, enter:
```

```
symrdb -type Oracle -tbs tbl_space1 list TABLE
```

To display a detailed listing of the table name EMP that resides in schema SCOTT, enter:

```
symrdb -type Oracle -schema SCOTT show TABLE EMP
```

To display a detailed listing of the table name SYS$TABLE that resides in tablespace tbl_space1, and to display the extent information in expanded mode and size in megabytes, enter:

```
symrdb -type Oracle -tbs tbl_space1
show TABLE SYS\\$TABLE -expand -mb
```

To define the database name HR as the default database name, enter:

```
setenv SYMCLI_RDB_NAME HR
```

To define the database login parameters to be user = sa, passwd = sa_pass, and service = local, enter:

```
setenv SYMCLI_RDB_CONNECT "sa/sa_pass@local"
```

To list all tablespaces that define the database named master, enter:

```
symrdb -type SQLServer -db master list TBS
```

To list all files that define the database named master, enter:

```
symrdb -type SQLServer -db master list FILE
```

To list all tables that reside in the database named master, enter:

```
symrdb -type SQLServer -db master list TABLE
```

To create a REGULAR device group named newdg with only the R1-BCV devices from the SQL Server database named master, enter:

```
symrdb -type SQLSERVER -db master rdb2dg newdg
-R1 -bcv -dgtype REGULAR
```

To create a composite group named newcg with the R1 and R1-BCV devices from the Oracle tablespace named tbl_space1, enter:

```
symrdb -type ORACLE -tbs tbl_space1 tbs2cg newcg
-cgtype RDF1 -R1
```

To start up an Oracle data manager with mode OPEN READ ONLY, enter:

```
symrdb startup -type ORACLE OPEN_READ_ONLY
```

To shut down an Oracle data manager with mode NORMAL, enter:

```
symrdb shutdown -type ORACLE NORMAL
```

To display statistics about a session of a specified database Oracle every 30 seconds for one hour, enter:

```
symrdb stats -type ORACLE -i 30 -c 120 -target SESSION

H:M:S      sessionID   memory
A              B        C

A  Time of day
B  Session ID
C  memory usage per second

NOTE: All the statistics values are per second.
```

symrdf

Performs a Symmetrix ping to all Symmetrix arrays
or a specified Symmetrix array.

Lists all or selected SRDF devices on a Symmetrix
array.

Performs Symmetrix SRDF control or set operations
on a group of device pairs using:
device group
composite group
storage group
rdf group
device file

Performs Symmetrix SRDF query or verify on a group
of device pairs using:
device group
composite group
storage group
rdf group
device file

Performs Symmetrix dynamic SRDF group operations
to add, modify, remove SRDF groups.

SYNOPSIS

```
symrdf -h

symrdf [-sid <SymmID>] [-i <Interval>] [-c <Count>] [-rdf]

   ping

symrdf [-sid <SymmID>] [-i <Interval>] [-c <Count>]
     [-offline] [-v]
     [-all]
     [-rdfg <GrpNum>] [-bcv | -nobcv] [-rdfa] [-resv]
     [-consistency] [-concurrent] [-cascade]
     [-diskless_rdf] [-metro] [-metrodr] [-unknown]
     [[-R1 | -R2 | -R21] |
      [-dynamic [-R1 | -R2 | -BOTH]]]
     [-half_pair] [-dup_pair] [-star_mode]
     [-star_sync_target] [-star_async_target]
     [-exempt] [-rdfa_wpace_exempt]
     [-rdfa_not_pace_capable]

   list [dev]

   list pd

symrdf -sid <SymmID> -rdfg <GrpNum> [-v]
     [-noprompt] [-i <Interval>] [-c <Count>]
     [-star]

   activate    <<-rdfa_dse [-both_sides]> |
                <-rdfa_pace | -rdfa_wpace | -rdfa_devpace
                 [-force]>>

   deactivate  <<-rdfa_dse [-both_sides]> | -rdfa_wpace |
                <<-rdfa_pace | -rdfa_devpace> [-symforce]>>

   msc_cleanup

   set rdfg    <[-limbo <Secs>]
                [-domino <State>]
```

```
                    [-autolink_recovery <State>]
                    [-hwcomp <State>]
                    [-swcomp <State>]>
                    [-both_sides]

        set rdfa    <[-cycle_time <CycleTime>]
                    [-priority <SessPriority>]
                    [-transmit_idle <State>]>
                    [-both_sides]

        set rdfa_pace
                   <[-dp_autostart <State>]
                    [-wp_autostart <State>]
                    [-delay <WpaceDelay>]
                    [-threshold <WpaceThreshold>]>
                    [-both_sides]

        set rdfa_dse
                   <[-autostart <State>]
                    [-threshold <DseThreshold>]
                    [-fba_pool <PoolName>]
                    [-ckd3390_pool <PoolName>]
                    [-ckd3380_pool <PoolName>]
                    [-as400_pool <PoolName>]>
                    [-both_sides]

        set label <GrpLabel>

Device Groups

     symrdf -g <DgName> [-v | -noecho] [-force] [-symforce]
          [-bypass] [-noprompt] [-i <Interval>] [-c <Count>]
          [-hop2 | -bcv [-hop2] | -all | -rbcv | -brbcv]
          [-rdfg <GrpNum>] [-star]

        deletepair

        establish [-full]

        failback  [-remote] [-rp]

        failover  [-establish | -restore [-remote]]

        half_deletepair

        half_movepair -new_rdfg <GrpNum>

        half_swap

        merge     [-rp]

        movepair -new_rdfg <GrpNum>

        restore   [-remote] [-rp] [-full]

        resume    [-remote] [-rp]

        split

        swap   [-refresh R1|R2]

        suspend

        update    [-remote] [-rp] [-until <InvalidTracks>]

        invalidate <R1|R2> [-nowd]
```

```
      not_ready  <R1|R2>

      ready      <R1|R2>

      refresh    <R1|R2> [-rp]

      rw_disable R2

      rw_enable     <R1|R2>

      write_disable <R1|R2>

   symrdf -g <DgName>
        [-hop2 | -bcv [-hop2] | -rbcv | -brbcv | -all]
        [-offline] [-i <Interval>] [-c <Count>]
        [-rdfg <GrpNum>]

      query [-rdfa | -summary]

      verify [-summary]
        [-enabled | -synchronized | -suspended |
         -susp_offline | -split | -failedover | -updated |
         -syncinprog | -updateinprog | -partitioned |
         -valid | -consistent [-noinvalids] |
         -acp_disk | -acp_wp | -asynchronous |
         -semisynchronous | -synchronous]

   symrdf -g <DgName> [-v] [-bypass] [-force] [-star]
        [-noprompt] [-i <Interval>] [-c <Count>]
        [-hop2 | -bcv [-hop2] | -rbcv | -brbcv | -all]
        [-rdfg <GrpNum>]

      set acp_skew <SkewVal>

      set domino <State>

      set mode <ModeVal> [skew <SkewVal>] [-consistent]

      set nr_if_invalid <State>

Device group operations specific to SRDF/A

   symrdf -g <DgName> [-v | -noecho] [-force] [-symforce]
        [-bypass] [-noprompt] [-i <Interval>] [-c <Count>]
        [-hop2 | -bcv [-hop2] | -all | -rbcv | -brbcv]
        [-rdfg <GrpNum>] [-star]

      disable

      enable

      failover [-immediate] [-establish | -restore [-remote]]

      half_movepair -new_rdfg <GrpNum> [-exempt]

      movepair -new_rdfg <GrpNum> [-exempt]

      split    [-immediate]

      suspend  [-immediate | -exempt]

      activate  <-rdfa_dse |
              -rdfa_pace | -rdfa_wpace | -rdfa_devpace |
              -rdfa_wpace_exempt>

      deactivate <-rdfa_dse |
              -rdfa_pace | -rdfa_wpace | -rdfa_devpace |
```

```
                -rdfa_wpace_exempt>

     symrdf -g <DgName>
          [-hop2 | -bcv [-hop2] | -rbcv | -brbcv | -all]
          [-i <Interval>] [-c <Count>] [-rdfg <GrpNum>]

        checkpoint

Device group operations specific to SRDF/Metro configuration

     symrdf -g <DgName> [-v | -noecho] [-force] [-symforce]
          [-bypass] [-noprompt] [-i <Interval>] [-c <Count>]
          [-rdfg <GrpNum>]

        deletepair [-keep <R1|R2>]

        establish [-full] [-use_bias]

        half_deletepair

        half_movepair -new_rdfg <GrpNum>

        half_swap

        movepair -new_rdfg <GrpNum> [-exempt | -keep <R1|R2>]

        query [-summary]

        restore  [-full] [-use_bias]

        set bias <R1 | R2>

        suspend  [-keep <R1 | R2 >]

        verify [-summary]
          [-enabled | -activeactive | -activebias | -suspended |
           -syncinprog | -partitioned |
           -valid]

Composite Groups

     symrdf -cg <CgName> [-v | -noecho] [-force] [-symforce]
          [-bypass] [-noprompt] [-i <Interval>] [-c <Count>]
          [-sid <SymmID> |
           -rdfg <SymmID:GrpNum <,GrpNum,...>> | all [,...] |
            name:<RdfGroupName>[,<RdfGroupName>,...]]
          [-hop2 | -bcv [-hop2] | -rbcv | -brbcv] [-star]

        deletepair

        establish  [-full]

        failback   [-remote] [-rp]

        failover   [-establish | -restore [-remote]]

        half_deletepair

        half_swap

        merge     [-rp]

        msc_cleanup

        restore  [-remote] [-rp] [-full]

        resume   [-remote] [-rp]
```

```
        split

        suspend

        swap    [-refresh <R1|R2>]

        update    [-remote] [-rp] [-until <InvalidTracks>]

        invalidate <R1|R2> [-nowd]

        not_ready  <R1|R2>

        ready      <R1|R2>

        refresh    <R1|R2> [-rp]

        rw_disable   R2

        rw_enable  <R1|R2>

        write_disable <R1|R2>

     symrdf -cg <CgName> [-offline] [-i <Interval>] [-c <Count>]
          [-hop2 | -bcv [-hop2] | -rbcv | -brbcv]
          [-sid <SymmID> |
           -rdfg <SymmID:GrpNum <,GrpNum,...>> | all [,...] |
           name:<RdfGroupName>[,<RdfGroupName>,...]]

        query  [-detail | -summary]

        verify [-summary]
          [-enabled | -synchronized | -suspended |
           -susp_offline | -split | -failedover | -updated |
           -syncinprog | -updateinprog | -partitioned |
           -valid | -consistent [-noinvalids] |
           -cg_consistent |
           -acp_disk | -acp_wp | -asynchronous |
           -semisynchronous | -synchronous]

     symrdf -cg <CgName> [-v] [-force]
          [-bypass] [-noprompt] [-i <Interval>] [-c <Count>]
          [-hop2 | -bcv [-hop2] | -rbcv | -brbcv]
          [-sid <SymmID> | -rdfg <SymmID:GrpNum <,GrpNum,...>> |
           name:<RdfGroupName>[,<RdfGroupName>,...]] [-star]

        set acp_skew <SkewVal>

        set domino <State>

        set mode <ModeVal> [skew <SkewVal>]

        set nr_if_invalid <State>

Composite group operations specific to SRDF/A

     symrdf -cg <CgName> [-v | -noecho] [-force] [-symforce]
          [-bypass] [-noprompt] [-i <Interval>] [-c <Count>]
          [-sid <SymmID> |
           -rdfg <SymmID:GrpNum <,GrpNum,...>> | all [,...] |
            name:<RdfGroupName>[,<RdfGroupName>,...]]
          [-hop2 | -bcv [-hop2] | -rbcv | -brbcv] [-star]

        failover [-immediate] [-establish | -restore [-remote]]

        split    [-immediate]
```

```
          suspend  [-immediate | -exempt]

          activate <-rdfa_dse |
                  -rdfa_pace | -rdfa_wpace | -rdfa_devpace |
                  -rdfa_wpace_exempt>

          deactivate  <-rdfa_dse |
                  -rdfa_pace | -rdfa_wpace | -rdfa_devpace |
                  -rdfa_wpace_exempt>

     symrdf -cg <CgName> [-i <Interval>] [-c <Count>]
          [-hop2]
          [-rdfg name:RdfGroupName]

          checkpoint

Composite group operations specific to SRDF/Metro configuration

     symrdf -cg <CgName> [-v | -noecho] [-force] [-symforce]
          [-bypass] [-noprompt] [-i <Interval>] [-c <Count>]
          [-sid <SymmID> |
           -rdfg <SymmID:GrpNum> | all |
            name:<RdfGroupName>]

          deletepair [-keep <R1|R2>]

          establish [-full] [-use_bias]

          half_deletepair

          half_swap

          query [-summary]

          restore  [-full] [-use_bias]

          set bias <R1 | R2>

          suspend  [-keep <R1 | R2 >]

          verify [-summary]
            [-enabled | -activeactive | -activebias | -suspended |
             -syncinprog | -partitioned |
             -valid]

Storage Groups

     symrdf -sg <SgName> -sid <SymmID> -rdfg <GrpNum>
          [-bypass] [-noprompt] [-i <Interval>] [-c <Count>]
          [-v | -noecho] [-force] [-symforce] [-star]
          [-hop2]

          createpair -type <R1|R2> -remote_sg <SgName>
            <-invalidate <R1|R2> | -establish | -restore [-rp] |
             -format [-establish]>
          [-hop2_rdfg <GrpNum>]
          [-rdf_mode <sync | acp_wp | acp_disk>]
          [-remote] [-nowd]

     symrdf -sg <SgName> -sid <SymmID> [-rdfg <GrpNum>]
          [-bypass] [-noprompt] [-i <Interval>] [-c <Count>]
          [-v | -noecho] [-force] [-symforce] [-star]
          [-hop2]

          deletepair

          establish [-full]
```

```
   failback  [-remote] [-rp]

   failover [-establish | -restore [-remote]]

   half_deletepair

   half_movepair -new_rdfg <GrpNum>

   half_swap

   merge [-rp]

   movepair -new_rdfg <GrpNum>

   restore  [-remote] [-rp] [-full]

   resume  [-remote] [-rp]

   split

   suspend

   swap [-refresh <R1|R2>]

   update [-remote] [-rp] [-until <InvalidTracks>]

   invalidate    <R1|R2> [-nowd]

   not_ready     <R1|R2>

   ready         <R1|R2>

   refresh       <R1|R2> [-rp]

   rw_disable     R2

   rw_enable     <R1|R2>

   write_disable <R1|R2>

symrdf -sg <SgName> -sid <SymmID> [-rdfg <GrpNum>]
     [-offline] [-i <Interval>] [-c <Count>]
     [-hop2]

   query [-summary]

   verify [-summary]
     [-enabled | -synchronized | -suspended |
      -susp_offline | -split | -failedover | -updated |
      -syncinprog | -updateinprog | -partitioned |
      -valid |
      -acp_disk | -acp_wp |
      -synchronous]

symrdf -sg <SgName> -sid <SymmID> [-rdfg <GrpNum>]
      [-force] [-v] [-bypass] [-noprompt]
      [-i <Interval>] [-c <Count>] [-star]
      [-hop2]

   set acp_skew <SkewVal>

   set domino <State>

   set mode <ModeVal> [skew <SkewVal>] [-consistent]

   set nr_if_invalid <State>
```

```
Storage group operations specific to SRDF/A

     symrdf -sg <SgName> -sid <SymmID> -rdfg <GrpNum>
          [-bypass] [-noprompt] [-i <Interval>] [-c <Count>]
          [-v | -noecho] [-force] [-symforce] [-star]
          [-hop2]

        createpair -type <R1|R2> -remote_sg <SgName>
          <-invalidate <R1|R2> | -establish | -restore [-rp] |
           -format [-establish]>
          [-hop2_rdfg <GrpNum>]
          [-rdf_mode <async>]
          [-remote] [-exempt] [-nowd]

     symrdf -sg <SgName> -sid <SymmID> [-rdfg <GrpNum>]
          [-bypass] [-noprompt] [-i <Interval>] [-c <Count>]
          [-v | -noecho] [-force] [-symforce] [-star]
          [-hop2]

        disable

        enable

        failover [-immediate] [-establish | -restore [-remote]]

        half_movepair -new_rdfg <GrpNum> [-exempt]

        movepair -new_rdfg <GrpNum> [-exempt]

        split    [-immediate]

        suspend  [-immediate | -exempt]

        activate   <-rdfa_dse |
                    -rdfa_pace | -rdfa_wpace | -rdfa_devpace |
                    -rdfa_wpace_exempt>

        deactivate <-rdfa_dse |
                    -rdfa_pace | -rdfa_wpace | -rdfa_devpace |
                    -rdfa_wpace_exempt>

     symrdf -sg <SgName> -sid <SymmID> [-rdfg <GrpNum>]
          [-offline] [-i <Interval>] [-c <Count>]
          [-hop2]

        query [-rdfa | -summary]

        verify [-summary]
          [-enabled | -suspended |
           -susp_offline | -split | -failedover | -updated |
           -syncinprog | -updateinprog | -partitioned |
           -valid | -consistent [-noinvalids] |
           -asynchronous]

     symrdf -sg <SgName> -sid <SymmID> [-rdfg <GrpNum>]
          [-i <Interval>] [-c <Count>]
          [-hop2]

        checkpoint

Storage group operations specific to SRDF/Metro configuration

     symrdf -sg <SgName> -sid <SymmID> -rdfg <GrpNum>
          [-bypass] [-noprompt] [-i <Interval>] [-c <Count>]
          [-v | -noecho] [-force] [-symforce]
```

```
      createpair -type <R1|R2> -remote_sg <SgName>
        -metro
        <-invalidate <R1 [-remote] | R2> |
         <<-establish | -restore [-remote] > [-use_bias]>>

      createpair -remote_sg <SgName> -metro -format

      createpair -type <R1|R2> -remote_sg <SgName>
        -metro -exempt

   symrdf -sg <SgName> -sid <SymmID> [-rdfg <GrpNum>]
        [-bypass] [-noprompt] [-i <Interval>] [-c <Count>]
        [-v | -noecho] [-force] [-symforce]

      deletepair [-keep <R1|R2>]

      establish [-full] [-use_bias]

      half_deletepair

      half_movepair -new_rdfg <GrpNum>

      half_swap

      movepair -new_rdfg <GrpNum> [-exempt | -keep <R1|R2>]

      restore  [-full] [-use_bias] [-remote]

      set bias <R1 | R2>

      suspend  [-keep <R1 | R2 >]

   symrdf -sg <SgName> -sid <SymmID> [-rdfg <GrpNum>]
        [-offline] [-i <Interval>] [-c <Count>]
        [-hop2]

      query [-summary]

      verify [-summary]
        [-enabled | -activeactive | -activebias | -suspended |
         -syncinprog | -partitioned |
         -valid]

Device Files

   symrdf -file <Filename> -sid <SymmID> -rdfg <GrpNum>
        [-bypass] [-noprompt] [-i <Interval>] [-c <Count>]
        [-v | -noecho] [-force] [-symforce] [-star]

      createpair -type <R1|R2>
        <-invalidate <R1|R2> | -establish | -restore [-rp] |
         -format [-establish]>
        [-rdf_mode <sync|semi|acp_wp|acp_disk|async>]
        [-g <NewDg>] [-remote] [-nowd]

      deletepair

      establish [-full]

      failback  [-remote] [-rp]

      failover [-establish | -restore [-remote]]

      half_deletepair

      half_movepair -new_rdfg <GrpNum>
```

```
          half_swap

          merge [-rp]

          movepair -new_rdfg <GrpNum>

          restore  [-remote] [-rp] [-full]

          resume  [-remote] [-rp]

          split

          suspend

          swap [-refresh <R1|R2>]

          update [-remote] [-rp] [-until <InvalidTracks>]

          invalidate    <R1|R2> [-nowd]

          not_ready     <R1|R2>

          ready         <R1|R2>

          refresh       <R1|R2> [-rp]

          rw_disable     R2

          rw_enable     <R1|R2>

          write_disable <R1|R2>

       symrdf -file <Filename> -sid <SymmID> -rdfg <GrpNum>
            [-offline] [-i <Interval>] [-c <Count>]

          query [-rdfa | -summary]

          verify [-summary]
            [-enabled | -synchronized | -suspended |
             -susp_offline | -split | -failedover | -updated |
             -syncinprog | -updateinprog | -partitioned |
             -valid | -consistent [-noinvalids] |
             -acp_disk | -acp_wp | -asynchronous |
             -semisynchronous | -synchronous]

       symrdf -file <Filename> -sid <SymmID> -rdfg <GrpNum>
             [-force] [-v] [-bypass] [-noprompt]
             [-i <Interval>] [-c <Count>] [-star]

          set acp_skew <SkewVal>

          set domino <State>

          set mode <ModeVal> [skew <SkewVal>] [-consistent]

          set nr_if_invalid <State>

   Device File operations specific to SRDF/A

       symrdf -file <Filename> -sid <SymmID> -rdfg <GrpNum>
            [-bypass] [-noprompt] [-i <Interval>] [-c <Count>]
            [-v | -noecho] [-force] [-symforce] [-star]

          createpair -type <R1|R2>
            <-invalidate <R1|R2> | -establish | -restore [-rp] |
             -format [-establish]>
            [-rdf_mode <async>]
```

```
          [-g <NewDg>] [-remote] [-exempt] [-nowd]

       disable

       enable

       failover [-immediate] [-establish | -restore [-remote]]

       half_movepair -new_rdfg <GrpNum> [-exempt]

       movepair -new_rdfg <GrpNum> [-exempt]

       split    [-immediate]

       suspend  [-immediate | -exempt]

       activate   <-rdfa_dse |
                   -rdfa_pace | -rdfa_wpace | -rdfa_devpace |
                   -rdfa_wpace_exempt>

       deactivate <-rdfa_dse |
                   -rdfa_pace | -rdfa_wpace | -rdfa_devpace |
                   -rdfa_wpace_exempt>

    symrdf -file <Filename> -sid <SymmID> -rdfg <GrpNum>
         [-i <Interval>] [-c <Count>]

       checkpoint

    symrdf -file <Filename> -sid <SymmID> -rdfg <GrpNum>
         [-noprompt] [-i <Interval>] [-c <Count>]
         [-v | -noecho] [-force]

       migrate <-setup |
                -replace <R1 -new_rdfg <GrpNum> | R2>>
                <-config <PairConfigType>>

Device File operations specific to SRDF/Metro configuration

    symrdf -file <Filename> -sid <SymmID> -rdfg <GrpNum>
         [-bypass] [-noprompt] [-i <Interval>] [-c <Count>]
         [-v | -noecho] [-force] [-symforce]

       createpair -type <R1|R2> -metro
         <-invalidate <R1 [-remote] | R2> |
          <<-establish | -restore [-remote] > [-use_bias]>>
         [-g <NewDg>]

       createpair -metro -format [-g <NewDg>]

       createpair -type <R1|R2> -metro
         -exempt [-g <NewDg>]

       deletepair [-keep <R1|R2>]

       establish [-full] [-use_bias]

       half_deletepair

       half_movepair -new_rdfg <GrpNum>

       half_swap

       movepair -new_rdfg <GrpNum> [-exempt | -keep <R1|R2>]

       query [-summary]
```

```
        restore  [-full] [-use_bias] [-remote]

        set bias <R1 | R2>

        suspend  [-keep <R1 | R2 >]

        verify [-summary]
          [-enabled | -activeactive | -activebias | -suspended |
           -syncinprog | -partitioned |
           -valid]
```

RDF Groups

RDF group operations specific to SRDF/Metro configuration

```
     symrdf -sid <SymmID> -rdfg <GrpNum>
          [-noprompt] [-i <Interval>] [-c <Count>]

        establish [-use_bias]

        query [-summary]

        verify [-summary]
          [-enabled | -activeactive | -activebias | -suspended |
           -syncinprog | -partitioned |
           -valid]
```

Dynamic Groups

```
     symrdf addgrp -label <GrpLabel> -rdfg <GrpNum>
          -sid <SymmID>
          -dir <Dir[:<Port>,...][,Dir[:<Port>,...][,...]>
          -remote_rdfg <GrpNum> -remote_sid <SymmID>
          -remote_dir <Dir[:<Port>,...][,Dir[:<Port>,...][,...]>
          [-noprompt] [-i <Interval>] [-c <Count>]
          [[-fibre] | [-gige] | [-farpoint]]
          [-link_domino] [-remote_link_domino]
          [-auto_link_recovery] [-remote_auto_link_recovery]
          [-link_limbo <Secs>] [-rem_link_limbo <Secs>]
          [-witness]
          [-vasa -async
           -sc_name <SC_Name> -remote_sc_name <SC_Name>]

     symrdf modifygrp <-add | -remove>
          <-rdfg <GrpNum> | -label <GrpLabel>>
          -sid <SymmID>
          [-dir <Dir[:<Port>,...][,Dir[:<Port>,...][,...]>]
          [-remote_dir <Dir[:<Port>,...][,Dir[:<Port>,...]
           [,...]>]
          [-noprompt] [-i <Interval>] [-c <Count>] [-star]
          [-witness]

     symrdf removegrp -sid <SymmID>
          <-rdfg <GrpNum> | -label <GrpLabel>>
          [-noprompt] [-i <Interval>] [-c <Count>] [-star]
          [-symforce]
          [-witness]
```

DESCRIPTION

The symrdf command invokes an SRDF operation on a group of remotely-mirrored standard devices. These operations can be performed on device groups, device files, composite groups, and storage groups.

These operations include establishing (data copy from the source side to the target side), splitting the remotely

mirrored pair, restoring (data copy from the target side to the source side), querying the state of the remotely mirrored pair, and setting the SRDF mode.

Both the establish and restore operations can be done fully (entire copy) or incrementally (only changed tracks are synchronized). By default, if the -full option is not specified, an incremental establish or restore is attempted. Note that you cannot perform an establish or restore if the remote pair state is not split or suspended.

Other operations include: creating, modifying and removing dynamic SRDF groups; creating, deleting and swapping SRDF device pairs; querying the state of SRDF pairs; verifying the state of SRDF pairs.

A composite group enabled for remote database consistency is called an SRDF consistency group.

Dynamic SRDF group operations consist of adding and modifying dynamic SRDF groups and creating and modiying SRDF device pairs.

ARGUMENTS

activate	Activates the SRDF/A DSE or SRDF/A write pacing feature.
addgrp	Creates a dynamic SRDF group.
checkpoint	Returns checkpoint complete status when the data in the current cycle is committed to the R2 side. Valid only for SRDF/A-capable devices that are participating in an active SRDF/A session. All the supplied devices must be in the same SRDF/A session.
createpair	Creates dynamic SRDF pairs based on devices specified in a device file. The Symmetrix ID specified is the R1 side by default, but can be the R2 side if the -R2 option is used.
deactivate	Deactivates the SRDF/A DSE or SRDF/A write pacing feature.
deletepair	Deletes dynamic SRDF pairing in the specified device group.
disable	Disables consistency protection for SRDF/A-capable devices.
enable	Enables consistency protection for SRDF/A-capable devices.
establish	Resumes remote mirroring and initiates a data copy from the source (R1) side to the target (R2) side. Depending on whether the operation is full or incremental, all or only the changed tracks are copied to the target (R2) side. Write disables the target devices to

their local hosts. Subsequently, for each SRDF pair, invalidates all or the required tracks for the target (R2) side. Finally, a full data copy is started from the source (R1) side to target (R2) side.

For this action to execute, the standard SRDF devices in the group must be split.

failback — Switches data processing back to the source (R1) side.

If the target (R2) is operational, write disables the devices on the target side to their local hosts, and resumes I/O traffic on the SRDF links. Then it write enables the devices on the source (R1) side to their local hosts.

failover — Switches data processing from the source (R1) to the target (R2) side.

If the source (R1) is operational, suspends I/O traffic on the SRDF links and write disables the devices on the source (R1) side to their local hosts. Then, it suspends traffic on the SRDF links, write enables the devices on the target side to their local hosts.

half_deletepair — Deletes one-half of the designated dynamic SRDF pair.

half_movepair — Moves one side of the dynamic SRDF pair from one SRDF group to another SRDF group.

half_swap — Swaps the SRDF personality of one-half of the designated dynamic SRDF pair. Source R1 devices become target R2 devices, and target R2 devices become source R1 devices.

invalidate — Invalidates the source (R1) device(s) or the target (R2) device(s) so that a full copy can be initiated from the remote mirror.

label — Modifies the label of an existing dynamic SRDF group.

list — Lists all SRDF devices.

merge — Merges the device track tables of the source (R1) side and the target (R2) side for one or all devices in a device group.

migrate — Migrates an existing R1 or R2 device to a new device in a new Symmetrix array.

modifygrp — Modifies an existing dynamic SRDF group.

movepair — Moves both sides of a dynamic SRDF pair from one SRDF group to another SRDF group.

msc_cleanup — Initiates an MSC cleanup operation.

not_ready	Sets the source (R1) devices or the target (R2) devices to be SRDF Not Ready to their respective local hosts.
ping	Pings one or more Symmetrix arrays. By default, only the remotely-connected Symmetrix arrays are pinged over SRDF links. If the -rdf option is specified, an SRDF-configured Symmetrix array attached locally is pinged over SRDF links. The default is to ping it locally over the I/O channel. The ping action returns a unique return code if all or some of the targeted Symmetrix arrays were successfully pinged.
query	Returns information about the state of the SRDF mirroring for all device pairs in a device group.
ready	Sets the source (R1) devices or the target (R2) devices to be SRDF Ready to their respective local hosts.
refresh	Marks the source (R1) devices or the target (R2) devices to refresh from the remote mirror.
removegrp	Removes a dynamic SRDF group.
restore	Resumes remote mirroring and initiates a data copy from the target (R2) side to the source (R1) side. Depending on the operation, all (full) or only changed tracks (incremental) are copied to the source (R1) side. Write disables the target devices to their local hosts. Subsequently, for each SRDF pair, invalidates all or the required tracks for the source (R1) side. Finally, a data copy is started from the target (R2) side to the source (R1) side. This action can only be executed if the standard SRDF devices in the group are in the Split pair state.
resume	Resumes I/O traffic on the SRDF links for the remotely-mirrored pairs in the group.
rw_disable	Read/write disables the source (R1) devices or the target (R2) devices to their local hosts.
rw_enable	Sets the source (R1) devices or the target (R2) devices to be read and write enabled to their local hosts.
set	Modifies the setting to the specified value.
split	Stops remote mirroring for the SRDF pairs in the group. For each SRDF pair, suspends I/O traffic on the SRDF links and write enables the target devices to their local hosts.

This action can only be executed if the remotely-mirrored standard devices in the group are synchronized.

suspend — Suspends I/O traffic on the SRDF links for the remotely mirrored pairs in the group.

swap — Swaps the SRDF personality of the designated SRDF devices. Source R1 devices become target R2 devices, and target R2 devices become source R1 devices.

update — Starts an update of the source (R1) side after a failover, and while the target (R2) side may still be operational to its local hosts.

verify — Verifies whether all SRDF device pairs in a device group or device file are in specific pair state, SRDF mode, or are enabled.

The verify action returns a unique return code if the verify criteria are not met.

write_disable — Write disables the source (R1) devices or the target (R2) devices to their local hosts.

KEYWORDS

acp_skew — Modifies the value of the skew factor for the set adaptive copy mode.

bias — Sets the bias to the R1 or R2 device. The device that has the bias set, will be exported as the R1.

When the SRDF link becomes Not Ready (NR), the bias device will be made accessible to the host and the non-bias device will be made not accessible to the host.

This action can only be executed if the SRDF devices in the group are in the ActiveBias SRDF pair state.

dev — Lists all SRDF devices that are configured on the Symmetrix arrays attached to this host. This is the default with the list command.

domino — Sets the domino mode for the SRDF pairs in a device group to enabled or disabled. Acceptable values are on (enabled) or off (disabled).

mode — Sets the SRDF mode for the SRDF pairs in a device group to a <ModeVal> of synchronous (sync), semi-synchronous (semi), asynchronous (async), adaptive copy disk mode (acp_disk), or adaptive copy write pending mode (acp_wp), or it turns

off the adaptive copy mode (acp_off).

nr_if_invalid — Identifies whether the R2 devices in the devices pairs are marked as Not Ready due to invalid tracks. Acceptable values are on (enabled) or off (disabled).

pd — Lists all SRDF devices visible to this host.

rdfa — Sets the SRDF/A attributes.

rdfa_dse — Sets the SRDF/A data set extension attributes.

rdfa_pace — Sets the SRDF/A pace attributes.

rdfg — Sets the SRDF/A group attributes.

SC_Name — Storage container name

OPTIONS

-acp_disk — Verifies whether the SRDF device pairs are operating in adaptive copy disk mode.

-acp_wp — Verifies whether the SRDF device pairs are operating in adaptive copy write pending mode.

-activeactive — Verifies whether the SRDF device pairs are in the ActiveActive SRDF pair state.

-activebias — Verifies whether the SRDF device pairs are in the ActiveBias SRDF pair state.

-add — Adds supporting SRDF directors to a dynamic SRDF group.

-all — When used with list, lists all SRDF mirrors of the selected devices and will also list SRDF devices which are VVOLs.

When used with a device group to perform an SRDF control or set operation, it targets the SRDF action at all devices in the device group: Standard SRDF devices and locally-attached BCV SRDF devices.

This option is only supported for list and device group operations.

-asynchronous — When used with verify, verifies whether the SRDF device pairs are operating in asynchronous SRDF mode.

When used with addgrp, identifies that the VASA SRDF group being created should be created in Asynchronous mode. Only allowed when -vasa is specified.

-as400_pool — Specifies a DSE pool name containing SAVE devices with an AS400 emulation.If no argument is provided, the currently associated as400 pool is removed from the SRDF group.

-auto_link_recovery

Specifies whether the auto_link_recovery feature is locally enabled or disabled. Acceptable values are on (enabled) or off (disabled).

-autolink_recovery
Specifies whether the autolink_recovery feature is locally enabled or disabled, or if enabled on both sides if -both_sides is specified. Acceptable values are on (enabled) or off (disabled).

-autostart Specifies whether SRDF/A DSE is automatically activated when an SRDF/A session is on (Enabled) or off (Disabled) for the SRDF group. Acceptable values are on (Enabled) or off (Disabled).

-bcv Targets the SRDF action at the device group's locally-associated BCV devices that are configured as SRDF BCV devices.

-both Lists all SRDF devices that are RDF1 or RDF2 capable, when used with -dynamic.

-both_sides Targets the SRDF control at both sides of the SRDF link.

-brbcv Targets the SRDF action at the device group's remotely-associated SRDF BCV devices that can be BCV paired with the remote mirrors of the locally-associated SRDF BCV devices.

-bypass Bypasses any existing Symmetrix exclusive locks during an SRDF operation.
WARNING: Only use this flag if you are certain no other SRDF operation is in progress at the local and/or remote Symmetrix arrays.

-c Specifies the number (count) of times to repeat the operation, displaying results appropriate to the operation at each iteration. Used with list, ping, query, and verify operations.

Specifies the number (count) of times to attempt to acquire an exclusive lock on the Symmetrix host database, or on the local and/or remote Symmetrix arrays.
For SRDF control and set operations.

The time to wait between operation iterations or between attempts to acquire a needed lock is specified by -i (interval).

If neither -c nor -i is specified, these operations fail if unable to acquire a requested lock.

If -c is not specified and -i is specified, the program loops continuously, repeating the specified list, ping, query, or verify operation, or until the specified SRDF control or set operation has acquired the locks it needs and can start.

-cascade Lists cascaded SRDF devices (RDF21) and SRDF devices paired with RDF21 devices. When used with -R1, lists RDF1 devices paired with RDF21 devices. When used with -R2, lists RDF2 devices paired with RDF21 devices.

-cg Specifies a composite group name.

-cg_consistent Verifies that CG is SRDF consistent.

-ckd3380_pool
Specifies a DSE pool name containing SAVE devices with a CKD3380 emulation. If no argument is provided the currently associated ckd3380 pool is removed from the SRDF group.

-ckd3390_pool
Specifies a DSE pool name containing SAVe devices with CKD3390 emulation. If no argument is provided, the currently associated ckd3390 pool is removed from the SRDF group.

-concurrent Lists concurrent SRDF devices (RDF11, RDF22) and SRDF devices paired with a concurrent SRDF device. When used with -R1, lists RDF11 devices and SRDF devices paired with a concurrent SRDF device. When used with -R2, lists RDF22 devices and RDF2 devices paired with a concurrent device.

-config Specifies the existing device pair configuration type. The only valid value is pair, which indicates the device pair is an R1-R2 pair that is not part of an SRDF concurrent or cascaded configuration.

-consistency Displays the SRDF consistency state when listing SRDF devices.

-consistent Verifies that the SRDF device pairs for a single SRDF group are in the R2 consistent SRDF pair state, when used with the verify command. If the -noinvalids option is specified, it also verifies the SRDF device pairs do not have invalid tracks on the R1 and R2 sides.

When used with set mode sync, transitions from async to sync mode.

-cycle_time The minimum time to wait before attempting an SRDF/A cycle switch.

-delay The maximum host I/O delay that the SRDF/A write pacing feature will cause. Valid range of values for wpace_delay is between 1 and 1000000 micro-seconds.

-detail Displays detailed information for a CG, with information specific to the enable mode.

-dir Lists the local Symmetrix directors

(separated by commas), such as, 1a, 1b, and so on.

-diskless_rdf — Lists diskless SRDF devices as well as devices paired with diskless SRDF devices. When used with -R1, lists RDF1 devices that are either diskless or that are paired with a diskless device. When used with -R2, lists RDF2 devices that are either diskless or are paired with a diskless device. When used with -R21, lists RDF21 devices that are either diskless or that are paired with a diskless device.

-domino — Specifies whether link domino is locally enabled or disabled, or enabled on both sides of an SRDF link when -both_sides is specified. Acceptable values are on (enabled) or off (disabled).

-dp_autostart — Specifies whether the SRDF/A device-level pacing feature is automatically on (Enabled) or off (Disabled) when an SRDF/A session is activated for the SRDF group. Acceptable values are on (enabled) or off (disabled).

-dup_pair — Lists devices whose partner is of the same SRDF personality.

-dynamic — Lists dynamic SRDF devices only. When used with -R1, lists RDF1 devices that are dynamic. When used with -R2, lists RDF2 devices that are also dynamic. When used with -both, lists SRDF devices that are RDF1 and RDF2 capable.

-enabled — Verifies that the SRDF device pairs are in the SRDF consistency state.

-establish — Begins a device copy. When used with createpair, a full copy from the source to the target is started, synchronizing the dynamic SRDF pairs in the device file. When used with failover, the dynamic SRDF device pairs swap personality and an incremental establish is initiated. When used with -format option, the dynamic SRDF pair(s) will be made read/write on the SRDF link without synchronization of the source and target after all tracks are cleared on what will become the R1 and R2 side.

-exempt — Allows devices to be added, removed, or suspended without affecting the state of the SRDF/A or SRDF/Metro session or requiring that other devices in the session be suspended. Used for an SRDF group supporting an active SRDF/A session or an active SRDF/Metro session.

When used with list operations, lists devices that are consistency exempt or that are paired with devices that are consistency exempt, and lists devices that are exempt within an SRDF/Metro session.

-failedover — Verifies the SRDF device pairs are

in the Failedover pair state.

-farpoint — Uses the FarPoint communication protocol.

-fba_pool — Specifies a DSE pool name containing SAVE devices with an FBA emulation. If no argument is provided, the currently associated FBA pool is removed from the SRDF group.

-fibre — Uses the Fibre Channel communication protocol.

-file — Specifies a device file for SRDF operations. The device file contains device pairs (SymDevnames) listing a pair on each line. R1 devices are listed in the first column and R2 devices are listed in the second column. When using this option, specify an SRDF group to which all devices in the first column belong.

For an R1 migration, you create a device file to pair SRDF devices with the new non-SRDF devices. This pairing is used to temporarily transfer data from the existing R1 devices to the devices that will eventually replace them in an SRDF pair. The first column of the device file contains the existing R1 devices paired with the new non-SRDF devices in the second column.

For an R2 migration, you create a device file to pair SRDF devices with the new non-SRDF devices, which will eventually replace the existing R2 devices. The first column of the device file contains the R1 devices that are paired with the new devices in the second column when migration is complete.

-force — Attempts to force the operation even though one or more devices in the device group may not be in the normal, expected SRDF state or SRDF mode for that operation.

-format — When used with createpair, no data synchronization is done between source and target dynamic SRDF pairs in the device file after all tracks are cleared on what will become the R1 and R2 side.

-full — Requests a full establish or restore operation. By default, if the -full option is not specified, an incremental establish or restore is attempted.

-g — Specifies the device group name. For createpair, the device group is created with the devices in the device file.

-gige — Uses the Gigabyte Ethernet protocol.

-h — Provides brief online help information.

-half_pair — Lists devices whose partner is not an SRDF device.

-hop2 Targets the SRDF action at the group's second-hop devices in a cascaded SRDF relationship. For example, in an RDF1 group, the action targets the R21->R2 pair of the R1->R21->R2 relationship.

-hop2_rdfg Specifies the SRDF group number for the second-hop. This can only be used when issuing a createpair -hop2 with an SG.

-hwcomp Specifies whether the hardware compression feature is on (enabled) or off (disabled). Compression minimizes the amount of data to be transmitted over an SRDF link. Acceptable values are on (enabled) or off (disabled).

-i Specifies the interval, in seconds, to wait, either between successive iterations of a list, ping, query, or verify operation, or between attempts to acquire an exclusive lock on the Symmetrix host database or on the local and/or remote Symmetrix arrays for SRDF control and set operations.

The default interval is 30 seconds.
The minimum interval is 5 seconds.
For passive actions the minimum interval is 15 seconds. Passive actions are actions that do not acquire an exclusive lock.

-immediate Applies only to SRDF/A-capable devices. Causes failover, split, and suspend commands to drop the SRDF/A session immediately.

-invalidate Marks the source (R1) devices or the target (R2) devices to invalidate for a full copy when an SRDF pair is created.

-keep Sets the winner side of the SRDF/Metro group to the R1 or the R2 side, as specified.

When the SRDF link becomes Not Ready (NR), devices on the winner side will be made accessible to the host and devices on the loser (non-winner) side will be made inaccessible to the host.

This option can only be used when the SRDF devices in the group are in the Active SRDF mode.

When used with movepair, this option can be used when moving devices out of the SRDF/Metro group but not when moving devices into the group.

-label Specifies a label for a dynamic SRDF group.

-limbo Specifies the local link limbo value if used with the set argument and the remote link limbo value if used with the -both_sides option.

This option is for advanced users only. The link limbo range is 0-120 seconds. This value specifies the length of time for Enginuity to wait from the point of link-down detection before actually updating the link status as down. If the link status is still sensed as Not Ready after the link limbo time expires, devices are then marked Not Ready to the link. The default time is 10 seconds.

-link_domino — Specifies whether the link domino feature is enabled or disabled. Acceptable values are on (enabled) or off (disabled).

-link_limbo — Specifies a local link limbo value if used with the addgrp argument.

This option is for advanced users only. The link limbo range is 0-120 seconds. This value specifies the length of time for Enginuity to wait from the point of link-down detection before actually updating the link status as down. If the link status is still sensed as Not Ready after the link limbo time expires, devices are then marked Not Ready to the link. The default time is 10 seconds.

-metro — When used with the createpair action, indicates the SRDF pairs will be created in an SRDF/Metro configuration.

When used with the list action, it will list devices that are in an SRDF/Metro configuration.

-metrodr — Lists the Symmetix devices that are identified as MetroDR devices.

-new_rdfg — Specifies a new SRDF group in which to move a dynamic SRDF pair.

When used with the migrate -replace R1 operation, specifies the SRDF group that connects the new R1 device to the existing R2 device.

-nobcv — Lists standard SRDF devices only (excludes SRDF BCV devices).

-noecho — Does not echo the progress status of the SRDF action to stdout.

-noinvalids — Used with the -consistent option to verify the SRDF device pairs are in the R2 Consistent pair state, and the device pairs do not have invalid tracks on the R1 and R2 sides.

-noprompt — Requests that prompts are not displayed after the command is entered. The default is to prompt the user for confirmation.

-nowd — Bypasses the check to ensure the target of the operation is not writable by the host.

This applies to the source (R1) devices for the invalidate R1 action and to the target (R2) devices for the invalidate R2 action.

It applies to the R2 devices for all createpair actions, as well as to the R1 devices for the createpair -invalidate R1 action.

-offline — Obtains data only from the configuration database on the host. No connections are made to any Symmetrix arrays. Available only for query and list actions.

-partitioned — Verifies whether the SRDF device pairs are in the Partitioned pair state.

-priority — The priority used to determine which SRDF/A sessions to drop if the cache becomes full. Valid range of values is 1-64.

-R1 — Lists the RDF1 (R1) devices only. When used with -dynamic, lists the dynamic RDF1 devices.

-R2 — Lists the RDF2 (R2) devices only. When used with -dynamic, lists the dynamic RDF2 devices.

-R21 — Lists the RDF21 (R21) devices only.

-rbcv — Targets the SRDF action at the device group’s locally-associated SRDF BCV devices that can be BCV-paired with the remote mirrors of the standard SRDF devices.

-rdf — Pings the specified Symmetrix array using the SRDF links, even though it may be attached locally to the host.

-rdf_mode — Sets the following SRDF modes for one or more SRDF pairs when issuing createpair: Synchronous (sync), Semi-synchronous (semi), Asynchronous (async), Adaptive Copy Disk mode (acp_disk), or Adaptive Copy Write Pending mode (acp_wp).

-rdfa — Lists or queries devices that are SRDF/A-capable.

-rdfa_devpace — Indicates that the operation affects the SRDF/A device-level write pacing feature.

-rdfa_dse — Indicates that the operation affects the SRDF/A Delta Set Extension (DSE) feature.

-rdfa_not_pace_capable — When used with list operations, lists SRDF pairs that cannot be write paced for any reason other than exemption from group-level SRDF/A Write Pacing.

-rdfa_pace — Indicates that the operation affects

both the group-level and the device-level components of the SRDF/A Write Pacing feature.

-rdfa_wpace Indicates that the operation affects the SRDF/A group-level Write Pacing feature.

-rdfa_wpace_exempt
When used with activate operations, excludes the specified devices from SRDF/A group-level Write Pacing.

When used with deactivate operations, clears a previously-set exempt state so that the specified devices can participate in SRDF/A group-level Write Pacing.

When used with list operations, lists devices that are exempt from SRDF/A group-level Write Pacing.

-rdfg Displays the SRDF devices belonging to the specified Symmetrix RA (SRDF) group number when used with list operations. When combined with other list options, displays devices that match those options within the specified SRDF group.

When used with controls, modify, and query, this option causes the operation to target a specific SRDF group.

When used with -sg createpair -hop2 this identifies the SRDF group associated with the SG. The -hop2_rdfg identifies the SRDF group to be used to create the new hop2 pair.

For the migrate operation, -rdfg represents the SRDF group used to pair the existing R1 device to the associated device in the second column of the file. When migrating the R1 device, this SRDF group is temporary.

For composite groups, the format is a comma-separated combination of Symmetrix IDs and RA group numbers (SID:GRP,SID:GRP), or a comma-separated list of predefined names in the form of name:RdfGroupName, RdfGroupName, and so on.

-refresh Marks the source (R1) devices or the target (R2) devices to refresh from the remote mirror.

-remote Requests a remote data copy flag with failback, failover, restore, update, and resume. When the concurrent link is ready, data is copied to the concurrent SRDF mirror. These operations require the remote data copy option, or the concurrent link to be suspended.

-remote_auto_link_recovery

Enables auto-link recovery remotely.

-remote_dir Specifies a comma-separated list of remote Symmetrix directors, such as 1a,1b, and so on.

-remote_link_domino
Enables link domino remotely.

-remote_link_limbo
Specifies a remote link limbo value (range is 0-120 seconds).

This option is for advanced users only.

This period specifies the length of time for Enginuity to wait from the point of remote-link-down detection before actually updating the remote link status as down. If the remote link status is still sensed as Not Ready after the link limbo time expires, devices are then marked Not Ready to the remote link. Default time is 10 seconds.

-remote_rdfg Specifies the SRDF group number for the remote Symmetrix array.

-remote_sc_name
Specifies the storage container name associated with the remote sid.

Only allowed when -vasa is specified.

-remote_sg Specifies the remote storage group name. When used with createpair -hop2 this is the remote storage group for the second-hop.

-remote_sid Specifies the remote Symmetrix array unique ID.

-remove Removes supporting SRDF directors from a dynamic SRDF group.

-replace Replaces an original R1 or R2 device with a new device on another Symmetrix array. This is the second step of the SRDF migrate procedure.

-restore Begins a full copy from the target to the source, synchronizing the dynamic SRDF pairs in the device file. When used with failover, the dynamic SRDF device pairs swap personality and an incremental restore is initiated.

-resv Lists SRDF devices that have SCSI reservations.

-rp Allows the operation even though one or more devices are tagged for RecoverPoint. When used with refresh, only allowed for refresh R1.

-sc_name Specifies the storage container name associated with the sid.

Only allowed when -vasa is specified.

-semisynchronous
Verifies whether the SRDF device pairs are operating in a semi-synchronous SRDF mode.

-setup Establishes a relationship and starts data synchronization between an existing R1 device on one Symmetrix array and a new device on another Symmetrix array. This is the first step of the SRDF migrate procedure.

-sg Specifies a storage group name.

-sid Specifies the unique Symmetrix ID.

For file operations, identifies the Symmetrix ID associated with the devices in the first column of the file.

-split Verifies the SRDF device pairs are in the Split pair state.

-star Specifies the action is targeted for devices in STAR mode.

-star_mode Lists devices currently in STAR mode.

-star_async_target
Allows the devices in async mode in an SRDF/Star environment to track changes between two target SRDF/Star sites.

-star_sync_target
Allows the devices in sync mode in a SRDF/Star environment to track changes between two target SRDF/Star sites.

-summary Displays a table showing the number of device pairs in each SRDF pair state.

When used with query operations, replaces the usual command output.

When used with the verify operation, includes the table with the command output.

When used with the -interval and/or -count options, also displays synchronization rate and estimated ime to completion for both the query and verify operations.

-suspended Verifies whether the SRDF device pairs are in the Suspended pair state.

-susp_offline Verifies whether the SRDF device pairs are in the Suspended pair state and the SRDF link Suspend state is OFFLINE.

-swcomp Specifies whether the software compression feature is enabled or disabled. Compression minimizes the amount of data to be transmitted over an SRDF link. Acceptable values are on (enabled) or off (disabled).

-symforce Requests the Symmetrix array force operation

be executed when normally it is rejected. Use extreme caution when using this option.

When used with removegrp action, removes one side of a dynamic SRDF group if the other side is not defined or is not accessible.

CAUTION: Use care when applying -symforce, as data could be lost or corrupted. Use of this option is not recommended, except in an emergency.

NOTE: To enable -symforce, a parameter called SYMAPI_ALLOW_RDF_SYMFORCE in the options file must be set to TRUE.

When used with -symforce, a split command executes on an SRDF pair, even when the pair is sync in progress or restore in progress. During the execution of an establish or restore command, -symforce prohibits the verification of valid tracks on the device at the source.

-synchronized Verifies whether the SRDF device pairs are in the Synchronized pair state.

-synchronous Verifies whether the SRDF device pairs are operating in synchronous SRDF mode.

-syncinprog Verifies whether the SRDF device pairs are in the SyncInProg pair state.

-threshold When used with rdfa_dse, it specifies the percentage of the Symmetrix array's write pending limit. Once the cache usage of all active SRDF/A groups in the array exceeds this limit, data tracks for this SRDF group start to spill over to disks. Acceptable values are between 20 and 100 integer values.

When used with rdfa_pace, it specifies the minimum percentage of the system write pending cache at which the Symmetrix array starts pacing host write I/Os for this SRDF group. Valid range of values is between 1 and 99.

-transmit_idle
If set to on (enabled), provides an extra level of protection so that the SRDF/A session does not drop when the SRDF link cannot transmit data. Acceptable values are on (enabled) or off (disabled).

-type Indicates the SRDF mirror type (R1 or R2) of the local devices when creating SRDF device pairs.

-unknown Lists the Symmetix devices that are defined in an RDF group which has an unknown identifier.

-until Specifies the number of invalid tracks to reach on the target side before the update action stops repeating every time the SRDF

pair becomes updated.

-updated — Verifies whether the SRDF device pairs are in the R1 Updated pair state.

-updateinprog — Verifies whether the SRDF device pairs are in the R1 UpdInProg SRDF pair state.

-use_bias — Used with createpair -establish/-restore, establish or restore actions, to indicate that SRDF/Metro configuration will use bias instead of witness protection.

-v — Provides a more detailed, verbose output.

-valid — Verifies whether the SRDF device pairs are in a valid pair state.

-vasa — Specifies the RDF group being created is a VASA SRDF group.

-witness — When used with addgrp it identifies the SRDF group as a witness SRDF group.

When used with removegrp or modifygrp it specifies the action is targeted for an SRDF group which is a witness SRDF group.

-wp_autostart — Specifies whether the SRDF/A write pacing feature is automatically on (enabled) when an SRDF/A session is activated for the SRDF group. Acceptable values are on (enabled) or off (disabled).

PARAMETERS

acp_disk — Sets the device pairs to adaptive copy disk mode.

acp_off — Turns off the adaptive copy mode for the device pairs.

acp_wp — Sets the device pairs to adaptive copy write pending mode.

all — All SRDF (RA) groups.

async — Sets the device pairs to asynchronous mode.

CgName — The composite group name.

CycleTime — The minimum time to wait before attempting an SRDF/A cycle switch.

DgName — The device group name.

Dir — Local or remote Symmetrix director.

DseThreshold — Specifies the percentage of the Symmetrix array’s write pending limit.

Filename — The device file name.

GrpLabel — The dynamic SRDF group label.

GrpNum — The SRDF (RA) group number.

ModeVal — Sets the SRDF mode for one or more SRDF pairs in a device group. Possible values are:

- sync
- semi
- acp_disk
- acp_wp
- acp_off
- async

Note: The skew <SkewVal> cannot be used with the async value. The -consistent option is only allowed with the sync value.

NewDg — Identifies the DG name in which to add the device pair after the pair is created.

off — Turns domino mode, hwcomp, swcomp, auto_link_recovery, transmit_idle, wp_autostart, dp_autostart, autostart or nr_if_invalid off.

on — Turns domino mode, hwcomp, swcomp, auto_link_recovery, transmit_idle, wp_autostart, dp_autostart, autostart or nr_if_invalid on.

PairConfigType — Identifies the pair configuration type for the migrate action. Possible values are:

- pair

PoolName — Specifies the name of a collection of SAVE devices with a particular emulation type to use for SRDF/A DSE.

Port — Local or remote Symmetrix director port.

R1 — Operation is targeted at the source (R1) side.

When used with set bias, indicates the bias will be set on the R1.

When used with suspend -keep, indicates that the R1 side will remain accessible to the host when the suspend completes.

R2 — Operation is targeted at the target (R2) side.

When used with set bias, indicates the bias will be set on the current R2, so that it will subsequently be reported as the R1; the current R1 will subsequently be reported as the R2.

When used with suspend -keep, indicates that the current R2 side will be the R1 when the suspend completes and its data will remain accessible to the host; the current R1 will be the R2 side when the suspend completes, and its data will not be accessible to the host.

RdfGroupName — Logical name associated with the SRDF

(RA) group(s).

Secs — Number of seconds to set for link limbo.

semi — Sets the device pairs into semi-synchronous mode.

SessPriority — The priority used to determine which SRDF/A sessions to drop if the cache becomes full.

SgName — The storage group name.

SkewVal — Sets the skew factor for the adaptive copy mode. Possible values range from 0 to 65,534 tracks. For devices larger than 2 GB, a value of 65,535 can be specified to target all the tracks of any given drive.

State — Specifies the state for various SRDF set operations. Possible values are:
- on
- off

SymmID — 12-character ID that specifies the Symmetrix array.

sync — Sets the device pairs into synchronous mode.

value — Adaptive copy skew value or link limbo value.

WpaceDelay — The maximum host I/O delay that the SRDF/A write pacing feature will cause.

WpaceThreshold — The minimum percentage of the system write pending cache at which the Symmetrix array will start pacing host write I/Os for this SRDF group.

RETURN CODES

Code #	Code Symbol
0	CLI_C_SUCCESS
1	CLI_C_FAIL
2	CLI_C_DB_FILE_IS_LOCKED
3	CLI_C_SYM_IS_LOCKED
18	CLI_C_ALREADY_IN_STATE The device or device group is already in the desired SRDF state or mode. Applicable only for SRDF control and SRDF set mode actions.
19	CLI_C_GK_IS_LOCKED All GateKeepers to the Symmetrix array are currently locked.
21	CLI_C_NEED_MERGE_TO_RESUME You must issue an SRDF merge track table before you can resume the SRDF links.

22 CLI_C_NEED_FORCE_TO_PROCEED
You can only proceed if you are certain, use the force flag.

23 CLI_C_NEED_SYMFORCE_TO_PROCEED
You can only proceed if necessary, use the symforce flag. NOTE: EMC DOES NOT RECOMMEND USING THIS FLAG EXCEPT IN AN EMERGENCY.

WARNING: Using symforce may result in DATA INTEGRITY PROBLEMS.

The following codes are returned by the ping action:

8 CLI_C_NOT_ALL_PINGED
Not all of the targeted Symmetrix arrays are successfully pinged.

9 CLI_C_NONE_PINGED
None of the targeted Symmetrix arrays are successfully pinged.

The following codes are returned by the verify action:

4 CLI_C_NOT_ALL_SYNCHRONIZED
Not all requested devices are in the Synchronized state.

5 CLI_C_NONE_SYNCHRONIZED
No requested devices are in the Synchronized state.

6 CLI_C_NOT_ALL_UPDATED
Not all requested devices are in the Updated state.

7 CLI_C_NONE_UPDATED
No requested devices are in the Updated state.

14 CLI_C_NOT_ALL_VALID
Not all requested devices are in a valid SRDF state (some devices are in the Invalid SRDF state).

15 CLI_C_NONE_VALID
No requested devices are in a valid SRDF state (all devices are in the Invalid SRDF state).

25 CLI_C_NOT_ALL_SPLIT
Not all requested devices are in the Split SRDF state (some devices are in the Split SRDF state).

26 CLI_C_NONE_SPLIT
No requested devices are in the Split SRDF state.

27 CLI_C_NOT_ALL_SYNCINPROG
Not all requested devices are in the SyncInProg SRDF state

(some devices are in the SyncInProg SRDF state).

28 CLI_C_NONE_SYNCINPROG
No requested devices are in the SyncInProg SRDF state.

31 CLI_C_NOT_ALL_SUSPENDED
Not all requested devices are in the Suspended SRDF state (some devices are in the Suspended SRDF state).

32 CLI_C_NONE_SUSPENDED
No requested devices are in the Suspended SRDF state.

33 CLI_C_NOT_ALL_FAILED_OVER
Not all requested devices are in the Failed Over SRDF state.

34 CLI_C_NONE_FAILED_OVER
No requested devices are in the Failed Over SRDF state.

35 CLI_C_NOT_ALL_UPDATEINPROG
Not all requested devices are in the R1 UpdInProg SRDF state.

36 CLI_C_NONE_UPDATEINPROG
No requested devices are in the R1 UpdInProg SRDF state.

37 CLI_C_NOT_ALL_PARTITIONED
Not all requested devices are in the Partitioned SRDF state.

38 CLI_C_NONE_PARTITIONED
No requested devices are in the Partitioned SRDF state.

39 CLI_C_NOT_ALL_ENABLED
Not all devices in the request are in the Enabled consistency state.

40 CLI_C_NONE_ENABLED
No devices in the request are in the Enabled consistency state.

45 CLI_C_NOT_ALL_SUSP_AND_OFFLINE
Not all devices in the request are in the Suspended state and the Offline link suspend state.

46 CLI_C_NONE_SUSP_AND_OFFLINE
None of the devices in the request are in the Suspended state and the Offline link suspend state.

70 CLI_C_NOT_ALL_CONSISTENT
Not all of devices in the request are consistent.

```
 71         CLI_C_NONE_CONSISTENT
            None of the devices in the
            request are consistent.

146         CLI_C_NOT_ALL_CONSISTENT_NOINVALIDS
            Not all of the devices in the
            request are consistent and have no
            invalid tracks.

147         CLI_C_NONE_CONSISTENT_NOINVALIDS
            None of the devices in the
            request are consistent and have no
            invalid tracks.

148         CLI_C_NOT_ALL_SYNCHRONOUS
            Not all of the devices in the
            request are in synchronous SRDF mode.

149         CLI_C_NONE_SYNCHRONOUS
            None of the devices in the
            request are in synchronous SRDF mode.

150         CLI_C_NOT_ALL_SEMISYNCHRONOUS
            Not all of the devices in the
            request are in semisynchronous SRDF mode.

151         CLI_C_NONE_SEMISYNCHRONOUS
            None of the devices in the
            request are in semisynchronous SRDF mode.

152         CLI_C_NOT_ALL_ASYNCHRONOUS
            Not all of the devices in the
            request are in asynchronous SRDF mode.

153         CLI_C_NONE_ASYNCHRONOUS
            None of the devices in the
            request are in asynchronous SRDF mode.

154         CLI_C_NOT_ALL_ACP_WP
            Not all of the devices in the
            request are in adaptive copy write pending
            SRDF mode.

155         CLI_C_NONE_ACP_WP
            None of the devices in the
            request are in adaptive copy write pending
            SRDF mode.

156         CLI_C_NOT_ALL_ACP_DISK
            Not all of the devices in the
            request are in adaptive copy disk SRDF mode.

157         CLI_C_NONE_ACP_DISK
            None of the devices in the
            request are in adaptive copy disk SRDF mode.

182         CLI_C_NOT_ALL_ACTIVEACTIVE
            Not all requested devices are
            in the ActiveActive state.

183         CLI_C_NONE_ACTIVEACTIVE
            No requested devices are in
            the ActiveActive state.

184         CLI_C_NOT_ALL_ACTIVEBIAS
            Not all requested devices are
            in the ActiveBias state.
```

```
185         CLI_C_NONE_ACTIVEBIAS
            No requested devices are in
            the ActiveBias state.
```

EXAMPLES

To create an R1 device group called ProdDB, enter:

```
symdg create ProdDB -type RDF1
```

To split all standard SRDF devices in device group ProdDB, enter:

```
symrdf -g ProdDB split
```

To establish all standard devices in group ProdDB, and initiate a full data copy from the source side (R1) to the target side (R2) for all the devices, enter:

```
symrdf -g ProdDB -full establish
```

To perform an incremental restore from the target (R2) side to the source side (R1) for the SRDF pairs in group ProdDB, enter:

```
symrdf -g ProdDB restore
```

To query information about all SRDF standard devices in device group ProdDB, enter:

```
symrdf -g ProdDB query
```

The following example creates a dynamic SRDF pair from a file called devices. The devices file contains Symmetrix device names that constitute the dynamic pairs. The local source Symmetrix is sid 810. Enter:

```
symrdf createpair -g ProdDB -file devices -sid 810
  -rdfg 2 -invalidate r2 -nop -type RDF1
```

Communication is through SRDF group 2. The -invalidate option indicates that the R2 devices are the targets that will be refreshed from the R1 source devices.

The device file syntax contains two columns. R1 devices are listed in the first column and R2 devices are listed in the second column as follows:

```
010A   00B7
010F   00BF
0106   00C5
```

To delete the SRDF pair, enter:

```
symrdf deletepair -g ProdDB -rdfg 2
```

symrecover

Provides the user interface to the EMC SRDF Automated Recovery tool.

SYNOPSIS

```
symrecover [-h]

symrecover [-env | -version]

symrecover start -g <DgName> | -cg <CgName>
                  -mode SYNC | ASYNC [-out <LogPath>]
     [-options <OptionFile>]
```

DESCRIPTION

The symrecover command provides the user interface to the EMC SRDF Automated Recovery tool. The various compound actions perform the necessary commands, in the proper order, to allow you to monitor and restart a single EMC SRDF/S or SRDF/A session.

This command can be run from either the R1 or the R2 side as long as the group being monitored is fully viewable from the host. Either -g or -cg must be specified.

PATH is a REQUIRED environment variable setting. The fully qualified path to the PERL binary directory that is shipped with Solutions Enabler must be added to the head of default system path. Optionally the monitor can be invoked by fully qualifying the perl executable.

ARGUMENTS

start	Starts the recovery session.

OPTIONS

-cg	Specifies the name of the composite group to monitor.
-env	Displays the default option settings. All other coded options are ignored. This option is ignored if coded in the options file.
-g	Specifies the name of the device group to monitor.
-h	Provides brief, online help information.
-mode	Specifies the type of SRDF session to monitor. Valid values are SYNC and ASYNC. There is no default and this option must be specified.
-options	The fully-qualified file containing these program options. Double quote characters are allowed to frame the option’s value but are not required. If quotes are used, leading and trailing blanks are preserved in all enclosed values.

Options with no CLI value can be assigned 0 for disable and 1 for enable. Comments are allowed using the "#" character anywhere on the line. Blank lines are permitted.

'g' and 'cg' can not be placed in the options file.

Option names are case sensitive.

Example:

```
#
# Option file for cg TestCG01
#

monitor_cycle_time = 180
     # Changing monitor
     # cycle time from 300
     # (default) to 180
     # seconds
# Currently installed symcli location
symcli_dir =
  C:\\program files\\tps\\emc\\symcli\\bin
#
monitor_only = 1
#
```

-out — Specifies an alternate fully-qualified directory location to use for the log file. If the output directory cannot be created or written to, a warning message is issued and monitoring continues, and messages are displayed on the console but are not written to the log file.

The default is to the log in the current working directory.

-version — Returns the installed symrecover version. This option is ignored if coded in the options file.

PARAMETERS

CgName — Specifies the user-defined composite group name.

DgName — Specifies the user-defined device group name.

LogPath — Defines an alternate directory location for the log file directory. Overrides the default directory.

OptionFile — Specifies the filename and path of the internal log file. If an alternate log file name is specified when starting a symreplicate session, the same log file name must be specified when issuing the stop, restart, query, show, and delete commands for the same symreplicate session.

RETURN CODES

Code #	Code Symbol

```
 0          CLI_C_SUCCESS

 1          CLI_C_FAIL
```

FILES

The option file is created by the user. It must conform to the following syntax:

```
   #Comment
   cascaded_monitor_both_hops=[0 | 1]

   email_server= <e_srvr_addr> /
      [email_addr_target= <e_addr1, e_addr2, ..., ...>
      [email_subject= <err_subject_string>]
      [email_log_level= <severity_level>]
      [email_frequency= <email_frequency>]

   goldcopy_location= <location>
   goldcopy_type= <copytype>
   goldcopy_state_startup= <copystate>
   goldcopy_state_post_restart= <copystate>
   goldcopy_resync_interval= <resynctime>
   goldcopy_max_wait= <maxwaittime>
   goldcopy_clone_list= <list>

   log_level= <severity_level>

   monitor_cycle_time= <cyc_time>
   monitor_only=[0 | 1] | run_once=[0 | 1]
         | run_until_first_failure=[0 | 1]

   rdfg <concur_def>

   restart_adcopy_resynch_threshold= <tracks>
   restart_attempt_pause= <time>
   restart_delay= <time>
   restart_group_on_startup
   restart_max_attempts= <attempts>
   restart_max_wait_adcopy_sync= <time>
   restart_max_wait_state_change= <statetime>
   restart_max_wait_warn_interval= <warntime>
   restart_rdfa_min_cycle_warn_interval= <cyclewarntime>
   restart_rdfa_min_cycle_warn_value= <warntime>
   restart_state_syncinprog_wait_time= <time>
   restart_state_transmit_warn_interval= <time>
   restart_state_transmit_wait_time= <transwaittime>
   restart_sync_type= <synctype>
   restart_window= <time>

   # End of Option File script
```

The following are the descriptions of the various option parameters available within an option file:

Option	Description
cascaded_monitor_both_hops	Values: 0 or 1 Specifies to monitor both hops of a cascaded SRDF configuration. This option is not enabled by default. This option requires the R1->R21 session to be in SYNC mode and R21->R2

session to be in ASYNC mode.

Note: If this option is enabled, the -mode option is ignored.

email_addr_source — Value: an email address, e_addr1

Specifies the email address to use in the 'from' field of all emails sent by symrecover. No checks are done about the validity of this email address. If this value is not set, then a default value is generated based on the system's hostname and current user.

email_addr_target — Value: a list of valid email addresses, e_addr1,e_addr2, ..., ...

Specifies the email notification address(es) to alert on errors. If any of the email_* options are specified, this option must also be specified to activate email alerts. Multiple comma-delimited addresses may be specified.

There is no default value.

email_server — Value: e_srvr_addr

Specifies the host target email server. If any of the email_* options are specified, then this option must also be specified to activate email alerts.

There is no default value.

email_subject — Value: err_subject_string

Specifies the email notification subject on errors. The default value is:

SymRecover Alert: Host [HostName] Group [DgName]

email_log_level — Value: severity_level

Specifies the severity level desired for the email alert-triggering message. Possible values are:

Value	Meaning
0	Off
1	Only errors are reported
2	Errors and Warnings are reported
3	Errors, Warnings, and Informational messages are reported
4	All messages are reported including all SYMCLI commands and responses.

Note: For each message that meets the particular logging level requirement, an

email is shipped with that message. It is highly recommended that at most this be set to either a 1 or a 2.

If the required email options (email_server and email_addr_target) are not specified, then the default value is 0. If they are specified, then the default value is 1.

email_frequency — Value: email_frequency

Specifies the email frequency desired. Possible values are:

Value	Meaning
per_log_msg	Email per log msg that meets logging level requirement.
per_monitor_loop	Email per monitor loop
per_restart	Email when error is detected, when session is restarted and finished.

default: per_restart.

goldcopy_location — Value: location

Specifies the location of the backup (gold copy). Possible values are:

Location	Meaning
NONE	No gold copy is desired. All other goldcopy_* options are ignored.
R2	A gold copy on the R2 side is desired.
ALL	A gold copy on the R21 side and R2 side is desired.

The default is R2 and this value is case insensitive.

goldcopy_type — Value: copytype

Specifies the type of goldcopy to create on the R2 side. Possible values are:

copytype	Meaning
none	No gold copy is desired.
bcv	A BCV gold copy on the R2 side is desired.
clone	A Clone gold copy on the R2 will be created.

The default is bcv and this value is not case sensitive.

Note: When bcv copytype is selected, BCVs must be paired with the R2 devices before starting symrecover.

Note: When clone copytype is selected, target devices must have a clone session with the R2 devices before starting symrecover.

goldcopy_state_startup
Value: copystate

Specifies the desired state of the R2 BCV gold copy upon routine startup. Possible values are:

copystate	Meaning
establish	The devices should be established (BCV only).
split	The devices should be split (BCV only).
activated	The devices should be activated (Clone only).
created	The devices should be in created state (Clone only).
none	The devices should be unchanged.

The default is none and this value is case insensitive.

Note: If the goldcopy type is BCV and the default state of the BCVs is establish, this may drop an SRDF/A session under certain conditions.

goldcopy_state_post_restart
Value: copystate

Following a successful SRDF/A session, restart or BCV resync, specifies the required state of the R2 gold copy. Possible values are:

copystate	Meaning
establish	The devices should be left established (BCV only).
split	The devices should be split (BCV only).
activated	The devices should be activated (Clone only).
created	The devices should be in created state (Clone only).

The default is split and this value is case insensitive.

Note: If the goldcopy type is BCV and the default state of the BCVs is establish this drop an SRDF/A session under certain conditions.

goldcopy_max_wait Value: maxwaittime

Specifies the maximum length of time in seconds for the program to wait for a group to finish synchronizing the standard devices with the goldcopies during a restart.

Possible values are 0 to maxint.

The default is 0 which is to wait forever.

goldcopy_clone_list
Value: list

If goldcopy_type is set to CLONE, this option tells symrecover which list within the DG or CG to use to find the CLONE devices.

Possible values are:

list	Meaning
tgt	Use the TGT list.
bcv	Use the BCV list.

goldcopy_resync_interval
Value: resynctime

Defines the amount of time in minutes when the gold copy BCV mirror or clone is automatically resynchronized. If the goldcopy_state_post_restart is split or activated, the resynchronization causes the BCV mirror to be established and then the split or clone to be recreated and activated. If the goldcopy_state_post_restart is establish, or created then resynchronization, this ensures the BCVs or Clones are in the requested state.

This action only takesplace during non-error periods.

Valid values are 0, and 15 to maxint.

Zero (0) indicates that the mirrors are never to be automatically synchronized outside of error producing events. The default is 0.

Note: If the goldcopy type is BCV, then the act of frequently synchronizing the R2 BCVs may drop an SRDF/A session under certain conditions.

monitor_cycle_time Value: cycletime

Defines the number of seconds to pause between monitor status scans.

The minimum value is 60 seconds; the maximum is 3600 seconds.

The default value is 300 seconds.

monitor_only Values allowed: 0 or 1

Specifies whether to monitor only the state of the specified group. No recovery actions are performed.

This option is not enabled by default.

Note: monitor_only, run_once, and run_until_first_failure are mutually exclusive options.

run_once Values allowed: 0 or 1

Specifies whether to check the status of the group once. If the group required recovery actions, it performs them. Exits after one check.

This option is not enabled by default.

This option ignores the setting of restart_max_attempts.

Note: monitor_only, run_once, and run_until_first_failure are mutually exclusive options.

run_until_first_failure
Values allowed: 0 or 1

Specifies whether to monitor the group until the first failure occurs, and then exits without performing any recovery action.

This option is not enabled by default.

This option ignores the setting of restart_max_attempts.

Note: monitor_only, run_once, and run_until_first_failure are mutually exclusive options.

rdfg Specifies the concurrent RDF definition for the group. This value is taken directly as specified and no data validation is done on it.

Monitoring RDF1 type DG/CG containing concurrent R1 devices or monitoring RDF2 type DG/CG containing concurrent R2 devices requires that the -rdfg option is specified.

This option is not set by default and non-concurrent RDF groups are assumed.

Note: If the group is a composite group and consistency is enabled, then this must be of the "name:" format and this value is case sensitive.

restart_adcopy_resynch_threshold
Value: tracks

Specifies the number of tracks outstanding that, during recovery, will trigger a switch over to SRDF/A.

The default value is 30000.

restart_attempt_pause
Value: time (seconds)

Specifies the wait time before an attempt is made to restart a failed session to allow for things to settle down. Then after the restart_attempt_pause is complete, symrecover restarts the overall monitor loop.

If there is still a problem, the restart failure count is incremented and a restart is attempted.

Valid values are 30 to 3600 seconds.

The default is 60 seconds.

restart_delay Value: time (seconds)

Specifies the wait time after an attempt is made to restart a failed session and the attempt itself fails.

Valid values are 0 (no delay, immediately restart) to maxint.

The default is 30 seconds.

restart_group_on_startup
On symrecover startup, if the group being monitored is not initially in a CONSISTENT state, symrecover, by default, considers that an error occurred and exits.

If this option is specified, symrecover attempts to recover the group on startup.

This option is not enabled by default.

restart_max_attempts
Value: attempts

Specifies the maximum number of restart attempts to perform within the restart_window interval. After this limit is reached, the program

terminates.

The valid range is from 0 to maxint.

The value of 0 means to infinitely attempt.

The default is 5 attempts.

restart_max_wait_adcopy_sync
Value: time (seconds)

Specifies the length of time in seconds that, during a restart, the program waits for a group to achieve the restart_adcopy_resync_threshold number of tracks pending.

Valid values are 0 to maxint.

The value of 0 means to infinitely wait.

The default is 0.

restart_max_wait_state_change
Value: statetime (seconds)

Specifies the length of time in seconds, during a restart, that the program waits for a group to change to a desired state once the change is requested.

Valid values are 0 to maxint.

The value of 0 means to infinitely wait.

The default is 0.

restart_max_wait_warn_interval
Value: warntime (seconds)

Specifies the length of time in seconds, while waiting for a state change to occur during a restart, for a progress warning message to be displayed.

Valid values are 0 and 30 to maxint.

The value of 0 means to wait forever.

The default is 600 seconds.

restart_rdfa_min_cycle_warn_interval
Value: cyclewarntime (seconds)

Specifies the length of time in seconds for a warning message to be repetitively displayed when the RDFA minimum cycle time exceeds the restart_rdfa_min_cycle_warn_value parameter.

Valid values are 30 to maxint.

The default is 600.

restart_rdfa_min_cycle_warn_value

Value: warntime (seconds)

Specifies the maximum value in seconds which a trigger can occur with a warning message, indicating that the RDFA minimum cycle time has exceeded this value.

Valid values are 0 and 30 to maxint.

The value of 0 means this feature is turned off.

The default is 0.

restart_state_syncinprog_wait_time
Value: time (seconds)

Specifies the maximum length of time in seconds, during a group syncinprog state, that a sleep is done before rechecking the group status.

Valid values are 30 to maxint.

The default is 120 seconds.

restart_state_transmit_warn_interval
Value: time (seconds)

Specifies the interval of time in seconds, while a group remains in a transmit idle state, that a warning message is generated.

Possible values are 0 to maxint.

The default is 300 seconds.

restart_state_transmit_wait_time
Value: transwaittime (seconds)

Specifies the maximum length of time in seconds during a group transmit idle state, that a sleep is done before rechecking the group status.

Valid values are 30 to maxint.

The default is 120 seconds.

restart_sync_type Value: synctype

Specifies the type of synchronization to use following the detection of a failed SRDF/A session. Possible values are:

synctype	Meaning
ADCOPY	Adaptive copy disk
SYNC	Synchronous mode
NONE	Intermediate track resynch stage is attempted. A direct re-establish using the existing SRDF session

mode is attempted.

The default is ADCOPY.

restart_window — Value: time (seconds)

Specifies the length of time in seconds, starting with the first failure, that begins the clock for counting all successive failures.

Any failures that occur within this timespan are considered grouped.

This window is used to determine the maximum number of restarts that are permitted per window of time.

The minimum value is 1800 seconds; the maximum is 86400 seconds.

The default is 3600 seconds.

log_level — Value: The desired logging level.

Possible values are:

Value	Meaning
0	Off.
1	Only Errors are reported.
2	Errors and Warnings are reported.
3	Errors, Warnings, and Informational messages are reported.
4	All messages are reported.

The default is 3.

symreplicate

Performs a coherent, recurrent, background copy of data through an SRDF/TimeFinder configuration.

SYNOPSIS

```
symreplicate [-h]

   start -g <DgName> | -cg <CgName>
     -options <OptionFile> [-log <LogFile>]
     [-sid <SymmID>] [-preaction <ScriptFile>]
     [-postaction <ScriptFile>] [-postcycle <ScriptFile>]
     [-steperror <ScriptFile>] [-foreground] [-noprompt]
     |-consistent] [-recover]
     [-setup [-optimize|-optimize_rag|-exact]]

   setup -g <DgName> | -cg <CgName>
     -options <OptionFile>
     [-log <LogFile>] [-sid <SymmID>]
     [-optimize|-optimize_rag|-exact]
     [-foreground] [-noprompt] [-recover]

   stop -g <DgName> | -cg <CgName> | -log <LogFile>
     [-sid <SymmID>] [-step] [-noprompt]

   restart -g <DgName> | -cg <CgName> | -log <LogFile>
     [-sid <SymmID>] [-options <OptionFile>]
     [-foreground] [-noprompt] [-recover]

   query -g <DgName> | -cg <CgName> | -log <LogFile>
     [-sid <SymmID>] [-i Interval] [-c Count]

   show -g <DgName> | -cg <CgName> | -log <LogFile>
     [-sid <SymmID>] [-args] [-devs] [-opts] [-all]

   list -sid <SymmID> [-sort <Field>]

   delete [-g <DgName> | -cg <CgName> | -log <LogFile>]
     -sid <SymmID>

   release -g <DgName> | -cg <CgName> | -log <LogFile>
     [-sid <SymmID>] [-force]

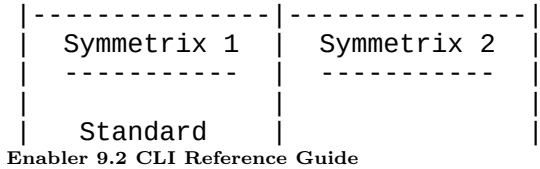
   stats -g <DgName> | -cg <CgName> | -log <LogFile>
     [-sid <SymID>] [-i Interval] [-c Count]
     [-cycle] [-itrks] [-all]
```

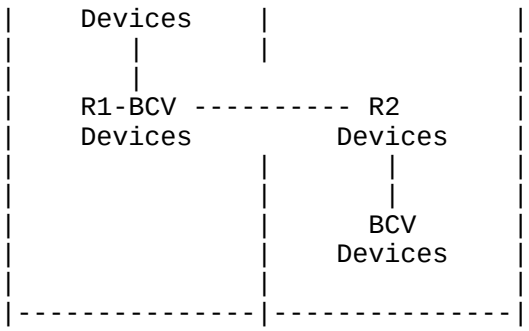
DESCRIPTION

The symreplicate command performs automated, incremental, coherent copies of data. By default, the symreplicate session is performed as a background process on Unix and Windows platforms. Two Symmetrix array configurations are supported:

1. SINGLE_HOP configuration
2. MULTI_HOP configuration

The symreplicate SINGLE_HOP configuration:





The single-hop configuration copies data from the standard devices on Symmetrix array 1 to the BRBCV devices on Symmetrix array 2. Then symreplicate incrementally establishes SRDF and BCV pairs, and differentially splits BCV pairs to reduce required data transfers.

Device locks are held during the entire symreplicate session. This keeps other applications from altering device states while symreplicate is active. For example, if a symreplicate session terminates unexpectedly due to a system crash, you may need to manually release the locks. For more information on releasing the locks, refer to the symdev manpage.

To set up this configuration, take any number of standard devices of the same type (R1, R2, or non-RDF), and create a device or composite group of the same type. Add devices to the group, associate an equal number of R1-BCV devices of matching sizes, and then associate an equal number of BRBCV devices, also of matching sizes.

The required SYMCLI command sequence is similar to the following:

```
symdg create newdg
symdg add dev 000 -g newdg -sid 0001
symdg add dev 001 -g newdg
 < . . . >
symbcv associate dev 1C0 -g newdg
symbcv associate dev 1C1 -g newdg
 < . . . >
symbcv associate dev 210 -g newdg -bcv -rdf
symbcv associate dev 211 -g newdg -bcv -rdf
 < . . . >
```

To start a symreplicate session, the following conditions must be met, or you must perform these actions using the setup command:

- Both sets of BCV pairs must have a pairing relationship.

- The local BCV pairs must be Established, the SRDF pairs must be Suspended, and the BRBCV pairs must be Split.

These conditions can be met through a manual process or symreplicate can put the devices into the initial state if you specify -setup on the command line.

To manually set up the devices, follow these steps, which are the same steps used by the symreplicate -setup option:

1) Split the BCV pairs.
   (Wait for any ongoing establish to complete...)
    symmir split -g newdg

2) Establish the SRDF pairs.
    symrdf establish -g newdg -bcv

3) Suspend the SRDF pairs.
   (Wait for the establish operation to complete...)
    symrdf suspend -g newdg -bcv

4) Establish the BCV pairs.
    symmir establish -g newdg

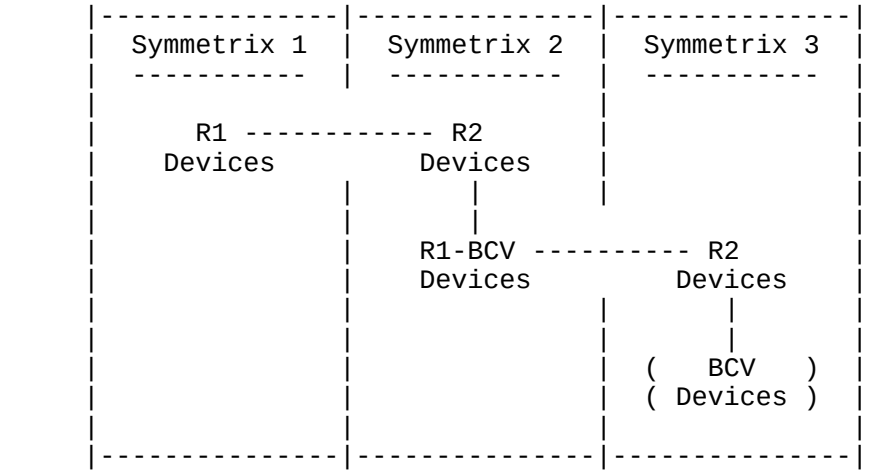
5) Establish the BRBCV pairs.
    symmir establish -g newdg -bcv -rdf

6) Split the BRBCV pairs.
    (Wait for the establish operation to complete...)
    symmir split -g newdg -bcv -rdf

You may have to use additional flags for the SYMCLI commands shown above (such as -full) if the BCV pairs do not have an existing pairing relationship. For more information on BCV pairing, refer to the symmir manpage.

Use the -preaction and -postaction options to specify scripts to run before and after step 1.

The symreplicate multi-hop configuration:



The multi-hop configuration copies data from the R1 devices on Symmetrix array 1 to the R2 devices on Symmetrix array 2 (and, by default, to BCV devices), which are then copied to devices on Symmetrix array 3. The symreplicate command then incrementally establishes the SRDF and BCV pairs, and differentially splits the BCV pairs to reduce required data transfers.

The second-hop BCV devices are used by default, but may be omitted by disabling the USE_FINAL_BCV option in the options file. For more information, refer to the FILES section below.

Device locks are held during the entire symreplicate session. This prohibits other applications from altering device states while symreplicate is active. For example, if a symreplicate session terminates unexpectedly due to a system crash, you may need to manually release the locks. For more information on releasing the locks,

refer to the symdev manpage.

To set up this configuration, create an R1 device or composite group, and add any number of R1 devices. Remotely associate an equal number of matching sized R1-BCV RBCV devices. If the second-hop BCVs are being used, these must also have a pairing relationship with the second-hop R2 devices.

The required SYMCLI command sequence is similar to the following:

```
symdg create newdg2 -type RDF1
symdg add dev 040 -g newdg2 -sid 0001
symdg add dev 041 -g newdg2
 < . . . >
symbcv associate dev 1A0 -g newdg2 -rdf
symbcv associate dev 1A1 -g newdg2 -rdf
 < . . . >
```

Note that the final BCVs, if used, are not a part of the group itself. It is useful during setup to use a device file to control the second-hop pairs. For more information on device files, refer to the symmir man page.

To start a symreplicate session, the following conditions must be met, or use -setup to perform these tasks:

- All BCV pairs must have a pairing relationship.

- The local SRDF pairs must be Synchronized, the BCV pairs must be Established, and the remote SRDF pairs must be Suspended.

- If the final BCVs are used, the BCV pairs on the second-hop Symmetrix array must also be in the Split state.

These conditions can be met through a manual process or symreplicate can put the devices into the initial state if you specify -setup on the command line.

To manually setup the devices, follow these steps, which are the same steps used by the symreplicate -setup option:

1) Split the first-hop BCV pairs.
   (Wait for any ongoing establish to complete...)
   symmir split -g newdg2 -rdf -remote

2) Establish the remote SRDF pairs.
   (This step is completed via the previous command.)

3) Suspend the remote SRDF pairs.
   (This step will be completed by the next command.)

4) Establish the first-hop BCV pairs.
   (Wait for the RDF establish to complete...)
   symmir establish -g newdg2 -rdf

5) Establish the second-hop BCV pairs.
   symmir establish -f second_hop_devs.txt

6) Split the second-hop BCV pairs.
   (Wait for the establish operation to complete...)
   symmir split -f second_hop_devs.txt

Note that steps 5 and 6 are only performed if the second-hop BCV pairs are used. The second_hop_devs.txt file is the device file. For more information on device files, refer to the symmir man page.

You may need to use additional flags for the SYMCLI commands shown above (such as -full) if the BCV pairs do not have an existing pairing relationship. For more information on BCV pairing, refer to the symmir manpage.

Use the -preaction and -postaction flags to specify scripts to run before and after step 1.

ARGUMENTS

delete	Deletes the symreplicate log files written to the SFS (Symmetrix File System).
list	Lists the symreplicate log files written to the SFS.
query	Displays the status of a symreplicate session.
release	Releases device locks held from a terminated symreplicate session.
restart	Restarts a terminated symreplicate session at the step where it stopped.
setup	Places all devices into the initial state, then stops.
show	Shows information from the symreplicate log file.
start	Begins a new symreplicate session.
stats	Displays cycle time and/or invalid track statistics.
stop	Terminates an existing symreplicate session.

OPTIONS

-all	Displays all available information, including -args, -devs, and -opts for a show action or "-itrks -cycle" for a stats action. This is the default for both actions.
-args	Shows information about command line arguments used to start the symreplicate session.
-c	Specifies a count to perform a specific action.
-cg	Specifies a composite group name.
-consistent	Consistently splits all the BCV pairs on the local Symmetrix array for a single-hop configuration, or on the Hop 1 remote Symmetrix array for a multi-hop configuration.

-cycle Displays cycle time statistics. Valid for a stats action only.

-devs Shows the devices that participate in the symreplicate session.

-exact Causes setup to pair all STDs and BCVs exactly as they appear in the group. Valid only for -setup.

-force Forces the SYMAPI server to allow an action that would ordinarily fail. Valid for a release action only.

-foreground Makes the symreplicate process run in the foreground (background is the default). Not available on Windows systems.

-g Specifies the device group name.

-h Provides brief, online help information.

-i Specifies the interval in seconds to repeat a specific action.
The default interval is 30 seconds if -c is used.
The minimum interval is 5 seconds.
For passive actions the minimum interval is 15 seconds. Passive actions are actions that do not acquire an exclusive lock.

-itrks Display invalid track statistics. Valid for a stats action only.

-log Specifies a log file name.

-noprompt Disables the system prompt for user confirmation.

-optimize Causes -setup to split all pairs and perform an optimized STD-BCV pairing within a Symmetrix array. Valid only for -setup.

-optimize_rag Causes -setup to split all pairs and perform an optimized STD-BCV pairing within RA Groups. Valid only for -setup.

-options Specifies an options file.

-opts Shows options from the symreplicate options file that were used to start the session.

-postaction Performs this action after the instant or consistent split. Information is provided through the following command line arguments:

Argument	Meaning
P1	Device group name

-postcycle Performs this action after each cycle. Information about the completed cycle is provided through the following

command line arguments:

Argument	Meaning
P1	Device group name.
P2	Current cycle number.
P3	Number of cycles (<NumCycles> from the option file, see below).
P4	Cycle duration in seconds.

The -postcycle option is not available on all platforms.

-preaction Performs this action before the instant or consistent split. Information is provided through the following command line arguments:

Argument	Meaning
P1	Device group name

-recover Tells symreplicate to recover the device locks, if possible. Makes sure that no other symreplicate session using the same devices is running when using -recover.

-setup Performs the steps necessary to place the devices in the initial state.

-sid Specifies the unique Symmetrix ID to where the symreplicate log file is written. The Symmetrix array must be running Enginuity version 5669 or higher.

-sort Specifies how the symreplicate log file names are sorted, either by name or by type. The default is to sort the log files by name.

-step Stops after the current symreplicate step completes, rather than waiting until the end of a cycle.

-steperror Performs this action when symreplicate encounters an error during normal cycling. This parameter takes effect after a symreplicate session was successfully launched. Diagnostic information about the error is provided through the following command line arguments:

Argument	Meaning
P1	Device or composite group name.
P2	Current cycle number.
P3	Number of cycles (<NumCycles> from the option file, see below).
P4	Cycle duration in seconds.
P5	Step number where the error occurred.
P6	Text message describing when the error occurred. Details about the error are written to the SYMAPI log file.

This parameter does not replace the need to check for a non-zero exit status from symreplicate. Errors that occur before the symreplicate session is launched do not cause the script to execute. The -steperror option is not available on all platforms.

PARAMETERS

CgName — Composite group name provided by user.

DgName — Device group name.

Field — Field to use when sorting the file names. Valid fields are name and type.

LogFile — Filename and path of an internal log file. If an alternate logfile name is specified when starting a symreplicate session, the same logfile name must be specified when issuing stop, restart, query, show, and delete commands for the same symreplicate session.

OptionFile — Name of the text file that contains the required parameters of the replicate actions.

ScriptFile — Filename of the pre-action or post-action script.

SymmID — 12-digit Symmetrix ID. See the -sid option above for restrictions. Also, if -sid is specified when starting a symreplicate session, the same Symmetrix ID must be specified when issuing stop, restart, query, show, and delete commands for the same symreplicate session.

RETURN CODES

Code #	Code Symbol
0	CLI_C_SUCCESS
1	CLI_C_FAIL

FILES

The option file is created by the user. It must conform to the following syntax:

```
#Comment
SYMCLI_REPLICATE_HOP_TYPE=<RepType>
SYMCLI_REPLICATE_CYCLE=<CycleTime>
SYMCLI_REPLICATE_CYCLE_OVERFLOW=<OvfMethod>
SYMCLI_REPLICATE_CYCLE_DELAY=<Delay>
SYMCLI_REPLICATE_NUM_CYCLES=<NumCycles>
SYMCLI_REPLICATE_USE_FINAL_BCV=<TRUE|FALSE>
SYMCLI_REPLICATE_LOG_STEP=<TRUE|FALSE>
SYMCLI_REPLICATE_GEN_TIME_LIMIT=<TimeLimit>
SYMCLI_REPLICATE_GEN_SLEEP_TIME=<SleepTime>
SYMCLI_REPLICATE_RDF_TIME_LIMIT=<TimeLimit>
SYMCLI_REPLICATE_RDF_SLEEP_TIME=<SleepTime>
```

```
    SYMCLI_REPLICATE_BCV_TIME_LIMIT=<TimeLimit>
    SYMCLI_REPLICATE_BCV_SLEEP_TIME=<SleepTime>
    SYMCLI_REPLICATE_MAX_BCV_SLEEP_TIME_FACTOR=<Factor>
    SYMCLI_REPLICATE_MAX_RDF_SLEEP_TIME_FACTOR=<Factor>
    SYMCLI_REPLICATE_PROTECT_BCVS=<Protection>
    SYMCLI_REPLICATE_TF_CLONE_EMULATION=<TRUE|FALSE>
    SYMCLI_REPLICATE_PERSISTENT_LOCKS=<TRUE|FALSE>
    SYMCLI_REPLICATE_CONS_SPLIT_RETRY=<NumRetries>
    SYMCLI_REPLICATE_R1_BCV_EST_TYPE=<EstablishType>
    SYMCLI_REPLICATE_R1_BCV_DELAY=<EstablishDelay>
    SYMCLI_REPLICATE_FINAL_BCV_EST_TYPE=<EstablishType>
    SYMCLI_REPLICATE_FINAL_BCV_DELAY=<EstablishDelay>
    SYMCLI_REPLICATE_ENABLE_STATS=<TRUE|FALSE>
    SYMCLI_REPLICATE_STATS_RESET_ON_RESTART=<TRUE|FALSE>

The allowed options are as follows:

Option Name     Description
--------------  -----------------------
SYMCLI_REPLICATE_HOP_TYPE
                <RepType> is required, and is one of:

                  <RepType>  Meaning
                  ---------  ---------
                  SINGLE     Single-hop configuration.
                  MULTI      Multi-hop configuration.

SYMCLI_REPLICATE_CYCLE
                <CycleTime> is the period to wait
                between copy operations, in minutes, or
                in hh:mm format. Defaults to 0. Either
                SYMCLI_REPLICATE_CYCLE or
                SYMCLI_REPLICATE_CYCLE_DELAY is
                required, though both may be set to 0.

SYMCLI_REPLICATE_CYCLE_OVERFLOW
                <OvfMethod> is a description of what to
                do if the cycle overruns the specified
                CycleTime. It is one of the following:

                  <OvfMethod>  Meaning
                  -----------  ---------
                  IMMEDIATE    Begin the next cycle
                               immediately. This is the
                               default.
                  NEXT         Skip this cycle and wait
                               for the next to begin.

SYMCLI_REPLICATE_CYCLE_DELAY
                <Delay> is the minimum time to wait
                between adjacent cycles. Even if a cycle
                overruns the specified CycleTime and
                OvfMethod is set to IMMEDIATE, if a
                Delay is specified, symreplicate still waits
                this long before beginning another cycle.
                Defaults to 0. Either
                SYMCLI_REPLICATE_CYCLE or
                SYMCLI_REPLICATE_CYCLE_DELAY is
                required, though both may be set to 0.

SYMCLI_REPLICATE_NUM_CYCLES
                <NumCycles> is the number of cycles to
                perform before exiting. A value of zero
                causes the action to cycle indefinitely.

SYMCLI_REPLICATE_USE_FINAL_BCV
                May be used to omit the final BCV copy
```

in the chain of data propagation for a multi hop configuration. By default, the final BCV is used. If this option is set to FALSE, the second-hop BCV devices will be omitted.

SYMCLI_REPLICATE_LOG_STEP

Causes symreplicate to write an entry to the SYMAPI log file after each step is completed. The entry shows the time the step ended and whether or not the step was successful.

SYMCLI_REPLICATE_GEN_TIME_LIMIT

Controls how long to retry recoverable errors that occur when a BCV or RDF control operation fails. An example of this type of error is failing to acquire a Symmetrix lock.

<TimeLimit>

Controls how long symreplicate retries operations. The timer applies only when an error occurs continuously, or when no data has flowed for the duration of the timer.

Therefore, it is possible to set a TimeLimit that is smaller than the actual amount of time that the operation requires. For example, if on average it takes an hour to establish your RDF devices, it is possible to set a 15 minute RDF time limit that only expires when no data is transferred between the devices for a period of 15 minutes. As long as data flows between the devices, the timer does not expire.

Time limits are specified using either of the following formats:

1. HH:MM - Specifies the number of hours and minutes
2. SSS - Specifies the number of seconds

If not specified, the following default time limit, specified as HH:MM, applies:

SYMCLI_REPLICATE_GEN_TIME_LIMIT=00:30

A time limit of zero (0) indicates that no time limit applies. This causes the operation to be retried indefinitely.

SYMCLI_REPLICATE_BCV_TIME_LIMIT

Controls how long to retry errors, or to wait for data to flow, while querying BCV devices to determine whether or not they have entered a specific state.

<TimeLimit>

See SYMCLI_REPLICATE_GEN_TIME_LIMIT.

If not specified, the following default time limit, specified as HH:MM, applies:

SYMCLI_REPLICATE_BCV_TIME_LIMIT=02:00

A time limit of zero (0) indicates that no time limit applies. This causes the operation to be retried indefinitely.

SYMCLI_REPLICATE_RDF_TIME_LIMIT

Controls how long to retry errors, or to wait for data to flow, while querying SRDF devices to determine whether or not they have entered a specific state.

<TimeLimit>

See SYMCLI_REPLICATE_GEN_TIME_LIMIT.

If not specified, the following default time limit, specified as HH:MM, applies:

SYMCLI_REPLICATE_RDF_TIME_LIMIT=04:00

A time limit of zero (0) indicates that no time limit applies. This causes the operation to be retried indefinitely.

SYMCLI_REPLICATE_GEN_SLEEP_TIME

During normal processing, symreplicate must wait for certain operations to complete before going on to the next step. The symreplicate action determines that an operation is complete when devices have entered a specific state. If the operation is not complete, symreplicate sleeps for a period of time, then checks the device state again.

<SleepTime>

Specifies the minimum time that symreplicate sleeps before checking to see if devices have entered a specific state, or retrying an operation when a recoverable error occurs.

When checking the device state, symreplicate calculates how long to sleep based on the number of invalid tracks and the rate at which data is moving. Therefore, the actual time that symreplicate sleeps may be greater than SleepTime. To set the maximum time that symreplicate sleeps, specify the corresponding factor.

SleepTimes are specified using the same format as for TimeLimits. However, unlike TimeLimits, a SleepTime must be greater than zero.

If not specified, the following default sleep time applies:

SYMCLI_REPLICATE_GEN_SLEEP_TIME=10

SYMCLI_REPLICATE_BCV_SLEEP_TIME
During normal processing, symreplicate must wait for certain BCV operations to complete before going on to the next step. The symreplicate action determines that a BCV operation is complete when devices have entered a specific state. If the operation is not complete, symreplicate sleeps for period of time, then checks the device state again.

<SleepTime>

See SYMCLI_REPLICATE_GEN_SLEEP_TIME

If not specified, the following default sleep time applies:

SYMCLI_REPLICATE_BCV_SLEEP_TIME=10

To set a maximum sleep time for BCV operations, use SYMCLI_REPLICATE_MAX_BCV_SLEEP_TIME_FACTOR

SYMCLI_REPLICATE_RDF_SLEEP_TIME
During normal processing, symreplicate must wait for certain SRDF operations to complete before going on to the next step. The symreplicate action determines that an SRDF operation is complete when devices have entered a specific state. If the operation is not complete, symreplicate sleeps for period of time, then checks the device state again.

<SleepTime>

See SYMCLI_REPLICATE_GEN_SLEEP_TIME

If not specified, the following default sleep time applies:

SYMCLI_REPLICATE_RDF_SLEEP_TIME=15

To set a maximum sleep time for SRDF operations, use SYMCLI_REPLICATE_MAX_RDF_SLEEP_TIME_FACTOR

SYMCLI_REPLICATE_MAX_BCV_SLEEP_TIME_FACTOR
Provides a way to specify the maximum time that symreplicate sleeps before checking again to see if BCV devices have entered a specific state.

<Factor>

The product of this value multiplied by the sleep time provides the maximum time that symreplicate sleeps.

Specifies a factor using a positive, non-zero integer.

If not specified, the following default factor applies:

SYMCLI_REPLICATE_MAX_BCV_SLEEP_TIME_FACTOR=3

By default, symreplicate sleeps between 10 and 30 seconds when checking on the state of BCV devices, up to a maximum time of 2 hours.

SYMCLI_REPLICATE_MAX_RDF_SLEEP_TIME_FACTOR

Provides a way to specify the maximum time that symreplicate sleeps before rechecking if the SRDF devices have entered a specific state.

<Factor>

The product of this value multiplied by the sleep time provides the maximum time that symreplicate sleeps.

Specifies a factor using a positive, non-zero integer.

If not specified, the following default factor applies:

SYMCLI_REPLICATE_MAX_RDF_SLEEP_TIME_FACTOR=4

By default, symreplicate sleeps between 15 and 60 seconds, up to a maximum of 4 hours, when checking on the state of SRDF devices.

SYMCLI_REPLICATE_PROTECT_BCVS

Specifies how symreplicate performs a TimeFinder establish operation when using two-way mirrored BCV devices. If specified, all mirrors of the BCV devices join with the mirrors of the standard devices.

<Protection> can be one of the following values:

<Protection>	Meaning
NONE	Performs a normal TimeFinder establish operation, which is the default.
LOCAL	Performs a protected BCV establish for the local devices only in a single-hop configuration.
FIRST_HOP	Performs a protected BCV establish for the first-hop devices only in a multi-hop configuration.
REMOTE	Perform a protected BCV establish for the remote devices only in a single-hop configuration.
SECOND_HOP	Performs a protected BCV establish for the second-hop devices only in a multi-hop configuration.

BOTH	Perform a protected BCV establish for both the local and remote devices (single-hop) or the first and second-hop devices (multi-hop).

SYMCLI_REPLICATE_TF_CLONE_EMULATION

Indicates TimeFinder clone emulation is enabled. By default, clone emulation is disabled. A value of TRUE indicates clone emulation is enabled.

SYMCLI_REPLICATE_PERSISTENT_LOCKS

Causes symreplicate to acquire the device locks for the session with the SYMAPI_DLOCK_FLAG_PERSISTENT attribute. The default is FALSE, indicating the persistent attribute is not used when acquiring the device locks for the session. A value of TRUE indicates to use the persistent attribute.

SYMCLI_REPLICATE_CONS_SPLIT_RETRY

Controls retry attempts when a consistent split operation fails because the timing window closed before the split operation completed.

<NumRetries>

Specifies the number of error recovery attempts to be made.

A default retry value of 3 is used if the SYMCLI_REPLICATE_CONS_SPLIT_RETRY option parameter is not specified when a consistent split (-consistent) is requested.

A retry value of 0 indicates that no retry attempts be made.

SYMCLI_REPLICATE_R1_BCV_EST_TYPE

Specifies the establish type for the local first-hop BCV devices.

<EstablishType>

Specifies the way the BCV establish operation is executed by TimeFinder. Specify one of the following values:

<EstablishType>	Meaning
SINGULAR	BCV devices are established one at a time; the next device is not established until the previous device is established.
SERIAL	BCV devices are established as fast as the establish requests are accepted by the

Symmetrix array.

PARALLEL — BCV device establish requests are passed in parallel to each of the servicing DA directors.

For establish types of SINGULAR and PARALLEL, an <EstablishDelay> is specified through the SYMCLI_REPLICATE_R1_BCV_DELAY option.

SYMCLI_REPLICATE_FINAL_BCV_EST_TYPE
Specifies the establish type for the remote second-hop BCV devices.

<EstablishType>

See SYMCLI_REPLICATE_R1_BCV_EST_TYPE

For establish types of SINGULAR and PARALLEL, an <EstablishDelay> can be specified through the SYMCLI_REPLICATE_FINAL_BCV_DELAY option parameter.

SYMCLI_REPLICATE_R1_BCV_DELAY
Controls the rate of issuing establish requests for the local first-hop BCV devices.

<EstablishDelay>

For an establish type of SINGULAR, this value denotes how long to wait between issuing establish requests.

For an establish type of PARALLEL, the delay value indicates how long to wait before passing the next establish request to an individual servicing DA director.

An establish delay of 0 to 30 seconds may be specified with a value of 0 being the default.

SYMCLI_REPLICATE_FINAL_BCV_DELAY
Controls the rate of issuing establish requests for the remote/second-hop BCV devices.

<EstablishDelay>

See SYMCLI_REPLICATE_R1_BCV_DELAY

SYMCLI_REPLICATE_ENABLE_STATS
May be used to enable or disable the gathering of statistics. By default, statistics gathering is enabled.

A value of FALSE indicates that statistics gathering is to be disabled.

SYMCLI_REPLICATE_STATS_RESET_ON_RESTART
May be used to reset statistics when a restart action is executed. By default

the statistics are not reset upon restart of a symreplicate session.

A value of TRUE indicates that statistics are to be reset when restarting a symreplicate session.

EXAMPLES

To start a session for device group newdg, using an option file named opt.txt, enter:

```
symreplicate -g newdg -options opt.txt start
```

The file opt.txt could contain the following:

```
#Copy the data in a single-hop setup...
SYMCLI_REPLICATE_HOP_TYPE=SINGLE

#...every 15 minutes...
SYMCLI_REPLICATE_CYCLE=15

#...or as often as possible, if 15 minutes is
#... not enough.
SYMCLI_REPLICATE_CYCLE_OVERFLOW=IMMEDIATE
```

To query the status of the above session, enter:

```
symreplicate -g newdg query
```

To terminate the above session, enter:

```
symreplicate -g newdg stop
```

To restart the terminated session, enter:

```
symreplicate -g newdg restart
```

SEE ALSO

symmir(1), symrdf(1), symdev(1), symreturn(1)

symreturn

Indicates a return value within pre-action and post-action scripts.

SYNOPSIS

```
symreturn [-h] [<return_code>]
```

DESCRIPTION

The symreturn command is the suggested method of exiting a command script used by the symmir -instant split command.

The command script is specified as an argument to the -preaction or -postaction option to the symmir command.

ARGUMENTS

none

OPTIONS

-h	Provides brief, online help information.

PARAMETERS

return_code	A return code can be supplied to indicate a success or failure result code from the script. If no return code is supplied, a success is assumed. The return code can be user-defined to indicate a unique error condition that might occur in a specific situation during script processing.

RETURN CODES

Code #	Code Symbol
0	CLI_C_SUCCESS
#	CLI_C_FAIL

EXAMPLES

To exit the script with a successful return code, enter:

```
symreturn 0
```

```
symrpi

        Perform a setup, control or list operation
        on a RecoverPoint Appliance Cluster.

        SYNOPSIS

             symrpi -h

             symrpi -sid <SymmID> -cluster <ClusterName>
                  [-i <Interval>] [-c <Count>] [-noprompt]

                environment <-setup [-repository] | -remove [-symforce] |
                             -expand>

                create -journal -cap <#> [-captype <cyl|mb|gb|tb>]
                  [-N <#>]

                create -repository

                delete -journal -devs <<SymDevStart>:<SymDevEnd> |
                  <SymDevName>[,<<SymDevStart>:<SymDevEnd> |
                  <SymDevName>>...]>

                protect -sg <SgName>

                unprotect <-sg <SgName> [-symforce] | -devs <<SymDevStart>:
                  <SymDevEnd> | <SymDevName> [,<<SymDevStart>:
                  <SymDevEnd> | <SymDevName>>...]>>

             symrpi [-sid <SymmID>] [-cluster <ClusterName>
                  [[-sg <SgName>] | [-dev_info [<journal | atdev | protected | reposito
ry | all>]]]
                  [-detail]]

                list


        DESCRIPTION

              The symrpi command performs setup, control and list
              operations a RecoverPoint Appliance Cluster.

        ARGUMENTS

             create         Create devices, tag them as RP_INTERNAL
                            and add them to a RecoverPoint Storage Group.

             delete         Delete journal device(s).

             environment    Setup, remove or expand the environment
                            required to integrate a RecoverPoint
                            Appliance Cluster to a storage array.

             list           List general information about all RP
                            clusters running on a storage array or detailed
                            information on a specific target RP cluster.

             protect        Place the devices associated with the storage
                            group under RecoverPoint replication.

             unprotect      Remove the devices associated with the storage
                            group from RecoverPoint replication.

        OPTIONS
```

-c	Used with control operations, specifies the number (count) of times to attempt to acquire an exclusive lock on the VMAX host database. If you do not specify this option and specify an interval (-i), the program will loop continuously to list or start the control operation.
-cap	Sets the device capacity to a specific value (in cylinders, megabytes, gigabytes, or terabytes). See the '-captype' switch to set the units used.
-captype	Sets the capacity units to a specific value (either 'cyl', 'mb', 'gb', or 'tb'). Default value is megabytes (mb).
-cluster	Used to specify the name of the RecoverPoint Appliance Cluster that is the target of the specified action.
-detail	Provide detailed information on a specified RecoverPoint cluster or RP-Protected storage group, including a list of RP-Protected devices that may need attention.
-devs	Specifies a set of Symmetrix device ranges and/or individual Symmetrix devices.
-dev_info	Used with cluster list operation, to list various types of RP devices in a RP environment.
-expand	Used with the environment action to expand the infrastructure required by the RecoverPoint Appliance Cluster on the target arrays. This process involves creating the supplemental RP storage groups and masking views on the array.
-i	Specifies the repeat interval, in seconds, to wait, either between control operation attempts to acquire an exclusive lock on the VMAX host database. The default interval is 30 seconds. The minimum interval is 5 seconds.
-journal	Used with the create and delete actions to specify that RecoverPoint Journal devices are to be created or deleted.
-N	Sets the number of devices to create.
-noprompt	Requests that prompts are not displayed after the command is entered. The default is to prompt the user for confirmation.
-remove	Used with the environment action to remove the infrastructure for the RecoverPoint cluster created by the -setup option.
-repository	Used with the environment setup or create action to indicate that a Repository device should be created as part of the operation.

-setup	Used with the environment action to create the infrastructure required by the RecoverPoint Appliance Cluster on the target arrays. This process involves creating the initial RP storage group on the array.
-sg	For the list operation, provide summary or detailed information on a specific RP-protected storage group.
-sid	Specifies the unique Symmetrix ID.
-symforce	Attempts to force the operation, bypassing error checks. Can be used with the "environment -remove" action to force cleanup operations to complete. Must be used with the "unprotect -sg" action to force removal of all RP resources associated with the storage group. The -symforce option should be used cautiously and only when the user understands the implications of the action.

PARAMETERS

ClusterName	The RecoverPoint Appliance Cluster name. Name must begin with an alphanumeric character and may contain embedded hyphens but no underscores.
Count	The number of times (count) to repeat.
Interval	The interval between repetitions, in seconds.
SgName	The storage group name.
SymDevName	The Symmetrix device name, unique per Symmetrix, such as 01C.
SymDevStart	The first Symmetrix device name in a sequence, such as 001C.
SymDevEnd	The last Symmetrix device name in a sequence, such as 00B6.
SymmID	The 12-digit ID of a VMAX or VMAX3 array.

symrslv

Displays detailed mapping information about
a disk storage object.

SYNOPSIS

```
symrslv [-h] [-version] [-kb | -blocks | -mb]

   pd <PdevName>
     [-no_extents | -expand | -pdev_extents]

   lv <LVolName> -g <VgName>
     [-no_extents | -expand | -pdev_extents]
     [-stripe_column] [-type <VgType>]

   file <FileName>
     [-no_extents | -expand | -pdev_extents |
      -collapse | -phys_collapse]

   dir <Directory>
     [-no_extents | -expand | -pdev_extents]

   fs <MountPoint>
     [-no_extents | -expand | -pdev_extents]
     [-nfs]

   object <ObjectName>

   identify <ObjectName>
```

DESCRIPTION

The symrslv command displays detailed logical-to-physical mapping information specific to a disk storage object. By default, it provides data relating to the physical extents of these objects. Currently, the supported disk storage objects are:

- Physical devices
- Logical volumes
- Regular files
- Directories
- File systems
- Object

Object restriction:
Because not all options apply to all supported objects, no options can be passed if keyword object is specified.

Note that Veritas Quick I/O devices (or the symbolic links to them) are treated like physical devices by symrslv and require the pd argument.

The collapse flag is applicable to files that reside on file systems that are mounted on a striped or RAID5 LVM mirror. This collapse is a logical collapse, meaning the data can be reconstructed with the meta data returned. The phys_collapse flag causes a physical collapse, which means that the data cannot be reconstructed with the meta data returned.

In the lists of mirror physical extents and mirror physical devices for the disk storage object, CLARiiON devices are distinguished from other device types by

a (C) indicator.

Volume group name length restriction:
The volume group name field is limited to 63 characters. This length restriction is inclusive of the name and absolute path of the volume group even if the user only specified the volume group name itself. The behavior is undefined if the length is exceeded.

Logical volume name length restriction:
The logical volume name field is limited to 63 characters. The behavior is undefined if the length is exceeded.

Note that network based file systems based on the NFS protocol is supported on Linux, Solaris, HP_UX and AIX. Network based file systems based on CIFS protocol is supported on Linux and Windows. Resolving file system to a physical device and extent level is not applicable for network based file systems.

ARGUMENTS

dir	Specifies a directory.
file	Specifies a file name.
fs	Specifies a file system mount point.
identify	Specifies that the object be only identified but not resolved. Only objects with its corresponding object name can be specified as an argument to identify. Options cannot be specified because not all options apply to all supported object types, and the object type is not known when object is specified in the command line.
lv	Specifies a logical volume name.
pd	Specifies a device physical name.
object	Specifies a generic object. In this case object could be either a directory, filename, file system mount point, logical volume name or a device physical name. Options cannot be specified because not all options apply to all supported object types, and the object type is not known when the object is specified in the command line.

OPTIONS

-blocks	Displays size information in 512-byte blocks.
-collapse	Logically collapses the extents.
-expand	Expands the extents, if possible. The default is to collapse the extents.
-g	Specifies a volume group name for use with the lv argument.
-h	Provides brief, online help information.

-kb Displays size information in Kilobytes.

-mb Displays size information in Megabytes. This is the default.

-nfs Displays information on mounted network based file systems.

-no_extents Excludes any extents information from displaying.

-pdev_extents Displays Pdev level extents only (does not expand extents to reflect an underlying meta device configuration).

-phys_collapse
Physically collapses the extents.

-stripe_column
For striped logical volumes, displays the extent's stripe column number.

-type Specifies the volume group type.

-version Displays SYMAPI build and runtime versions.

PARAMETERS

Directory The directory name. For a directory that is also a mount point, use a trailing slash at the end of the directory name to obtain the directory information.

FileName The file name.

LVolName The logical volume name.

MountPoint The file system mount point, (for example, d:). For a directory that is also a mount point, do not use a trailing slash at the end of the directory name to obtain the file system.

ObjectName The directory, file name, logical volume name, MountPoint, or PdevName.

PdevName The device physical (host) name for the device, such as /dev/rdsk/c2t0d2s2.

VgName The logical volume group name.

VgType The volume group type. Supported values are:

Group Type	
DEFAULT	LINUX_LVM
AIX_LVM	LINUX_POOL
AIX_VXVM	LINUX_VXVM
AS400_LVM	ORACLE_ASM
DYNIX_SVM	SUN_SOLSTICE
EMC_PVM	SUN_VXVM
HP_LVM	WIN_LDM
HP_VXVM	WIN_VXVM

RETURN CODES

Code #	Code Symbol
0	CLI_C_SUCCESS
1	CLI_C_FAIL

EXAMPLES

To display physical extent information in kilobytes about the physical device /dev/rdsk/c2t0d2s2, enter:

```
symrslv -kb pd /dev/rdsk/c2t0d2s2
```

To display an expanded listing of physical extent information in blocks about the logical volume lvol1 in volume group ProdVG, enter:

```
symrslv -blocks -expand -g ProdVG lv lvol1
```

To display physical extent information in megabytes about the file accounts, enter:

```
symrslv -mb file accounts
```

To display physical extent information in blocks about the directory /usr, enter:

```
symrslv -blocks dir /usr
```

To display physical extent information in kilobytes about the file system whose mount point is /, enter:

```
symrslv -kb fs /
```

To display physical extent information about object /usr, enter:

```
symrslv object /usr
```

To identify object /usr, enter:

```
symrslv identify /usr
```

Notes:
- If object were to be passed as an argument to symrslv, user should not specify dir, file, fs, lv, or pd as an argument. The keyword object also does the type resolution.

- On the AS400 platform, only the lv and pd arguments are supported.

Lists the ports visible from a given director and the logical unit numbers (LUNs) visible behind a given remote port.

Lists the remote RDF directors and Symmetrix arrays visible from a given Symmetrix array and RDF director.

Shows iSCSI sessions between iSCSI initiators in an initiator group and SE director ports on a given array.

Performs ping to Remote IP Address from IP Address configured on SE/RE director.

SYNOPSIS

```
symsan [-h]

   list <-sanports | -sanluns -wwn <SanPort> >
        -sid <SymmID>
        <-dir <# | ALL> | -DX <# | ALL> | -FA <# | ALL>>
        -port <# | ALL> [-detail]

   list -sanrdf
        -sid <SymmID>
        -dir <# | ALL> [-port <# | ALL>]

symsan -sid <SymmID>

   show -iscsi_sessions
        -ig <InitiatorGroupName>
        -SE <# | ALL> [-port <# | ALL>]

symsan -sid <SymmID> [-i <Interval>] [-c <Count>]

   ping -SE <#> -network_id <NetId>
        [-ip_address <IPAddress>]
        -remote_ip_address <IPAddress>

   ping -RE <#> -port <#> -remote_ip_address <IPAddress>
```

DESCRIPTION

The symsan command lists the ports visible from a given director and the LUNs visible behind a given remote port.

It can show iSCSI sessions between iSCSI initiators in an initiator group and SE director ports along with session state.

It includes the ability to ping iSCSI Initiator IP Address from the IP Address configured on the specified SE director and network ID. The symsan can also be used to ping IP Interface configured on remote RE director port from the IP Address configured on the specified RE director and port in SRDF environment.

ARGUMENTS

list	Lists all sessions for a specified Symmetrix array, or for all Symmetrix arrays.

show — Shows detailed information about iSCSI sessions between iSCSI initiators in an IG and SE director ports on a given array.

OPTIONS

-c — Specifies the number (count) of times to repeat the operation, displaying results appropriate to the operation at each iteration. Used with ping operation.

-detail — Provides additional information if availiable. This includes the array ID for port scans and the extended LUN World Wide Name (WWN) field in LUN scans.

-dir — Specifies a DX or FA director when scanning for ports and specifies an RDF director when scanning for remote RDF directors and Symmetrix arrays.

-DX — Specifies the DX director.

-FA — Specifies the FA director.

-h — Provides brief, online help information.

-i — Specifies the interval, in seconds, to wait, between successive iterations of a ping operation.

The default interval is 30 seconds. The minimum interval is 5 seconds.

-ig — Specifies the initiator group to show iSCSI sessions for.

-iscsi_sessions — Shows iSCSI sessions on the supplied SE director port.

-ip_address — Specifies a valid IPv4 or IPv6 address configured on SE director and NetID.

-network_id — Specifies the network id for iSCSI IP interface.

-port — Specifies the port on the local director.

-remote_ip_address — Specifies a valid IPv4 or IPv6 address of host initiator or gateway.

-RE — Specifies the RE director.

-sanluns — Lists the specific device LUNs visible to the supplied director/port, behind the specified remote port WWN.

-sanports — Lists the ports visible to the supplied director/port.

-sanrdf — Lists the remote RDF directors and Symmetrix arrays visible to the supplied

RDF director.

-SE Specifies the SE director.

-wwn Specifies the SAN WWN to use for listing device LUNs behind a given remote port.

PARAMETERS

A local director or port number.

IPAddress A valid IPv4 or IPv6 address is expected.

NetId The network_id/namespace used for isolating IP routing tables in iSCSI environment.

SanPort The port WWN.

SymmID The 12-digit Symmetrix array ID.

RETURN CODES

Code #	Code Symbol
0	CLI_C_SUCCESS
1	CLI_C_FAIL

Examples

To list ports visible from a given local port, enter:

```
symsan list -sanports -sid 123 -dir 3B -port 0
```

To list LUNs visible from behind a port, enter:

```
symsan list -sanluns -sid 123 -dir 3B -port 0
       -wwn 0123456789ABCDEF
```

To show iSCSI sessions on a specific SE director port, enter:

```
symsan -sid 123 show -iscsi_sesions
       -ig test_ig
       -SE 3F -port 7
```

To show iSCSI sessions on all ports of a SE director, enter:

```
symsan -sid 123 show -iscsi_sesions
       -ig test_ig
       -SE 3F -port ALL
```

To show iSCSI sessions on all SE director ports for an array, enter:

```
symsan -sid 123 show -iscsi_sesions
       -ig test_ig
       -SE ALL -port ALL
```

To ping remote IP Address from SE director:Network ID, enter:

```
symsan -sid 123 ping -i 20 -c 4
       -remote_ip_address 10.87.133.189 -SE 1H -network_id 4
       -ip_address 10.228.205.196
```

To ping remote IP Address configured on RE director:port from Local RE director:port, enter:

```
symsan -sid 123 ping -i 20 -c 3
       -rem 1:0:0:0:0:ffff:6f6f:6f7c -RE 2F -port 4
```

symsg

Performs operations on storage groups located in a specified Symmetrix array.

SYNOPSIS

```
symsg -h

symsg [-sid <SymmID>] [-i <Interval>] [-c <Count>]
   [-v [-mb | -gb| -tb]]

   list [-offline]

   list -by_port -demand
     [-pg <PgName> | -dir <# [-p <#|ALL>] |ALL>]

   list -by_pg -demand [-pg <PgName>]

   list [-detail [-by_sl | -by_srp]]

   list < [-sl] [-policy -type <snap> [-effective]] >

symsg -sid <SymmID> [-i <Interval>] [-c <Count>] [-v]

   create <SgName>
     [-bw_max <MBperSec>]
     [-iops_max <IOperSec>]
     [-dynamic <NEVER | ALWAYS | ONFAILURE>]
     [-sl <SLName> [-wl <WorkloadName>]]
     [-srp <SRPName>] [-nocompression]

   convert -cascaded <SgName> <ChildSgName>
     [-host_IO <on_parent | on_child>]

   convert -standalone <SgName>
     [-host_IO <keep_parent | keep_child>]

   delete <SgName> [-force]

   export <SgName> [-file <FileName>] [-offline]

   exportall [-file <FileName>] [-offline]

   import <SgName> [-file <FileName>]

   importall [-file <FileName>]

   rename <OldSgName> <NewSgName>

   sg2cg <SgName> <CgName> [-bcv | -vdev | -tgt]
     [-R1 | -R2 | -R21 | -noRDF]
     [-apidb | -rdf_consistency]

   sg2dg <SgName> <DgName> [-bcv | -vdev | -tgt]
     [-R1 | -R2 | -R21 | -noRDF]

   show <SgName> [-offline] [-mb | -gb | -tb]

symsg -sg <SgName> -sid <SymmID> [-i <Interval>]
     [-c <Count>] [-v] [-celerra] [-rp] [-ckd]

   add dev <SymDevName>

   add sg <SgName1>[,<SgName2>,<SgName3>,.,<SgNamen>]
```

```
   copy dev <SymDevName> <DestSgName>

   move dev <SymDevName> <DestSgName> [-force]

   remove dev <SymDevName> [-force]

   remove sg <SgName1>[,<SgName2>,<SgName3>,.,<SgNamen>]

symsg -sg <SgName> -sid <SymmID> [-i <Interval>]
     [-c <Count>] [-v] [-celerra] [-rp] [-ckd]
     [-SA <# | ALL>] [-p <#>] [-N <#>]
     [-cap <#> [-captype <mb> | <cyl>]]
     [-devs <<SymDevStart>:<SymDevEnd> | <SymDevName>
       [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]> |
      -file <DeviceFileName> [-tgt] ]

   addall [pd | devs]

   copyall <DestSgName>

   moveall <DestSgName> [-force]

   rmall [-force]

symsg -sg <SgName> -sid <SymmID> [-i <Interval>]
     [-c <Count>]

   add -policy <PolicyName> -type <snap>

   remove -policy <PolicyName> -type <snap>

symsg -sg <SgName> -sid <SymmID> [-i <Interval>]
     [-c <Count>]

   set <[-bw_max <MBperSec> | NOLIMIT ]
        [-iops_max <IOperSec> | NOLIMIT ]
        [-dynamic <NEVER | ALWAYS | ONFAILURE>]>
        [-sl <SLName> [-wl <WorkloadName>] |-nosl]
        [-srp <SRPName> | -nosrp]
        [-compression | -nocompression]

symsg -sg <SgName> -sid <SymmID>

   merge <SrcSgName>

   split <TgtSgName> -view_name <MvName>
     [-devs <<SymDevStart>:<SymDevEnd> | <SymDevName>
       [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]]

symsg -sg <SgName> -sid <SymmID> [-i <Interval>]
     [-c <Count>] [-v] [-noprompt] [-rp] [-star]
     [-celerra] [-skip -lockid <lockNum>]

   ready [-metro]

   not_ready [-metro]

   rw_enable

   write_disable

   hold

   unhold [-symforce]

   compress [-stop]
```

```
   uncompress [-stop]

   bind -pool <PoolName>

   unbind

   rebind -pool <PoolName>

   allocate [-persistent]

   allocate -stop

   free [-all]

   free [-all] -stop

   reclaim [-persistent]

   reclaim -stop

   set -persistent

   unset -persistent

   set -orm < system | on | off >

   set -gcm [-symforce]

   unset -gcm [-symforce]

   reset -identity

symsg -sg <SgName> -sid <SymmID> [-i <Interval>]
     [-c <Count>] [-v] [-noprompt] [-star]
     [-skip -lockid <lockNum>]

   pin

   unpin

   host_active [-force]
```

DESCRIPTION

The symsg command performs operations specific to storage groups: creating new storage groups, adding devices or other storage groups to a storage group, copying, moving and removing devices in a storage group, removing storage groups from a storage group, importing storage groups, exporting storage groups, deleting storage groups, renaming storage groups, converting storage groups between cascaded and standalone and listing and showing information about a storage group.

The symsg command also performs the following control actions on all of the devices in a storage group: write_disable, rw_enable, ready, not_ready, hold, unhold, pin, unpin, compress, uncompress, set, unset, bind, unbind, rebind, allocate, free and reclaim.

The export argument creates a group file (ASCII text) and the import argument reads a file to import a single storage group. The file contains as many device description lines or storage group description lines as there are devices or storage groups defined in the Storage Group detailed. There cannot be both device and storage group description lines in the same file. In

addition, there may also be Host I/O limit maximum bandwidth, maximum IOPS or dynamic distribution description lines. Any lines that are blank or have a pound sign (#) in the first column are ignored.

```
<SymDevName>
. . .
<SymDevName>
B <MBperSec>
I <IOperSec>
D <DynDistribution>
L <SLName>
R <SRPName>
W <WorkloadName>
P <compression enabled>

   or

S <StorageGroupName>
. . .
S <StorageGroupName>
B <MBperSec>
I <IOperSec>
D <DynDistribution>
L <SLName>
R <SRPName>
W <WorkloadName>
```

The exportall argument creates a group file (ASCII text) and the importall argument reads a file to import all storage groups. The file contains as many storage group records as there are storage groups defined. Each record can contain as many device description lines or storage group description lines as there are devices or storage groups defined in the storage group being detailed. There cannot be both device and storage group description lines in the same record. In addition, there may also be Host I/O limit maximum bandwidth, maximum IOPS or dynamic distribution description lines. Any lines that are blank or have a pound sign (#) in the first column are ignored.

Group files contain device or storage group parameters in the following formats:

```
<StorageGroupName>
<SymmID>
<SymDevName>
. . .
<SymDevName>
B <MBperSec>
I <IOperSec>
D <DynDistribution>
L <SLName>
R <SRPName>
W <WorkloadName>
P <compression enabled>

<StorageGroupName>
<SymmID>
S <StorageGroupName>
. . .
S <StorageGroupName>
B <MBperSec>
I <IOperSec>
D <DynDistribution>
L <SLName>
R <SRPName>
```

W <WorkloadName>

ARGUMENTS

add/addall	Adds single or multiple devices or storage groups to a storage group. Associates a Snapshot Protection Policy with an SG.
allocate	Allocates storage in the thin pool.
bind	Binds the thin device(s) to the thin pool.
compress	Starts data compression on thin device(s). When combined with the -stop option, data compression is stopped.
convert	Converts a storage group between cascaded and standalone.
copy/copyall	Copies devices from the <SgName> to the <DestSgName>.
create	Creates a storage group.
delete	Deletes a storage group. Deletion of a storage group is not allowed if the storage group is contained in a Masking View or associated with a FAST policy.
export	Creates a text file that details the members of an existing storage group. The storage group can later be recreated from this file using the import command.
exportall	Creates a text file that details the members of all existing storage groups. The storage groups can later be recreated from this file using the importall command.
free	Frees storage in the thin pool.
hold	Creates a hold on all available devices from an existing device group. When a hold is placed on a device, TimeFinder operations are blocked.
host_active	Sets the host active mode on device(s). The device(s) must be in a host passive mode for this operation to succeed.
import	Creates a storage group from data contained in a text file previously created using the export command.
importall	Creates storage groups from data contained in a text file previously created using the exportall command.
list	Lists storage groups or with optional parameters produces Host IO Limit demand reports.
merge	Merges the source and target storage group without disrupting the host connectivity. The source storage group must be a

	standalone storage group and the target storage group can be either standalone or cascaded storage group.
move/moveall	Moves the device(s) and deletes them from the <SgName> before adding them to the <DestSgName>.
not_ready	Sets the device(s) to Not Ready. The device must be in a User Ready status for this operation to succeed.
pin	Sets the device(s) to a user-pinned state. User-pinned devices are not moved by the FAST controller, but can be moved with Optimizer or Symmigrate.
ready	Sets the device(s) to Ready. The device must be in a User Not Ready status for this operation to succeed.
rebind	Rebinds the device(s) to the thin pool.
reclaim	Reclaims storage from the thin pool.
remove	Removes a single device or a single or multiple storage groups from a storage group. Disassociates a Snapshot Protection Policy from an SG.
rename	Renames the ASCII name of a storage group.
reset	Sets the device to its original identity when combined with -identity option.
rmall	Removes multiple devices from a storage group. If an optional range of devices is specified, only those devices included in the range are removed. If no range of devices is provided, all of the devices are removed.
rw_enable	Sets the device(s) to Read and Write Enabled to the local hosts.
set	Sets the persistent indicator for allocations on thin device(s) when combined with the -persistent option. Or sets Host IO Limit on the specified storage group when combined with the -bw_max, -iops_max or -dynamic options. Sets the Optimized Read Miss mode when combined with the -orm option. Sets GCM mode when combined with -gcm.
sg2cg	Adds selected members of a storage group to a target composite group. If the composite group does not exist, it is created. If none of the optional device types are specified, the default is to add standard devices.
sg2dg	Adds selected members of a storage group to a target device group. If the device

group does not exist, it is created. If none of the optional device types are specified, the default is to add standard devices.

show — Shows detailed information about storage groups.

split — Splits the source standalone or cascaded storage group without disrupting the host connectivity.

unbind — Unbinds device(s) from the thin pool.

uncompress — Starts data decompression on thin device(s). When combined with the -stop option, data decompression is stopped.

unhold — Releases devices that were previously set to the hold state.

unpin — Removes the device(s) from the user-pinned state.

unset — Clears the persistent indicator for allocations on thin device(s) when combined with the -persistent option. Clears GCM mode when combined with -gcm.

write_disable — Sets the devices to Write Disabled to their local hosts.

KEYWORDS

dev — Indicates a single Symmetrix device name.

devs — Indicates multiple Symmetrix device names.

pd — Indicates a physical (host) device name.

sg — Indicates a list of storage group names.

OPTIONS

-all — Used with the free operation in order to specify that all allocations associated with the indicated devices are to be freed, regardless of whether data has been written or not.

-apidb — Overrides the options file setting and stores the RDF CG in the SYMAPI database only.

-bcv — Specifies that only local BCV devices are taken from the storage group and added to the device group, via usage of the sg2dg command, or to the composite group, via usage of the sg2cg command.

-bw_max — Specifies the Host IO Limit maximum bandwidth in MB per second to be set on the storage group. The maximum bandwidth will be set to unlimited if NOLIMIT is specified.

-by_pg — Specifies that the Host IO Limit demand report is being requested for port group

information.

-by_port
Specifies that the Host IO Limit demand report is being requested for port information.

-c
Specifies the number (count) of times to attempt to acquire an exclusive lock on the Symmetrix host database.

The time to wait between attempts to acquire a needed lock is specified by -i (interval).

If neither -c nor -i is specified, operations will fail if they are unable to acquire a requested lock.

-cap
Sets a minimum device size to the selection criteria of devices.

-captype <mb | cyl>
Specifies the units of capacity, either megabytes or cylinders. The default if not specified is MB.

-cascaded
Used with the convert operation to select that a standalone storage group be converted to a cascaded storage group.

-celerra
Allows operations on Celerra FBA devices.

-ckd
Allows operations on CKD devices.

-compression
Enables compression on SG.

-orm
Allows setting the Optimized Read Miss mode to system default, on, or off for the specified devices.

-demand
Specifies a Host IO Limit demand report.

-devs
Specifies multiple range(s) of Symmetrix device names.

-dir
Specifies the director number(s) for which the Host IO Limit demand report is being requested.

-dynamic
Specifies the Host IO Limit dynamic distribution setting for the storage group.

-effective
Displays the Snapshot Protection Policies that are associated and enbaled for the Storage Group. For a child Storage Group, this could be a Snapshot Protection Policy on its parent.

-file
Specifies a filename to use for the import/importall or export/exportall operations.

-file
Specifies a filename to use as input.

-force
Forces the operation on the storage group.

-gcm
Allows setting or clearing the device GCM

mode.

-h Provides online help.

-host_IO When converting a storage group to cascaded, it is used to select were an existing Host IO Limit is set.

When converting a storage group to standalone it is used to select which Host IO Limit to keep.

-i Specifies the interval time, in seconds, to wait between attempts to acquire an exclusive lock on the Symmetrix host database.
The default interval is 30 seconds.
The minimum interval is 5 seconds.
For passive actions the minimum interval is 15 seconds. Passive actions are actions that do not acquire an exclusive lock.

-identity Used with the reset command, the original identity of the device(s) is restored.

-iops_max Specifies the Host IO Limit maximum IOPs in I/Os per second. The maximum IOPs will be set to unlimited if NOLIMIT is specified.

-lockid Specifies the lock holder ID for preserving the target locks on the control operation. The lock number ID must be a hexadecimal number.

-N Sets a number of devices to add, remove, or move.

-nocompression Disables the compression on SG.

-noprompt Disables the confirmation prompt.
The default is to prompt the user for confirmation before executing the operation.

-noRDF Copies non-SRDF devices only.

-nosl Specifies there is no Service Level or removes the Service Level (SL) set on the SG. This will also remove any workload that was assigned for the Service Level. If the SG has an SRP, then the SG will get an activity-based Service Level. Otherwise, there will be no Service Level or SRP for the SG and the SG will no longer be FAST Managed.

-nosrp Specifies there is no SRP or removes the SRP set on the SG. If the SG has a Service Level (SL) set the system default SRP for the emulation type will be used with the SG. Otherwise, there will be no SRP or SL for the SG and the SG will no longer be FAST Managed.

-offline Obtains information from the Symmetrix host configuration database.

-p Specifies the front-end (SCSI or Fibre)

director port number to only select devices that are primarily visible through this director port. By default, all ports are selected.

-persistent Used with set or unset it specifies operation on the persistent indicator. Used with allocate or reclaim it specifies the use of persistent storage.

-policy Specifies the Snapshot Protection Policy to be associated with the SG. Displays the Snapshot Protection Policies associated with the SG.

-pool Specifies a Thin Pool Name.

-pg Specifies that the Host IO Limit demand report is being requested for the port(s) in the specified port group.

-R1 Copies RDF1 (R1) devices only.

-R2 Copies RDF2 (R2) devices only.

-R21 Copies RDF21 (R21) devices only.

-rdf_consistency
Allowing created CG to be enabled for RDF consistency.

-metro When specified with ready and not_ready identifies the devices being controlled are part of an RDF/Metro configuration.

-rp Specifies that the action is targeted for devices tagged for RecoverPoint.

-SA Specifies the front-end (SCSI or Fibre) director number to only select devices that are primarily visible through this director. Alternatively, if ALL (the default) is specified, all devices satisfying any other selection criterion will be selected.

-sid Specifies the unique Symmetrix ID.

-sg Specifies the name of the storage group.

-skip Skips the device locks action for control operations. Requires the -lockid option.

-sl Specifies the Service Level name to be set on the SG. Or display the Service Level set on the SG.

-srp Specifies a SRP name to be set on SG.

-standalone Used with the convert operation to select that a cascaded storage group be converted to a standalone storage group.

-star Indicates that the action is targeted for devices in STAR mode.

-stop Specifies that the compress, uncompress,

allocate, free or reclaim operation will be stopped.

-symforce — Requests the Symmetrix array force the operation to be executed when normally it is rejected. Use extreme caution when using this option.

-tgt — Allows the user to specify that only local TGT devices are taken from the storage group and added to the device group, via usage of the sg2dg command, or to the composite group, via usage of the sg2cg command.

When used in conjunction with -file, specifies that devices are only taken from the second column of DeviceFileName.

-type — Specifies the Snapshot Protection Policy.

-vdev — Allows the user to specify that only local VDEV devices are taken from the storage group and added to the device group, via usage of the sg2dg command, or to the composite group, via usage of the sg2cg command.

-v — Provides a more detailed, verbose listing.

-view_name — Specifies the target masking view name for the split operation.

-wl — Specifies a workload name to be set on SG.

PARAMETERS

ALWAYS — The Host IO Limits for the storage group are always dynamically redistributed.

ChildSgName — The child storage group name.

DestSgName — The name of the storage group to use as the destination of a copy/copyall or move/moveall operation.

DeviceFileName — The name of the file where devices are listed. The device file can contain devices (SymDevNames) separated by new lines or device pairs listing a pair each line (the source device, followed by a target device).

Examples of a device file format:

Example 1:

```
0026  0029
0015  0016
```

Example 2:

```
0001
0002
```

DynDistribution
The Host IO Dynamic Distribution setting

FileName — The specified output or input of export, exportall, import, and importall commands, respectively.

IOperSec — The number of I/Os per second. Valid values are between 100 and 2000000 IO/sec, in units of 100 IO/sec.

MBperSec — The number of Mega Bytes per second. Valid values are between 1 and 100000 MB/sec.

NEVER — The Host IO Limits for a storage group are never dynamically redistributed (static).

NewSgName — The new storage group name.

OldSgName — The original storage group name.

ONFAILURE — The Host IO Limits for the storage group are dynamically redistributed only upon failure of a Front-End Port.

PolicyName — The Snapshot Protection Policy name.

SgName — The storage group name.

SLName — The name of the Performance Service Level.

SrcSgName — The source storage group name for the merge operation.

SRPName — The name of the SRP.

SymDevName — The Symmetrix device name.

SymDevEnd — The last Symmetrix device name in a sequence, such as 00B6.

SymDevStart — The first Symmetrix device name in a sequence, such as 001C.

SymmID — The 12-digit ID of the Symmetrix array.

TgtSgName — The target storage group name or child storage group name for the split operation.

WorkloadName — The name of the Workload.

RETURN CODES

Code #	Code Symbol
0	CLI_C_SUCCESS
1	CLI_C_FAIL

EXAMPLES

To create a Symmetrix storage group named mysg_1 on Symmetrix array ID# 59866000123, enter:

```
symsg -sid 123 create mysg_1
```

To delete Symmetrix storage group storgrp_f and all contained devices on Symmetrix array ID# 59866000123, enter:

```
symsg -sid 123 delete storgrp_f -force
```

To rename Symmetrix storage group mysg_1 to storgrp_a on Symmetrix array ID# 59866000123, enter:

```
symsg -sid 123 rename mysg_1 storgrp_a
```

To add a single device to storage group storgrp_a, enter:

```
symsg -sid 123 -sg storgrp_a add dev 30
```

To add all devices that are primarily visible from this host to storage group storgrp_a, enter:

```
symsg -sid 123 -sg storgrp_a addall pd
```

To add a range of physical devices to storage group storgrp_a, enter:

```
symsg -sid 123 -sg storgrp_a addall pd -devs 30:3F
```

To add all devices listed in a file named my_storgrp_b.txt to a Symmetrix storage group named storgrp_b on Symmetrix array ID# 59866000123, enter:

```
symsg -sid 123 addall -file my_storgrp_b.txt
      -sg storgrp_b
```

To add storage groups storgrp_1 and storgrp_2 to storage group storgrp_a, enter:

```
symsg -sid 123 -sg storgrp_a add sg storgrp_1,storgrp_2
```

To copy a device from storage group storgrp_a to storgrp_b, enter:

```
symsg -sid 123 -sg storgrp_a copy dev 30 storgrp_b
```

To move a device from storage group storgrp_a to storgrp_b, enter:

```
symsg -sid 123 -sg storgrp_a move dev 30 storgrp_b
```

To copy multiple devices from storage group storgrp_a to storgrp_b, enter:

```
symsg -sid 123 -sg storgrp_a copyall
      -devs 30:3F,40 storgrp_b
```

To move multiple devices from storage group storgrp_a to storgrp_b, enter:

```
symsg -sid 123 -sg storgrp_a moveall
      -devs 30:3F,40 storgrp_b
```

To list all storage groups on Symmetrix array ID# 59866000123, enter:

```
symsg -sid 123 list
```

To list all Symmetrix storage groups in detailed format, enter:

```
   symsg list -v
```

To show all devices in storage group storgrp_a, enter:

```
   symsg -sid 123 show storgrp_a
```

To export a Symmetrix storage group named storgrp_c on Symmetrix array ID# 59866000123 to a file named my_storgrp_c, enter:

```
   symsg -sid 123 export storgrp_c -file my_storgrp_c
```

To import a Symmetrix storage group named storgrp_c to Symmetrix array ID# 59866000123 from a file named my_storgrp_c, enter:

```
   symsg -sid 123 import storgrp_c -file my_storgrp_c
```

To convert a Symmetrix storage group named storgrp_c to a device group named ProdDG, enter:

```
   symsg -sid 123 sg2dg storgrp_c ProdDG
```

To remove a single device from storage group storgrp_a, enter:

```
   symsg -sid 123 -sg storgrp_a remove dev 30
```

To remove all devices from storage group storgrp_a, enter:

```
   symsg -sid 123 -sg storgrp_a rmall
```

To remove multiple devices from storage group storgrp_a, enter:

```
   symsg -sid 123 -sg storgrp_a rmall -devs 31:35,37,40:43
```

To remove storage groups sgrp_1 and sgrp_2 from storage group storgrp_a, enter:

```
   symsg -sid 123 -sg storgrp_a remove sg sgrp_1,sgrp_2
```

```
symsnap

        Performs TimeFinder/Snap control operations on
        a device group, composite group, or on devices
        in a device file.

        SYNOPSIS

             symsnap -h

             symsnap -g <DgName> [-v] [-noprompt] [-i <Interval>]
                  [-bcv | -rdf | -rbcv | -hop2] [-tgt]
                  [-c <Count>] [-force] [-star]
                  [-preserveTGTLocks -lockid <LockNum>]

                create [-exact] [-skip]
                  [-svp <PoolName> | -duplicate] [-concurrent]

                activate [-consistent [-both_sides]]
                  [-concurrent] [-duplicate]
                  [-preaction <ScriptFile>]
                  [-postaction <ScriptFile>]
                  [-not_ready] [-skip]

                duplicate [-consistent] [-exact] [-concurrent]
                  [-preaction <ScriptFile>]
                  [-postaction <ScriptFile>]
                  [-not_ready] [-skip]

                recreate [-skip] [-concurrent]

                terminate [-symforce] [-skip] [-restored]
                  [-duplicate]

                establish [-full] [-exact] [-svp <poolname>]
                  [-consistent [-both_sides]] [-concurrent]
                  [-preaction <ScriptFile>]
                  [-postaction <ScriptFile>]
                  [-not_ready] [-skip]

                restore [-full] [-not_ready]

             symsnap -g <DgName> [-offline] [-i <Interval>] [-c <Count>]
                  [-bcv | -rdf | -rbcv | -hop2]

                query [-multi] [-restore | -changed] [-attach] [-pools]
                  [-summary] [-mb | -gb | tb]

                verify [-created | -copied | -copyonwrite |
                  -restinprog | -restored | -failed | -recreated]
                  [-force] [-concurrent] [-summary]

             symsnap -g <DgName> [-v] [-noprompt] [-i <Interval>]
                  [-c <Count>]
                  [-bcv | -rdf | -rbcv | -hop2]

                attach

                detach

             symsnap -cg <CgName> [-v] [-noprompt] [-i <Interval>]
                  [-bcv | -rdf | -rbcv | -hop2] [-tgt]
                  [-c <Count>] [-force] [-star] [-sid <SymmID> |
                   -rdfg <SymmID>:<GrpNum>[,<GrpNum>,...]|<all>[,...] |
                   name:<RDFName>[,<RDFName>,...]>]

                create [-exact] [-skip]
```

```
     [-svp <PoolName> | -duplicate] [-concurrent]

   activate [-consistent [-both_sides]]
     [-preaction <ScriptFile>]
     [-postaction <ScriptFile>]
     [-not_ready] [-skip] [-concurrent]
     [-duplicate]

   duplicate [-consistent] [-exact]
     [-preaction <ScriptFile>]
     [-postaction <ScriptFile>]
     [-not_ready] [-skip] [-concurrent]

   recreate [-skip] [-concurrent]

   terminate [-symforce] [-skip] [-restored]
     [-duplicate]

   establish [-full] [-exact] [-svp <poolname>]
     [-consistent [-both_sides]]
     [-preaction <ScriptFile>]
     [-postaction <ScriptFile>]
     [-not_ready] [-skip] [-concurrent]

   restore [-full] [-not_ready]

symsnap -cg <CgName> [-offline] [-i <Interval>]
     [-c <Count>]
     [-bcv | -rdf | -rbcv | -hop2] [-sid <SymmID> |
     -rdfg <SymmID>:<GrpNum>[,<GrpNum>,...]|<all>[,...] |
      name:<RDFName>[,<RDFName>,...]>]

   query [-multi] [-restore] [-attach] [-sid <SymmID>]
     [-changed] [-pools] [-summary] [-mb | -gb | tb]

   verify [-created | -copied | -copyonwrite |
      -restinprog | -restored | -failed | -recreated]
     [-force] [-concurrent] [-summary]

symsnap -cg <CgName> [-v] [-noprompt] [-i <Interval>]
     [-c <Count>] [-sid <SymmID>]
     [-bcv | -rdf | -rbcv | -hop2]
     [-rdfg <SymmID>:<GrpNum>[,<GrpNum>,...]|<all>[,...] |
      name:<RDFName>[,<RDFName>,...]>]

   attach

   detach

symsnap -sid <SymmID> <-file <DeviceFileName> [-noprompt] |
     -noprompt 'redirect stdin'> [-v] [-force]
     [-i <Interval>] [-c <Count>]
     [-preserveTGTLocks -lockid <LockNum>] [-star]

   create [-skip] [-svp <PoolName> | -duplicate]

   activate [-consistent] [-duplicate]
     [-preaction <ScriptFile>] [-postaction <ScriptFile>]
     [-not_ready] [-skip]

   duplicate [-consistent]
     [-preaction <ScriptFile>] [-postaction <ScriptFile>]
     [-not_ready] [-skip]

   recreate [-skip]

   terminate [-symforce] [-skip] [-restored] [-duplicate]
```

```
   establish [-full] [-svp <PoolName>] [-consistent]
     [-preaction <ScriptFile>] [-postaction <ScriptFile>]
     [-not_ready] [-skip]

   restore [-full] [-not_ready]

symsnap -sid <SymmID> <-file <DeviceFileName> |
     'redirect stdin'>
     [-i <Interval>] [-c <Count>]

   query [-multi] [-restore] [-attach] [-pools] [-changed]
     [-summary] [-mb | -gb | tb]

   verify [-created | -copied | -copyonwrite |
     -restinprog | -restored | -failed | -recreated]
     [-force] [-summary]

symsnap -sid <SymmID> <-file <DeviceFileName> [-noprompt] |
     -noprompt 'redirect stdin'> [-v]
     [-i <Interval>] [-c <Count>]

   attach

   detach

symsnap [-sid <SymmID>] [-i <Interval>] [-c <Count>]
        [-offline] [-mb | -gb | tb]

   list
```

DESCRIPTION

The symsnap command performs snap operations on a device group, composite group, or on devices in a device file.

These operations include creating and activating a source device with a target device, terminating the snap session, and querying the state of the device pair.

You can perform all of these operations on a group or a device file.

Before you can create a copy session between a target device and a source device, the target device must be associated with the device group and the target device must be the same size as the source device.

ARGUMENTS

activate	Activate an internal copy session with the devices in the device or composite group and one or more target devices associated with the group. While the operation is in progress, the state of the device pair is Copy on Write. If the source device is completely written to, the state changes to Copied.
attach	When a device create is issued, attaches a virtual device (VDEV) target to a source device as the preferred target device for pairing.
create	Creates an internal snap session with the devices that are in the device group with

one or more target devices that are associated with the group.

While the operation is in progress, the state of the device pair is CreateInProg. When the operation completes, the state changes to Created.

detach — Detaches a VDEV target device from the source device so that it is no longer the preferred target device of the source device.

duplicate — Creates and activates a duplicate snap session in a single operation. This is equivalent to performing create -duplicate followed by activate -duplicate.

establish — Creates and activates an internal snap session with the devices in the group and one or more target devices associated with the group. Specifying this argument without the -full option performs a recreate followed by an activate operation.

list — Lists snap sessions.

query — Returns snap state information about all device pairs in a group or device file.

recreate — Recreates the snap session on an existing VDEV. This only applies to sessions that have been activated.

restore — Restores a VDEV to another device. After the restore operation, the target of the restore is left in a Ready state unless the -not_ready option is used.

terminate — Terminates (stops) the existing internal copy session between the specified source and target devices in a device group.

verify — Verifies, by default, whether all device pairs in a group are in the CopyOnWrite state.

KEYWORDS

SRCDEVS — Specifies to use the path names from the standard devices being controlled.

VDEV — Specifies a Symmetrix virtual device.

OPTIONS

-attach — Displays target attachment information for the standard device(s) in the device group.

-bcv — Uses BCV devices as the source devices and VDEV devices as the target devices. You can only use this option with device or composite groups.

-both_sides — Activates all locally and remotely associated VDEV pairs in an SRDF group.

-c	Specifies the number (count) of times to display or to acquire an exclusive lock on the Symmetrix host database. If this option is not specified and an interval (-i) is specified, the command will loop continuously to display or to start the mirroring operation.
-cg	Applies a composite group name to the command.
-changed	When specified with the query argument, this option shows the number of tracks that were written to either the source device or to the virtual device.
-concurrent	When specified with the verify argument, this option verifies the standard device and multiple target devices. When specified with active commands, this option performs the operation on a pair with an additional target device.
-consistent	Causes the source and target pairs to be consistently activated.
-copied	Verifies that the snap device pair(s) are in the Copied state.
-copyonwrite	Verifies that the snap device pair(s) are in the CopyOnWrite state.
-created	Verifies that the snap device pair(s) are in the Created state.
-duplicate	For create, activate, and terminate actions indicates that the action is to be performed on a VDEV to VDEV pair.
-exact	Pairs devices in the exact order in which the source and target devices were added to the device group.
-failed	Verifies that the snap device pair(s) are in the Failed state.
-file	Applies a device file to the command. The device file contains device pairs (by device number) listing a pair per each line (the source device first, a space, and the VDEV target device last within each line entry). A Symmetrix ID is required for this option. -f is synonymous with -file.
-force	Attempts to force the operation even though one or more paired devices in the device group may not be in the normal, expected state(s) for the specified operation.
-full	Performs a full restore. This option is used with the restore command.
-g	Applies a device group name to the command.
-gb	Displays counts in gigabytes.
-h	Provides brief online help information.

-hop2 Performs the specified action on the Symmetrix array two hops away.

-i Specifies the repeat interval in seconds to display or to acquire an exclusive lock on the Symmetrix host database.
The default interval is 30 seconds.
The minimum interval is 5 seconds.
For passive actions the minimum interval is 15 seconds. Passive actions are actions that do not acquire an exclusive lock.

-lockid Specifies the lock holder ID for preserving the target locks on the control operation.

-mb Displays counts in megabytes.

-multi Applies to a query operation in a multi-target environment to show all targets that are paired with source devices. Devices are listed in chronological order.

-noprompt Requests to not return a prompt after you enter a command. The default is to prompt for confirmation.

-not_ready Performs the snap control operation but leaves the target device(s) Not Ready. That is, each target device will be set Not Ready prior to the operation completing.

-offline Specifies that the Symmetrix array data connection is offline from the host in-memory database.

-pools Used with query to display pool names for each session.

-postaction Executes the script argument after a snap session has been activated.

-preaction Executes the script argument before a copy session has been activated.

-preserveTGTLocks
Prevents the action from taking out device locks on the target devices. The target devices must already be locked by the same lock holder ID.

-rbcv Uses the RBCV devices as the source devices and the RVDEV devices as the target devices. This is used only with device or composite groups.

-rdf Performs the action on the remote Symmetrix array.

-rdfg This option causes the command to perform the requested action on a subset of the composite group defined by one or more Symmetrix/RA group combinations supplied as the argument to -rdfg.
This argument is a comma separated list in the form:
SID:GrpNum,SID:GrpNum,...

or a comma-separated list of predefined names in the form:
name:Name,Name,...
GrpNum may be specified as "all" to use all of the RA Groups on a Symmetrix.

-recreated — Verifies that the snap device pair(s) are in the Recreated state.

-restinprog — Verifies that the snap device pair(s) are in the RestInProg (restore) state.

-restore — Shows the VDEV where the snap pair was restored.

-restored — Verifies that the snap device pair(s) are in the Restored state.

-sid — Supplies the unique Symmetrix ID.

-skip — Skips the source locks action. This option will not lock the source devices if all of the specified source devices are either locked or are unlocked.

-star — Targets the action at devices in STAR mode.

-summary — Shows device state summary.

-svp — Filters the list based on the supplied PoolName.

-symforce — Forces the operation to execute when normally it would be rejected. On terminate, it causes the Symmetrix array to stop a snap session. IMPORTANT: Use extreme caution with this option.

-tb — Displays counts in terabytes.

-tgt — Specifies to use VDEVs associated as TGT devices for snap targets. This uses local TGT devices, remote RTGT devices, and two hop2TGT devices.

-v — Provides a more detailed, verbose listing.

PARAMETERS

CgName — Composite group name.

Count — Number of iterations to execute before exiting.

DgName — Device group name.

DeviceFileName — Device Filename.
The device file contains device pairs (SymDevNames) listing a pair of devices on each line (the source device first, a space, followed by the target device name on each line. Comments are prefixed with #.

GrpNum — RDF (RA) group number.

Interval — Interval between polls, in seconds.

LockNum — Hexadecimal value of the lock holder ID.

Name — Logical name associated with the RDF (RA) group(s).

PoolName — Name of the SAVE device pool.

ScriptFile — Full pathname of a script file to be executed.

RETURN CODES

Code #	Code Symbol
0	CLI_C_SUCCESS
1	CLI_C_FAIL
2	CLI_C_DB_FILE_IS_LOCKED
18	CLI_C_ALREADY_IN_STATE The device or device group is already in the desired snap state.
19	CLI_C_GK_IS_LOCKED All gatekeepers to the Symmetrix array are currently locked.
22	CLI_C_NEED_FORCE_TO_PROCEED Requires the force flag to proceed.
23	CLI_C_NEED_SYMFORCE_TO_PROCEED Requires the symforce flag to proceed. CAUTION: Extreme caution should be exercised when using this option.

Return codes for symsnap verify

12	CLI_C_NOT_ALL_RESTORED Not all source devices are in the Restored state.
13	CLI_C_NONE_RESTORED No source devices are in the Restored state.
29	CLI_C_NOT_ALL_RESTINPROG Not all source devices are in the RestInProg state.
30	CLI_C_NONE_RESTINPROG No source devices are in the RestInProg state.
55	CLI_C_NOT_ALL_COPIED Not all source devices are in the Copied state.
56	CLI_C_NONE_COPIED No source devices are in the Copied state.
60	CLI_C_NOT_ALL_CREATED Not all source devices are in the Created state.

61 CLI_C_NONE_CREATED
No source devices are in the Created state.

66 CLI_C_NOT_ALL_COPYONWRITE
Not all source devices are in the CopyOnWrite state.

67 CLI_C_NONE_COPYONWRITE
No source devices are in the CopyOnWrite state.

EXAMPLES

To create the device group ProdDB as a REGULAR device group, enter:

```
symdg create ProdDB
```

To define the device group ProdDB as the default device group, enter:

```
setenv SYMCLI_DG ProdDB
```

To create a snap of the source devices in group ProdDB with target devices (associated with the group), enter:

```
symsnap create -g ProdDB
symsnap activate -g ProdDB
```

To query information about all paired devices in device group ProdDB, enter:

```
symsnap query
```

symsnapvx

Performs TimeFinder/SnapVX control, list,
and verify operations on a device list,
a list of ranges, a device group (DG),
composite group (CG), or storage group (SG).
Also performs symmetrix wide list operations.

SYNOPSIS

```
symsnapvx -h

symsnapvx -g <DgName> -name <SnapshotName>
     [-rdf | -hop2]
     [-v] [-noprompt] [-force] [-star]
     [-i <Interval>] [-c <Count>]

   establish [-secure <-delta <delta_time> | -absolute <date_time>> |
      -ttl <-delta <delta_time> | -absolute <date_time>>]
     [-both_sides]
     [-preaction <ScriptFile>] [-postaction <ScriptFile>]

symsnapvx -g <DgName>
     -snapshot_name <SnapshotName>
       [-generation <GenerationNumber> | -snapid <SnapsetID>]
     [-rdf | -hop2]
     [-v] [-noprompt] [-force] [-star]
     [-i <Interval>] [-c <Count>]

   restore
     [-preaction <ScriptFile>][-postaction <ScriptFile>]
     [-remote]

   terminate [-restored [-symforce]]
             [[[-keep_count <Count>] |
               [-older_than -snapid <SnapsetID>] | -all ] -force]

   rename -name <NewSnapshotName>

   set secure
     <-delta <delta_time> | -absolute <date_time>>

   set ttl
     <-delta <delta_time | NONE> | -absolute <date_time>>

   set <persistent | nopersistent>

symsnapvx -g <DgName> -sid <SymmID> [-noprompt] [-v]
   terminate -all_snapshot_names -force

symsnapvx -g <DgName>
     -snapshot_name <SnapshotName>
       [-generation <GenerationNumber> | -snapid <SnapsetID>]
     [-rdf | -hop2]
     [-v] [-noprompt] [-force] [-star]
     [-i <Interval>] [-c <Count>]

   link [-copy [-remote] ] [-exact]

   relink [-copy [-remote]] [-exact]

   unlink [-symforce]

   set mode <copy | nocopy>
```

```
symsnapvx -g <DgName>
     -snapshot_name <SnapshotName>
       [-generation <GenerationNumber> | -snapid <SnapsetID>]
     [-rdf | -hop2]
     [-force] [-summary [-mb | -gb | -tb]]
     [-i <Interval>] [-c <Count>]

   verify [-established | -estinprog]

   verify [-restored | -restinprog] [-defined]

   verify [-linked [-defined] [-by_tgt]]

   verify [[-copyinprog | -copied [-destaged]] [-by_tgt]]

symsnapvx -g <DgName>
     [-snapshot_name <SnapshotName>
        [-generation <GenerationNumber> | -snapid [<SnapsetID>]]]
     [-failed] [-detail [-last_n <count>] [-mb | -gb | -tb]]
     [-rdf | -hop2]
     [-i <Interval>] [-c <Count>] [-offline]

   list [[-bgdefinprog] [-secured] [-manual] [-policy] [-cloud]
         [-zdp] [-expired] [-persistent]] |
        [-linked [-by_tgt]] [-restored]

symsnapvx -g <DgName>
     [-snapshot_name <SnapshotName>
       [-generation <GenerationNumber> | -snapid [<SnapsetID>]]]
     [-rdf | -hop2]
     [-mb | -gb | -tb]
     [-i <Interval>] [-c <Count>] [-offline]

   list -summary

symsnapvx -sid <SymmID> -name <SnapshotName>
     <-file <DeviceFileName> [-noprompt] |
      -noprompt 'redirect stdin' |
      -devs <<<SymDevStart>:<SymDevEnd> | <SymDevName>>
            [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]
            [-noprompt]>
     [-v] [-force] [-star]
     [-i <Interval>] [-c <Count>]

   establish [-secure <-delta <delta_time> | -absolute <date_time>> |
      -ttl <-delta <delta_time> | -absolute <date_time>>]
     [-preaction <ScriptFile>] [-postaction <ScriptFile>]

symsnapvx -sid <SymmID>
     <-file <DeviceFileName> [-noprompt] |
      -noprompt 'redirect stdin' |
      -devs <<<SymDevStart>:<SymDevEnd> | <SymDevName>>
           [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]
           [-noprompt]>
     -snapshot_name <SnapshotName>
       [-generation <GenerationNumber> | -snapid <SnapsetID>]
     [-v] [-force] [-star]
     [-i <Interval>] [-c <Count>]

   restore
     [-preaction <ScriptFile>] [-postaction <ScriptFile>]
     [-remote]

   terminate [-restored [-symforce]]
             [[[-keep_count <Count>] |
               [-older_than -snapid <SnapsetID>] | -all ] -force]
```

```
   rename -name <NewSnapshotName>

   set secure
     <-delta <delta_time> | -absolute <date_time>>

   set ttl
     <-delta <delta_time | NONE> | -absolute <date_time>>

   set <persistent | nopersistent>

symsnapvx -sid <SymmID> <-file <DeviceFileName> |
      -devs <<<SymDevStart>:<SymDevEnd> | <SymDevName>>
           [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]
   terminate -all_snapshot_names -force [-noprompt] [-v]

symsnapvx -sid <SymmID>
     <-file <DeviceFileName> [-noprompt] |
      -noprompt 'redirect stdin' |
      <-devs <<<SymDevStart>:<SymDevEnd> | <SymDevName>>
         [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]>
       -lndevs <<<SymDevStart>:<SymDevEnd> | <SymDevName>>
         [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]>
       [-noprompt]>>
     -snapshot_name <SnapshotName>
       [-generation <GenerationNumber> | -snapid <SnapsetID>]
     [-v] [-force] [-star]
     [-i <Interval>] [-c <Count>]

   link [-copy [-remote]] [-exact]

   relink [-copy [-remote]] [-exact]

   unlink [-symforce]

   set mode <copy | nocopy>

symsnapvx -sid <SymmID>
     <-file <DeviceFileName> |
      'redirect stdin' |
      -devs <<<SymDevStart>:<SymDevEnd> | <SymDevName>>
         [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]> |
      -lndevs <<<SymDevStart>:<SymDevEnd> | <SymDevName>>
         [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]>
     -snapshot_name <SnapshotName>
       [-generation <GenerationNumber> | -snapid <SnapsetID>]
     [-force] [-summary [-mb | -gb | -tb]]
     [-i <Interval>] [-c <Count>]

   verify [-established | -estinprog]

   verify [-restored | -restinprog] [-defined]

   verify [-linked [-defined] [-by_tgt]]

   verify [[-copyinprog | -copied [-destaged]] [-by_tgt]]

symsnapvx -sid <SymmID>
     -file <DeviceFileName> |
     'redirect stdin' |
     -devs <<SymDevStart>:<SymDevEnd> | <SymDevName>
         [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]> |
     -lndevs <<SymDevStart>:<SymDevEnd> | <SymDevName>
         [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]>
     [-snapshot_name <SnapshotName>
        [-generation <GenerationNumber> | -snapid [<SnapsetID>]]]
     [-failed] [-detail [-last_n <count>] [-mb | -gb | -tb]]
     [-i <Interval>] [-c <Count>] [-offline]
```

```
   list [[-bgdefinprog] [-secured] [-manual] [-policy] [-cloud]
         [-zdp] [-expired] [-persistent]] |
        [-linked [-by_tgt]] | [-restored]

symsnapvx -sid <SymmID>
     -file <DeviceFileName> |
     'redirect stdin' |
     -devs <<SymDevStart>:<SymDevEnd> | <SymDevName>
         [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]> |
     -lndevs <<SymDevStart>:<SymDevEnd> | <SymDevName>
         [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]>
     [-snapshot_name <SnapshotName>
        [-generation <GenerationNumber> | -snapid [<SnapsetID>]]]
     [-mb | -gb | -tb]
     [-i <Interval>] [-c <Count>] [-offline]

   list -summary

symsnapvx -sid <SymmID>
     [-failed] [-detail [-last_n <count>] [-mb | -gb | -tb]]
     [-i <Interval>] [-c <Count>] [-offline]

   list [[-bgdefinprog] [-secured] [-manual] [-policy] [-cloud]
         [-zdp] [-expired] [-persistent]] |
        [-linked [-by_tgt]] | [-restored]

symsnapvx -sid <SymmID> -sg <SgName> -name <SnapshotName>
     [-v] [-noprompt] [-force] [-star]
     [-i <Interval>] [-c <Count>]

   establish [-secure <-delta <delta_time> | -absolute <date_time>> |
      -ttl <-delta <delta_time> | -absolute <date_time>>]
     [-both_sides]
     [-preaction <ScriptFile>] [-postaction <ScriptFile>]

symsnapvx -sid <SymmID> -sg <SgName>
     -snapshot_name <SnapshotName>
       [-generation <GenerationNumber> | -snapid <SnapsetID>]
     [-v] [-noprompt] [-force] [-star]
     [-i <Interval>] [-c <Count>]

   restore
     [-preaction <ScriptFile>] [-postaction <ScriptFile>]
     [-remote]

   terminate [-restored [-symforce]]
             [[[-keep_count <Count>] |
               [-older_than -snapid <SnapsetID>] | -all ] -force]

   rename -name <NewSnapshotName>

   set secure
     <-delta <delta_time> | -absolute <date_time>>

   set ttl
     <-delta <delta_time | NONE> | -absolute <date_time>>

   set <persistent | nopersistent>

symsnapvx -sg <SgName> -sid <SymmID> [-noprompt] [-v]
   terminate -all_snapshot_names -force

symsnapvx -sid <SymmID> -sg <SgName> -lnsg <SgName>
     -snapshot_name <SnapshotName>
       [-generation <GenerationNumber> | -snapid <SnapsetID>]
     [-v] [-noprompt] [-force] [-star]
```

```
     [-i <Interval>] [-c <Count>]

   link [-copy [-remote]] [-exact]

   relink [-copy [-remote]] [-exact]

   unlink [-symforce]

   set mode <copy | nocopy>

symsnapvx -sid <SymmID> [-sg <SgName> | -lnsg <SgName>]
     -snapshot_name <SnapshotName>
       [-generation <GenerationNumber> | -snapid <SnapsetID>]
     [-force] [-summary [-mb | -gb | -tb]]
     [-i <Interval>] [-c <Count>]

   verify [-established | -estinprog]

   verify [-restored | -restinprog] [-defined]

   verify [-linked [-defined] [-by_tgt]]

   verify [[-copyinprog | -copied [-destaged]] [-by_tgt]]

symsnapvx -sid <SymmID> -sg <SgName>
     [-snapshot_name <SnapshotName>
       [-generation <GenerationNumber> | -snapid [<SnapsetID>]]]
     [-failed] [-detail [-last_n <count>] [-mb | -gb | -tb]]
     [-i <Interval>] [-c <Count>] [-offline]

   list [[-bgdefinprog] [-secured] [-manual] [-policy] [-cloud]
        [-zdp] [-expired] [-persistent]] |
       [-linked [-by_tgt]] | [-restored]

symsnapvx -sid <SymmID> -sg <SgName> [-detail]
     [-snapshot_name <SnapshotName> [-snapid <SnapsetID>]]

   list -aggregate [-manual] | [-policy]

symsnapvx -sid <SymmID> -lnsg <SgName>
     [-snapshot_name <SnapshotName>]
       [-generation <GenerationNumber> | -snapid [<SnapsetID>]]
     [-failed] [-detail [-last_n <count>] [-mb | -gb | -tb]]
     [-i <Interval>] [-c <Count>] [-offline]

   list -linked -by_tgt

symsnapvx -sid <SymmID> -sg <SgName>
     [-snapshot_name <SnapshotName>]
       [-generation <GenerationNumber>]
     [-mb | -gb | -tb]
     [-i <Interval>] [-c <Count>] [-offline]

   list -summary

symsnapvx -cg <CgName> -name <SnapshotName>
     [-sid <SymmID> |
      -rdfg <SymmID>:<GrpNum>[,<GrpNum>,...]|<all>[,...] |
     name:<RdfGroupName>[,<RdfGroupName>,...]>]
     [-rdf | -hop2] [-v] [-noprompt] [-force] [-star]
     [-i <Interval>] [-c <Count>]

   establish [-secure <-delta <delta_time> | -absolute <date_time>> |
      -ttl <-delta <delta_time> | -absolute <date_time>>]
     [-both_sides]
     [-preaction <ScriptFile>] [-postaction <ScriptFile>]
```

```
symsnapvx -cg <CgName>
     -snapshot_name <SnapshotName>
       [-generation <GenerationNumber> | -snapid <SnapsetID>]
     [-rdf | -hop2] [-v] [-noprompt] [-force] [-star]
     [-i <Interval>] [-c <Count>]
     [-sid <SymmID> |
      -rdfg <SymmID>:<GrpNum>[,<GrpNum>,...]|<all>[,...] |
      name:<RdfGroupName>[,<RdfGroupName>,...]>]

   restore
     [-preaction <ScriptFile>][-postaction <ScriptFile>]
     [-remote]

   terminate [-restored [-symforce]]
             [[[-keep_count <Count>] |
               [-older_than -snapid <SnapsetID>] | -all ] -force]

   rename -name <NewSnapshotName>

   set secure
     <-delta <delta_time> | -absolute <date_time>>

   set ttl
     <-delta <delta_time | NONE> | -absolute <date_time>>

   set <persistent | nopersistent>

symsnapvx -cg <CgName> -sid <SymmID> [-noprompt] [-v]
   terminate -all_snapshot_names -force

symsnapvx -cg <CgName>
     -snapshot_name <SnapshotName>
       [-generation <GenerationNumber> | -snapid <SnapsetID>]
     [-rdf | -hop2] [-v] [-noprompt] [-force] [-star]
     [-i <Interval>] [-c <Count>]
     [-sid <SymmID> |
      -rdfg <SymmID>:<GrpNum>[,<GrpNum>,...]|<all>[,...] |
      name:<RdfGroupName>[,<RdfGroupName>,...]>]

   link [-copy [-remote]] [-exact]

   relink [-copy [-remote]] [-exact]

   unlink [-symforce]

   set mode <copy | nocopy>

symsnapvx -cg <CgName>
     -snapshot_name <SnapshotName>
       [-generation <GenerationNumber> | -snapid <SnapsetID>]
     [-rdf | -hop2]
     [-force] [-summary [-mb | -gb | -tb]]
     [-i <Interval>][-c <Count>]
     [-sid <SymmID> |
     -rdfg <SymmID>:<GrpNum>[,<GrpNum>,...]|<all>[,...] |
      name:<RdfGroupName>[,<RdfGroupName>,...]>]

   verify [-established | -estinprog]

   verify [-restored | -restinprog] [-defined]

   verify [-linked [-defined] [-by_tgt]]

   verify [[-copyinprog | -copied [-destaged]] [-by_tgt]]

symsnapvx -cg CgName
     [-snapshot_name <SnapshotName>
```

```
          [-generation <GenerationNumber> | -snapid [<SnapsetID>]]]
       [-failed] [-detail [-last_n <count>] [-mb | -gb | -tb]]
       [-rdf | -hop2]
       [-sid <SymmID> |
       -rdfg <SymmID>:<GrpNum>[,<GrpNum>,...]|<all>[,...] |
        name:<RdfGroupName>[,<RdfGroupName>,...]>]
       [-i <Interval>] [-c <count>] [-offline]

   list [[-bgdefinprog] [-secured] [-manual] [-policy] [-cloud]
         [-zdp] [-expired] [-persistent]] |
        [-linked [-by_tgt]] | [-restored]

symsnapvx -cg CgName
     [-snapshot_name <SnapshotName>
        [-generation <GenerationNumber> | -snapid [<SnapsetID>]]]
     [-rdf | -hop2]
     [-sid <SymmID> |
     -rdfg <SymmID>:<GrpNum>[,<GrpNum>,...]|<all>[,...] |
      name:<RdfGroupName>[,<RdfGroupName>,...]>]
     [-mb | -gb | -tb]
     [-i <Interval>] [-c <count>] [-offline]

   list -summary
```

DESCRIPTION

The symsnapvx command performs TimeFinder/Snapvx operations on a device list, a list of device ranges, a device group (DG), composite group (CG), or storage group (SG).

These operations are establishing, restoring, linking, and terminating snapshots. In addition, they allow listing the state of snapshots on devices or on the entire Symmetrix.

All SnapVX operations may be performed on a group or individual device basis. For group operations all devices must have been previously associated with the group.

ARGUMENTS

establish	Creates and activates a Snapvx snapshot.
link	Presents the snapshot data on the target device. With the -copy option the data is copied to the target.
list	Lists all snapshots in a device file, device ranges, DG, CG, or SG filtered by snapshot_name, generation, restored, linked, or failed. Output options are detail, summary, last_n, mb, gb and tb.
relink	Removes the current link to target device and presents a different snapshot. The relink command may also be used to relink to the same snapshot. This has the effect of refreshing the point-in-time copy on the link target when it's been modified by host writes. With the -copy option performs a differential copy to the target.
rename	Changes the name assigned to a snapshot.
restore	Copies the point-in-time data from a snapshot back to the original source device. When possible the restore operation will

automatically determine whether this is done incrementally (copying changed tracks only) or if a complete copy is required. The snapshot must be fully Established to perform a restore.

set mode — Changes the copy mode of a link to Copy or NoCopy.

set persistent — Set the persistent flag on a policy snapshot that will not be terminated by count based auto termination when creating new policy snapshot on a device.

set nopersistent — Clears the persistent flag set on a policy snapshot.

set secure — Converts a snapshot to a secure snapshot and sets a secure expiration time for a snapshot.

Secure snapshots cannot be terminated until they expire. Use extreme caution with this argument.

Secure snapshots may only be terminated after they expire or by customer-authorized EMC support. Please refer to Knowledgebase article 498316 for additional information.

set ttl — Sets a time to live for a snapshot as either number of days from now (-delta) or a date (-absolute).

terminate — Removes an existing Snapvx snapshot. A secure snapshot or a snapshot that has a link target or a restore session cannot be terminated.

unlink — Removes the current link to target device(s).

verify — Verifies whether one or more devices are in the requested state.

KEYWORDS

copy — Sets the link copy mode to perform background copy to the target device(s).

name — Indicates an RDF group’s logical name within a CG.

nocopy — Sets the link copy mode to not perform background copy to the target device(s).

OPTIONS

-absolute — Specifies a date and time for the snapshot time to live and secure expiration time in the form MM/DD/YYYY[:hh] (month/date/year[:hour]) with reference to the host time. The specified date may be up to 400 days in the future. The specified hours can be set to any value between 0 and 23. It has to be at least one hour in the future from current time. It will be set as close to

the specified hour as possible.

-aggregate Reports only snapset aggregate summary on a SG.

-all Terminate all snapshots on the devices for a given snapshot name.

-all_snapshot_names
Terminate all snapshots on the devices

-bgdefinprog Reports only the snapshots that were created on a target device when background define process was in progress and the define is not complete yet.

-both_sides Performs the operation on both locally and remotely associated snapshots. Only valid with DG, CG or SG.

-by_tgt Used with list -linked to display by link target or verify -linked, -copyinprog, or -copied to verify by link target.

-c Specifies the number (count) of times to Display or verify.

Specifies the number (count) of times to attempt to acquire an exclusive lock on the Symmetrix host database during control operations.

If you do not specify this option and specify an interval (-i), the program will loop continuously to list or verify or start the control operation.

-cg Applies a composite group name to the command.

-copied Verifies that the link(s) are in the Copied state. The -destaged switch may be used to verify that all tracks have been physically written to the link target.

-copy Used with link and relink to specify copy mode.

-copyinprog Verifies that the link(s) are in the CopyInProg state.

-cloud Reports only cloud snapshots

-delta Specifies the time to live and secure expiration time in the form Days[:hh] (days[:hour]). The specified days can be set to any value between 1 and 400. The specified hours can be set to any value between 0 and 23.

-defined Used with verify -linked, -restored and -restinprog to wait until all tracks have been defined.

-destaged Used with verify -copied to wait until all tracks have been fully destaged and

copied to the target before indicating in-state.

-detail — Used with list to get an expanded display that provides all information.

-devs — Specifies the ranges of Symmetrix source devices.

-established — Verifies that the snapshot is in the Established state.

-estinprog — Verifies that the snapshot is in the EstInProg state.

-exact — When specified, pairs source and link devices in their ordinal positions within the selection. When not set uses the source and link device selections as a pool that pairs by best match.

-expired — Only report expired snapshots.

-failed — Used with list to display only failed snapshots or links.

-file — Applies a device file to the command. The device file contains a list of devices or device pairs (SymDevnames) listing a device or pair per each line. For single devices, one device is entered per line. For device pairs, one pair is entered per line (the source device first, a space, and the target device). Device files may include comment lines that begin with the pound sign (#). A Symmetrix ID is required for this option. -f is synonymous with -file.

-force — Attempts to force the operation even though one or more devices may not be in the normal, expected state(s) for the specified operation.

-g — Applies a device group name to the command.

-gb — Displays counts in gigabytes.

-generation — Uniquely identifies snapshots of the same name and device. The generation number is incremented by one for each snapshot on a device and they are ordered by timestamp. If omitted the operation will default to generation zero (the most recent) for control operations.

-h — Provides brief online help information.

-hop2 — Performs the specified action on the Symmetrix array two hops away. Only valid with DG or CG.

-i — Specifies the repeat interval, in seconds, to wait, either between successive iterations of a list or verify operation or between control operation attempts to acquire an exclusive lock on the Symmetrix host

database.

The default interval is 30 seconds.
The minimum interval is 5 seconds.
For passive actions the minimum interval is 15 seconds. Passive actions are actions that do not acquire an exclusive lock.

When used with the list or verify action, the number of seconds specified indicates the interval of time (in seconds) to repeat the verify command before the verify action finds and reports the pairs in state

-keep_count Used with terminate action to terminate policy snapshots by only keeping the most recent keep_count number of snapshots.
Allowed with force option only.

-last_n Limits list output items to the number of last_n per device. Used with list to limit the number of generations to display for each device. Only applies to the detailed displays.

-linked When used with the list action displays the links associated with the selected devices. When used with the verify action verifies that nocopy links are in the linked state.

-lndevs Specifies the ranges of Symmetrix target devices.

-lnsg Applies an SG name to the command for target devices.

-mb Displays counts in megabytes.

-name For establish this is the user supplied name for a new Snapvx snapshot.

-noprompt Requests to not return a prompt after a command is entered. The default is to prompt for confirmation.

-offline Specifies that the Symmetrix data connection is offline and the operation will use the host in-memory database.

-older_than Used with terminate action to terminate all the policy snapshots with the SnapsetID lower(older) than the specified SnapsetID. Allowed with force option only.

-persistent Reports only persistent snapshots

-postaction Executes the script argument after a snapshot has been established or restored.

-preaction Executes the script argument before a snapshot has been established or restored.

-secure Creates a secure snapshot and sets the secure expiration time.

Secure snapshots cannot be terminated until they expire. Use extreme caution with this

option.

Secure snapshots may only be terminated after they expire or by customer-authorized EMC support. Please refer to Knowledgebase article 498316 for additional information.

-secured — Reports only secured snapshot.

-policy — Reports only policy snapshot

-snapid — Uniquely identifies snapshots of the same name and device. When used with list action, displays SnapsetID(absolute generation number) instead of the relative generation number of the snapshots. If omitted the operation will default to displaying or using the relative generation(generation zero being the most recent).

-rdf — Performs the action on the remote Symmetrix array. Only valid with DG or CG.

-rdfg — Performs the requested action on a subset of the CG defined by one or more Symmetrix/RA Group combinations supplied as the argument to -rdfg.

-remote — Acknowledges that the data will be propagated to the remote mirror of the RDF device.

-restinprog — With the verify command, verifies that the snapshot(s) are in the RestoreInProgress state.

-restored — With the verify command, verifies that the snapshot(s) are in the restore state. With the terminate command, terminates restore(s). With the list command, displays restore and restore in progress snapshots only.

-snapshot_name — The name of an existing snapshot used to identify the snapshot to be operated on.

-sid — Applies the command to the specified Symmetrix ID. Use this option with the -file or -sg option to select the Symmetrix array on which to perform the operation, or specify this option with -cg option to restrict the operation to a single Symmetrix array.

-sg — Applies an SG name to the command for source devices.

-star — Targets the action at devices in STAR mode.

-summary — Shows a summary of snapshot and link states for list or verify.

-symforce — Forces the operation to execute when normally it is rejected. Use extreme caution with this option. If used when a link is copy in progress or when a restore is restore in progress, this will cause an incomplete

copy and data on the copy target would not be usable.

-tb Displays counts in terabytes.

-ttl Applies a time to live when establishing a new snapshot.

-v Provides a more detailed, verbose listing.

-zdp Reports only zDp snapshots.

PARAMETERS

CgName The composite group name.

Count The number of times (count) to repeat.

DeviceFileName The device file name. The device file contains a list of devices or device pairs (SymDevNames).

DgName The device group name.

GenerationNumber
The generation number of the snapshot.

GrpNum The RDF group number.

Interval The interval between polls, in seconds.

NewSnapshotName
The new snapshot name.

NONE Removes the time to live set on a snapshot.

RdfGroupName The logical name associated with the RDF group(s) in a CG.

SnapsetID The snapsetID (absolute generation number) of the snapshot.

SnapshotName The snapshot name.

ScriptFile The full pathname of a script file to execute.

SymDevEnd The last Symmetrix device name in a range.

SymDevStart The first Symmetrix device name in a range.

SymmID The 12-digit ID of the Symmetrix array.

SgName The storage group name.

RETURN CODES

Code #	Code Symbol
0	CLI_C_SUCCESS
1	CLI_C_FAIL
2	CLI_C_DB_FILE_IS_LOCKED
18	CLI_C_ALREADY_IN_STATE The device or device group is already in the

desired Copy state.

19 CLI_C_GK_IS_LOCKED
All gatekeepers to the Symmetrix array are currently locked.

22 CLI_C_NEED_FORCE_TO_PROCEED
Requires the force flag to proceed.

23 CLI_C_NEED_SYMFORCE_TO_PROCEED
Requires the symforce flag to proceed.
CAUTION: Extreme caution should be exercised when using this option.

Return codes for symsnapvx verify

12 CLI_C_NOT_ALL_RESTORED
Not all devices are in the 'Restored' state.

13 CLI_C_NONE_RESTORED
None of the devices are in the 'Restored' state.

18 CLI_C_ALREADY_IN_STATE
The Device(s) is (are) already in the desired state or mode.

29 CLI_C_NOT_ALL_RESTINPROG
NOT all of the pairs are in the 'RestInProg' state.

30 CLI_C_NONE_RESTINPROG
NONE of the pairs are in the 'RestInProg' state.

162 CLI_C_NOT_ALL_ESTABLISHED
Not all devices are in the 'Established' state.

163 CLI_C_NONE_ESTABLISHED
None of the devices are in the 'Established' state.

164 CLI_C_NOT_ALL_ESTINPROG
Not all devices are in the 'Estinprog' state.

165 CLI_C_NONE_ESTINPROG
None of the devices are in the 'Estinprog' state.

166 CLI_C_NOT_ALL_LINKED
Not all devices are in the 'Linked' state.

167 CLI_C_NONE_LINKED
None of the devices are in the 'Linked' state.

168 CLI_C_NOT_ALL_DEFINED
Not all devices are in the 'Defined' state.

169 CLI_C_NONE_DEFINED
None of the devices are in the 'Defined' state.

170	CLI_C_NOT_ALL_CIPLINKED Not all linked devices are in the 'Copyinprog' state.
171	CLI_C_NONE_CIPLINKED None of the linked devices are in the 'Copyinprog' state.
172	CLI_C_NOT_ALL_COPIEDLINKED Not all linked devices are in the 'Copied' state.
173	CLI_C_NONE_COPIEDLINKED None of the linked devices are in the 'Copied' state.
174	CLI_C_NOT_ALL_DESTAGED Not all devices are in the 'Destaged' state.
175	CLI_C_NONE_DESTAGED None of the devices are in the 'Destaged' state.

```
symstar

        Performs SRDF/Star control operations
        on a composite group.

        SYNOPSIS

             symstar -h

             symstar show <CgName> [-detail]

             symstar list [-c <Count>] [-i <Interval>]
                  [-full] [-offline] [-local]

             symstar -cg <CgName> [-noprompt] [-i <Interval>]
                  [-c <Count>] [-wkload <SiteName>]
                  [-opmode <concurrent | cascaded>]

                cleanup -site <SiteName> [-force] [-v]

                configure -add recovery_rdf_pairs

                configure -reset rdf_mode

                connect -site <SiteName> [-force] [-v] [-full]
                  [-keep_data <SiteName> [-remote]]

                disable [-force] [-v]

                disconnect -site <SiteName> [-force] [-trip] [-v]

                enable [-force] [-v]

                halt [-reset] [-force] [-v]

                isolate -site <SiteName> [-force] [-v]

                protect -site <SiteName> [-force] [-v]

                query [-detail] [-offline]

                reconfigure -path <SrcSiteName>:<TgtSiteName>
                  -site <TgtSiteName>
                  [-remove <SrcSiteName>:<TgtSiteName>]
                  [-full] [-reset] [-force] [-v]

                recover [-force]

                reset -site <SiteName> [-force] [-v]

                switch -site <SiteName> [-keep_data <SiteName>] [-force]
                  [-full] [-v]

                unprotect -site <SiteName> [-force] [-v]

             symstar -cg <CgName> [-i <Interval>] [-c <Count>]
                  [-noprompt]

                verify -site <SiteName> -connected | -disconnected |
                  -halted | -haltfail | -haltstarted | -isolated |
                  -pathfail [-cleanreq] | -pathfailinprog | -protected

                verify -protected | -tripped | -trip_inprogress |
                  -unprotected

             symstar -cg <CgName> [-noprompt]
```

```
   buildcg -site <SiteName> [-update]

   setup -options <OptionsFile> [-distribute]
     [-site <SiteName>]
     [-opmode <concurrent | cascaded>]

   setup -options <OptionsFile> -reload_options

   setup -remove [-force]

symstar -cg <CgName> [-i <Interval>] [-c <Count>]
     [-noprompt] [-v]
     -sid <SymmID>
     -devs <<SymDevStart>:<SymDevEnd> | <SymDevName>
       [,<<SymDevStart>:<SymDevEnd> | <SymDevName>>...]> |
       -file <FileName>
     -stg_rdfg <GrpNum>[,<GrpNum>]
     -cg_rdfg <CgGrpNum>[,<CgGrpNum>]
     [-stg_r21_rdfg <R21GrpNum> -cg_r21_rdfg <CgR21GrpNum>]

   modifycg -add [-force]

   modifycg -remove
```

DESCRIPTION

The symstar command provides query and composite control operations to manage an SRDF/Star environment.

ARGUMENTS

buildcg	Reads the internal definition for the SRDF/Star configuration and creates the matching host composite group (CG).
cleanup	Cleans up stale information after a disaster failure (loss of WorkloadSite).
configure	Changes the SRDF configuration.
connect	Makes an SRDF connection and starts the data flow.
disable	Disables SRDF/Star consistency protection.
disconnect	Suspends the SRDF data flow.
enable	Enables SRDF/Star consistency protection.
halt	Write disables devices and synchronizes SRDF data to remote sites.
isolate	Isolates the SyncTargetSite or AsyncTargetSite from the SRDF/Star replication. The RDF2 devices of the isolated site are made read/write enabled to their hosts.
list	Displays information about all SRDF/Star configurations with an SRDF/Star definition file that is present either locally or on a locally-attached Symmetrix array.
modifycg	Moves devices between the staging area and the SRDF/Star CG, and updates the CG definition to reflect the changes.
protect	Waits for SRDF data to be synchronized

to the remote SyncTargetSite or AsyncTargetSite and turns on SRDF consistency protection.

query — Displays the status of the SRDF/Star configuration.

reconfigure — Changes the SRDF/Star replication data path.

rdf_mode — Specifies that the SRDF mode for the connected sites in the CG be reset to adaptive copy. If the CG is configured with an R22 device, the recovery SRDF pairs are also reset to adaptive copy mode.

recover — Recovers the failed modify add or modify remove operation and places the SRDF/Star CG into a known state.

reset — Cleans up stale information after a transient failure (loss of connectivity) to the SyncTargetSite or AsyncTargetSite.

setup — Reads and validates the host composite group that manages the SRDF/Star replication.

Builds an internal representation of the composite group and saves it in the following directories:

- /var/symapi/STAR/def (for UNIX) or
- %Program Files%\\EMC\\SYMAPI\\Star\\def (for Windows)

show — Displays the contents of the internal definition for the SRDF/Star configuration.

switch — Performs the necessary operations to start the workload at the SyncTargetSite or AsyncTargetSite.

unprotect — Disables SRDF consistency protection for the devices on the SRDF links to the remote SyncTargetSite or AsyncTargetSite.

verify — Verifies a given site or checks if SRDF/Star is in a desired state.

KEYWORDS

cascaded — Sets the mode of operation for the SRDF/Star configuration to cascaded.

concurrent — Sets the mode of operation for the SRDF/Star configuration to concurrent.

recovery_rdf_pairs — Configures the existing SRDF/Star configuration to incorporate R22 devices for the recovery SRDF pairs.

OPTIONS

-add — Specifies the element of configuration to add.

-c — Specifies the number (count) of times to display or to acquire an exclusive lock on

the Symmetrix host database, the local Symmetrix array, and the remote Symmetrix arrays. If this option is not specified and an interval (-i) is specified, the display shows continuously, or until the SRDF/Star operation starts.

-cg Identifies the name of the host composite group.

-cg_rdfg The SRDF group(s) within the SRDF/Star CG in which to add or remove devices. For a concurrent SRDF/Star CG, two SRDF groups must be specified, separated by a comma. These SRDF groups are associated with the SRDF groups in the -stg_rdfg option. This association is based on their order in this option and -stg_rdfg.

-cg_r21_rdfg
The SRDF group connecting the R21 and R2 Symmetrix arrays of a cascaded SRDF/Star CG. It is only valid for operations involving cascaded R1 devices. This SRDF group is associated with the SRDF group specified in the -stg_r21_rdfg option.

-cleanreq Verifies the site is in the pathfail state and needs cleaning.

-connected Verifies the site is in the connected state.

-detail Includes extended information in the output when used with the query and show commands.

-devs Specifies the ranges of Symmetrix devices to add or remove.

-disconnected
Verifies the site is in the disconnected state.

-distribute
Performs an automatic SRDF/Star definition file distribution. This form of setup does not disrupt an active protected SRDF/Star setup.

-file Specifies the filename containing the list of devices to act upon. Only the SymDevName specified in the first column of each line is used.

-force Allows the action to proceed even if the SRDF/Star environment is not currently in the proper state for that action. When used with the switch action, this lets you specify the -keep_data SiteName option if the current state of that site’s data is not consistent. When used with the setup -remove action, this removes all SRDF/Star metadata associated with the SRDF/Star CG, even when the CG is not defined in the symapi database.

-full Used by reconfigure, switch, and connect. Performs a full SRDF resynchronization if SRDF incremental resync is not available. Used by the list action to display full names instead of abbreviations.

-h Provides brief, online help information.

-halted Verifies the site is in the halted state.

-haltfail Verifies the site is in the haltfail state.

-haltstarted
Verifies the site is in the haltstarted state.

-i Specifies the repeat interval in seconds to display or to acquire an exclusive lock on the Symmetrix host database, the local Symmetrix array, and the remote Symmetrix arrays.
The default interval is 30 seconds.
The minimum interval is 5 seconds.
For passive actions the minimum interval is 15 seconds. Passive actions are actions that do not acquire an exclusive lock.

-isolated Verifies the site is in the isolated state.

-keep_data Identifies which site's data is retained when used with the switch and connect action. If you switch to the SyncTargetSite and choose to keep the data of the AsyncTargetSite, the SRDF devices are reconfigured to make a new R1->R2 pairing. For the connect action, an SRDF establish or restore operation is performed, depending on which site's data is retained. By default, the workload site data is retained.

-local Lists only the locally-defined CGs. Available only for the list action.

-noprompt Requests no prompt before performing action.

-offline Obtains data from the configuration database on the host. No connections are made to any Symmetrix arrays. Available only for query and list actions.

-opmode Specifies the mode of operation (concurrent or cascaded).

Required when initially setting up an SRDF/Star configuration with R22 devices.

Not allowed when initially setting up an SRDF/Star configuration without R22 devices.

-options Identifies the name of the file containing the options to use for this SRDF/Star configuration.

-path Specifies the sites on which the new SRDF pairs are created when the reconfigure action is issued.

-pathfail Verifies the site is in the pathfail

state.

-pathfailinprog
Verifies the site in in the pathfailinprog state.

-protected Verifies the site is in the protected state. If -site is not specified, verifies that SRDF/Star is in the protected state.

-reload_options
Reads the specified options file to update the SRDF/Star definition file when using the setup action. Do not change any SITE_NAME values with this option.

-remote Indicates the remote data copy flag. Used with the connect action when keeping remote data and the concurrent link is ready. Data is also copied to the concurrent SRDF mirror. Not required if the concurrent link is suspended.

-remove For the reconfigure action, specifies the sites on which the SRDF pairs are removed.

For the setup action, specifies that all SRDF/Star mode settings for all SRDF groups be set to off if the CG is defined in the symapi database, and to remove all SRDF/Star metadata associated with the group.

For the modifycg action, indicated to remove the specified devices from the SRDF/Star CG to the staging area.

-reset Performs a reset action on the path when the reconfigure action is issued. When used with the halt action, allows the application to be restarted at the same site after the halt command has completed or failed. When used with the configure action, specifies the element of the reset operation.

-sid Specifies the unique Symmetrix ID.

-site Specifies the SiteName to apply the given action.

-stg_r21_rdfg
For modifycg operations, indicates the SRDF group comprising the staging area at the R21 Symmetrix array when the configuration is cascaded. It is required for an add or remove operation when the setup is cascaded. This SRDF group is associated with the SRDF group in the -cg_r21_rdfg option.

-stg_rdfg For the modifycg operations, indicates the SRDF group(s) comprising the staging area. For a concurrent CG, two groups must be specified, separated by a comma. These SRDF groups are associated with the SRDF groups in the -cg_rdfg option. This association is based on their order in this option and -cg_rdfg.

-trip Transitions the site to pathfail state when used

with disconnect action.

-tripped Verifies SRDF/Star is in the tripped state.

-trip_inprogress
Verifies SRDF/Star is in the trip_inprogress
state.

-unprotected
Verifies the site is in the unprotected
state. If -site is not specified, verifies
SRDF/Star is in the unprotected state.

-update Allows the updating of the existing host
composite group from the STAR definition file.

-v Provides a more detailed, verbose listing.

-wkload Specifies the current workload site name
if symstar fails to determine the current
workload site name.

PARAMETERS

CgGrpNum The SRDF (RA) group number in the CG.

CgName The name of the host composite group.

Count The number of times (count) to repeat.

FileName The name of the file that contains a list. Only
the SymDevName specified on the first column
of each line is used.

GrpNum The SRDF (RA) group number.

Interval The repeat interval in seconds.

OptionsFile The name of a file containing the symstar
options.

R21GrpNum The SRDF (RA) group number of the R1 mirror of
the R21 device.

SiteName The user-specified name for the
SyncTargetSite or AsyncTargetSite.

SrcSiteName The name of the source (R1) site in a path.

SymDevEnd The last Symmetrix device name in a sequence,
such as 00B6.

SymDevName The Symmetrix device name , unique per
Symmetrix array, such as 001C.

SymDevStart The first Symmetrix device name in a sequence,
such as 001C.

SymmID The 12-digit ID of the Symmetrix array.

TgtSiteName The name of the target (R2) site in a path.

RETURN CODES

Code #	Code Symbol

```
0       CLI_C_SUCCESS
The action was successful.

1       CLI_C_FAIL
The action was unsuccessful.

18      CLI_C_ALREADY_IN_STATE
The system is already in the desired state.

22      CLI_C_NEED_FORCE_TO_PROCEED
The system is not in the proper state
to execute this procedure. Use -force
to execute this procedure anyway.

90      CLI_C_NONE_VERIFIED
The system is not in desired state.

191      CLI_C_FEATURE_NOT_AVAILABLE
The feature is not available in this version.
FILES

The options file is created by the user. It must
conform to the following syntax:
SYMCLI_STAR_WORKLOAD_SITE_NAME            = <Wname>
SYMCLI_STAR_SYNCTARGET_SITE_NAME          = <Sname>
SYMCLI_STAR_ASYNCTARGET_SITE_NAME         = <Aname>
SYMCLI_STAR_ADAPTIVE_COPY_TRACKS          = <Numtracks>
SYMCLI_STAR_ACTION_TIMEOUT                = <Numseconds>
SYMCLI_STAR_TERM_SDDF                     = <YES/NO>
SYMCLI_STAR_ALLOW_CASCADED_CONFIGURATION  = <YES/NO>
SYMCLI_STAR_SYNCTARGET_RDF_MODE           = <ACP/SYNC>
SYMCLI_STAR_ASYNCTARGET_RDF_MODE          = <ACP/ASYNC>

The supported options include:

Option            Description
---------------  ---------------------
SYMCLI_STAR_WORKLOAD_SITE_NAME
Value: <Wname>

<Wname> is the name for the site where
the concurrent RDF1 devices are local.

The default value is SITE_A.

SYMCLI_STAR_SYNCTARGET_SITE_NAME
Value: <Sname>

<Sname> is the name for the
synchronous target site. It must match
the name assigned to the SRDF/S RDF
groups in the composite group.

The default value is SITE_B.

SYMCLI_STAR_ASYNCTARGET_SITE_NAME
Value: <Aname>

<Aname> is the name for the
asynchronous target site. It must
match the name assigned to the SRDF/A
RDF groups in the composite group.

The default value is SITE_C.

SYMCLI_STAR_ADAPTIVE_COPY_TRACKS
Value: <Numtracks>
```

<Numtracks> is the number of invalid tracks to reach before transitioning out of adaptive copy mode and setting the mode to SRDF/S or SRDF/A.

The default value is 30000.

SYMCLI_STAR_ACTION_TIMEOUT
Value: <Numseconds>

<Numseconds> is the maximum number of seconds to wait to achieve consistency protection or R2_Recoverable STAR protection, or for devices to reach <Numtracks> invalid tracks while syncing RDF devices.

The default value is 18000.

SYMCLI_STAR_TERM_SDDF
Allowed values: YES or NO

If set to YES, SDDF sessions are terminated on both the SYNC and ASYNC target sites at the time of symstar disable.

If set to NO, SDDF sessions on both SYNC and ASYNC target sites are not terminated (deactivated instead) at the time of symstar disable.

SYMCLI_STAR_ALLOW_CASCADED_CONFIGURATION
Allowed values: YES or NO

If set to YES, cascaded configurations are allowed.

In cascaded configurations, the data flows from the workload site to the synchronous target site and then to the asynchronous target site.

SYMCLI_STAR_SYNCTARGET_RDF_MODE
Valid values: ACP or SYNC

If set to ACP, the SRDF mode between the workload site and the synchronous target site is transitioned to adaptive copy mode at the end of 'symstar unprotect'. If set to SYNC, the SRDF mode between the workload site and synchronous target site remains in synchronous mode at the end of 'symstar unprotect'.

The default value is ACP.

SYMCLI_STAR_ASYNCTARGET_RDF_MODE
Valid values: ACP or ASYNC

If set to ACP, the SRDF mode between the workload site and the asynchronous target site is transitioned to adaptive copy mode at the end of 'symstar unprotect'. If set to ASYNC,

the SRDF mode between the workload
site and asynchronous target site
remains in asynchronous mode at the
end of 'symstar unprotect'.

The default value is ACP.

EXAMPLES

To create the composite group for SRDF/Star protection,
enter:

```
symcg create MyStar -type RDF1 -rdf_consistency
```

To add the concurrent SRDF devices to the MyStar composite
group where the SRDF/A group is 30 and the SRDF/A group
is 31, enter:

```
symcg -cg MyStar addall dev -sid 63 -rdfg 31
```

To assign the SRDF group names in the MyStar composite
group, enter:

```
symcg -cg MyStar -rdfg 63:30 set -name synctgt
symcg -cg MyStar -rdfg 63:31 set -name asynctgt
```

To assign the recovery SRDF groups in the MyStar composite
group, enter:

```
symcg -cg MyStar -rdfg 63:30 set -recovery_rdfg 20
symcg -cg MyStar -rdfg 63:31 set -recovery_rdfg 21
```

To create the SRDF/Star definition file, enter:

```
symstar -cg MyStar setup -option options.file
```

To connect the synchronous target site, enter:

```
symstar -cg MyStar connect -site synctgt
```

To protect the synchronous target site, enter:

```
symstar -cg MyStar protect -site synctgt
```

To connect the asynchronous target site, enter:

```
symstar -cg MyStar connect -site asynctgt
```

To protect the asynchronous target site, enter:

```
symstar -cg MyStar protect -site asynctgt
```

To enable the SRDF/Star protection for the composite group,
enter:

```
symstar -cg MyStar enable
```

To verify synctgt is in a protected state, enter:

```
symstar -cg MyStar verify -site synctgt -protected
```

To change the flow of data from:

workload site to synctgt and workload site to asynctgt

to:

```
workload site to synctgt to asynctgt,

Enter:

symstar -cg MyStar reconfigure -path synctgt:asynctgt
-site asynctgt

To change the flow of data from:

workload site to synctgt to asynctgt.

to:

workload site to synctgt and workload site to asynctgt,

Enter:

symstar -cg MyStar reconfigure -path workload:asynctgt
-site asynctgt
```

From Solutions Enabler v9.0 and above, the symstat CLI is no longer supported. The symstat CLI is only supported up to Solutions Enabler v8.4.

symtier

Allows you to create and manage storage tiers
that can be used for FAST policies.

SYNOPSIS

```
symtier -sid <SymmID> [-i <Interval>] [-c <Count>]

   create -name <TierName>
      <-tgt_raid1 |
       -tgt_raid5 -tgt_prot <3+1 | 7+1> |
       -tgt_raid6 -tgt_prot <6+2 | 14+2>>
      -technology <EFD | FC | SATA>
      -inc_type dynamic

   create -name <TierName>
      <-tgt_raid1 |
       -tgt_raid5 -tgt_prot <3+1 | 7+1> |
       -tgt_raid6 -tgt_prot <6+2 | 14+2>>
      -technology <EFD | FC | SATA>
      -inc_type static
      [-dsk_grp <<DiskGroupID>[,<DiskGroupID>...] |
                 name:<DiskGroupName>[,<DiskGroupName>...]>]

   create -name <TierName>
      <-tgt_unprotected | -tgt_raid1 |
       -tgt_raid5 -tgt_prot <3+1 | 7+1> |
       -tgt_raid6 -tgt_prot <6+2 | 14+2>>
      <-technology <EFD | FC | SATA>>
      [-external]
      -vp
      [-pool <<PoolName>[,<PoolName>...]>]

   modify -tier_name <TierName>
      -technology <EFD | FC | SATA>

   delete -tier_name <TierName> [-force]

   rename -tier_name <TierName> -name <NewTierName>

   add -tier_name <TierName> [-propagate]
      -dsk_grp <<DiskGroupID>[,<DiskGroupID>...]> |
                <name:<DiskGroupName>[,<DiskGroupName>...]>

   add -tier_name <TierName>
      -pool <<PoolName>[,<PoolName>...]>

   remove -tier_name <TierName> [-propagate]
      -dsk_grp <<DiskGroupID>[,<DiskGroupID>...]> |
                <name:<DiskGroupName>[,<DiskGroupName>...]>

   remove -tier_name <TierName>
      -pool <<PoolName>[,<PoolName>...]>

symtier [-sid <SymmID>] [-v] [-offline]

   list [-dp | -vp [-ckd | -fba]]
        [-technology <EFD | FC | SATA> | -external]

symtier -sid <SymmID> [-offline]

   show -tier_name <TierName>
```

DESCRIPTION

The symtier command provides the ability to create,

delete and modify the storage tiers. The tiers can be added to FAST policies.

ARGUMENTS

add	Adds disk groups or thin pools to the storage tier.
create	Creates a storage tier with the specified attributes.
delete	Deletes the storage tiers.
list	Lists storage tier names and details.
modify	Modify tier attributes.
remove	Removes disk groups or thin pools from the storage tier.
rename	Renames the storage tier.
show	Shows detailed information about the storage tier.

OPTIONS

-c	Specifies the number (count) of times to try the requested action. If this option is not specified, and an interval (-i) is specified, the process will attempt -c number of times waiting for the database lock.
-ckd	Specifies the operation to be on CKD emulation tiers.
-dp	Specifies the operation to be on disk group-provisioned tiers.
-dsk_grp	Specifies the disk groups be added to the tier. The list can contain either the disk group IDs or the disk group names.
-external	Indicates that the tier will contain externally provisioned VP pools.
-fba	Specifies the operation to be on FBA emulation tiers.
-force	Allows a non-empty tier to be deleted.
-h	Provides brief, online help information.
-i	Specifies the repeat interval for retrying the requested action. This option indicates how often to attempt to get the needed resources to start a new session. The default interval is 30 seconds. The minimum interval is 15 seconds.
-inc_type	Specifies if the storage tier is static or dynamic. A static tier only includes the specified disk groups. A dynamic tier includes all disk groups that match the tier specifications, including any new disk groups which were created after the tier was created.

-name Specifies the name of the tier being created or the new name if the tier is being renamed.

-offline Displays the Symmetrix devices from the Symmetrix configuration database without refreshing the data from the Symmetrix array.

-pool Specifies the names of the thin pools to be included, added, or removed from the VP tier.

-propagate Allows the changes to be propagated to all necessary tiers to prevent partial overlap of disk groups.

-sid Restricts the selection criterion to the specified Symmetrix array ID.

-technology Specifies the drive type. The supported types are EFD, FC, or SATA.

-tier_name Specifies the storage tier name.

-tgt_prot Applies to devices with -raid5 or -raid6 only. Further filters raid devices based on protection types, 3+1, 7+1, 6+2 or 14+2.

-tgt_raid1 Indicates that the target protection type is RAID-1.

-tgt_raid5 Indicates that the target protection type is RAID-5.

-tgt_raid6 Indicates that the target protection type is RAID-6.

-tgt_unprotected
Indicates that the target protection type is unprotected.

-v Provides a more detailed, verbose listing.

-vp Specifies the operation to be on virtual provisioned tiers.

PARAMETERS

Count The number of iterations to execute before exiting.

DiskGroupID The disk group ID.

DiskGroupName
The disk group name.

Interval The interval between polls, in seconds.

NewName The new storage tier name.

PoolName The thin pool name.

SymmID The 12-digit ID of the Symmetrix array.

TierName The storage tier name.

RETURN CODES

Code #	Code Symbol
0	CLI_C_SUCCESS
1	CLI_C_FAIL
2	CLI_C_DB_FILE_IS_LOCKED
19	CLI_C_GK_IS_LOCKED All Gatekeepers to the Symmetrix array are currently locked.

EXAMPLES

To create a static disk group-provisioned storage tier, enter:

```
symtier -sid 207 create -name PrimeTier -tgt_raid1
   -inc_type static -technology EFD -dsk_grp 1
```

To create a dynamic disk group-provisioned storage tier, enter:

```
symtier -sid 207 create -name PrimeDBTier -tgt_raid5
   -tgt_prot 3+1 -inc_type dynamic -technology EFD
```

To create a virtually-provisioned storage tier, enter:

```
symtier -sid 207 create -name VPTier -tgt_raid1
   -technology EFD -vp
```

To delete a storage tier, enter:

```
symtier -sid 207 delete -tier_name PrimeTier
```

To add a disk group to a disk group-provisioned storage tier, enter:

```
symtier -sid 207 add -dsk_grp 2 -tier_name PrimeTier
```

To add a thin pool to a virtually-provisioned storage tier, enter:

```
symtier -sid 207 add -tier_name VPTier -pool
   AddPool
```

To remove a disk group from a disk group-provisioned storage tier, enter:

```
symtier -sid 207 remove -dsk_grp 1
   -tier_name PrimeTier
```

To remove a thin pool from a virtually-provisioned storage tier, enter:

```
symtier -sid 207 remove -pool RemPool
   -tier_name VPTier
```

To rename a storage tier, enter:

```
symtier -sid 207 rename -tier_name PrimeDBTier
   -name PrimeTierR1
```

symtw

Defines time windows for FAST, FAST VP, and Optimizer.

SYNOPSIS

```
symtw -h

symtw -sid <SymmID> -inclusive [-noprompt]
      -type <move_dp | move_vp | perf | all>

   add    -days <DayList>
          -start_time <Time> -end_time <Time>

   remove <-days <DayList>
           -start_time <Time> -end_time <Time>

symtw -sid <SymmID> -exclusive [-noprompt]
      -type <move_dp | move_vp | perf | all>

   add    -start_day <DateTime> -end_day <DateTime>

   remove -start_day <DateTime> -end_day <DateTime>

symtw -sid <SymmID> <<-inclusive> <-exclusive>> [-noprompt]
      -type <move_dp | move_vp | perf | all>

   rmall

symtw [-sid <SymmID>] [-offline]

   list [-type <move_dp | move_vp | perf>]

   list -summary [-date <Date>]

symtw -sid <SymmID> [-noprompt]

   convert [-date <Date>] [-force]
```

DESCRIPTION

The symtw command defines time windows to control FAST, FAST VP, and Optimzier.

Three types of time windows can be added and removed. The list command displays all the defined time windows.

ARGUMENTS

add	Adds a new time window.
convert	Converts the legacy symoptmz time window definitions to the symtw enhanced time window definitions.
list	Lists the time window information for a given Symmetrix array, or for all Symmetrix arrays.
remove	Removes a time window from the Symmetrix array.
rmall	Removes all time windows that match the specified type.

KEYWORDS

all	Indicates all time window types for both

inclusive and exclusive time windows. When combined with the add operation, the specified time window will be added for all three window types. When combined with the remove operation, any time window of any type that matches will be removed.

move_dp — Indicates the disk group provisioning time window type.

move_vp — Indicates the virtual provisioning time window type.

perf — Indicates the performance time window type.

OPTIONS

-date — Indicates the week which includes the start date when generating or displaying the composite time windows. The week always runs from Sunday to the following Saturday.

-days — Indicates the days in a week that the time window applies.

-end_day — Indicates the end date and time of the exclusive time window.

-end_time — Specifies the ending time of day for the time window. Valid values are from 00:00 to 24:00 in 30 minute increments. The ending time has to be within the day boundary. The time 00:00 represents midnight AM and 24:00 represents midnight PM. For the inclusive time window only.

-exclusive — Indicates the time windows that will not allow the operation to be executed. The exclusive time window will supersede all inclusive time windows.

-force — Attempts to force the convert operation to happen even though the symoptmz time windows cannot be translated to the symtw time windows.

-inclusive — Indicates the time windows that will execute operations.

-noprompt — Requests that no prompts are returned after the command is entered. The default is to prompt for confirmation.

-offline — Displays information from the Symmetrix configuration database without refreshing the data from the Symmtrix array.

-sid — Specifies the unique Symmetrix ID.

-start_day — Indicates the start date and time of the exclusive time window.

-start_time — Specifies the starting time of day for the time window. Vaild values are from 00:00 to 24:00 in 30 minute increments. The starting time must be within the day boundary. The time 00:00 represents midnight AM and 24:00

represents midnight PM. For the inclusive time window only.

-summary — Displays a calendar view of all defined time windows.

-type — Indicates the type of the time window. Valid values are move_dp, move_vp, and perf.

PARAMETERS

Date — The date in the format of MMDDYYYY.

DateTime — The date and time in the format of MMDDYYYY:HHMM. The time of day values are in 30 minute increments. The valid values for minutes are 0 and 30.

DayList — Any comma-separated combination of Mon, Tue, Wed, Thu, Fri, Sat, and Sun.

SymmID — The 12-digit ID of the Symmetrix array.

Time — The time of day in the format of HH:MM. The valid values are 00:00 to 24:00 for each day in 30 minute increments. The time 00:00 represents midnight AM and 24:00 represents midnight PM.

RETURN CODES

Code #	Code Symbol
0	CLI_C_SUCCESS
1	CLI_C_FAIL
2	CLI_C_DB_FILE_IS_LOCKED
19	CLI_C_GK_IS_LOCKED All GateKeepers to the Symmetrix array are currently locked.

EXAMPLES

To add a new disk group provisioning time window, enter:

```
symtw add -sid 123 -type move_dp -inclusive
      -days Mon,Tue,Fri
      -start_time 18:00 -end_time 24:00
```

To remove a disk group provisioning time window, enter:

```
symtw remove -sid 123 -type move_dp
      -days Mon,Tue,Fri
      -start_time 08:00 -end_time 12:00
```

symvg

Displays information for logical volume groups (vg) that are defined by the platform's logical volume manager.

SYNOPSIS

```
symvg [-h] [-type <VgType>]

   list [-v]

   show <VgName>

   deport <VgName> [-newvg <NewVgName>] [-host <hostid>]
     [-mapfile <Filename>] [-overwrite]

   import <VgName> [-newvg <NewVgName>]
     [-mapfile <Filename>] [-cluster] [-clear] [-persistent]

   rescan

   create  <VgName> [-p PartitionSize] <PdevName...>

   destroy <VgName>

   adddev  <VgName> <PdevName...>

   rmdev   <VgName> <PdevName...>

   recover <VgName>

symvg [-h] [-v] [-type <VgType>] [-force] [-sid <SymmID>]
     [-rdfg <GrpNum>] [-R1 | -R2] [-bcv | -nobcv | -vdev]

   vg2dg <VgName> <DgName>
     [-dgtype [REGULAR | RDF1 | RDF2 | ANY]]

   vg2cg <VgName> <CgName>
     [-cgtype [REGULAR | RDF1 | RDF2 | ANY]]
     [-apidb | -rdf_consistency]
```

DESCRIPTION

The symvg command displays information and provides for the provisioning of logical volume groups that are defined on the host system.

Additionally, the user can convert the devices of a specified volume group to a device group or composite group.

In the list of physical device members for a volume group, CLARiiON devices are distinguished from other device types by a (C) indicator.

Import, deport, rescan and the provisioning operations such as create, destroy, adddev, rmdev, and recover are supported only on specific logical volume managers.

Volume group name length restriction:
The volume group name field is limited to 63 characters. This length restriction is inclusive of the name and absolute path of the volume group even if the user only specified the volume group name itself. The behavior is undefined if the

length is exceeded.

When mapping objects in the ASM volume manager, three environment variables are required in order to contact the ASM instance:

Environment Variable	Description
SYMAPI_ASM_HOME	Oracle Home of ASM instance
SYMAPI_ASM_SID	Oracle Sid of ASM instance
SYMCLI_ASM_CONNECT	username/password of ASM instance

ARGUMENTS

adddev — Extends a volume group by adding the specified devices to the volume group.

create — Creates a volume group using the specified devices.

deport — Deports a specified volume group so that it can be imported later.

destroy — Destroys a volume group.

import — Imports a specified volume group into the system.

list — Lists all of the volume groups that were defined for this host. The behavior is undefined if the volume group name is more than 63 characters.

recover — Recovers a failed volume group. This operation is currently supported for SUN_VXVM, HP_VXVM, AIX_VXVM, WIN_LDM, WIN_VXVM, LINUX_VXVM, EMC_PVM and OSF1_LSM.

rescan — Rescans all of the volume groups. This operation is currently supported only for the Logical Disk Manager (LDM) volume groups and Veritas VXVM volume groups on Windows platforms, Linux LVM volume groups on Linux platforms, and HP LVM volume groups on HP-UX platforms.

rmdev — Removes the specified devices from the volume group.

show — Shows information about a volume group.

vg2cg — Converts the specified volume group into a composite group.

vg2dg — Converts the specified volume group into a device group.

OPTIONS

-apidb — Creates the composite groups in the SYMAPI database only.

-bcv — Associates only BCV devices to the target group.

-cgtype — Specifies a composite group type.

-clear Imports a volume group and clears the host ID on the volume group. This flag is only for Veritas volume managers.

-cluster Imports a Windows VxVM volume group as a cluster.

-dgtype Specifies a device group type.

-force Attempts to force the operation even though one or more devices in the volume group may already be part of another device group or composite group.

-h Provides brief, online help information.

-host Specifies the host ID of the host on which the deported volume group can be imported.

-mapfile Specifies the filename where volume group information is stored when an import or deport is performed.

-newvg Specifies a new volume group name for the volume group.

-nobcv Adds standard devices only to the target group. The default behavior is to add both standard and BCV devices.

-overwrite Used in conjunction with the -mapfile option to overwrite an existing mapfile if set.

-p Specifies the partition size in megabytes. This option is only valid for AIX_LVM type LVM.

-persistent Imports the volume group by configuring physical devices in persistent DSF format. This option is only valid for HP LVM on HPUX 11.31.

-R1 Adds R1 devices to the target device group.

-R2 Adds R2 devices to the target device group.

-rdf_consistency
Creates a composite group and enables it for SRDF consistency protection after devices are added to it.

-rdfg Selects SRDF devices that belong to a specified Symmetrix SRDF (RA) group number.

-sid Specifies a unique Symmetrix ID.

-type Specifies the volume group type.

-v Provides a more detailed, verbose listing.

-vdev Adds VDEVs to the target group.

PARAMETERS

CgName The composite group name.

DgName	The device group name.
Filename	The file name where the volume group information is stored when an import or deport operation is performed.
GrpNum	The SRDF (RA) group number.
hostid	The host identification number.
NewVgName	The new logical volume group name.
PartitionSize	The partition size for a device in MBs.
PdevName	A fully-qualified device path of a character device.
SymmID	The 12-digit ID of the Symmetrix array.
VgName	The logical volume group name.
VgType	The volume group type. Values are:

Group Type	
DEFAULT	LINUX_LVM
AIX_LVM	LINUX_POOL
AIX_VXVM	LINUX_VXVM
AS400_LVM	ORACLE_ASM
DYNIX_SVM	SUN_SOLSTICE
EMC_PVM	SUN_VXVM
HP_LVM	WIN_LDM
HP_VXVM	WIN_VXVM

RETURN CODES

Code #	Code Symbol
0	CLI_C_SUCCESS
1	CLI_C_FAIL

EXAMPLES

To list all the default logical volume groups in a list format, enter:

```
symvg list
```

To list all the SunOS Veritas VxVM logical volume groups in a list format, enter:

```
symvg list -type SUN_VXVM
```

To create a composite group named newcg with the R1 and R1-BCV devices from the volume group named thisvg, enter:

```
symvg vg2cg thisvg newcg -cgtype RDF1 -R1
```

To create a REGULAR device group named newdg with only the R1-BCV devices from the volume group named thisvg, enter:

```
symvg vg2dg thisvg -R1 -bcv newdg -dgtype REGULAR
```

To create an ANY device group named newdg from the volume group named thisvg enter:

```
symvg vg2dg thisvg newdg -dgtype ANY
```

To deport a volume group named testvg out of the system, enter:

```
symvg deport testvg
```

To deport a volume group named testvg so that it can be imported on a host named foo, enter:

```
symvg deport testvg -host foo
```

Note: The previous option is available only with VxVM on HP and SUNOS platforms.

To import a volume group named testvg into the system, enter:

```
symvg import testvg
```

To import on an AIX LVM volume group named aixtestvg, enter:

```
symvg import aixtestvg  -mapfile hdisk22
```

Note: In the previous example, the -mapfile option specified the device name hdisk22, which existed as part of the volume group aixtestvg when the volume group was deported.

To create a volume group named testvg on the host system, enter:

```
symvg create testvg /dev/rdsk/c0t0d0s2
```

To create a volume group named aixtestvg of type AIX LVM with a partition size of 16MB, enter:

```
symvg create aixtestvg -p 16 /dev/rhdisk40
```

To add a device to the volume group named testvg, enter:

```
symvg adddev testvg /dev/rdsk/c0t0d1s2
```

To remove two devices from the volume group named testvg, enter:

```
symvg rmdev testvg /dev/rdsk/c0t0d1s2 /dev/rdsk/c0t5d5s2
```

To remove the volume group named testvg from the system, enter:

```
symvg destroy testvg
```

Volume group options for import/deport operations:

Type	action	VgName	-newvg	-mapfile	-cluster	-persistent
AIX_LVM	I	M	O	M	NA	NA
AIX_LVM	D	M	NA	NA	NA	NA
EMC_PVM	I	O	O	M	NA	NA
EMC_PVM	I	M	O	O	NA	NA
EMC_PVM	D	M	NA	O	NA	NA

HP_LVM	D	M	NA	M	NA	NA
HP_LVM	I	M	NA	M	NA	O
HP_VXVM	I/D	M	O	NA	NA	NA
NT_LDM	I	M	O	NA	O	NA
NT_LDM	D	M	NA	NA	NA	NA
OSF1_LSM	I/D	M	O	NA	NA	NA
SUN_VXVM	I/D	M	O	NA	NA	NA
AIX_VXVM	I/D	M	O	NA	NA	NA
LINUX_VXVM	I/D	M	O	NA	NA	NA
LINUX_LVM	I/D	M	NA	M	NA	NA
SUN_SOLSTICE	I/D	M	NA	NA	NA	NA

Legend for the abbreviations used in the table:

D - Deport
I - Import
M - Mandatory
NA - Not Applicable
O - Optional

Notes:

1. For an import operation for AIX LVM, the -mapfile option is used to specify a device name that existed as part of the volume group.

2. The output of the symvg show command can report incorrect device status. Whenever the device status is reported incorrectly, it is recommended that the user run symcfg sync and then run symvg show to report the correct device status.

3. On Windows platforms, import and deport operations are supported for Veritas VxVM 2.7 and higher. Provisioning operations such as create, destroy, adddev, rmdev are supported for Veritas VxVM 3.0 and higher.

4. With Veritas volume managers on all host operating systems, the deport operation on volume groups named rootdg is not allowed.

5. The recover action is not supported for the following volume managers:
   - AIX LVM on the AIX platform.
   - HP LVM on the HP-UX platform.
   - Native LVM on the Linux platform.
   These are the default LVMs for their respective platforms.

6. On the HP-UX platform with HP-UX LVM, with a volume group containing an EMC Symmetrix disk, you can only add Symmetrix disks with the same attributes to the volume group. For example, if a volume group contains an EMC Symmetrix disk of a 2-way-mir type, only Symmetrix disks of a 2-way-mir type can be added to this volume group.

7. On the Solaris platform with SUN_SOLSTICE volume manager, a volume group named solstice cannot be created.

Options file

The Options file contains parameters that can be set to critically change the default behavior of SYMCLI operations, SYMAPI calls, and their control actions. It can be used to set certain global restrictions as well as customize and streamline command line coding to your specific environment.

Note

These parameters are intended for experienced SYMCLI or SYMAPI users and are not a prerequisite for normal use. Modifying these parameters can cause unwanted restriction of features or adversely effect the operation of the array. This file should be protected so that only authorized users can make changes.
The options file is located in the SYMAPI configuration directory.

Directors	System
/var/symapi/config	Unix
C:\Program files\EMC\Symapi\config	Windows
your_specific_installation_directory	OpenVMS, AS/400, MVS

The following table provides the description and default values for each option.

Option Name	Description	=<optionalvalue | defaultvalue>
SYMAPI_ACC_ADMIN_VIA _SERVER	Enables/disables the client SYMAPI/SYMCLI access control commands for prepare, release, and commit actions to execute at the SYMAPI server.	DISABLE | ENABLE
SYMAPI_ACC_DISPLAY_V IA_SERVER	Enables/disables the client SYMAPI/SYMCLI access control display commands for list and show actions to execute at the SYMAPI server.	DISABLE | ENABLE
SYMAPI_ALLOW_CELERRA _DEV_CTRL	Allows controls on Celerra devices.	ENABLE | DISABLE
SYMAPI_ALLOW_CG_ENAB LE_FROM_R2	When set to DISABLE this option will not allow enabling a CG from R2 side.	ENABLE | DISABLE
SYMAPI_ALLOW_R2_GT_R 1_ODD_CYL	When set to DISABLE this option will not allow an R2 larger than R1 configuration if the R1 is an odd cylinder size and is on an array running Enginuity 5876 and the R2 is on an array running HYPERMAXOS. If SYMAPI_RDF_CREATEPAI R_LARGER_R2 is set to DISABLE, this option file setting is ignored.	ENABLE | DISABLE
SYMAPI_ALLOW_RDF_SYM FORCE	Indicates whether users can specify -symforce when performing RDF control operations.	TRUE | FALSE
SYMAPI_ALLOW_SCRIPTS _VIA_SERVER	Specifies whether to allow pre-action and post-action scripts for TimeFinder commands to be run by the SYMAPI server.	ENABLE | DISABLE
SYMAPI_ALTERNATE_ACC ESS_ID	This option applies to Symmetrix Access Control. It controls whether a host's Access ID is generated based on the	ENABLE | DISABLE

-	hardware and operating environment of the host or based on a random number or a user-chosen passphrase DISABLE: The host access ID is generated based on the hardware and operating environment of the host. This mode cannot be used on Intel processors or hosts whose networking signatures may be non-unique. ENABLE: The host access ID is generated based on a random number or on a user-chosen passphrase. This mode must be used especially on Intel processors, but may be used on all platforms.	-
SYMAPI_APPREG_AUTO_EXPIRATION	Specifies whether to expire entries in the application registration table, based on their age.	DISABLE | ENABLE
SYMAPI_APPREG_EXPIRATION_PERIOD	Sets the number of days after which an entry can be expired.	15 - 365 | 90
SYMAPI_BCV_ESTAB_INC_TO_FULL	Controls whether an incremental BCV establish will automatically be converted to a full establish if needed.	ENABLE | DISABLE
SYMAPI_BCV_SINGULAR_INTERVAL	When SYMAPI_DEFAULT_BCV_ESTABLISH_TYPE is set to SINGULAR, the SYMAPI inserts a pause between the control of each pair. The pause can be set to any value between 0 and 30 seconds.	0 to 30 | 0
SYMAPI_CG_TIMEOUT	Sets the time-out period during which the host will attempt to trip a failed consistency group.	10 - 120 | 30
SYMAPI_CG_TIMEOUT_ACTION	Specifies which action to take for an SRDF consistency group time out.	FAIL | RETRY

SYMAPI_CLIENT_RETRIEVE_LOG	This option only applies to CLI commands run in Client/Server mode. If set to ENABLE, Solutions Enabler log file messages written at the Server while the CLI executed will be copied back to the Client's log file when the command completes. If more than 100 log messages were written, only the last 100 will be retrieved. A number of simple formatting changes are made to the message fields (e.g.: Date/Time, Process ID) before they are written to the Client's log file.	ENABLE | DISABLE
SYMAPI_CLIENT_SIDE_ACCESS_ID	This option applies to Symmetrix Access Control. It specifies whether the client access ID is sent to the server during client/server operations. ENABLE: If set to ENABLE, the client sends its access id to the server. If the server is using SYMAPI_USE_ACCESS_ID =CLIENT or ANY, the server applies the client's access id to all management operations. DISABLE: If set to DISABLE, the client does not send its access ID to the server. If the server expects an access ID from the client, operations will fail. See also the option SYMAPI_USE_ACCESS_ID .	ENABLE | DISABLE
SYMAPI_CLONE_COPY_ON_WRITE	Sets clone nocopy mode to copy-on-write (ENABLE) or copy-on-access (DISABLE).	ENABLE | DISABLE
SYMAPI_CLONE_LARGER_TARGET	When set to DISABLE this option will not allow snapshots to a target device that is larger than the source device.	ENABLE | DISABLE
SYMAPI_COLLAPSE_STRIPED_META_EXTENTS	Specifies whether mapping commands can be entered without having to type the	DISABLE | ENABLE

-	-collapse option in every command.	-
SYMAPI_COMMAND_SCOPE	Sets the scope of the device selection process. ENABLED limits the operation to the devices within the scope of the command selection type. DISABLED performs the operation on the devices within the scope of the command selection type plus any additional devices associated by session and/or state.	ENABLE | DISABLE
SYMAPI_CTRL_VIA_SERV ER	Blocks the client SYMAPI/SYMCLI control commands from executing at the SYMAPI server.	DISABLE | ENABLE
SYMAPI_DATED_LOGFILE _NAME	Enables/disables the creation of dated SYMAPI log files.	DISABLE | ENABLE
SYMAPI_DB_FILE_COMPR ESSION	Minimizes the overall database file size by compressing the file.	DISABLE | ENABLE
SYMAPI_DB_FSYNC_MODE	When writing the database, force its contents out to disk. Setting this option to false may provide a slight performance improvement, at the cost of a small possibility of the database file becoming corrupt following a host crash.	DISABLE | ENABLE
SYMAPI_DEFAULT_BCV_E STABLISH_TYPE	Specifies how the DA director processes the establish action on BCV and standard devices.	SINGULAR | SERIAL | PARALLEL
SYMAPI_DEFAULT_BCV_R ESTORE_TYPE	Sets the default behavior for a BCV restore operation.	SINGULAR | SERIAL | PARALLEL
SYMAPI_DEFAULT_BCV_S PLIT_TYPE	Sets the default behavior for a BCV split operation.	INSTANT | REGULAR

SYMAPI_DEFAULT_RDF_M ODE	Specifies the default rdf_mode for createpair operations.	SYNC | ACP
SYMAPI_DEFAULT_SNAP_ TERM_TYPE	Sets the way a DA director processes the terminate action on Snap devices.	PARALLEL | SERIAL
SYMAPI_ENABLE_DEVICE _RESERVATIONS	Specifies whether to enable device reservations.	FALSE | TRUE
SYMAPI_ENFORCE_DEVIC E_RESERVATIONS	Specifies whether to enforce a device reservation.	FALSE | TRUE
SYMAPI_ENHANCED_USER _AUTHENTICATE	Enables enhanced (KERBEROS) user authentication.	ENABLE | DISABLE
SYMAPI_FIPS	This option specifies the status of FIPS. If enabled SE will use cryptographic algorithms that meet the FIPS requirements. If disabled the standard SE cryptographic algorithms will be used. If the SYMAPI_SECURITY_LEVE L option is set but not set to SECURE this option is ignored.	ENABLE | DISABLE
SYMAPI_GNS_CS_STALE_ DATA_TIMEOUT	Sets the timeout period, in seconds, for the client's in-memory GNS group information before the server is polled for updates.	1 - 15 | 1
SYMAPI_GNS_MIRRORED_ GROUP_CONTROL	Allows group modifications to device groups that were created by a remote GNS daemon via remote SRDF mirroring.	ENABLE | DISABLE
SYMAPI_IO_DRAIN_TIME OUT	Controls the amount of time, in seconds, that PowerPath will wait for incoming I/Os to finish before responding that the I/Os are suspended.	5 - 120 | 60

SYMAPI_IO_THAW_INTER VAL	Controls the amount of time, in seconds, that PowerPath will suspend I/Os before automatically restarting them.	5 - 120
SYMAPI_LOGFILE_DATE_ FORMAT	Changes the date format in the log entries.	FORMAT2 | FORMAT1
SYMAPI_LOGFILE_FORMA T	Controls the presence of optional fields within log file records: pid include Process tid include Thread userid include User ID activityid include Activity ID	userid | activityid | pid | tid
SYMAPI_LOGFILE_RETEN TION	Sets the maximum number of days to retain a log file, after which the log file is deleted.	0, 6 - 1825
SYMAPI_OPENSSL_CIPHE R_SUITE	Specifies the cipher string used by storsrvd to determine which TLS ciphers will be allowed during TLS handshake negotiation. The cipher string must be valid, and can be tested with the following command: openssl ciphers -v <CIPHER_STRING> Any ciphers not supported in TLSv1.2 or later will be ignored.	Valid OpenSSL cipher string - TLSv1.2 and later ONLY
SYMAPI_OPENSSL_ELLIP TIC_CURVE_LIST	Specifies the Elliptic Curve string used by storsrvd to determine which Elliptic Curves will be available for ECDH/ECDHE key exchanges. This must specify one or more valid Elliptic Curves supported by OpenSSL with TLSv1.2 ciphers.	Valid OpenSSL Elliptic Curve string
SYMAPI_POOL_ALLOW_MI X_TYPE	Allows devices with different disk characteristics in a device pool (Snap/Thin).	ENABLE | DISABLE
SYMAPI_POOL_DRAIN_TH RESHOLD	This option can be set to change the Pool used space threshold that is validated when disabling	0 - 100

-	or draining a save or data device. If the used space of a pool will exceed this threshold after the drain or disable request is satisfied, then the request will be blocked. The allowable values for this setting are from 0 to 100 percent. The default setting is 90.	-
SYMAPI_RCOPY_GET_MOD IFIED_TRACKS	Enables/disables the calculation of modified tracks.	TRUE | FALSE
SYMAPI_RCOPY_SESSION _LIMIT	This option requires Enginuity 5771 or higher. Specifies how many sessions are allowed at any one time on this array.	0 - 1024 | 512
SYMAPI_RDF_CHECK_R2_ NOT_WRITABLE	When enabled, verify that the R2 devices are not writable by the host for createpair operations other than createpair -invalidate <R1|R2>.	ENABLE | DISABLE
SYMAPI_RDF_CREATEPAI R_LARGER_R2		ENABLE | DISABLE
SYMAPI_RDF_RW_DISABL E_R2	Causes the R2 device to be set to read/write disabled, or not-ready, on the RA during establish, restore, failback, or createpair-establish operations. Not applicable if an SRDF/Metro configuration	ENABLE | DISABLE
SYMAPI_SECURITY_ALT_ CERT_FILE	Any valid simple file name.	Any valid simple file name.
SYMAPI_SECURITY_ALT_ KEY_FILE	Any valid simple file name.	Any valid simple file name.
SYMAPI_SECURITY_LEVE L	Specifies the session security level. Should be SECURE on all platforms where Solutions Enabler supports SSL; NONSECURE otherwise. Possible values	SECURE | ANY | NONSECURE

-	are: SECURE: Accept secure sessions only. NONSECURE: Accept non-secure sessions only (client and server do not initialize secure socket library). ANY: Accept both types. Refer to the Solutions Enabler release notes.	-
SYMAPI_SERVER_DEBUG_ SCOPE	This option applies to Solutions Enabler debugging. It expresses the client's desire regarding where debug data will be collected. Even though the client may indicate server side debugging, the server configuration may prevent such collection. CLIENT: debug settings only apply on the client side SERVER: debug settings only apply on the server side BOTH: debug settings apply on both client and server	CLIENT | SERVER | BOTH
SYMAPI_SNAPVX_LARGER _TARGET	When set to DISABLE this option will not allow linking or relinking snapvx snapshots to a target device that is larger than the snapshot.	ENABLE | DISABLE
SYMAPI_SNAP_CONTROL_ INTERVAL	When operating in SERIAL mode, spaces the snap or clone create and terminate operations by inserting a pause between the control of each pair. The pause can be set to any value between 0 and 30 seconds.	0 - 30 | 0
SYMAPI_SNAP_COUNT_MO DIFIED_TRACKS	Sets snap to return a count of the changed tracks. This option is ignored when the request is made remotely. This option is ignored for Enginuity 5977 and above.	TRUE | FALSE
SYMAPI_SNAP_PERSISTE NT_RESTORE	Specifies whether snap restores are performed as persistent restores.	DISABLE | ENABLE

SYMAPI_SYMDEVNAME_WI DTH	This option can be set to specify the default string width for Symmetrix device numbers returned in API structures and displayed in CLI screens. Device numbers will be left-padded with zeros (0) as needed to fill the specified width. Device numbers in CLI displays will also be left-padded with zeros (0) as needed, and also left-justified and right-padded with spaces ( ) as needed to maintaint consistent column alignment, regardless of the specified width. Note: Valid option settings are 3, 4 or 5 (default). Values less than 3 will be normalized to 3 and values greater than 5 will be normalized to 5.	3 - 5
SYMAPI_TF_CHECK_ONLI NE_CKD	Enables CKD device online check.	ENABLE | DISABLE
SYMAPI_TF_COUNT_MODI FIED_TRACKS	Sets TimeFinder to return a count of the changed tracks. This option is ignored when the request is made remotely.	TRUE | FALSE
SYMAPI_TF_MULTI_ESTA B_REST	Controls whether TimeFinder uses the multi-instant establish and restore feature.	DISABLE | ENABLE
SYMAPI_TF_NOT_READY_ BCVS	Controls whether a user Not-Ready occurs on the BCV devices before a restore or an establish command.	ENABLE | DISABLE
SYMAPI_TF_RDF_SUSPEN D	Causes TimeFinder to suspend the RDF link of an R1-BCV prior to an establish or restore command. Normally, this is done by Enginuity.	DISABLE | ENABLE
SYMAPI_UNIQUE_ACL_PA RTITION_IDS	For AIX and HP-UX hosts. Determines how partition IDs are generated. ENABLE	DISABLE | ENABLE

-	generates unique IDs for each partition running on the machine. DISABLE uses a single unique ID for all partitions running on the machine.	-
SYMAPI_USE_ACCESS_ID	This option applies to Symmetrix Access Control. It specifies whether to use the access ID generated on client or server. Used only on the server side during client/server operations. Possible values are: CLIENT: The client access ID is used for every command performed. If a client access ID is not available the command will fail. See also SYMAPI_CLIENT_SIDE_A CCESS_ID. SERVER: The server access ID is used for every command performed. ANY: If the client access ID is available it is used for every command performed. If it is not available then the server access ID is used.	CLIENT | SERVER | ANY
SYMAPI_USE_GNS	Stores and maintains, in a common repository, SYMAPI device group (DG) and composite group (CG) definitions across Symmetrix arrays that are visible to all locally-attached hosts.	ENABLE | DISABLE
SYMAPI_USE_RDFD	Allows the creation of RDF_CONSISTENCY composite groups, to be managed by the RDF daemon.	ENABLE | DISABLE
SYMAPI_WAIT_FOR_BCV_ BG_SPLIT	Sets the default behavior for the BCV split operation to wait for the background split to be complete before returning your call.	TRUE | FALSE
SYMAPI_WAIT_FOR_BCV_ SYNCH	Sets the default behavior for the BCV establish operation to wait for the	TRUE | FALSE

-	establish operation to complete before returning your call.	-
SYMAPI_WAIT_ON_LOCKE D_GK	Specifies whether to wait when a locked gatekeeper device is encountered.	ENABLE | DISABLE

Daemon Options file

The Daemon Options file contains parameters that can be set to control the behavior of the various Solutions Enabler daemons. Each daemon reads this file as it starts and applies any settings within it that apply.

Note

These parameters are intended for experienced SYMCLI or SYMAPI users. In most cases, the default values used by the daemons will be sufficient. This file should be protected so that only authorized users can make changes.

Lines in this file can have one of the following formats:

NAME = VALUE	Set the parameter NAME for all daemons that understand this parameter.
stororad:NAME = VALUE	Set the parameter NAME for only the stororad daemon.
stororad*:NAME = VALUE	Set the parameter NAME for all daemons whose name begins with storora. The asterisk (*) is the wildcard.

Option values that are very long can extend to the next line by placing a backslash at the very end of the line. For example:

stororad:NAME = VALUE1 : VALUE2 : VALUE3 : \

VALUE4 : VALUE5

The daemon options file is located in the SYMAPI configuration directory.

Directors	System
/var/symapi/config	Unix
C:\Program files\EMC\Symapi\config	Windows
your_specific_installation_directory	OpenVMS, AS/400, MVS

The following tables provides the description, allowed values, and default values for each option.

Common daemon parameters

These common parameters apply to the core Solutions Enabler daemons:

storapid [Base Daemon]
storgnsd [GNS Daemon]
storrdfd [RDF Daemon]
storevntd [Event Daemon]
storstpd [STP Daemon]
storwatchd [Watchdog Daemon, UNIX only]
storsrvd [SYMAPI Server Daemon]
storvwmd [Witness Manager Daemon]
storvwlsd [Witness Lock Service Daemon]

Parameter	Allowed Values	Default Value	Description
AUTORESTART	enable | disable	enable	Specifies whether a watchdog mechanism (UNIX: storwatchd daemon) should be used to automatically restart a daemon if it crashes.
LOG_LEVEL	error | info | debug | verbose | warning	warning	Specifies a severity-based control over logging volume. Messages that are issued with a severity equal to or exceeding the level specified will be recorded in the log file. Do not use debug or verbose without direction from EMC Customer Support.
LOGFILE_TYPE	dated | wrap	wrap	Specifies which style of log file the daemon will use. Two styles are supported: wrap: Two log files are maintained: stor????.log0 and and stor????.log1. Logging alternates between these--switching to the other file each time the maximum size specified by the LOGFILE_SIZE parameter is reached. The modified time on the files can be used to determine the current one. dated: A separate log file is used for

-	-	-	each day: stor????-YYYYMMDD.log. (e.g.: storapid-20041130.log). There are no size limits on these files. Log files are written to the following directory: UNIX: /var/symapi/log Windows: c:\\Program Files\\EMC\\SYMAPI\\log The dated LOGFILE_TYPE is recommended for storstpd.
LOGFILE_SIZE	A number, in KB units	1000 (1000-KB)	For LOGFILE_TYPE=wrap style log files, how large each log file is allowed to grow before wrapping to the alternate file.
LOGFILE_RETENTION	A number of days, greater than zero.	3 (days)	For LOGFILE_TYPE=dated style log files, log files more than this many days old will be automatically deleted.
LOGFILE_PERMS	r | n | rw	rw	By default, daemon log files allow anyone to read or write them (UNIX mode of rwxrwxrwx). This option can be set to override that behavior, as follows: rw: Anyone can read or write (default). r: The owner (root) can read/write, others can read. n: The owner can read/write, no one else can access.
SECURE_DIRECTORY_PATH	One or more semicolon (On non-Window platforms a	None	Controls which directories the daemon is permitted to read from, write arbitrary files to, or execute a script from as directed to by SE applications. For example, this applies to backup/restore and script files that storsrvd can be asked to read, write or

-	-	-	execute. On Windows, this is a list of directories separated by a semicolon, ';'. Also, either forward or back slashes ('/','\') can be used when specifying the directory name. For example: storsrvd:secure_directory_path = c:\Temp\dir1;c:/Users/SE On non-Windows platforms, this is a list of directories separated by a semicolon, ';', or colon, ':'. Also, only a forward slash ('/') can be used when specifying the directory name. For example: storsrvd:secure_directory_path = /tmp/dir1;/opt/dir2;/users/se If this parameter is not set any directory can be used.
IBMI_JOBQ_NAME	Any name of 10 characters long.	*JOBD	Specifies the job queue in which this job is placed. *JOBD: The submitted job is placed on the job queue named in the specified job description. name : Specify the name of the job queue
IBMI_JOBQ_LIB_NAME	Any name of 10 characters long.	*LIBL	Specifies the library in which the job queue is located. *LIBL : All libraries in the library list for the current thread are searched until the first match is found. *CURLIB: The current library for the thread is used to locate the job, queue. If no library is specified as the current library for the thread, the QGPL library is used., name : Specify the name of the library

-	-	-	where the job queue is located.
IBMI_ALLOW_SUBSYSTEM_START	yes | no	yes	Specifies whether or not it is allowed to automatically start the subsystem specified in the jobq that defined by the daemon options IBMI_JOBQ_NAME and IBMI_JOBQ-LIB_NAME. When set to yes the subsystem defined in jobq above will be started. When the jobq name is set to *libl the subsystem and jobq will need to be located in a library that is in the library list or otherwise the subsystem cannot be started.

storsrvd [SYMAPI server daemon] parameters

This daemon handles remote SYMAPI client connections over TCP/IP to storage arrays. Storsrvd provides the same services as the former symapisrv command.
Refer to the storsrvd(3) man page for additional details.

Parameter	Allowed Values	Default Value	Description
LOG_FILTER	SERVER | CONTROLS | SESSION | APIREQ	None	Specifies category-based control over logging volume, by selecting which types of storsrvd numbered log messages are recorded in the storsrvd log files. Each log message issued by storsrvd is associated with a category, expressed by the filter names shown below. The numbered messages will appear in the storsrvd log files and on a terminal if one is available. Numbered messages issued by storsrvd use a message identifier that precedes the message text: ANR0001I SYMAPI server ready to accept SECURE connections. The value to be specified for LOG_FILTER consists of a series of keywords separated by a commas (,) or plus signs {+). SERVER: Records high level events of the server: initialization, termination, and other events not directly related to a SYMAPI session. SESSION: Records arrival and termination of SYMAPI sessions. APIREQ: Records arrival and completion of SYMAPI function requests. CONTROLS: Records

-	-	-	activity during execution of management requests from stordaemon or the system console in the z/OS environment. If LOG_FILTER specifies either SESSION or APIREQ it is most useful to set LOG_LEVEL to info.
LOG_SHOW_MSGID	disable | enable	enable	Displays message identifiers in the log messages issued by storsrvd.
LOG_SHOW_CATEGORY	enable | disable	disable	Displays the category (see LOG_FILTER above) in the output log messages, preceding the message identifier.
PORT	1 - 65534 | 2707	2707	Specifies the TCP/IP port on which storsrvd will listen for connections from remote SYMAPI clients. Generally, numbers less than 1000 are reserved for well known daemon servers such as telnet, ftp, and http. If this parameter is changed, the storsrvd must be restarted for the new value to take effect.
SECURITY_ALT_CERT_FILE	Any valid simple file name.	symapisrv_cert.pem	Specifies the name of an alternate certificate file to be used by the server instead of the certificate generated at installation, and will be paired with the file specified in the SECURITY_ALT_KEY_FILE option. If this option is used, the SECURITY_ALT_KEY_FILE option must also be used to specify an alternate key file. The certificate file

-	-	-	must be located in the /var/symapi/config/cert directory. Do not specify the directory in the value. If this parameter is changed, the storsrvd must be restarted for the new value to take effect.
SECURITY_ALT_KEY_FILE	Any valid simple file name.	symapisrv_key.pem	Alternate key file. Specifies the name of an alternate private key file to be used by the server instead of the key generated at installation, and will be paired with the file specified in the SECURITY_ALT_CERT_FILE option. If this option is used, the SECURITY_ALT_CERT_FILE option must also be used to specify an alternate certificate file. The key file must be located in the /var/symapi/config/cert directory. Do not specify the directory in the value. If this parameter is changed, the storsrvd must be restarted for the new value to take effect.
SECURITY_CLT_SECURE_LVL	MUSTVERIFY | NOVERIFY | VERIFY	VERIFY	Client certificate verification level. (This feature is not available on z/OS). This option controls the verification of the client certificate by the server. The following values may be specified: MUSTVERIFY: Indicates that the server will only accept communications from a version of the client that can send a certificate to be verified. NOVERIFY: Indicates that the server will not

-	-	-	verify the client certificate. VERIFY: Indicates that the server will verify a client certificate if the version of the client can send a certificate. THIS OPTION IS IGNORED IF SECURE COMMUNICATIONS ARE NOT ESTABLISHED.
MAX_SESSIONS	0 - 100	100	Maximum SYMAPI sessions to the server This option will take precedence over the options file option SYMAPI_MAX_CLIENTS. This specifies the global high-water mark of all sessions allowed by the server, without respect to source host or user. When the next new session will cause the current number of sessions to exceed this number, it will be refused. This parameter can be changed and reloaded while the server is running with the CLI stordaemon action storsrvd -cmd reload.
MAX_SESSIONS_PER_HOST	A positive number | NOLIMIT	NOLIMIT	Maximum SYMAPI sessions per host to the server This option specifies the high-water mark for concurrent sessions from any specific host. When the next new session from the source host will cause the current number of sessions from that host to exceed this number, the session will be refused. When setting a numeric value, it should be <= the MAX_SESSIONS value. If it is set greater than MAX_SESSIONS, then the maximum

-	-	-	number of sessions for any specific host will be restricted to MAX_SESSIONS. If the value is set to NOLIMIT, then host sessions are only restricted by the MAX_SESSIONS value. This parameter can be changed and reloaded while the server is running with the CLI stordaemon action storsrvd -cmd reload.
MAX_SESSIONS_PER_USER	A positive number | NOLIMIT	NOLIMIT	Maximum SYMAPI sessions per user to the server This option specifies the high-water mark for concurrent sessions from any specific user. When the next new session from the source user will cause the current number of sessions from that user to exceed this number, the session will be refused. When setting a numeric value, it should be <= the MAX_SESSIONS value. If it is set greater than MAX_SESSIONS, then the maximum number of sessions for any specific user will be restricted to MAX_SESSIONS. If the value is set to NOLIMIT, then user sessions are only restricted by the MAX_SESSIONS value. This parameter can be changed and reloaded while the server is running with the CLI stordaemon action storsrvd -cmd reload.
PERMIT_SYMAPI_DEBUG	none | client | server	server	clients, server, or not at all. When the value is 'none', debug logging is suspended, regardless of the settings of commonly used debug

-	-	-	flags configuration. When the value is 'server', the server's debugging configuration will be in effect, but settings from clients are ignored. When the value is 'client', settings sent to the server from the client are respected.
SECURITY_CERT_ALLOW_WILDCARDS	ENABLE | DISABLE	DISABLE	Specifies if wildcards are allowed in the Common Name of a client certificate. The following values may be specified: ENABLE: Indicates that the server will accept wildcards in the Common Name of a client certificate. DISABLE: Indicates that the server will reject any host with wildcards in the Common Name of a client certificate.
OPENSSL_CIPHER_SUITE	Valid OpenSSL cipher string - TLSv1.2 and later ONLY	AESGCM:-aNULL:-kRSA:-kDH:@STRENGTH	Specifies the cipher string used by storsrvd to determine which TLS ciphers will be allowed during TLS handshake negotiation. The cipher string must be valid, and can be tested with the following command: openssl ciphers -v <CIPHER_STRING> Any ciphers not supported in TLSv1.2 or later will be ignored.
SECURITY_DH_PARAM_FILE	Any valid simple file name	None	Specifies the name of an alternate Diffie-Hellman parameters file to be used by the server instead of the default internal parameters. The parameter file must be located in the /var/symapi/config/cert directory. Do not

-	-	-	specify the directory in the value. If this parameter is changed, the storsrvd must be restarted for the new value to take effect.
OPENSSL_ELLIPTIC_CURVE_LIST	Valid OpenSSL Elliptic Curve string	P-521:P-384:P-256	Specifies the Elliptic Curve string used by storsrvd to determine which Elliptic Curves will be available for ECDH/ECDHE key exchanges. This must specify one or more valid Elliptic Curves supported by OpenSSL with TLSv1.2 ciphers.

storapid [Base Daemon] parameters

This daemon facilitates I/O accesses to Symmetrix storage arrays.
Refer to the storapid(3) man page for additional details.

Parameter	Allowed Values	Default Value	Description
USE_ALL_GKS	disable | enable	enable	Specifies whether or not the storapid is free to use all available gatekeepers, if the syscall volume is high enough. When disabled, the storapid restricts itself to using no more than 75% of the available gatekeepers. If this parameter is changed, the storapid must be restarted for the new value to take effect.
GK_USE	dedicated_only | legacy	legacy	If set to 'dedicated_only', then the base daemon restricts itself to using only dedicated gatekeepers. This ensures that data devices will never be used as gatekeepers without the need to maintain the gkselect or gkavoid file. Note, even though VCM/ACLX devices can be used as gatekeepers, they are not classified as dedicated gatekeepers. If set to 'legacy', then the base daemon may temporarily use a data device as a gatekeeper if the syscall load is high enough. This parameter can be changed and reloaded while the base daemon is running with the CLI stordaemon action storapid -cmd reload.
INQUIRY_TIMEOUT	A number of seconds.	900	Specifies how long priority inquiry results are to remain in memory before

-	-	-	expiring, and new data is retrieved from the host and arrays. A value of -1 indicates the data never expires. A value of zero indicates the data always expires.
BACKGROUND_AUDIT_LOG	disable | enable	disable	Writes audit log entries on behalf of the requesting client, and returns control back to the client while the entry is being written. An audit log entry is preserved on disk until the entry is written to the Symmetrix array. This ensures no entries are lost if the base daemon is stopped and restarted before all entries are written. This parameter can be changed and reloaded while the base daemon is running with the CLI stordaemon action storapid -cmd reload.
SINGLE_GK_POLICY	pool | close	close	If the base daemon only has one gatekeeper device, this parameter tells the base daemon what to do with the device after each use. A value of pool tells the daemon to put it back into the gatekeeper pool. A value of close tells the daemon to close the device after each use. This parameter is useful for device masking on hosts that only see a single device. Setting the value to close will slow down all commands, so it is recommended that this setting only be used for as short a time as possible. An example of how to use this setting follows: 1.

-	-	-	Edit the daemon_options file and add: storapid:SINGLE_GK_POLICY = close 2. Reload storapid settings with this command: stordaemon action storapid -cmd reload 3. Perform all the device masking commands. 4. Edit the daemon_options file and set: storapid:SINGLE_GK_POLICY = pool 5. Reload the storapid settings. 6. Verify that the setting is back to pool: stordaemon action storapid -cmd show_gk_stats
INTERNODE_LOCK_INFORMATION_EXPORT	disable | enable	enable	Makes SEL and DEL lock information available for all base daemon hosts attached to the Symmetrix array for use with available LockShow API functions. This parameter can be changed and reloaded while the base daemon is running with the CLI stordaemon action storapid -cmd reload.
PARALLEL_INQUIRY_SIZE	0, multiples of 2	6	Indicates the number of threads to use when the base daemon is performing an inquiry. The device list is scattered across those threads to potentially improve performance. Devices that fail to respond within the DEVICE_INQUIRY_TIMEOUT period get abandoned, with inquiry data not provided to the requesting API client for those devices. A value of zero keeps the inquiry process as a single list, executed on the calling client's thread. To help

-	-	-	determine the optimal setting for the host use the CLI stordaemon action storapid -cmd scan -inquiry This parameter can be changed and reloaded while the base daemon is running, with the CLI stordaemon action storapid -cmd reload. This value can be viewed and set with the CLI stordaemon getvar and setvar commands.
DEVICE_INQUIRY_TIMEOUT	A number of seconds, >= 1	60	Specifies how long a thread in the parallel inquiry feature waits for a device to respond before abandoning it during the inquiry process. The thread may or may not complete, depending on why the device is not responding. If a device is eventually made to respond by other means and the nature of the OS does not provide any indication to the waiting request in the base daemon, the base daemon will need to be restarted. This parameter can be changed and reloaded while the base daemon is running, with the CLI stordaemon action storapid -cmd reload. This value can be viewed and set with the CLI stordaemon getvar and setvar commands.
BACKGROUND_DISCOVERY	disable | enable	enable	Performs a discovery of select syscall data on behalf of the requesting API client that initiated a discovery. These syscalls are issued in parallel with other syscalls being issued by the client. Syscall

-	-	-	data is kept in cache until the client requests the data, and then is discarded. This parameter can be changed and reloaded while the base daemon is running, with the CLI stordaemon action storapid -cmd reload.

storgnsd [Group Name Service (GNS) Daemon] notes

This daemon supports GNS (Group Name Services) on a host.
Refer to the storgnsd(3) man page for additional details.

Parameter	Allowed Values	Default Value	Description
AUTORESTART	disable | enable	enable	Specifies whether to use a watchdog mechanism to monitor and automatically restart storgnsd if it crashes.
GNS_DEVICE_POLL_INTERVAL	A number, in seconds.	15	Specifies how frequently storgnsd polls for changes made to groups from other hosts. The smaller this value is, the more quickly a host will recognize group changes made from other hosts.
GNS_PPATH_POLL_INTERVAL	A number, in seconds.	60	Specifies how frequently storgnsd checks to see whether PowerPath or RDF consistency group updates are necessary.
GNS_REMOTE_MIRROR	enable | disable	disable	Specifies whether storgnsd should automatically mirror RDF1 and RDF2 type DG and CG group definitions for use on remote (SRDF) Symmetrix arrays. Limitations include: - REGULAR type and type ANY groups are not mirrored. - Mirrors will not be created if the remote Symmetrix array is also directly attached to the local host. (Refer to the GNS_SYMAVOID_LOCAL option below.) - For CG groups, BCV type devices are not mirrored.

GNS_RMTARR_UPDATE_IN TERVAL	A number, in seconds.	60	Specifies how frequently storgnsd should attempt to propagate device groups over to remote (SRDF) Symmetrix arrays. This is only relevant if the GNS_REMOTE_MIRROR option is set to enable.
GNS_DB_BACKUP	enable | disable	enable	Specifies whether storgnsd should automatically backup both GNS databases for local and global groups. These backups, maintained on the local disk, can be used to restore local GNS group repository and GNS group repository on a Symmetrix array. It is recommended that this option be enabled on a subset of hosts running GNS daemons. Because of its effect on performance, it usually does not make sense to have more than 2-4 GNS daemons managing these backup copies--assuming that they can see all the relevant Symmetrix arrays.
GNS_DB_BACKUP_INTERV AL	A number, in hours.	6	Specifies how frequently storgnsd should attempt to perform GNS databases backup for both local and global groups. This is only relevant if the GNS_DB_BACKUP option is set to enable.
GNS_SYMAVOID	A comma separated list of Symmetrix IDs. Note that	None	Specifies a set of Symmetrix arrays, both local and (SRDF) remote, that GNS should NOT manage. Specifically, GNS will not examine these arrays for

-	-	-	group definitions and will fail if an attempt is made to define a group containing devices on one of these. Also, mirroring of groups will not occur to remote Symmetrix arrays if they are in this list.
GNS_SYMINCLUDE	A comma separated list of Symmetrix IDs. Note that	None	Specifies the only Symmetrix arrays, both local and (SRDF) remote, that GNS will manage. Specifically, GNS will only examine these arrays (and no others) for group definitions and will only allow groups to be defined that contain devices on one of these. Also, mirroring of groups will not occur to remote Symmetrix arrays if they are NOT in this list. If both GNS_SYMAVOID and GNS_SYMINCLUDE are supplied, the effect is cumulative. Only arrays specified by GNS_SYMINCLUDE but not in GNS_SYMAVOID will be managed.
GNS_SYMAVOID_LOCAL	A comma separated list of Symmetrix IDs. Note that	None	Specifies a set of local Symmetrix arrays that GNS should NOT manage. Specifically, if an array listed here can be accessed both locally and remotely (by RDF) from this host, the local connection will be ignored. This can be used to enable automatic group mirroring where the remote array is also local to the host (refer to the gns_remote_mirror option above).

storevntd [Event Daemon] notes

This daemon acts as the clearing house for Solutions Enabler events on a host.
Refer to the storevntd(3) man page for additional details.

Parameter	Allowed Values	Default Value	Description
SYMM_POLL_INTERVAL	A number of seconds.	60	Specifies how often, in seconds, to poll Symmetrix arrays for events that need to be delivered.
EVENT_LISTEN_PORT	An IP port number.	0 - means to let the OS pick an unused port	Which IP port to use when listening for events being forwarded from remote hosts. By default, an unused port picked by the OS is used. This option might be needed if the presence of a firewall requires the use of a particular port.
LOG_EVENT_TARGETS	One or more (space separated) of: snmp, file, system,	none	Controls whether events should be automatically logged. One or more of the following may be supplied: file: Events are written to a file on disk. snmp: Events are mapped into SNMP traps. system: Events are written to the Event Log (Windows) or the local syslog service (otherwise). syslog: Events are sent directly to a remote syslog server, bypassing any local syslog service. Some of these possible targets can be configured by options that are described below.
LOG_SYMMETRIX_EVENTS	see above	No events are logged.	Specifies events that are to be automatically logged.

-	-	-	Refer to the LOG_EVENT_TARGETS option above. This option consists of a records separated by a semicolon. Typically, each record will be placed on a line by itself. Each of these records in turn consists of a number of comma-separated fields. [sid=nnnnn,] CAT[, ...] [,ignore] [,tgt=xxx] sid=nnnnnnnnnnn Specifies a Symmetrix ID. By default, all known Symmetrix arrays will be monitored. CAT Specifies the event(s) to be monitored. This can be either the name of an event category or a numerical event ID. This is the only field that is required. One or more values (comma separated) may be present. Supported categories are: status events array subsystem or array_subsystem checksum diagnostic environmental device pool or device_pool service processor or service_processor srdf system or srdf_system srdf link or srdf_link srdfa session or srdfa_session srdf consistency group or srdf_consistency_gro up director device disk comp=xxxx "comp=aaa,bbb,ccc" Certain events apply to specific sub-components within the array: a device (5 digit hexadecimal), disk, pool. This field specifies that only events for the specified component

-	-	-	(or components) should be delivered. If more than one component is being present, the entire field must be enclosed in double quotes. e.g., "comp=00004,00010,00100" thresh_fatal=nn thresh_critical=nn thresh_major=nn (equivalent to: thresh=nn) thresh_minor=nn thresh_warn=nn thresh_info=nn thresh_norm=nn Certain events correspond to numerical quantities of some sort. A threshold is associated with each severity level, and an event is generated at that severity when the event's value exceeds the associated threshold. These fields can be used to override the default threshold values controlling when an event is delivered. One example of this is the event that indicates the percentage of space used for a pool. These fields can be set to control when events are to be generated. e.g.: thresh_critical=96, thresh_major=80, thresh_warn=60 ignore If present, these event(s) are not to be logged, even if they are matched by a different record. tgt=xxx If present, these event(s) are sent to only the specified target, which must be one of: snmp, file, system, syslog. The specified target must also be present in the default target list

-	-	-	given by the LOG_EVENT_TARGETS option. sid=222222222222, device pool ;\ sid=333333333333, device, "comp=00101,00102,00104" ;\ array subsystem , tgt=file
LOG_EVENT_FILE_NAME	file name, without any suffix	events	For event logging to a file (LOG_EVENT_TARGETS contains 'file'): the base name of the file that is used. This file is created within the standard log file directory. UNIX: /var/symapi/log Windows: c:\\Program Files\\EMC\\SYMAPI\\log Depending on the setting of the LOG_EVENT_FILE_TYPE option, a suffix will added to this name.
LOG_EVENT_FILE_TYPE	dated | wrap	wrap	For event logging to a file (LOG_EVENT_TARGETS contains 'file'): the type of log file that is used. Two styles are supported: wrap: Two log files are maintained: ????.log0 and and ????.log1. Logging alternates between these--switching to the other file each time the maximum size specified by the LOG_EVENT_FILE_SIZE parameter is reached. The modified time on the files can be used to determine the "current" one. dated: A separate log file is used for each day: ????-YYYYMMDD.log. (e.g., events-200411) There are no limits on how large these files can become.

LOG_EVENT_FILE_SIZE	A number, in KB	1000 (1000-KB)	Specifies how large each log file is allowed to grow before wrapping to the alternate file. This option is for event logging to a file (LOG_EVENT_TARGETS contains 'file'): if LOG_EVENT_FILE_TYPE == wrap.
LOG_EVENT_FILE_RETENTION	A number of days, greater than zero.	3 (days)	Indicates that log files more than this many days old will be automatically deleted. This option is for event logging to a file (LOG_EVENT_TARGETS contains 'file'): if LOG_EVENT_FILE_TYPE == dated.
LOG_EVENT_FILE_PERMS	n | rw | r	r	Specifies the permissions to be applied to new log files that are created. This option is for event logging to a file (LOG_EVENT_TARGETS contains 'file'). Possible values: rw: Anyone can read or write. r: The owner (root) can read/write, others can read. n: The owner can read/write, no one else can access.
LOG_EVENT_SYSLOG_HOST	A host name or IP address.	None - a value is required.	Specifies the host on which the remote syslog server is running. This option is for event logging to syslog (LOG_EVENT_TARGETS contains 'syslog').
LOG_EVENT_SYSLOG_PORT	A decimal port number.	514	Specifies the port to which the remote syslog server is listening. This option is for event logging to syslog

-	-	-	(LOG_EVENT_TARGETS contains 'syslog').
SNMP_TRAP_CLIENT_REGISTRATION	IP,port,filter,state,snmp version- with no spaces	none	Provides a list of target IPs and ports to send SNMP traps to, when the LOG_EVENT_TARGETS option specifies snmp. Format is: IP,port,filter,state,snmp version where filter represents the trap sending filtering levels as defined in the fcmgmt MIB, and state represents the start up row state in the trap_client_registration table in the fcmgmt MIB. snmp version is either v1 or v3. If none specified v1 will be considered as default. Multiple entries should be on their own line, delineated with a backslash (\) character on the preceding line. This parameter can be changed and reloaded while the event daemon is running, with the CLI stordaemon action storevntd -cmd reload. 55.66.77.88,162,10,ACTIVE,v3 \ 11.33.55.77,162,10,ACTIVE
SNMP_MANAGEMENT_URL	<ip|DNS host name><:port>	none	Provides launch capability of a management application, such as SMC or ECC, from within a third party management framework. This parameter can be changed and reloaded while the event daemon is running, with the CLI stordaemon action storevntd -cmd reload.

TEST_MODE	disable | enable	disable	Provides the capability to run storevntd in test mode. In this mode, faults can be artificially injected to generate events without stressing the Symmetrix.

storstpd daemon level options

These options control how the daemon behaves.

Parameter	Allowed Values	Default Value	Description
DMN_RUN_SPA	disable | enable	enable	Determines if storstpd starts the necessary components for supporting the SPA application. Note: SPA support is not available on the service processor and the setting of this option is ignored.
DMN_RUN_RTC	disable | enable	enable	Determines if storstpd starts the necessary components for real-time metrics collection.
DMN_ROOT_LOCATION	Full file path specification.	UNIX: /var/symapi/stp	Specifies a full file/path location that storstpd will use as its root directory for file creation. Note: This location MUST exist or will revert to default location. Note: If storstpd runs as a privileged user, the directory named here must also be specified in the storstpd:secure_directory_path option. If storstpd is not run as a privileged user, there is no harm in specifying it in the secure_directory_path, but it is not validated when running as a non-privileged user. Note: if storstpd runs as non-root in Unix (stordaemon setuser is used to change ownership) and this option is changed, the stordaemon setuser command must be reissued to set permissions properly after this change.

DMN_SYMMIDS	A comma-separated list of Symmetrix IDs; shortcuts	None (all SymmIDs)	Specifies a set of Symmetrix arrays for which the storstpd should collect statistics. If not present, storstpd will collect statistics for all local SymmIDs found.
DMN_REMOTE_MODE	enable | disable	disable	Indicates that storstpd will establish a remote connection, instead of operating locally. A valid DMN_REMOTE_ADDRESS and DMN_REMOTE_PORT must be provided. This setting is ignored when running on a service processor.
DMN_REMOTE_ADDRESS	A valid IP address.	None	Specifies a valid IP address used to establish a remote SYMAPI connection. This setting is ignored if DMN_REMOTE_MODE is disabled.
DMN_REMOTE_PORT	A valid IP port.	2707	Specifies a valid IP port used to establish a remote SYMAPI connection. This setting is ignored if DMN_REMOTE_MODE is disabled.
DMN_RETENTION_DAYS	0-365	30	Sets the current retention policy, in days.
DMN_DISK_SPACE_THRESHOLD	1 - 99	80	Tells storstpd not to allow disk space consumption to rise above the threshold. Must be an integer from 1 (percent) to 99 (percent).

Environment Variables

SYMCLI provides environment variables that can be preset to streamline a command line session. These variables can be set to common argument values for a series of associated commands, which eliminates repeated key strokes for during command line execution.

To view a list of environment variables that can be set for a given SYMCLI session, enter:

```
symcli -env
```

To view the environment variables currently set, enter:

```
symcli -def
```

To The following table provides the description and default values for each option.

Variable Name	Description
SYMCLI_ACCESS_PIN	Must specify your Access PIN when using either the COMMIT, PREPARE, RELEASE keywords of SYMACL. If this value is not set then you are prompted for the Access PIN.
SYMCLI_ASM_CONNECT	When mapping Oracle ASM, this must be set to connection information for the ASM instance. Format: user/passwd@service
SYMCLI_BCV_DELAY	Specifies the delay in seconds between establish operations when a SYMCLI_BCV_EST_TYPE of SINGULAR or PARALLEL is specified. Delay value can range from 0 to 30 seconds. The default value is 0.
SYMCLI_BCV_EST_TYPE	Specifies the BCV pair establish type, that SYMAPI uses when establishing a BCV pair. Can be set to SINGULAR, SERIAL, or PARALLEL.
SYMCLI_BCV_PAIR_POLI CY	Specifies the BCV pair cancel policy, that SYMCLI uses when establishing a new BCV pair and the maximum number of BCV pairs has been reached. Can be set to CANCEL_OLDEST, CANCEL_NEWEST or DONT_CANCEL. The default is to automatically cancel the oldest BCV pair.
SYMCLI_CG	Can be set as the default composite group name.
SYMCLI_CLONE_COPY_MO DE	Specifies the mode in which Clone sessions are created. Can be set to NOCOPY_NODIFF, COPY_NODIFF, PRECOPY_NODIFF, COPY_DIFF, PRECOPY_DIFF, or VSE_NODIFF. SYMCLI_CLONE_COPY_ON _WRITE: Changes clone nocopy mode when clone session is activated. ENABLED sets clone nocopy mode to copy_on_write. DISABLED sets clone nocopy mode to copy_on_access.
SYMCLI_CLONE_EMULATI ON	Specifies whether TimeFinder commands should be mapped to Clone commands by default. Can be set to ENABLED or DISABLED. The default is DISABLED. For Enginuity 5874 and above all TimeFinder uses Clone Emulation and this setting is ignored.
SYMCLI_CLONE_LARGER_ TGT	Can be set to ENABLED to allow the creation of Clone sessions where the target device is larger than the source device. The default is to block this type of operation.
SYMCLI_CLONE_PAIR_PO LICY	Specifies the CLONE terminate policy, that SYMCLI uses when establishing a new clone and the maximum number of clones has been reached. Can be set to TERM_OLDEST or DONT_TERM. The default is

-	DONT_TERM which causes the clone operations to fail.
SYMCLI_COMMAND_SCOPE	Sets the scope of the device selection process. ENABLED limits the operation to the devices within the scope of the command. DISABLED performs the operation on the devices within the scope of the command plus any additional devices associated by session and/or state. The default is DISABLED.
SYMCLI_CONNECT	Can be set to specify the SYMAPI Server connection information.
SYMCLI_CONNECT_TYPE	Can be set to specify the SYMAPI Server connection type. Valid values are: LOCAL, REMOTE, and REMOTE_CACHED. The default is LOCAL when SYMCLI_CONNECT is not set. When it is set, the default is REMOTE.
SYMCLI_CTL_ACCESS	Can be set to either EXCLUSIVE or PARALLEL to specify how to obtain a lock on the symapi database file before starting a Symmetrix control operation. The default is to obtain an EXCLUSIVE lock.
SYMCLI_DB_FILE	Can be set to the host's Symmetrix database pathname.
SYMCLI_DG	Can be set as the default device group name.
SYMCLI_FILE	Can be set to the default file for various operations.
SYMCLI_FULL_NAME	Can be set to 1 to preserve the complete name.
SYMCLI_FULL_PDEVNAME	Can be set to 1 to preserve the complete pdevname.
SYMCLI_GENERATOR_FIL E	Specifies a file to which to write a log of all active commands (BCV, SRDF, and Snap).
SYMCLI_GROUP_DB	Can be set to the host's group database pathname for offline access. Used in conjunction with SYMCLI_OFFLINE.
SYMCLI_INQ_DATA_CACH ED	Can be set to 1 to use the cached inquiry data.
SYMCLI_LOCKID	Specifies the lock holder ID for commands that require the lock ID.

SYMCLI_LOG	Can be set to specify a non-default logging pathname.
SYMCLI_MANPATH	Can be set to specify a non-default location for the symcli online help files. If set, it should be a complete specification of the path to the directory/folder containing those files.
SYMCLI_MAP_PRESERVE_ CASE	Can be set to 1 to preserve case of output fields on Windows.
SYMCLI_MAX_BCV_PAIRS	By default, SYMCLI allows up to 8 BCV pairs to be incrementally paired with a standard device. This variable can be used to set a new maximum (other than 8) in the range 1 to 16.
SYMCLI_META_HEAD_CON TROL	Changes operations on Meta devices to support Mixed Meta. ENABLED sets operations on Meta devices to control by Meta Heads only. DISABLED performs operations on Meta Heads and all Meta Members.
SYMCLI_MODE	Can be set to specify compatibility mode for SYMCLI utilities output. Valid modes include 'V84', 'V90', 'V91', 'V92'
SYMCLI_MULTI_VIRTUAL _SNAP	Specifies if more than 16 Snap sessions can be created on the same SRC device. Can be set to ENABLED or DISABLED. The default is DISABLED.
SYMCLI_NOLOGGING	Can be set to 1 to disable logging.
SYMCLI_NOPROMPT	Can be set to 1 to disable prompting.
SYMCLI_OFFLINE	Can be set to 1 for offline access.
SYMCLI_OSM_VERSION	Can be set as the version reported by the SYMCLI OSM SRDF Compatibility Mode.
SYMCLI_OUTPUT_MODE	Can be set to specify output mode for SYMCLI utilities output. Valid modes are limited to 'Standard', 'XML', and 'XML_ELEMENT'.
SYMCLI_PAGINATE	Can be set to FALSE to force inhibit of <press any key> message.
SYMCLI_PDEV_FILE	Can be set to specify the pathname for a pdev definitions file. These definitions replace the pdev's previously discovered or defined.

SYMCLI_RCOPY_COPY_MO DE	Specifies the mode in which Rcopy sessions are created. Can be set to COPY_DIFF, NOCOPY_DIFF, COPY_NODIFF, NOCOPY_NODIFF, PRECOPY_DIFF or PRECOPY_NODIFF.
SYMCLI_RDB_CONNECT	Can be set as the default relational database Connection information. Format: user/passwd@service
SYMCLI_RDB_NAME	Can be set as the default relational database name.
SYMCLI_RDB_TYPE	Can be set as the default relational database type.
SYMCLI_RDFG_CONSISTE NCY	Can be set to ENABLED to provide consistency verification on the RDF Group level when performing operations using consistency technology.
SYMCLI_REMOVE_SYMS	Can be set to 1 to cause discover to remove from the symapi database, any Symmetrix record and its dependent devices and device groups, when the symm is no longer reachable.
SYMCLI_RETURN_MODE	Prints return code mnemonics and error strings concluding the execution of SYMCLI commands. Possible values are DEFAULT and FORMATTED. DEFAULT is equivalent to an unset SYMCLI_RETURN_MODE FORMATTED enables the printing of return code mnemonics and error strings.
SYMCLI_SCHEMA_NAME	Can be set as the default relational database schema name.
SYMCLI_SG	Can be set as the default storage group name.
SYMCLI_SID	Can be set as the default Symmetrix ID.
SYMCLI_SKIP_ON_FAILU RE	Can be set to 1 to cause discover to skip loading devices from any Symmetrix units when errors are encountered when loading from those units.
SYMCLI_SNAPVX_LARGER _TGT	Can be set to DISABLED to block linking snapshots to the target device that is larger than the source device. The default is to allow this type of operation.
SYMCLI_SNAP_PAIR_POL ICY	Specifies the SNAP terminate policy, that SYMCLI uses when establishing a new snap and the maximum number of snaps has been reached. Can be set to

-	TERM_OLDEST or DONT_TERM. The default is DONT_TERM which causes the snap operations to fail.
SYMCLI_SVP	Can be set as the default savedev pool name.
SYMCLI_TBS_NAME	Can be set as the default relational database table space name.
SYMCLI_UPPERCASE	Can be set to 1 to convert names that the user enters to uppercase.
SYMCLI_VERBOSE	Can be set to 1 to enable verbose mode for SRDF, BCV, Snap, and Clone control operations as well as for all symbcv, symcg, symdg, and symsg operations.
SYMCLI_VG	Can be set as the default logical volume group name.
SYMCLI_WAIT_ON_DB	Can be set to 1 to cause the SYMCLI to block when it needs to exclusively lock the symapi database file. The process will wait until the lock on the database file can be obtained. The default is to NOT wait on the lock.
SYMCLI_WAIT_ON_GK	Can be set to 1 to cause the SYMCLI to block when getting information from the Symmetrix if all GateKeeper devices are currently busy. The process will wait until a GateKeeper becomes available.
SYMCLI_XML_SCHEMA	Can be set to specify a URL to the XML Schema document describing the output of SYMCLI in XML mode. It is advised that the schema is placed in a public location and this variable set to point to it. Without this variable set, no mention of a schema will occur. Note that this setting does nothing in non-XML mode.